



AURAK

الجامعة الأمريكية في رأس الخيمة
AMERICAN UNIVERSITY OF RAS AL KHAIMAH

03

LIFE-CHANGING EDUCATION WITH A GLOBAL REACH



AURAK CATALOG

ACADEMIC YEAR 2022-2023

Message from the President

Dear AURAK Students,

It is my pleasure to welcome you to AURAK.

At the American University of Ras Al Khaimah (AURAK), we are very proud to be a public, non-profit institution of higher education. This means that we care more about our students and their learning, growth, and success than we do about profit.

Our mission is to give you, our students, all that you require to succeed in your professional lives. As a liberal arts university, however, we aim to give you far more than merely knowledge and learning in your respective fields of study. We guarantee to enrich your lives by broadening your horizons with a well-rounded education. To accomplish this, our students are required to supplement their curriculum with general education courses outside of their chosen majors. These courses are specifically designed to produce global citizens who are successful, progressive, innovative, and will passionately and ethically contribute to the betterment of their communities and countries. We want to produce citizens who will leave legacies to their children and for all future generations to come, especially those in the Middle East.

In our goal to give our students all this and more, AURAK depends upon a highly qualified and committed support staff and faculty from around the world who know how to best serve our students.

We are delighted with your interest in AURAK. Should you need any further information, please browse our website, or feel free to contact us by e-mail, phone, or visit our campus.

We truly want you to achieve your goals and dreams, and we are committed to doing our very best to facilitate your success in achieving those goals and dreams.

Sincerely,

Prof. Hassan Hamdan Al Alkim

President, AURAK

AMERICAN UNIVERSITY OF RAS AL KHAIMAH

ACADEMIC CATALOG

2022-2023

Effective September 1, 2022

Board of Trustees

The Board of Trustees is the governing board for American University of Ras Al Khamiah. Members of the Board are:

H.E. Mr. Mohammed Omran Al Shamsi – Chairman

Mr. Salem Ali Al Sharhan

Dr. Naceur Jabnoun

Dr. Ateeq Abdul Aziz Jaka Al Mansouri

Prof. Winfred Thompson

Dr. Rashed Iqbal

Mr. Jean Diab

Dr. Pablo Tetter

Professor Hassan Hamdan Al Alkim – Ex-officio Member

University Administrative Officers

Professor Hassan Hamdan Al Alkim – President

Professor Stephen Wilhite – Senior Vice President for Academic Affairs
and Student Success/Provost

Mr. Basheer Daoud – Vice President of Operational Excellence &
Financial Management

Dr. Ayman Al Khanbouli – Chief Advancement Officer

Mr. Mohammed Hettini – Chief Marketing and Communications Officer

Ms. Amanda Fiona Forte – Chief Strategy and Excellence Officer

Professor Hariclea Zengos – Dean School of Arts & Sciences

Professor Joseph Wallis – Dean School of Business

Prof. Khalid Hussain - Dean School of Engineering

Dr. Mohammed Al Zarooni – Associate Provost for Research &
Community Service

Mr. Kevin Scott Konecny – Associate Provost of Student Affairs

Mr. Mohamed Darwish – Director of the Saqr Library

Ms. Rania Hafiz Hussein Ali – University Registrar

Table of Contents

ACADEMIC CALENDAR 2022 – 2023	1
ABOUT AMERICAN UNIVERSITY OF RAS AL KHAIMAH	4
AURAK ORGANIZATIONAL CHART.....	7
AURAK CAMPUS MAP	8
ACADEMIC PARTNER INSTITUTIONS.....	9
STUDENT RESOURCES AND SUPPORT SERVICES.....	11
COST OF ATTENDANCE – TUITION & FEES	23
SCHOLARSHIPS AND FINANCIAL AID	29
STUDENT RIGHTS AND RESPONSIBILITIES	31
STUDENT ATTENDANCE POLICY.....	39
ACADEMIC INTEGRITY	40
STUDENT RECORDS	45
UNDERGRADUATE ADMISSIONS.....	47
UNDERGRADUATE ACADEMIC POLICIES	55
ACADEMIC PROGRAMS.....	72
GENERAL EDUCATION COMPONENT	74
SCHOOL OF ARTS AND SCIENCES	81
BACHELOR OF ARTS IN MASS COMMUNICATION	81
BACHELOR OF SCIENCE IN BIOTECHNOLOGY.....	89
SCHOOL OF BUSINESS	99
BACHELOR OF SCIENCE IN BUSINESS ADMINISTRATION	99
SCHOOL OF ENGINEERING.....	115
BACHELOR OF ARCHITECTURE.....	115
BACHELOR OF ARTS IN INTERIOR DESIGN	120
BACHELOR OF SCIENCE IN ARTIFICIAL INTELLIGENCE	125
BACHELOR OF SCIENCE IN CHEMICAL ENGINEERING	131
BACHELOR OF SCIENCE IN CIVIL AND INFRASTRUCTURE ENGINEERING	137
BACHELOR OF SCIENCE IN COMPUTER ENGINEERING.....	143
BACHELOR OF SCIENCE IN COMPUTER SCIENCE	149
BACHELOR OF SCIENCE IN ELECTRICAL AND ELECTRONICS ENGINEERING	156
BACHELOR OF SCIENCE IN MECHANICAL ENGINEERING	161
BACHELOR OF SCIENCE IN PETROLEUM ENGINEERING.....	166
ADMISSION TO GRADUATE STUDIES	174
GRADUATE ACADEMIC POLICIES	180
ACADEMIC PROGRAMS.....	190
EXECUTIVE MASTER OF BUSINESS ADMINISTRATION	190
MASTER OF BUSINESS ADMINISTRATION	195
MASTER OF EDUCATION: EDUCATIONAL LEADERSHIP & POST GRADUATE DIPLOMA IN EDUCATION	199
MASTER OF SCIENCE IN ENGINEERING PROJECT MANAGEMENT	204

MASTER OF SCIENCE IN SUSTAINABLE AND RENEWABLE ENERGY	207
FACULTY LIST.....	212
COURSE DESCRIPTIONS	218

ACADEMIC CALENDAR 2022 – 2023

Fall 2022			
Month	Day	Week	Activity/Event
August	1-28		Placement Tests/Continuing Student Registration
August	1-28		New Student Course Registration
August	8		Joining Date for New Faculty
August	22		Returning Date for Continuing Faculty
August	22-26		Faculty Professional Development Workshops, Including New Faculty Orientation
August	22-28		Continuing Advising & Registration for Returning Students
August	25		New Student Orientation
August	29		First Day of Class
August	29-Sept 4	Week 1	
September	5-11	Week 2	
September	9		Graduation Date for Summer 2022
September	11		AURAK Admissions Application Deadline
September	11		Last Day to Add/Drop
September	12-18	Week 3	
September	19-25	Week 4	
September	25		Census Date for Fall
September	26-Oct 2	Week 5	
October	3-9	Week 6	
October	9		Prophet's Birthday*
October	10-16	Week 7	
October	17-23	Week 8	
October	23		Mid-Term Grades Report Due
October	24-30	Week 9	
October	30		Last Date to Withdraw a Course with a Grade of "W"
October	31-Nov 6	Week 10	
October	31		Advisement Begins for Spring 2023
November	7-13	Week 11	
November	7		Early Registration for Spring Semester Begin(s)
November	10		Commencement Ceremony (2021-2022)
November	14-20	Week 12	
November	21-27	Week 13	
November	28-Dec 4	Week 14	
December	1		UAE Commemoration Day
December	2-3		UAE National Day
December	5-11	Week 15	
December	11		Last Day of Classes
December	12-20	Week 16	Final Examination Period

December	23		Grades Available
December	21-Jan 8		Winter Break
January	25		Graduation Date for Fall 2021
Spring 2023			
January	2-8		Continuing Advising & Registration for Returning Students
January	2		Returning Date for Continuing Faculty and Joining Date for New Faculty
January	2-6		Online Faculty Professional Development Workshops, Including New Faculty Orientation
January	2		Placement Tests Begin
January	6		New Student Orientation & New Student Registration Begins
January	9		First Day of Class
January	9-15	Week 1	
January	16-22	Week 2	
January	22		AURAK Admissions Application Deadline
January	22		Last Day to Add/Drop
January	23-29	Week 3	
January	30-Feb 5	Week 4	
January	5		Census Date for Spring
February	6-12	Week 5	
February	13-19	Week 6	
February	20-26	Week 7	
February	27-Mar 5	Week 8	
March	5		Mid Term Grades Report Due
March	6-12	Week 9	
March	12		Last Date to Withdraw from a Course with a Grade of "W"
March	13-19	Week 10	
March	20-26	Week 11	
March	27-April 9		Spring Break
April	10		Classes Resume
April	10-16	Week 12	
April	17-23	Week 13	
April	17		Fall Advisement Begins
April	21-23		Eid Al Fitr*
April	24-30	Week 14	
April	24		Summer Registration Begins – Early Registration for Fall Semester Begins
April	24-May 7		Student Course Evaluations
May	1-7	Week 15	
May	7		Last Day of Classes
May	8-17		Last Day of Classes
May	20		Grades and Student Course Evaluation Results Available

June	25	Graduation Date for Spring 2023
June	26	Faculty Summer Break Begins
Summer Session 2023 (6 weeks)		
May	22-29	Course Registration
May	29	Classes Begin
June	4	Last Day to Add/Drop
June	7	Census Date
June	25	Last Date to Withdraw from a Course with a Grade of "W"
June	27-30	Eid Al Adha*
July	16	Last Day of Summer Session
July	18	Islamic New Year*
July	19	Grades Available
September	9	Graduation Date for Summer 2022
Internship Session 2023 (8 weeks)		
May	22-29	Internship Course Registration
May	29	Internship Begins
June	4	Last Day to Add/Drop
June	7	Census Date
June	27-30	Eid Al Adha*
July	9	Last Date to Withdraw from a Course with a Grade of "W"
July	30	Internship Ends
August	2	Grades Available
September	9	Graduation Date for Summer 2023
2022 - 2023 Commencement		
November	23	Commencement Ceremony (2022-2023)

Note: All Islamic Lunar Holidays will be fixed per announcement by the relevant Ministry.

ABOUT AMERICAN UNIVERSITY OF RAS AL KHAIMAH

Licensure and Accreditation:

The American University of Ras Al Khaimah (AURAK), located in the Emirate of Ras Al Khaimah, is officially licensed from 1 August 2009 to 15 September 2026 by the Ministry of Education of the United Arab Emirates to award degrees/qualifications in higher education.

AURAK is accredited by the Southern Association of Colleges and Schools Commission on Colleges (SACSCOC) to award baccalaureate and masters, degrees. Questions about the accreditation of AURAK may be directed in writing to the Southern Association of Colleges and Schools Commission on Colleges at 1866 Southern Lane, Decatur, GA 30033-4097, by calling (404) 679-4500, or by using information available on SACSCOC's website (www.sacscoc.org).

AURAK has earned accreditation through the Commission for Academic Accreditation of the Ministry of Education for all programs offered. In addition, the School of Engineering has the following program accreditations:

BS in Civil and Infrastructure Engineering is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>

BS in Computer Engineering is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>

BS in Computer Sciences is accredited by the Computing Accreditation Commission of ABET, <http://www.abet.org>

BS in Electronics and Communications Engineering is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>

BS in Industrial Engineering is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>

BS in Mechanical Engineering is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>

BS in Chemical Engineering is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>

BS in Electrical Engineering is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>

BS in Petroleum Engineering is accredited by the Engineering Accreditation Commission of ABET, <http://www.abet.org>

AURAK is a member of AACSB International—The Association to Advance Collegiate Schools of Business.

Location and Contact Information:

AURAK maintains an extensive campus located on American University of Ras Al Khaimah Road in the northern Emirate of Ras Al Khaimah. The university may be contact by mail at PO Box 10021; Ras Al Khaimah, UAE or by phone at +971 7 2210 900.

History:

With the support of His Highness Sheikh Saud bin Saqr Al Qasimi, Crown Prince of Ras Al Khaimah, the university was initially established as George Mason University, Ras Al Khaimah campus (GMU-RAK) in 2005. After George Mason University decided to discontinue its operations in Ras Al Khaimah in 2009, the Crown Prince decreed the establishment of the American University of Ras al Khaimah (AURAK) in the same location. Students who had enrolled in GMU-RAK were given the opportunity to continue at AURAK, transfer to GMU's main campus in Virginia, USA, or transfer to another institution.

Following the short-term presidencies of Professor Sharon Siverts (2009-10) and Dr. Shaukat Mirza (2010-11), AURAK's Board of Trustees appointed Professor Hassan Hamdan Al Alkim as Vice Chancellor in August 2011. Subsequent changes in the organizational structure saw Professor Al Alkim named as President in 2014.

Under Professor Al Alkim's leadership, the university has continued to develop new academic programs, attract new faculty and staff, and significantly increase its student enrollment. This rapid change has required an adaptable organizational structure to accommodate the effective and efficient delivery of programs and services. Consequently, in 2013, Vice President positions were added resulting in the organization of units in three divisions under the President, Senior Vice President Academic Affairs and Student Success and Vice President of Operational Excellence and Financial Management.

Today, AURAK's strategy focuses on cultivating excellence within the existing curricular offerings while improving policies and procedures across the institution. Student enrollment is projected to grow to 2,000 students by the end of 2024. It is also expected that, with an increasingly strong global economy, AURAK will capture the attention of educators, economists, business entrepreneurs, and other professionals who want to take advantage of the programs and services the institution offers, meeting national and regional workforce needs and earning the university regional and global recognition.

Vision:

Through inspired teaching, research, creative work, and community engagement, AURAK sets a new standard in the Gulf Region for student-centered excellence that empowers graduates to change the world.

Mission:

To be a leading institution in the Gulf Region, the American University of Ras Al Khaimah:

- Provides a transformational, student-centered learning experience based on the North American model of higher education enriched by attention to the culture of the region.
- Engages students, faculty, and staff in innovative undergraduate and graduate programs and high-impact research.
- Prepares future leaders and entrepreneurs through community outreach and creative initiatives involving local, regional, and global partners.

The American University of Ras Al Khaimah pursues this mission as an independent, public, state-owned, non-profit, coeducational institution.

Strategic Goals:

The following strategic goals guide all aspects of AURAK's operation:

- Implement exemplary governance and organizational processes that ensure student excellence;
- Attract talented students from around the world through a student-centered culture of success;
- Offer leading-edge academic programs and community engagement initiatives that meet local, regional, and international needs and generate high-impact research;
- Enhance faculty and staff excellence through improved recruitment, retention, and professional development; AND
- Provide stellar support services that contribute to the institution's excellence and maximize benefits to university stakeholders.

Values:

The following values guide all aspects of AURAK's operation:

Integrity:

To model adherence to a code of honesty, transparency, and accountability.

Excellence and Continuous Improvement:

To set high expectations for every unit and every constituent of the university.

Collaborative Engagement:

To ensure continual involvement of students, faculty, and staff in activities that contribute to the success of the institution and to the solution of local, regional, and global problems.

Diversity, Inclusiveness, and Cultural Authenticity:

To facilitate, through respectful dialogue, social and multicultural understanding that contributes to the success of all.

Scholarship and Research:

To advance knowledge through discovery, dissemination, application, and community engagement.

Empowerment:

To celebrate the power of inspired teaching to ignite transformational learning that shapes students into agents of positive change.

Academic Freedom:

To cultivate an environment of critical thought and enquiry, tolerance for divergent views and beliefs, and an open and respectful exchange of ideas.

Innovation and Entrepreneurship:

To encourage creative, enterprising ideas and the translation of those ideas into sustainable initiatives.

Global Interdependence:

To promote engagement with the international community in ways that encourage constituents of the university to see themselves as citizens of the world.

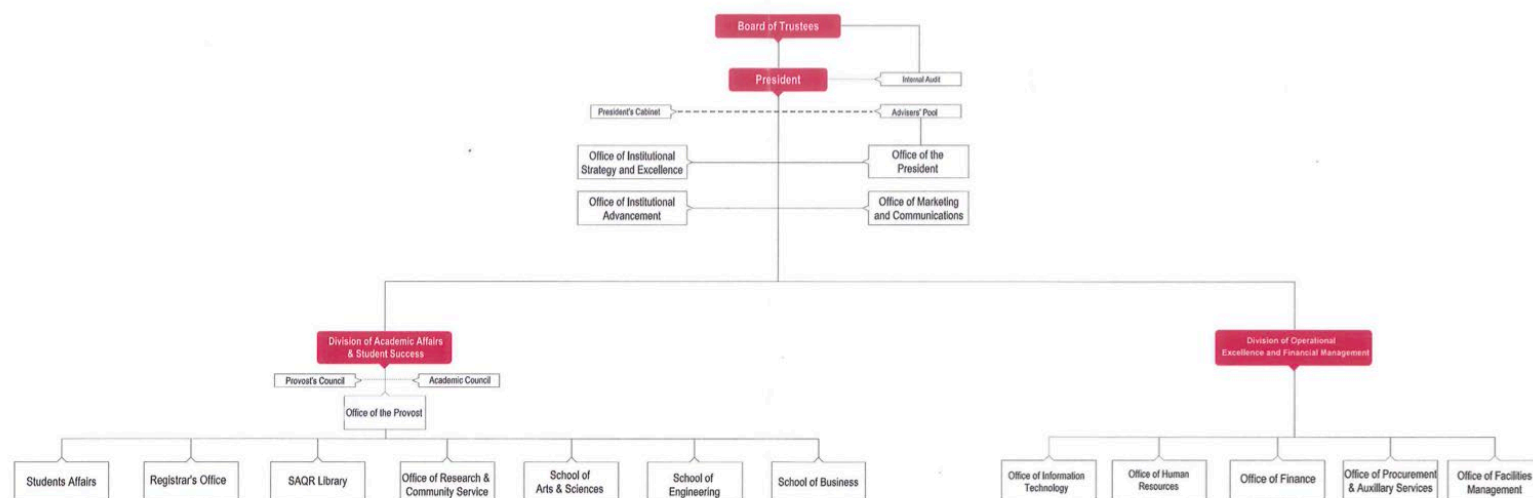
Sustainability:

To nurture an appreciation of the shared responsibility of humankind to preserve planet Earth for future generations.

AURAK Organizational Chart



AURAK ORGANIZATIONAL CHART



Board Approval


 H.E. Mr. Mohammed Omran Al Shamsi
 Chairman of the Board of Trustees

AURAK CAMPUS MAP



AURAK

الجامعة الأمريكية في رأس الخيمة
AMERICAN UNIVERSITY OF RAS AL KHAIMAH

Campus Map



- | | | |
|----------------|------------------------------|---------------------|
| 1 Building A | 8 Building G | 15 Store |
| 2 Building B | 9 Building H | 16 Workshop |
| 3 Building C | 10 Building J (Saqr Library) | 17 Basketball Court |
| 4 Building D | 11 Building K | 18 Tennis Court |
| 5 Building E | 12 Dormitories | 19 Football Pitch |
| 6 Building F | 13 Mosque | 20 Parking |
| 7 Building F 2 | 14 Gymnasium | |

ACADEMIC PARTNER INSTITUTIONS

Al Farabi-Kazakh National University, Republic of Kazakhstan

Appalachian State University, USA

Aspen Institute (Stevens Initiatives), USA

Coastal Carolina University, USA

Coventry University, UK

Hong Kong University of Science and Technology, Hongkong

Incheon National University – INU, South Korea

Knowledge Exchange Institute, USA

Management and Science University, Malaysia

Narxoz University, Republic of Kazakhstan

RAK Medical & Health Sciences University, RAK, UAE

Sudan University of Science & Technology (SUST), Sudan

Sungkyunkwan University Business School, South Korea

Temple University, USA

University of Texas Arlington USA

University at Albany, State University of New York, USA

University of Barcelona, Spain

University of Nebraska at Kearney, USA

Universiti Teknologi Petronas, Malaysia

University of Wyoming, USA

Wayne State University (WSU), USA

Windesheim University of Applied Sciences, Netherlands



STUDENT RESOURCES AND SUPPORT SERVICES

STUDENT RESOURCES AND SUPPORT SERVICES

AURAK provides a range of student resources and support services. The following narratives outlines six major areas of delivery: 1) university student resources and services, 2) the Saqr Library, 3) the Office of Student Affairs, 4) the Office of the Registrar 5) the Office of Research and Community Service, and 6) the university bookstore.

UNIVERSITY STUDENT RESOURCES AND SERVICES

Computer Facilities and Resources

AURAK provides computing resources to support the needs of the students and faculty with their academic work. This includes Wi-Fi coverage throughout the campus, audio/video system in classrooms/laboratories equipped with overhead projectors or interactive touch panels, and a dedicated email account for each student, faculty, and staff member.

Laboratories

- ***Engineering Laboratories***

AURAK has well equipped labs aligned with international standards for all of their programs. The electronic labs focus on electronics, communications, signal processing, digital systems, microprocessor, and microcontrollers. The laboratories are fully equipped with standard and specialized instrumentation including: dual trace digital storage oscilloscopes, dual trace cathode ray oscilloscope, frequency analyzers, digital sweep function generators, general function generators, digital multi meters, DC power supplies, electrical and electronic circuits trainer, digital circuits trainers, communication circuits trainers, different types breadboard modules, and temperature-controlled soldering stations.

- ***Science Laboratories***

To support the laboratory-based modules of the Biotechnology Program, there are dedicated biotechnology, chemistry, and physics laboratories. In these labs, student learning is facilitated through practical exercises. The Julphar Biotechnology Laboratory, housing cell culture and molecular biology facilities, has been functional since September 2012. The biotechnology and chemistry laboratories are fully equipped with standard and specialized instrumentation, including: digital balances, centrifuges, pH-meters, spectrophotometers, spectrofluorometer, upright and inverted microscopes, laminar flow cell culture cabinets, an ultralow freezer, PCR thermocycler, gas chromatography, rota evaporator, refractometer, incubators, shaking platforms, UV transilluminator, nucleic acid and protein electrophoresis equipment, and protein blotting apparatus, etc.

- ***Mass Communication Laboratory***

The Al Ain Media Lab and Studio at the American University of Ras Al-Khaimah aims to provide its expanding student body with access to advanced technological resources necessary for the implementation of media and communication-related projects. The lab accommodates up to sixteen Macintosh desktop computers with software applications, such as Final-Cut Pro for video editing and Photoshop for image manipulation. The studio includes high-definition digital video cameras and professional lighting kits with sound-mixing capabilities.

Mosque (Masjid)

The mosque, located adjacent to the campus beside Building D and the residence halls, is open for prayers at all times. The mosque is easily accessible from the campus. There are separate male and female prayer rooms within the mosque.

Parking

Parking services are offered to AURAK members on campus. Students will find information on “AURAK Parking Rules and Regulations” displayed on signs throughout the campus. Students must park only in the student designated parking, and they must display the appropriate parking permit on their vehicle. Permits can be obtained from the Office of Human Resources and Administrative Services.

Students are expected to drive carefully and follow parking rules on campus. This includes allowing pedestrians the right of way; not parking on fire lanes, curbs, special needs parking areas, or designated parking areas for university vehicles; parking within the white lines; and maintaining speeds between fifteen to twenty km/hour. Students who drive recklessly through the university parking lot and/or campus are subject to disciplinary action.

Photocopy Facilities

A coin-operated photocopier is available in the library. When using a photocopier, students are reminded that AURAK campus adheres to UAE copyright laws.

Security

The university provides twenty-four hours security services. Security guards monitor the entire campus premises including the student residence halls and all university buildings to ensure the safety and security of the campus community. CCTV cameras are installed in all campus buildings.

Student Center

The Student Center, located in Building A is the heart of the student community, serving students and the campus community by offering several ways to relax and socialize. The Student Center, open every day of the week from 8:00 am to 8:00 pm, provides wireless internet access, pool tables, table tennis, a foosball table, video games, and chess. If you would like more information, please contact the Department of Student Life located in Building D.

Food Services

Various restaurants and food serving facilities are available either on campus or immediately off campus. These services are available to ensure the health and wellbeing of AURAK’s student population.

THE SAQR LIBRARY

The spacious 3-floor new library building with 35,000 sq. ft. facility comprises the library stacks, Reading Areas, Individual and Group Study Rooms, Staff Work Areas, a Computer Laboratory for library applications, Meeting Rooms, Training Room, Conference Rooms, Auditorium, Faculty Reserved Room, and Special Collection Room. AURAK's SAQR Library plays a significant role in supporting student academic success. The library continuously upgrades information resources and services that support the instructional and research activity of the university. The library collection consists of many different types of library materials and information sources, either in print or electronic formats.

The SAQR Library is a key participant in the academic development of the university and actively responds to the needs of the learning community. The spacious 5,350 sq. ft. facility provides room for the library stacks, places for reading, faculty, staff, and student work areas, and a computer laboratory for library applications. Additionally, the SAQR library has sixteen individual private study carrels, a group study room, and a conference room.

The SAQR Library facilitates access to information through the sharing of resources at national, regional, and international levels, providing circulation, reference, Internet, printing, scanning services, and online access catalog services. The hours of operation, posted on the AURAK website and at the library entrance, provide students with optimal time for study and research.

SAQR Library hours of operation provides generous opportunities for students to utilize services and resources. Hours of operation vary depending on the university academic schedule and may be found on the AURAK library webpage and at the library entrance.

In addition to textbooks and reserve collections, SAQR Library houses a wide range of information sources including reference collections, periodicals, multimedia presentations, and electronic databases.

Library Circulation Services

Borrowing Service

Borrowing privileges are provided to all registered students and currently employed faculty and staff. A university ID card is required to check-out circulating items. Students, faculty, and staff have the privilege of borrowing books based on the following policy:

Loan Period

- Students – ten items for three weeks
- Faculty – twenty items for one semester
- Staff – three items for three weeks

Renewals

Circulating items may be renewed online or at the Library Circulation Desk. The borrowers may renew items twice in succession unless the item has been requested by another user.

Fines

All borrowers pay overdue fines for late return of the borrowed items. Days on which the library is closed are not counted. The current overdue fine is:

AED 10.00 per day per item (circulating and reserve materials)

Lost and/or Damaged Items

Replacement fines for lost or damaged library materials are the actual price of the material.

Reference Services

Upon request, the library provides reference services for students, faculty, staff, and library visitors to use and identify proper resources for their work. Patrons are encouraged to contact the librarian on duty for help in using reference services and the print and online resources of books and journals.

Internet Service

Workstations within the library lab have internet access, as well as access to common productivity software applications. The library is serviced by wireless internet to facilitate the use of laptops. The wireless network will allow users to access both the library catalog, as well as, the electronic databases inside and outside the university campus.

Online Public Access Catalog (OPAC)

The library catalog contains all library holdings and serves as a search guide for easy retrieval of library materials. The Online Public Access Catalog allows users to browse all existing library resources of the university online. The catalog can be accessed through the library webpage within the university homepage.

Printing and Scanning Services

Students can use the networked printer located in the library computer lab through their accounts (IDs and Password). The library also provides a scanner and a coin-operated self-service photocopier.

Library General Rules and Regulations

The Saqr Library exists to serve students. Accordingly, some rules are in place to ensure that all students are provided a positive learning environment. The library usage rules are as follows:

- Silence should be maintained in the library.
- An identification card must be presented for borrowing of library materials.
- Eating, drinking, and other activities that distract other readers are strictly prohibited.
- Online chatting and playing games are not allowed in the library computer laboratory.
- The use of mobile phones inside the library is strictly prohibited.
- Patrons should not leave valuables in the library. The library is not responsible for the loss of personal items or possessions.

- Patrons should not reshelved used library materials.
- Patrons should handle all library materials with care.
- Patrons should keep the library clean.

THE OFFICE OF THE REGISTRAR

The Office of the Registrar is committed to enhancing a culture of high academic performance and enhancing the student experience. The Registrar's team holds specialized expertise in supporting admissions and registration, maintaining student records, certifying degrees, and overseeing the university catalog. The Office of the Registrar is located in Building D.

THE OFFICE OF STUDENT AFFAIRS

The Office of Student Affairs (OSA) provides services and facilities necessary to promote the academic, professional, and social well-being of AURAK students and encourages students to become actively involved in campus life. Out-of-classroom learning experiences play a significant role in students' professional and personal development by enhancing, their social, leadership, and organizational skills. OAS offers comprehensive programs, orientation, activities, workshops, and initiatives that revolve around student needs, requests, and suggestions. OSA, in cooperation with the Student Government Association (SGA), organizes a wide range of activities for students throughout the year. These activities are carefully planned with the purpose of exposing students to a diverse variety of cultural, social, and educational experiences. In addition to their entertainment value, these activities enhance students' creativity, collegiate, humanitarian, and social skills.

There are a number of traditional activities that OSA conducts on an annual or semester basis. These include the UAE National Day Celebration, Global Day, IFTAE Dinner, start and end of semester celebrations, and academic and student recognition award ceremonies. These activities are supplemented by cultural events – museum and theatre trips, art exhibitions, hiking trips, dhow cruises, movie nights, and beach barbecues. AURAK's very active sports program completes the comprehensive repertoire of student activities. Tournaments and leagues are organized in most sports for both men and women.

Academic Recovery Programs and Counseling

Academic Recovery Programs and Counseling assist students in both academic and personal success. Students may request the free services from the point of entry into the university through graduation. Staff are available to provide one-on-one attention to assist in meeting students' unique needs and challenges. Services include individual counseling, the loan of self-help DVD's, workshops, educational abilities testing, long-term and short-term disability services, and assisting students who are experiencing challenges both personally and academically. Services provided include: 1) personal counseling, 2) academic counseling, and 3) learning support services.

AURAK provides personal and academic counseling services for individual students requiring assistance with a variety of issues. All counseling services are free and are provided to students enrolled in three or more credit hours. These services include counseling, loan of self-help DVDs, workshops, and, when necessary, referral services to qualified professionals and organizations. The service provides students the opportunity to discuss matters causing them concern or stress, with the objective of providing assistance to and support for the student so that the student can reach his or her highest potential, personally and academically. All services will be provided in a confidential, private setting.

Enrollment Management Department/Admissions Services

The Enrollment Management Department/Admissions Services team leads a comprehensive, institution-wide, research-driven, systemic process committed to providing educational access for the local, regional, and international community. Admissions supports AURAK's Strategic Plan and aims to place AURAK as the first institution of choice for the pursuit of quality post-secondary education. The purpose of Admissions and the Enrollment Management Department is to achieve the university's desired enrollment objectives, engage top-quality student applicants, and guide the delivery of a superior student experience to ensure student success and satisfaction with the University. Accordingly, the admissions team works with marketing, recruitment, and retention of students. Admissions Services also provide important information resources for students, providing them with an introduction, clarification, and guidance to the university environment, policies, and procedures.

The Enrollment Management Department exists to help students begin the academic journey. Services are available to provide incoming applicants the highest level of service and support. These services include guidance to UAE Nationals, UAE expatriates, and international students. The Department of Admissions is located in Building D.

Financial Aid and Scholarships

Financial Aid and Scholarships exists to dismantle those financial obstacles standing between students and their dreams. Every student is important to us. The University actively recruits the quality of student capable of raising the institution to a higher level. To facilitate the student's entry into the university, AURAK offers a host of scholarships. Students are encouraged to explore the many options available to assist them in achieving their educational goals. For questions related to financial and scholarships visit the Admissions team in Building D to meet with an Admissions staff member.

Athletics

We believe that engaging in physical activity plays a significant role in promoting a healthy physical condition, self-confidence, a high quality of life, and serving as a focal point for more opportunities to sustain community interaction and building social networks.

AURAK offers athletic scholarships to qualified, outstanding student-athletes., subject to terms and conditions of the AURAK Athletic Scholarship Program.

Gym and Sports Complex

The Sports Complex, located on the west side of the campus includes sports courts, fields, a running track and two gyms available for students, faculty, and staff. There are separate gym facilities for men and women located on the main floor of the Sports Complex building. The hours of the Sports Complex operation during the academic year are 8:00 am to 8:00 pm. AURAK students and staff living in the residence halls are eligible to use the gyms free of charge. For

others the fee is 500 AED monthly. Any questions about the facilities or gym memberships should be directed to the Department of Student Life within the Office of Student Affairs on the first floor of Building D or call 8864.

Wellness Education and First Aid Station

AURAK contracts with a medical provider to deliver an on-campus Wellness Education and First Aid station located in Building D on the first floor. It is open weekly from 8:00am to 4:00 pm, Monday through Friday when the university is open.

Health Insurance

AURAK requires all full-time non-Emirati students (local and internationals) to have a health insurance policy with adequate coverage. All undergraduate and graduate students are charged an insurance premium in the first semester of full-time enrollment each academic year. Students who begin in the spring semester will pay a prorated portion of the annual premium. Those who provide evidence of comparable coverage in the UAE, through an existing health insurance policy, can request a waiver of coverage through a review process.

Students seeking the insurance waiver submit their insurance card and declaration of coverage document listing the limits of coverage provided under their existing policy. The declaration page submitted must specify the upper AED limit and include the extent of coverage provided in the policy. These two documents must be uploaded on the AURAK EUMS portal by accessing the following link: <https://eums.aurak.ac.ae/SisStudentPortal/InsuranceForm>

Confirmation of comparable health insurance coverage must be received by the last day of the class drop/add period. Refunds are not granted for proof of insurance that is submitted after the last scheduled day to drop/add classes each semester. **Please note:** *Submitting proof of coverage does not guarantee that the policy will qualify for approval.* The University will evaluate the limits of coverage in the existing policy and compare it to the policy offered by the University to make the determination regarding comparable coverage.

Questions about the university's health insurance policy should be directed to the Student Life Department via email (student.life@aurak.ac.ae) or phone (+971 7 246 8864).

Student Clubs & Student Life Activities

The Department of Student Life facilitates student growth through a variety of clubs. It supports activities to complement courses such as academic, civic, social, or recreational clubs for fun and leisure. Membership in these clubs not only helps students make new friends and meet people with common interests, but it also helps them to acquire important skills such as leadership, organization, teamwork, and event planning. All clubs work very closely with the SGA. All student clubs must be authorized and registered annually with the Department of Student Life. The university encourages all students to establish or join clubs to explore or further their interests.

Student Government Association (SGA)

The SGA represents the student body, provides an opportunity for students to demonstrate leadership, and serves as a liaison between the students and the university administration. The SGA also plays an important role in planning and sponsoring student activities and events. The representative of the SGA serves on the different committees in the university.

The SGA at AURAK follows a constitution, which defines SGA's goals and objectives and establishes the roles and responsibilities of each SGA member. The constitution also defines the policies and procedures to follow for planning and organizing university events. The SGA constitution also establishes the minimum qualifications for students who want to run for office. Further information about the SGA can be obtained by contacting the Department of Student Life.

International Learning Opportunities: International Student Exchange and Study Trips

The Office of Student Affairs also supports visiting study abroad and exchange students to our campus as well as assisting current AURAK students with plans to study abroad/exchange. The study abroad and exchange programs provide students rich experiences to encourage tolerance, acceptance, and discovery. Students participating in the study abroad or exchange programs have the opportunity to build character, adaptability, increase knowledge of international issues, and foster multicultural understanding needed to successfully navigate an ever-changing global society. The Office of Student Affairs also assists in broadening AURAK's outreach and world standing by assisting in the development of university linkages and partnerships that enhance the academic and study abroad/exchange experience for the campus community.

Center for Learning Support Services, Advising, and eLearning

AURAK's Center for Learning Support Services, Advising, and eLearning operates to ensure that every AURAK student is provided the opportunity to maximize his or her academic success. Specifically, the Center seeks to help students achieve their academic goals by: 1) overseeing the Math and Science Center, the Writing Center and Peer Tutoring Center, 2) supporting the first-year experience, and 3) guiding the improvement of AURAK's advising and educational experiences.

The Center for Learning Support Services, Advising, and eLearning facilitates the application of the latest pedagogical theories and research to enhance knowledge instruction and acquisition at AURAK for all students, faculty, and staff. The center strives to identify and respond to the dynamic nature of student needs and learning environments. Students who request academic assistance are provided free tutoring by qualified peer tutors in various subject areas.

The Center for Learning Support Services, Advising, and eLearning, located in Building D, provides qualified peer tutors to aid students in a wide array of subjects including English, mathematics, biology, chemistry, physics, and a variety of computer courses. Tutoring services provided are at no cost to the students. All students are strongly encouraged to take advantage of this key academic support service. Students interested in tutoring are asked to take visit the Center or contact the Director of this Center.

Disability Services

Students with verified disabilities may request support through the Office of Student Affairs. The university will support students with special needs where resources are available. Requests are confidential and will be handled with complete discretion. The Office of Student Affairs will verify disabilities and determine the needed accommodations to provide equal access to educational facilities, programs, services, and activities. With the student's written consent, the departmental staff will communicate with appropriate faculty, staff, and other individuals regarding needed accommodations.

OFFICE OF INSTITUTIONAL ADVANCEMENT

The Office of Institutional Advancement (OIA) assists in facilitating international learning opportunities for students. Study abroad and student exchange programs provide students with opportunities to grow both personally and academically and provide a means for students to develop the multicultural perspectives needed to actively participate in a global society. OIA also serves as a liaison between AURAK and recognized educational institutions around the world, promoting student opportunities and increasing the university's standing and global recognition through educational partnerships.

International Learning Opportunities

The OIA is responsible for supporting visiting study abroad and exchange students to our campus as well as assisting current AURAK students with their plans to study abroad/exchange. The study abroad and exchange programs provide students rich experiences which encourage tolerance, acceptance, and discovery. Students participating in the study abroad or exchange programs have the opportunity to build character, adaptability, increase knowledge of international issues, and foster multicultural understanding needed to successfully navigate an ever-changing global society. The OIA also assists in broadening AURAK's outreach and world standing by assisting in the development of university linkages and partnerships that enhance the academic and study abroad/exchange experience for the campus community.

Career Services

AURAK is committed to providing students with a wide range of career services. Students visit with a career counselor who can assist in self-awareness exploration (values, skills, abilities, and personal style), career exploration (job descriptions, educational requirements, salary range, and advancement possibilities), and employment and employability skills (writing cover letters, curriculum vitas/resume, interviewing skills, and employee decorum). Students access job listings, internship opportunities, and career fairs.

Alumni Affairs

The OIA oversees AURAK's Alumni Affairs. AURAK celebrates the success of its distinguished alumni and encourages each alumnus to stay in contact with the University. AURAK's alumni are also encouraged to utilize the University's student support services and the Saqr Library.

OFFICE OF RESEARCH AND COMMUNITY SERVICE

The Office of Research and Community Service promotes research and community engagement activities among AURAK's faculty and students. The Office also supervises AURAK's research-oriented programs and makes available small seed grants to faculty on a competitive basis. These seed grants often support students engaged in research under a research assistant contract. The Office promotes a culture of research and scholarship. Students are encouraged to attend sponsored events in order to broaden their knowledge and gain awareness of the different research activities of the AURAK faculty.

The Office of Research and Community Service manages the Ras Al Khaimah Research Innovation Center (RAKRIC) and the Center of Executive and Professional Education (CEPE). RAKRIC is a state-of-the-art research and development center specializing in sustainable system solution development related to solar technologies. RAKRIC accommodates for several senior research projects of AURAK's students under the supervision of the faculty and with the assistance of the RAKRIC staff. CEPE provides professional training courses and English language training/testing including the EmSAT, TOEFL, and IELTS tests.

COST OF ATTENDANCE – TUITION & FEES

TUITION FEES for Fall 2022 - Spring 2023

PLEASE NOTE: AURAK reserves the right to increase tuition fees and other charges up to a maximum of 10% per annum.

Undergraduate Program	Less than 12 Credit Hours	12-16 Credit Hours	More than 16 Credit Hours	Payment Terms
All undergraduate programs except otherwise mentioned for specific programs	2,000 Per Credit Hour	24, 150	1,575 per Credit Hour	Per Semester
Architecture Engineering	2,000 per Credit Hour	25,200	1,575 per Credit Hour	Per Semester
Petroleum Engineering	2,500 Per Credit Hour	30.975	1,935 per Credit Hour	Per Semester
Chemical Engineering	2,500 Per Credit Hour	30.975	1,935 per Credit Hour	Per Semester
Artificial Intelligence	2,100 per Credit Hour	25,350	1.650 per Credit Hour	Per Semester
Biotechnology	2,100 per Credit Hour	25.350	1.650 per Credit Hour	Per Semester
Computer Engineering	2,100 per Credit Hour	25,350	1.650 per Credit Hour	Per Semester
Computer Science	2,100 per Credit Hour	25,350	1.650 per Credit Hour	Per Semester
Electrical and Electronics Engineering	2,100 per Credit Hour	25.350	1.650 per Credit Hour	Per Semester

Tuition fees for Summer shall apply on per credit hour basis as per below table

Undergraduate Programs – Tuition Fee (Summer)	Fee in AED
All undergraduate programs except where mentioned otherwise for specific programs	1,575 per Credit Hour
Architecture Engineering	1,575 per Credit Hour
Petroleum Engineering	1,575 per Credit Hour
Chemical Engineering	1,935 per Credit Hour
Artificial Intelligence	1,935 per Credit Hour
Biotechnology	1,650 per Credit Hour
Computer Engineering	1,650 per Credit Hour
Computer Science	1,650 per Credit Hour
Electrical and Electronics Engineering	1,650 per Credit Hour

Undergraduate Program – Non Tuition Fees in AED	Fall/Spring	Summer	Additional Information
Application	750	750	Application Submission, Non-refundable
Admission	750	750	Issuance of Admission Letter, Non-refundable
Technology Fees	1000	400	Per semester
Service Fees	500	500	Per Semester
Health Insurance	Actual Price	Actual Price	Subject to Market Price
Transportation	1500	1500	Per Semester (Optional)
Medical Services	100	100	Per Semester

Other Fees – Undergraduate & Graduate – in AED	Amount	Additional Information
Visa Fees	2,000	Non-refundable, due at the time of admission
Visa Change Status	900	Non-refundable, due at the time of admission
Visa Security Deposit	2,000	Due at the time of admission, Refundable after Visa cancellation
Security Deposit	1,500	One time – Refundable – if no damages
Graduation Fee	1,500	Non-refundable
Late Payment (if tuition and fees are not settled by due date)	500	Applicable to first payment, if not paid by due date
Late Registration Fee	1,000	Per Semester
Cheque Returned Fee	500	Per Returned Cheque
Transcript – Official	100	Per Transcript
Transcript – True	50	Per Transcript
ID Card Replacement Fee	100	Per Instance
Official Letter Fee	50	Per letter
Gymnasium	500	Per Month – Non-refundable
Library Books overdue Fine	10	Per day, Per item
Library Books Lost/Damage Fine	Actual Cost	Per item

Tuition and Fees for graduate studies apply as per the below tables

(PLEASE NOTE: AURAK reserves the right to increase tuition fees and other charges up to a maximum of 10% per annum.)

Graduate Programs – Tuition Fees		
Program	Amount (AED)	Payment Terms
Master of Business Administration (MBA – 36 total credit hours)	2,520	Per Credit Hour
Executive Master of Business Administration (EMBA -36 total credit hours)	3,150	Per Credit Hour
Master of Education in Educational Leadership (33 total credit hours)	2,100	Per Credit Hour
Master of Engineering Project Management (MEPM 36 total credit hours)	2,730	Per Credit Hour
Master in Sustainable and Renewable Energy (36 total credit hours)	3,000	Per Credit Hour
PGDE in Educational Leadership (24 total credit hours)	2,100	Per Credit Hour

Graduate Program – Non Tuition Fees	Fall/Spring Fees in AED	Summer Fees in AED	Due by:
Application	1,000	1,000	Application Submission, Non-refundable
Admission	2,000	2,000	Issuance of Admission Letter, Non-refundable
Technology	750	400	Per Semester Non-Refundable
Service Fees	500	500	Per Semester
Health Insurance	Actual Cost	Actual Cost	Subject to Market Price
Transportation	1,500	1,500	Optional
Graduation Fee	1,500	1,500	One Time, Non-Refundable
Medical Service	100	100	Per Semester

The Housing Accommodations for both undergraduate and graduate students are outlined in the following table.

Housing and Accommodation			
Accommodation Type (Amount in AED)	Fall 2022 & Spring 2023	Summer 2023	Payment Terms
Housing (Large Room) Single Occupancy	11,000	2,400	Per Semester
Housing (Small Room) Single Occupancy	10,000	2,200	Per Semester
Housing (Large Room) Shared Occupancy	6,500	1,600	Per Semester
Housing (Small Room) Shared Occupancy	5,500	1,400	Per Semester
Security Deposit	2,000	2,000	Refundable

Housing and Accommodation

*VAT shall apply wherever applicable as per UAE Law and Federal Tax Authority (FTA) guidelines.

Payment Policies and Procedures

Tuition and all other fees must be paid by the first day of classes unless other arrangements are made with the Office of Finance deadlines published on University's website. Fees can be paid by cash, check, wire transfer, or credit card.

Students who do not pay their fees by the specified deadline will be put on financial hold. Students on financial hold are not allowed to register for classes or take final exams and will not be issued official transcripts until all fees are paid.

Students are responsible for full tuition payment for all courses in which they are registered unless their registration is canceled for nonpayment; canceled administratively due to academic suspension, dismissal, or termination; or the course is dropped before the tuition liability begins.

Refund Policies

Tuition fees are refundable subject to the following rate for the Fall and Spring semesters:

Withdraw by	% of Refund
The end of the First Week of the semester	100%
The end of the Second Week of the semester	75%
The end of the Third Week of the semester	50%
The end of the Fourth Week of the semester	25%
Beyond the end of the Fourth Week of the semester	No Refund

Please check the university Academic Calendar for dates of Drop/Add.

Tuition fees are refundable subject to the following rate for the Summer terms:

Withdraw by	% of Refund
The end of the First Week of the term	50%
Beyond the end of the First Week of the term	No Refund

Payment by Installment

If full fees cannot be paid immediately, installments can be arranged with the Office of Finance. Deferment of tuition and fees is approved only if the following conditions are met:

- The first installment shall be paid through cash, current dated cheque, credit card or bank/wire transfer and the remaining three installments through post-dated cheques (PDCs).

- The remaining installments must be paid within the same semester.
- If any installment is not paid on a due date, the student shall be placed on financial hold.
- For Fall and spring, registration shall be approved only if the first installment of 25% is paid and PDCs are received for remaining three installments.
- For Summer, installments shall not be allowed, and students must settle full payment before the start of classes.
- Any students who require a deviation to the above payment plan may visit cash counters in building D or send an email at student.finance@aurak.ac.ae

For detailed guidelines on payment plan, please refer to AURAK website.

Failure to attend a course (no show) does not result in a refund of charges. Students not planning to attend must drop their registration during drop/add period. Students reported as “no show” in class will be withdrawn from the class without a refund. These students shall be placed on a drop/delete hold and will not be able to register in future semesters unless the fees are paid in full in advance.

For detailed guide on payment deadlines please refer the University website payment guide section.

Payment Plan Arrangement Procedure

Students who wish to apply for a payment plan should submit a proposed payment plan through the online portal following the instructions provided. For Fall and Spring semesters students may submit a payment plan consisting of maximum four (4) installments including the first installment of 25%. All installments pertaining to a semester must be cleared within the same semester.

The mode of payment which may be by cheque, cash, credit card, wire transfer or online payment. Please note: the provision of post-dated cheques will be permitted on condition that the dates on each cheque is within the applicable semester.

Payment and Plan Violations

Where the student does not meet a scheduled payment, the University will place the student on a financial hold and block their access to the portal. **This may also result in the student not being able to attend the exams.**

For any clarification, please visit the Office of Finance or send an email at: student.finance@aurak.ac.ae.

Financial Responsibilities

The following outline the financial responsibilities relating to attendance at AURAK.

- Students are responsible for maintaining current addresses with AURAK and for activating and checking their personal AURAK e-mail accounts.
- Registration shall not be considered complete unless all outstanding balances from the prior term are paid in full.
- By registering for classes, students accept responsibility for charges for the entire semester.
- Failure to receive a reminder bill confirming charges does not waive the requirement for

payment when due.

- Students are responsible for dropping unwanted courses by the drop dates and using the drop and withdrawal procedures published in each term's Schedule of Classes. Full or partial tuition liability may apply.
- Non-returning students are responsible for submitting a written withdrawal to the Associate Provost of Academic Affairs. Penalties may apply.
- Students may not attend classes without completing the registration process.

Residence Halls

AURAK residence halls are located on campus. These residence halls are managed by the university's Student Residential Life team under the supervision of the Office of Student Affairs. EDRAK. For information about AURAK's residence halls, please contact the Office of Student Affairs, studentaffairs@aurak.ac.ae

Issues Relating to AURAK Representatives

AURAK utilizes representatives to assist and recruit applicants from countries external to the UAE. These representatives are held to the same standards as all AURAK employees. Applicants should never pay money to any representative. All funds are to be paid to the Office of Finance for which official university receipts will be issued. **Should an issue ever arise in dealing with any AURAK representative, concern should be addressed with the Office of Student Affairs for investigation and resolution as appropriate.**

SCHOLARSHIPS AND FINANCIAL AID

The university provides scholarships and tuition reductions to **qualified, fulltime students** of **all nationalities** as a means of providing educational opportunity and of promoting academic excellence. These scholarships and tuition reductions are intended to encourage qualified students from Ras Al Khaimah, the region and overseas to join the university's academic programs.

Categories of Scholarships and Financial Aid

The following institutional scholarships will be in effect for students joining AURAK beginning in AY 2022-2023 and onward. The awards will be determined by the following standards. All of the scholarships and grants described below will be awarded to eligible admitted students upon their signing a formal Agreement of Undertaking that will be subject to renewal for up to four years (five years if admitted to the Bachelor of Architecture), contingent on the students meeting the performance expectations for continuation of the scholarship or discount.

Recipients are only eligible for one institutional scholarship/discount.

Scholarship Performance Expectations

Performance expectations for qualification and maintenance of the indicated awards described below are based on UAE standards or equivalents. The EmSAT is the recommended standard for establishing the required levels of English proficiency. Recognized alternatives to the EmSAT are acceptable for demonstrating the required level of English language proficiency.

Academic Achievement Scholarships

New **fulltime undergraduate students** are eligible to apply for a four-year Academic Achievement Scholarship (five-year if enrolling in the Bachelor of Architecture) if they meet the following criteria. All academic achievement scholarships are offered on a competitive basis subject to application and institutional review.

Scholarship	High School Percentage	Scholarship Amount	Basis for Award	EmSAT Requirement	CGPA Requirement	Required Monthly Hours of Service
Presidential Scholarship	98% or Higher	75%	Competitive	1675	3.50	8 hours
Dean's Scholarship	95% or Higher	40%	Competitive	1550	3.25	5 hours
AURAK Scholarship	90% or Higher	25%	Competitive	1400	3.00	4 hours

AURAK Sibling and Partner Scholarships

AURAK values members of the AURAK family. The sibling (or siblings) of an existing, fully-enrolled student in any AURAK undergraduate program is entitled to the AURAK Sibling Scholarship if the sibling is enrolled at the University during the same semester. Upon the presentation of a letter from a recognized AURAK partner organization (Formal Memorandum of Understanding (MOU)),

the employees and their immediate family members are eligible to receive the AURAK Partner Scholarship as specified in the official MOU.

External Scholarships

External scholarships are available for UAE Nationals from the Sheikh Saqr Program for Government Excellence (SSPGE) and the Telecommunications Regulatory Authority (TRA). External scholarships are available for students of all nationalities from RAK Charity. All external scholarships are subject to the selection criteria of the indicated entities.

Applying for a Scholarship or Discount

To apply for a tuition reduction discount, students must submit application via the online portal available at <https://eums.aurak.ac.ae/>.

Obtaining Additional Information Concerning Scholarships and Financial Aid

Information relative to the process for applying for Scholarships and Financial Aid can be obtained by contacting the Office of Enrollment Management. **All Financial Aid and Scholarships are overseen by the Financial Aid and Scholarship Committee.**

STUDENT RIGHTS AND RESPONSIBILITIES

AURAK adheres to the following values that underlie the philosophy and goals of the university:

- The importance of personal integrity.
- The right of every individual to be treated with respect and dignity.
- The freedom of intellectual inquiry in the pursuit of truth.
- The freedom of speech and the open exchange of ideas.
- The acceptance and appreciation of diversity with regards to race, gender, religion, age, disability, and ethnicity.
- The recognition that community service is an important component of the intellectual development of students.

STUDENT RIGHTS

AURAK students have the right to pursue their education in a free and fair manner so long as they maintain their eligibility to remain as a member of the student community by meeting its academic and behavioral standards. It is the responsibility of students to observe the regulations imposed by the university for the governance of the academic community.

- Students have a right to expect AURAK to maintain a climate conducive to thinking and learning. University teaching should reflect consideration for the dignity of students and their rights as persons. Students have a right to be treated with courtesy and respect.
- Students have a right to reasonable notice of the general content of the course, what will be required of them, and the criteria upon which their performance will be evaluated. Students have a right to have their performance evaluated promptly, conscientiously, without prejudice or favoritism, and consistently by the criteria stated at the beginning of the course.
- Students have a right to perform student evaluations considered in the retention and promotion reviews of faculty members.
- Students have a right to be free from discrimination and sexual harassment. Campus policy prohibits discrimination and harassment of a student because of his/her race, color, religion, national origin, sex, age, or status as an individual with a disability.
- Students have a right to privacy and confidentiality subject to reasonable campus rules and regulations, right to protect against unauthorized disclosures of confidential information contained in their educational records. Students have a right to examine the information contained in their educational records.
- Students have a right to form student organizations for any lawful purpose consistent with the goals and mission of AURAK.
- Students have the right to excellent faculty, academic technology, classrooms, libraries, presentations and other resources necessary for the learning process.

Student Responsibilities

AURAK students are required to abide by the expectations within the Code of Student Conduct, maintain current contact information with the University, including permanent and local addresses, telephone numbers, student ID number, and legal name (as written in their passport). Each student must also maintain the University e-mail account assigned at the time of admission. Students are responsible for official communications directed to AURAK e-mail accounts. Students are responsible for providing a copy of their University ID when requested by AURAK administration, faculty staff members and/or security guards and to comply with directives of a university official.

Student Academic Grievance Policy and Procedures

Academic grievance include grievance students may have against a faculty member, procedures implemented in class, or related academic issues. All students have the right to express their dissatisfaction. If the student has a disagreement with the faculty, grades, or the classroom the grievance is a Student Academic Grievance. To appeal a grade, the Grade Appeal Form, obtained from the Office of the Registrar and filled out by the student is the most expeditious and efficient way to address a grade appeal with a professor. The following is the process for a student academic grievance.

If any student wishes to file a formal grievance, these steps apply:

1. The grievant must file a formal written complaint with the Associate Provost for Student Affairs.
2. The Associate Provost for Student Affairs will ascertain if the complaint falls within the scope of an action appropriate for a grievance and will attempt to resolve the complaint informally before instituting the official grievance process.
3. If such action is deemed appropriate, the Associate Provost for Student Affairs will convene a committee of a minimum of three people including faculty, administration, and student representative(s) within three working days.
4. The committee will review the complaint, interview the complainant and any witnesses cited in the complaint.
5. The Committee will recommend a cause of action to the Associate Provost for Student Affairs within seven working days.
6. The Associate Provost for Student Affairs will forward to the student the recommendation of the Committee or a modification of its recommendation deemed appropriate after consultation with the committee within three working days.
7. If the student does not agree with the decision, the student has the right to file an appeal with the Provost. The Provost will review the decision and render a final decision in the matter within three working days. The Provost is the final authority to whom appeal may be made.
8. If a grievance is with the Provost, then the student may submit the grievance in writing to the President to initiate the process.

Student Non-Academic Grievance Policy and Procedures

All students have the right to express their dissatisfaction and misconduct issues are handled under the non-academic grievance policy and procedures. Incidents that occur outside the classroom or disagreements with the staff fall under the non-academic grievance policy. If a student wishes to file a grievance, these steps apply.

1. The grievant must file a formal written complaint with the Associate Provost for Student Affairs
2. The Associate Provost for Student Affairs will ascertain if the complaint falls within the scope of an action appropriate for a grievance and will attempt to resolve the complaint informally before instituting the official grievance process.
3. If such action is deemed appropriate, the Associate Provost will convene a committee of a minimum of three people including faculty, administration, and student representative(s) within three working days.
4. The committee will review the complaint, interview the complainant and any witnesses cited in the complaint.
5. The committee will recommend a cause of action to the Associate Provost for Student Affairs within seven working days.
6. The Associate Provost for Student Provost will forward to the student the recommendation of the committee or a modification of its recommendation deemed appropriate after consultation with the committee within three working days.
7. If the student does not agree with the decision, the student has the right to file an appeal with the Provost. The Provost will review the decision and render a final decision in the matter within three working days. The Provost is the final authority to whom appeal may be made.
8. If a student's grievance is with the Associate Provost for Student Affairs, it may be submitted to the Provost to begin the process.
9. If a grievance is with the VPAASS or other Vice-President, the student may submit the grievance in writing to the President to begin the process.

Apart from formal reporting of grievances, there are multiple Complaint/Suggestion Boxes installed throughout the University. If students do not wish to directly contact the Faculty, Advisor or any other Staff, they are advised to write their concern and drop it in one of these boxes. Any complaint/suggestion dropped in this box comes directly to the Office of the President, and confidentiality is ensured. Student must include their ID number when placing grievances in the suggestion box.

Obtaining Additional Information

The Office of Student Affairs is available to answer any questions that may arise concerning AURAK's grievance policies and procedures. Please feel free to visit should you require additional information.

STUDENT RESPONSIBILITIES

AURAK students are expected to conduct themselves according to AURAK policy. The following policies are of particular interest in maintaining these expectations.

AURAK Communication Policy

AURAK students are required to maintain current contact information with the university, including permanent and local addresses, telephone numbers, student ID number, and legal name (as written in their passport). Each student must also maintain the university e-mail account assigned at the time of admission. Students are responsible for official communications directed to AURAK e-mail accounts.

Student Publication Policy

Any student publication at AURAK will be required to conform to the norms of responsible journalism and avoid libelous, indecent, or harassing material. The same publication policies will apply when deciding what material can be included on the web site, or any other social media, developed and managed by the students. All the publications should particularly keep in mind the cultural norms of Ras al Khaimah and the UAE.

Students should get all articles, stories, or any other related materials reviewed by the editor to ensure the correctness of the English language. Students should take prior approval from the Associate Provost of Student Affairs before distributing, posting, or publishing any printed/online material in the name of AURAK.

Guidelines for Using AURAK Name, Logo, and Mark

Students may use the AURAK name in association with their university-sanctioned activities. Recognized student groups producing merchandise for sale that incorporates AURAK's name, marks, or logos must comply and obtain permission from the Associate Provost of Student Affairs. Students may use the name of a school, department, or other AURAK programs outside the university (other than on a resume) only with approval of the appropriate authorized personnel.

STUDENT CONDUCT

The American University of Ras Al Khaimah adopts the following Student Code of Conduct to ensure that students conduct themselves appropriately at all times as members of the University community. Students will:

- Attend classes regularly and punctually and complete assignments on time.
- Participate actively in class discussions and other course-related classroom activities.
- Be knowledgeable of and plan appropriately to meet the requirements for course and program completion.
- Adhere to the published test or examination rules and regulations.
- Demonstrate sensitivity to the cultural and religious norms and customs of the UAE and ensure that their conduct is in line with those cultural expectations.

- Display principles of integrity at all times, including those associated with academic integrity, ethics, and honesty.
- Treat others as they would themselves wish to be treated.
- Show proper respect to fellow students as well as to AURAK employees and representatives. This includes security personnel. Respectful behavior includes refraining from disruptive behavior such as:
 - Talking during speeches or classes when the speaker or instructor is talking
 - Arriving late for classes or leaving early
 - Receiving or initiating telephone calls on mobile phones during classes
 - Sending text messages during class
 - Making negative or rude comments during class about other students, faculty members, or the opinions of other students
- Have knowledge of and observe the regulations of AURAK.
- Use University property/facilities responsibly.
- Provide security guards with personal identification (ID card) and appropriate documentation when requested.
- Seek permission for using someone else's property, and refrain from using another person's name and/or ID number for any reason.
- Plan and organize all extra-curricular activities on the AURAK campus in collaboration with the Department of Student Life, obtaining all necessary permissions.
- Obtain permission from the Department of Student Life to organize events in the name of AURAK.
- Obtain permission from the Department of Student Life to distribute leaflets or journals, post notices, or collect signatures on the premises of AURAK.
- Use the internet responsibly, observing all applicable IT policies.
- Refrain from the possession, use, or sale of illegal substances or materials.
- Refrain from exchanging any gifts of significant monetary value with members of the faculty, staff, or administration.
- Refrain from carrying mobile phones or any other electronic devices inside classrooms in which examinations are being administered.

Students should note that during the Covid pandemic (or similar occurrences) the university reserves the right to impose protective measures to secure a safe environment for students, staff, faculty, and administrators. Failure to comply with any of these protective measures constitutes a violation of the student code of conduct. Repeated debate with the issuance of these protective measures also constitutes a violation of the student code of conduct.

Student Behavioral Misconduct

The Associate Provost for Student Affairs is responsible for the administration of disciplinary procedures related to student behavioral misconduct. Reports of alleged student violations of university regulations may be submitted by any individual to the Associate Provost of Student Affairs. Allegations of violations of the AURAK Code of Conduct are accepted for investigation only when the apparent infractions are observed on AURAK property, at AURAK student events and activities off campus or at other locations at which AURAK provides services. Infractions of UAE or Ras Al Khaimah laws occurring off campus shall be the sole concern of the civil authorities, except when such actions: 1) directly affect the health, safety, or security of AURAK community, 2) affect AURAK's pursuit of its educational goals, or 3) occur as a direct result of an AURAK-related activity.

All student behavioral misconduct will, if possible, be resolved by the Associate Provost for Student Affairs through mediation or through verbal and written warning. For examples of behavioral misconduct, please see the *AURAK Student Handbook*.

The resolution of cases of behavioral misconduct by the Associate Provost may involve issuance of a verbal or written warning. However, the Associate Provost may judge that the conduct violation requires the student's probation, suspension or dismissal. In this situation, the case will be referred to the Disciplinary Committee, following any immediate action warranted by the Associate Provost. The following summarizes possible violations and disciplinary action to be taken by the Disciplinary Committee.

Student Behavioral Misconduct	Possible Disciplinary Action to be taken by the Disciplinary Committee
Inappropriate Student Conduct Level 1	Disciplinary Written Warning Disciplinary Probation
Inappropriate Student Conduct Level 2	Disciplinary Probation Dismissal
Damage to University Property	Orders by committee to pay actual repair or replacement costs – Failure to pay may lead to additional disciplinary actions and/or Registration Hold, Transcript Hold Disciplinary Probation Dismissal

For all levels of behavioral misconduct the Disciplinary Committee may remove the privilege of participating in or representing the university in extracurricular activities, including athletics, as well as the privilege of running for or holding office in any student organization that is allowed to use university facilities or receives university funds, or other restrictions as determined by the Disciplinary Committee.

Academic Integrity Violations

Behavior that violates the Student Academic Integrity Policy is addressed through specific procedures associated with that policy.

Release of Student Course Evaluation Results

Each semester AURAK conducts course evaluations. Students are assured of anonymity under Board Policy, which does not allow release of the results until all grades are posted. Should any student feel that anonymity of the course evaluation has been violated, a grievance may be filed with the Associate Provost of Student Affairs.

POLICIES RELATING TO STUDENT CONDUCT

Drug and Alcohol Policy

The AURAK campus adheres to the laws of the UAE concerning alcohol and drug use. Any student found using drugs will face automatic dismissal. Any student found drinking alcohol or having it in their possession on campus will be automatically dismissed.

Individuals with Disabilities Policy

The campus is committed to providing equal access to employment and educational opportunities for people of determination. AURAK recognizes that people of determination may need reasonable accommodations to have equally effective opportunities to participate in or benefit from campus educational programs, services, and activities and to have equal employment opportunities. The campus will adhere to all applicable UAE federal and state laws, regulations, and guidelines with respect to providing reasonable accommodations as necessary to afford equal employment opportunity and equal access to programs for qualified people of determination. Students who need reasonable accommodations for a disability should contact the Office of Student Affairs.

Non-discrimination Policy

AURAK's non-discrimination policy prohibits discrimination based on race, color, religion, national origin, disability, sexual orientation, gender, or age. AURAK is committed to providing faculty, staff, and students with an environment in which they may pursue their careers or studies free from discrimination.

Statement Relating to Sexual Assault

The following policy applies equally to all members of the AURAK community: students, faculty, administrators, staff, contract employees, and visitors.

AURAK and the AURAK campus is committed to providing an institutional environment where all people may pursue their studies, careers, duties, and activities in an atmosphere free of threat of unwelcome and unwanted sexual actions. It strongly condemns sexual offenses, will not tolerate sexual offenders, and supports those who have been victimized.

The campus will respond promptly, fairly, and decisively to all reports of sexual assault. Members of the campus community accused of sexual assault will be subject to campus disciplinary procedures when the alleged incident has occurred, on or off-campus, and affects the learning

environment or operations of the campus. For those seeking assistance, please contact the Office of Student Affairs.

Statement Relating to Sexual Harassment and Associated Grievances

The university is committed to providing an institutional environment in which all persons may pursue their studies, careers, duties, and activities in an atmosphere free of threat of unwelcome and unwanted sexual advances. The university strongly condemns sexual offenses, will not tolerate offenders, and supports those who have been harassed. Students who believe they have been the victim of sexual harassment should report the incident to the Office of Student Affairs. This policy applies equally to all members of university community: students, faculty, administrators, staff, contract employees, and visitors.

Unwelcome sexual advances, requests for sexual favors, and other verbal or physical conduct of a sexual nature constitute harassment when:

- The submission to or rejection of such conduct is made either explicitly or implicitly a term or condition of an individual's academic performance or employment.
- The submission to or rejection of such conduct by an individual is used as the basis for decisions about academic evaluation, employment, promotion, transfer, selection for training, performance evaluation, or selection for academic awards or benefits.
- The conduct has the purpose or effect of creating an intimidating, hostile, or offensive educational or work environment or substantially interferes with a student's academic or an employee's work performance.

Students who believe they have been the victim of sexual harassment should immediately report the incident to the Associate Provost of Student Affairs. Every effort will be taken to provide protection should any student feel threatened in any way.

Statement Relating to Stalking

Stalking is prohibited and will not be tolerated. Stalking is defined as behaviors or activities, which, when taken as a whole, give the victim reason to believe that his or her safety is at risk or affects the learning experience or participation at AURAK. This policy applies to students, faculty, staff, contractors, and visitors, and applies to behaviors and activities occurring both on and off campus. Victims of stalking are encouraged to report incidents to the Office of Student Affairs.

Pets on Campus

No pets, except those assisting people of determination, are permitted on campus at any time. Pets on campus grounds must be on a leash and under supervision at all times.

Facilities Management & Maintenance Policy - Smoke-free Environment

Smoking is not permitted in any building on the AURAK campus including the student residences. There are designated areas for smoking outside the buildings.

STUDENT ATTENDANCE POLICY

AURAK'S Student Code of Conduct requires that students attend classes regularly and punctually, complete assignments on time, and participate actively in class discussions and other course-related classroom activities. This section of the catalog outlines the University's expectations of regular student attendance in class meeting in all courses offered by the institution.

AURAK students are expected to:

- Attend all learning and teaching sessions associated with the courses in which they are enrolled.
- Notify their course instructors in advance that they will be absent from scheduled class sessions.
- Obtain prior permission from their instructors for planned absences of two or more consecutive class sessions during the semester.
- Provide a medical certificate or other corroborating evidence to explain an absence based on a health or medical condition.

Unsatisfactory student attendance includes failure to regularly attend class meetings without providing a satisfactory reason to instructors for absence and failure to observe protocols for online course attendance (e.g., having camera turned on). Unsatisfactory student attendance also includes persistent late arrival at, or early departure from, class meetings. At their discretion, instructors may count students absent if they arrive late for or depart early from scheduled class meetings as long as the information is included in the syllabus.

Where students fail to attend 10% of the scheduled classes for the semester without the University's permission, students will receive a "non-attendance warning," (NAW) and will be required to provide satisfactory explanation for their non-attendance. Where students fail to attend 15% of the scheduled classes for the semester without the University's permission, a second NAW will be issued. With each NAW issued, a formal report on the student's non-attendance is made to the student's academic advisor and to the course instructor.

Where students fail to attend 20% of the scheduled classes for the semester, students will receive a notice that they have been withdrawn from the course with a grade of W if the 20% non-attendance is recorded prior to the deadline for withdrawing with a grade of W. If the 20% non-attendance is recorded after the deadline for withdrawing with a grade of W, students will receive a notice that they have failed the course and will receive an "F (Fail)" or "U (Unsatisfactory)" grade, as appropriate.

All students must adhere to AURAK expectations regarding class attendance.

ACADEMIC INTEGRITY

AURAK HONOR CODE

To promote a stronger sense of mutual responsibility, respect, trust, and fairness among all members of the AURAK community and with the desire for greater academic and personal achievement, we, the student members of the university community, have set forth this honor code:

“As a member of the AURAK family, I pledge to be reliable, responsible, honorable, and embrace only the success and recognition I earn and deserve. I will act with integrity and honesty and will strive daily to become the best version of myself – a decent human being, a conscientious student, a good citizen, and a successful professional.”

The role of the Honor Code and associated Academic Integrity Policy is to protect the academic integrity of the university, encourage consistent ethical behavior among students, and foster a climate of honorable academic achievement. The Honor Code is an integral part of university life and students are responsible, therefore, for understanding and abiding by the code’s provisions. While a student's commitment to honesty and personal integrity is assumed and expected, this Code and associated policy and procedures provides clarity of expectations. To maintain an academic community according to these standards, students and faculty follow the reporting policy and procedures herein.

At the beginning of each semester, faculty members have the responsibility to explain the Honor Code and their expectations for students in each class with special focus on plagiarism as well as specifics related to anticipated coursework and assignments. They must also explain the extent to which aid from others, if any, is permitted on academic work. Students should request an explanation of any aspect of an instructor’s expectations regarding the Honor Code that they do not fully understand. Students have an obligation not only to follow the code themselves, but to encourage respect for and compliance with provisions of the code by other students. This includes the student obligation to report potential violations by fellow students and to understand that not doing so may also violate the Honor Code.

Faculty members are responsible for maintaining the integrity of the learning and evaluation process. Faculty members are expected to actively proctor exams, and should enforce the prohibition on possession of mobile phones during exams. Faculty may also take any additional action they believe is warranted to assure academic integrity.

While the internet and easy access to information and knowledge is commonplace, the University is committed to students understanding and demonstrating the ethical foundational values of scholarship. For all academic assignments, project work, and presentations, students need to ensure that due acknowledgement is given to the source of any information which they incorporate in their work.

Expectations

Cheating, plagiarism, and all other forms of academic fraud are unacceptable; they are serious violations of university policy. AURAK expects all students to be familiar with university policies

on academic integrity. The university will not accept a claim of ignorance—either of the policy itself or of what constitutes academic fraud—as a valid defense against such a charge.

Violations of Academic Integrity

Violations of academic integrity constitute academic fraud. Academic fraud consists of any action that serves to undermine the integrity of the academic process or that gives the student an unfair advantage, including:

- Inspecting, duplicating or distributing test materials without authorization.
- Cheating, attempting to cheat, or assisting others to cheat – relevant here is the prohibition on being in possession of a mobile telephone or similar electronic device during a test or examination. In case such devices are found with a student, the student will be deemed to have attempted to cheat and will be subject to disciplinary action under this policy.
- Altering work after it has been submitted for a grade.
- Plagiarizing.
- Using or attempting to use anything that constitutes unauthorized assistance.
- Fabricating, falsifying, distorting, or inventing any information, documentation, or citation.

Each student's program may have on record additional specific acts particular to a discipline that constitutes academic fraud. These specific acts are specified in relevant handbooks or course syllabi.

Statement on Plagiarism

One of the most common violations of academic integrity is plagiarism. Plagiarism can be intentional or unintentional. However, since each student is responsible for knowing what constitutes plagiarism, unintentional plagiarism is as unacceptable as intentional plagiarism and commission of it will bring the same penalties. In many classes, faculty members will provide their definitions of plagiarism. In classes where a definition is not provided, students will be held to the definition of plagiarism that follows:

Definition of Plagiarism. Plagiarism—submitting the work of others as one's own—is a serious offense. In the academic world, plagiarism is theft. Information from sources—whether quoted, paraphrased, or summarized—must be given credit through specific citations. When a student paraphrases a work, it is still necessary to cite the original source. Merely rearranging a sentence or changing a few words is not sufficient. The citation style should be appropriate for the discipline and should clearly indicate the beginning and ending of the referenced material. All sources used in the preparation of an academic paper must also be listed with full bibliographic details at the end of the paper, as appropriate in the discipline.

Faculty and Student Responsibilities

- Every student, faculty member, and administrator is responsible for upholding the highest standards of academic integrity. Every member of the AURAK community shall honor the spirit of this policy by refusing to tolerate academic fraud.
- When expectations for a course are not addressed in this policy, it is the responsibility of the instructor to provide students with additional guidelines for what constitutes “authorized” and “unauthorized” assistance.
- It is the responsibility of every student to seek clarification if in doubt about what constitutes “authorized” and “unauthorized” assistance. In cases involving collaborative work, all students within the collaborative group may be held responsible for violating the code if any member of the group receives, accepts, or utilizes “unauthorized” assistance.
- Students are required to obtain permission prior to submitting work, any part of which was previously or will be submitted in another course. The instructor has the option of accepting, rejecting, or requiring modification of the content of previously or simultaneously submitted work. A student who suspects that a violation of academic integrity has occurred should report that violation to the dean or to the Office of the Provost. In this report, the student should describe any action taken, such as talking with the person involved or with a faculty or staff member. Every effort will be made to preserve the anonymity of the student reporting the incident; however, confidentiality cannot be guaranteed.

Resolution at the Faculty/Student Level

Where a faculty member suspects a student of a possible violation of academic integrity, the faculty member writes formally to the student, using the Student Academic Integrity Violation Submission Form, with a copy of the communication sent to the Academic Integrity Officer. The communication provides the student with five working days to respond, prior to the faculty member making a final determination regarding an academic integrity violation.

Taking into account any response provided by the student, the faculty member decides, following the five-day period for the student’s response, if the student has breached academic integrity and whether a penalty is warranted. The faculty member then prepares a report detailing the alleged academic fraud, with a copy sent to the:

- Dean of the School in which the affected course is located.
- Dean of the School in which the student is enrolled.
- Student
- Academic Integrity Office (as Office of Record).

The suggested penalty for academic fraud in any course is failure in the course. However, faculty members may impose lesser penalties. Penalties available to faculty members include:

- Formal warning – appropriate only for early assignments in first-year undergraduate courses.
- Reduction in grade for the assignment or assessment (test) up to and including a grade of zero for the assignment or assessment – a reduction in grade less severe than a grade of zero is appropriate only for early assignments/assessments in first-year undergraduate courses.
- Reduction in the grade for the course up to and including a grade of F for the course. A reduction in grade for an assignment or assessment is likely to result in a reduction in the grade for the course. However, in instances in which the assignment or assessment is weighted such that a grade of zero does not result in a grade of F for the course, the faculty member may impose the penalty of a grade of F in the course. A faculty member can also choose to impose this penalty for an egregious act of academic dishonesty, such as attempting to steal an examination or submitting the work of another person as one's own.

Review by the Academic Review Board

All reports of violations of academic integrity are reviewed by the Academic Integrity Officer to determine whether there is a record of previous academic integrity violations and if the violation warrants a penalty different from that imposed by the faculty member. If the student has multiple academic integrity violations or the Academic Integrity Officer determines the violation under review warrants a penalty different from that imposed by the faculty member, the case may be referred to the Academic Review Board.

Penalties for academic integrity violations available to the Academic Review Board include:

- **An F grade in the course.** An F grade assigned for a course as a result of academic fraud will replace any other grade previously posted for the course.
- **Academic Dishonesty Probation.** When a student is found to have a second academic integrity violation but the offenses are not sufficiently egregious to warrant dismissal, a student will be placed on Academic Dishonesty Probation for a designated period of time, e.g., until graduation.
- **Academic Dishonesty Suspension.** Any academic integrity violation by a student while on Academic Dishonesty Probation will result in suspension from the university for at least one semester and possibly up to one academic year. Students suspended for academic dishonesty cannot transfer into AURAK any credits earned at another institution during the suspension. To qualify for readmission to AURAK following suspension, a suspended student will be asked to complete online training related to academic integrity. Readmission is not assured, but will be based on approval by the Academic Integrity Officer in consultation with the Dean of the School in which the student is enrolled and the Provost.
- **Academic Dishonesty Dismissal.** Any additional instance of academic dishonesty following Academic Dishonesty Suspension will result in Academic Dishonesty Dismissal. A student dismissed for academic dishonesty will not be readmitted to the university.

Appeal to the Academic Review Board

Students may appeal a decision on academic integrity made by either a faculty member or the Provost to the Academic Review Board within ten (10) working days of notice of the decision. For decisions made by the Academic Review Board, the student may request reconsideration of the decision by the Academic Review Board within ten (10) working days of notice of the decision. Grounds for appeal are limited to:

- failure to follow university procedure for cases of alleged academic integrity violations;
- presentation of evidence not considered when the original decision was reached.

At an appeal before the Academic Review Board, the student has the opportunity to be heard and the right to produce witnesses or introduce evidence subject to the reasonable discretion of the Academic Review Board. The student may also be accompanied by a representative of his or her choosing who may not participate in the appeal.

The results of all appeals to the Academic Review Board, permitted in the Academic Integrity Policy, shall be final. The Academic Review Board will communicate its final decision to the student, the dean of the school in which the student is pursuing a program of study, the Provost, and the Department of Registrar.

Any matter submitted to a faculty member or the university for decision or review under this policy shall be decided in a timely manner. The parties shall use their best efforts to conclude the proceedings within the semester in which the alleged offense has occurred or the appeal is received. However, any delay in the proceedings shall in no way operate as a waiver of the university's right to assess any or all of the sanctions permitted under the Academic Integrity Policy.

STUDENT RECORDS

CUSTODY OF RECORDS

All transcripts and other documents submitted from other institutions at time of admission, or later, are the property of AURAK, and are part of the student record that is under the custody of the AURAK Registrar. The university is not required to provide copies of these documents. Transcripts submitted to AURAK for admission or credit transfer cannot be returned to the student or forwarded to other institutions.

Students Privacy Rights

Students have the right to inspect and review information contained in their educational records. The university is not required to provide copies of these documents. The university allows, nevertheless, copies to be made of specific documents included in a student's record. Copies of such documents will only be provided upon submission of a signed request from the student concerned.

Academic Transcripts

A permanent record reflecting the academic achievements of each student who registers at the university (i.e., transcript) is maintained by the Office of the Registrar. Official transcripts are available on request.

Records on Academic Integrity Code Violations

The retention of records on academic integrity code violations is governed by the following:

All records pertaining to the infringement of the code are maintained by the student's school. If the student does not graduate from AURAK, the records are retained for five years after the student's last registration. If the student graduates from AURAK, these records are destroyed by the school upon the student's graduation.

Official Verifications and Certifications

All certificates related to a student's status at AURAK must be requested from the Office of the Registrar; a nominal fee applies.

RELEASE OF INFORMATION

Access to a student's education records is available to authorized officials of AURAK for academic purposes, such as recording grades, attendance, advising, or review of academic progress.

The confidentiality of student information is maintained, and periodic reviews are made in order to check the efficacy of the system. Student records are not disclosed to any third party unless a student gives permission for this in writing. The only exceptions regarding release of student information include the release of information to official accreditation bodies conducting a review or to law enforcement officials conducting pertinent individual case reviews.



UNDERGRADUATE PROGRAMS

UNDERGRADUATE ADMISSIONS

AURAK seeks to admit qualified students who wish to become part of an innovative global academic institution based on the American model. It is the goal of AURAK to admit those students most likely to benefit from its academic programs and diverse student body. Applications are welcome from all students without regard to race, color, religion, national origin, disability, sex, or age. To be considered for admission, each applicant must present sufficient evidence of appropriate qualifications as outlines by AURAK Board Policy and AURAK Presidential Procedure. The factors considered include, but are not limited to, scores on school leaving certificates and on standardized exams, English proficiency, and level of difficulty and appropriateness of coursework, essays, and recommendations. An offer of admission is valid only for the semester for which the student applies but may be extended at the discretion of the university. ***As AURAK must comply with the expectations of licensure and accreditation, the university reserves the right to modify admissions criteria at any time by action of the Board of Trustees or the President.***

When reviewing applications, the Department of Admissions considers the academic rigor of the stream of study.

AURAK Academic Program	Admission Criteria Based on High School Performance and EmSAT Test Scores			
BA in Mass Communication BS in Computer Science BS in Artificial Intelligence Bachelor of Architecture BA in Interior Design	UAE Curriculum	Advanced (Scientific)	70%	
		General (Literary)	70%	
	Non UAE Curriculum	The University Recognizes all the other certificates and converts their grades to the equivalent grade.		
BS in Biotechnology	UAE Curriculum	Advanced (Scientific)	70%	
		General (Literary)	70%	Successful completion of the qualifying subjects: Physics, Biology and Chemistry at AURAK.
	Non UAE Curriculum	The University Recognizes all the other certificates and converts their grades to the equivalent grade.		
All Business Programs	UAE Curriculum	Advanced and Elite (Scientific)	70%	
		General (Literary)	75%	
	Non UAE Curriculum	The University Recognizes all the other certificates and converts their grades to the equivalent grade.		
	EmSAT requirements are: 1) 1100 for English proficiency 2) 600 for Mathematics No conditional admission is granted for applicants with less than an EmSAT English proficiency score of 1100. Personal interview is also required.			
All Engineering Programs	UAE Curriculum	Elite Track	75%	
		Advanced Track	80%	

AURAK Academic Program	Admission Criteria Based on High School Performance and EmSAT Test Scores	
		Or Equivalent in Standardized International Systems
	Non UAE Curriculum	The University Recognizes all the other certificates and converts their grades to the equivalent grade.
	EmSAT requirements are: 1) 1100 for English proficiency 2) 900 for Mathematics 3) 800 for Physics	

Secondary School Completion

For admission, applicants must provide a Secondary School Certificate confirming completion of 12 years of secondary school at a Ministry of Education School, or an equivalence certificate issued by the Ministry of Education for private schools.

English Language Proficiency

The language of instruction at AURAK is English, therefore prospective undergraduate students must demonstrate a reasonable level of proficiency when admitted to a degree program. AURAK accepts the following assessments at the indicated levels as evidence of English proficiency.

Type of English Proficiency Exam	Minimum Score
EmSAT	1100
TOEFL Paper-based	500
TOEFL iBT	61
TOEFL CBT	173
IELTS Academic	5.0

Students may be required to take the EmSAT to evidence associated scores in English, mathematics, physics, and/or Arabic language.

Institutional TOEFL scores from other institutions are not valid except if the TOEFL was completed in AURAK or AMIDEAST offices in the UAE. International TOEFL scores should be sent directly to AURAK's Department of Admissions or ordered online through the official ETS website at www.ets.org. AURAK's TOEFL school code is 4574.

IELTS (Academic) scores are valid for admission to the university, but IELTS (General) scores are not. The exam can be conducted in any of the test centers listed on www.ielts.org. Once the IELTS result is received, a copy of the results must be provided to the Department of Admissions by the student.

Transfer Admission Requirements

AURAK will award credit transfer where a student is transferring from:

- a UAE institution recorded in the National Register of Licensed Higher Education Institutions (HEIs) (or approved registry), or
- other organizations in the UAE approved by the Commission for Academic Accreditation, or
- recognized institutions of higher learning located outside the UAE, which are eligible for transfer admission.

Students will only be eligible for the above credit transfer arrangements, where they:

- Provide valid evidence that they meet the English language competency scores for full admission.
- Are in good academic standing and have a CGPA of 2.0 on a 4.0 scale, or equivalent.

Transfer students not in good academic standing at their previous institution, but who meet the English language competency scores for full admission may receive credit transfer to a program that is not related to the program that they previously studied.

In all cases, students seeking credit transfer may be admitted as per the Ministry of Education requirements, and credit transfer will be restricted to:

- Related courses where students have earned a grade of C (i.e., 2.0 on a 4.0 scale (or equivalent)).
- Courses that deliver similar outcomes.
- A maximum of fifty-percent (50%) of the credit for the receiving program.
- Courses other than graduation projects or thesis.

Where credit transfer is awarded for courses, it will only be awarded once for a given course and there will be no duplication of credit for a course. In addition, credit awarded by credit transfer will not contribute to the calculation of the CGPA.

AURAK will not award credit for experiential or other non-classroom based learning unless the student has undertaken a challenge examination and the process for awarding such credit has been pre-approved by the Commission for Academic Accreditation.

Other Transfer Requirements

The Department Chair/Program Coordinator, with the assistance of faculty specializing in the field covered by courses, will review the application and supporting documentation to identify whether credit transfer can be applied for the courses. The following will be applied:

- The courses for which credit transfer is being sought must have the same number of credits and equivalent learning outcomes as the AURAK courses.
- The courses for which credit transfer is being sought must be at the same level (upper or lower).
- Credit cannot be given for two courses which are substantially the same course.

- For Undergraduate credit transfer will only be permitted for up to 50% of the program credits and only courses graded as “C” or above are considered for transfer credit.
- For Graduate Courses credit transfer will only be permitted for up to 25% and only courses graded as “B” or above are considered for credit transfer.
- Graduation projects and thesis are not eligible for transfer credit.

The Department Chair/Program Coordinator will complete an Evaluation Form setting out which courses and credits are recommended for transfer credit, and will forward this to the Dean of School who will validate the appropriateness of the transfer credit recommended by the Department Chair/Program Coordinator.

Where there is disagreement the Dean of School will discuss with the Department Chair/Program Coordinator but the Dean of School decision is final. The Dean of School will sign the evaluation form with the final decision and notify the Department of Admissions and the Registrar.

The Registrar will finalize the number of credits to be transferred and will ensure that the:

- Transferred courses are listed on the student’s transcript courses.
- Transferred credits and grades are not used in calculating the cumulative grade point average (CGPA).

The Admissions Department will notify the transfer applicant of the final decision.

The Registrar will place the documentation related to the transfer credit on the student’s file.

Application Procedures (For Both Freshman and Transfer Applicants)

Prospective students can apply online or in person. The online application is available on <http://apply.aurak.ac.ae>

The following documents are required with every application:

- Passport copy
- Copy of health card
- Copy of valid Emirates ID (UAE residents only)
- Health History Form (Completed and signed by a physician)
- Four (4) recent passport-size photographs
- Exemption letter from the National and Reserve Service Authority (UAE male applicants between the ages of 18 and 30 only)
- Birth Certificate
- Family Book (UAE nationals only)
- Army Exemption/Completion Letter (UAE nationals only)

Attestation of Documents

Admitted students are required to submit officially attested education certificates by the deadline mentioned in their admission letter. Students with certificates obtained outside the UAE **may** be given an extension until the end of the first semester of their studies subject to the approval of the Manager of Admissions. Subsequent extensions are subject to approval of the Associate Provost of Student Affairs. Failure to provide appropriate attested education certificates will lead to expulsion from the institution. AURAK transcripts are not provided until all admission requirements are satisfied. For more information regarding attestations, contact the Enrollment Management Department: admissions@aurak.ac.ae.

Admission Interview

The Enrollment Management Department reserves the right to require an applicant to attend an admissions interview as part of the application process when required by the prospective program or there is a need for additional information. Interviews may take place on campus, in person, by digital means, or by phone where appropriate.

Admission Categories

Full Admission

Applicants must meet all outlined requirements to obtain full admission status. To qualify for full admission a new undergraduate student must hold a UAE valid secondary stage completion certificate (SSCC) or attested equivalent along with a detailed statement of the grades for each subject. Applicants may be granted full admission on the basis of unofficial copies of such documents but must provide the official copies of required documents within two semesters following initial enrollment or face withdrawal from the program of study.

Conditional Admission

Applicants considered marginal may be conditionally admitted subject to a recommendation from the program coordinator requesting exceptional consideration.

Applicants lacking the following may also be conditionally admitted:

- English language proficiency.
- Required EmSat score in Mathematics or Physics for the desired program of study.

During the initial semester of study as conditionally admitted students, students may take no more than 12 semester credits of foundational coursework corresponding to the missing EmSAT score(s) and appropriate General Education course work. The foundational coursework will not count toward the undergraduate degree or the students' CGPA.

Admission of Visiting Students

A visiting student is one who is not formally admitted to AURAK and who is not participating in a formal exchange agreement between AURAK and a partner university. A visiting student is typically enrolled in a degree program at another university. This student may take courses at AURAK for transfer to the student's home institution. Visiting students are advised to check with their home institutions to confirm the transferability of AURAK credits.

Applicants seeking visiting student status must submit to AURAK's Department of Admissions the following documents:

- Copy of high school certificate
- An official letter from the university allowing the student to enroll for the course(s) at AURAK
- English proficiency scores (EmSAT, TOEFL, IELTS or Equivalent).
- Passport copy, Emirates ID, and residence visa (If Applicable)
- Four (4) passport-size photographs
- Proof of acceptable health insurance or purchase AURAK health insurance

Admission Requirements Subject to Program Admission Specifications as Outlined by the Ministry of Education

From time to time, the Ministry of Education establishes expectations for admission to specific programs. These expectations are communicated by MOE Letters of Reform. In all cases, AURAK conforms to the program admission standards outlined in MOE Letters of Reform.

Application Deadlines

The application deadline for fall admission is notified through newspapers or through university notice boards for freshmen and transfer applicants. Admission decisions for first year students are usually made after receipt of the first-semester grades of the senior year of secondary school and all appropriate test results. Admission is contingent upon satisfactory completion of in-progress course work and graduation from the secondary school. Transfer decisions are made as files become complete.

AURAK Admissions Application Deadline for the Fall 2022 Semester is **11 September 2022**.

AURAK Admissions Final Application Deadline for the Spring 2023 Semester is **22 January 2023**.

Admission deadlines are subject to extension by the Department of Admissions where circumstances warrant.

Note: International students must apply two weeks in advance of the stated deadline to meet visa timeline requirements.

Acceptance of Admission Offer

Students will be required to confirm their acceptance of an admission offer by completing the enrollment confirmation form in the admission packet and submitting it to AURAK with the enrollment deposit, admission fee, visa and health insurance fee (if applicable), and residence hall fee (if applicable).

Application for a Second Bachelor's Degree

Existing and former AURAK students may apply to undertake a second degree at AURAK provided such degree will enable the completion of at least thirty (30) credit hours beyond the first degree undertaken at the institution. For existing students, this will only be permitted where they have met all the requirements of their first degree.

New Student Placement Exams

An English language placement exam is administered to entering students who have not yet provided appropriate assessment results from standard English proficiency exams. The local English language placement exam is utilized for placement in the appropriate level English course. AURAK uses a math placement exam to determine which mathematics course new students may take. Students who were awarded transfer credit for a math course taken at another institution are not required to take the math placement exam.

Scores from the EmSAT can be utilized for placement into mathematics, physics, and English courses.

Student Residence Visa

The Human Resources Department arranges student visas under the sponsorship of AURAK. Visas must be renewed every year. Students on an AURAK visa who discontinue their studies or transfer to another university must submit their passport for visa cancellation prior to their departure.

The visa process takes approximately two weeks during which students cannot leave the country as they must hand over their passports temporarily. The visa application requires:

- Eight (8) passport photos
- Passport copy
- Copy of the secondary school certificate
- Visa Request Form
- Payment of visa fee and security deposit for visa

All student applicants for a residence visa must undergo medical tests conducted by the Preventive Health Department of the RAK Government and fingerprinting by the Police Department. The medical tests include a blood test and an X-ray. Failure to pass either one of the medical tests leads to deportation to country of origin.

Right to Withdraw Offer

AURAK may on the approval of the Registrar withdraw offers of admission if an applicant fails to satisfy all requirements, or if it is determined that admission was obtained through the use of falsified, altered, or embellished information (or similar). In the case of withdrawal of admission from a matriculated student, credit earned may be withheld.

Readmission

Students who have missed two or more consecutive semesters (excluding Summer Semester) of enrollment at AURAK will be required to apply for re-admission on the following basis:

- A student returning after any absence during which he or she studied at another institution without prior written permission of the Vice President of Academic Affairs and Student Success (VPAASS) or designee will be treated as a **transfer student**.

- A student who failed to demonstrate good academic standing during their previous studies at AURAK, will be required to reapply for **full admission** in order to be considered for admission.
- A student who was in good academic standing during the previous studies at AUKAK, may be readmitted on completion of a Re-Enrollment Form available through the Office of the Registrar.

UNDERGRADUATE ACADEMIC POLICIES

Each student is responsible for knowing AURAK's rules, regulations, requirements, and academic policies. The Student Handbook, the University Catalog, and the institutional website are repositories of policy statements. Corrections, changes, or interpretations may be communicated by other means, including electronically. Any student in doubt about an academic matter should consult the Office of the Registrar or their assigned faculty advisor. Students are subject to the university's stated policies regarding patents and copyrights.

Academic Advising

AURAK is committed to helping students achieve success in their course of study. Ensuring that students receive appropriate academic advising is part of that commitment.

New students are first advised during a required orientation program in which they learn about university policies and procedures and receive a current university catalog and a student handbook. At the initial meeting with their faculty advisor during orientation, each student is informed of program procedures, course of studies, and the importance of understanding the information in the university catalog. Advisors also answer specific questions concerning the academic program.

Each student is responsible for his or her progress toward degree and for meeting all requirements for graduation. The advisement process is available to assist the student in completing his or her studies, but responsibility for doing so lies ultimately with each student.

Faculty advisors are required to post office hours for advising that coincide with students' availability. The faculty advisor is to maintain advising files for all advisees that contain a log of meetings with a brief description of major issues discussed or advice given. Any information contained in these files shall remain confidential.

Students on probation are required to meet with their advisor on a regular basis to assess their progress and to report their use of AURAK's various resources for academic support. Students violating AURAK's attendance policy or doing poorly at mid-term are also required to meet with their advisors.

If students have concerns, comments, or recommendations about their educational experiences at AURAK, they should contact the dean of the school in which their degree program is located.

Courses and Course Schedules

Each course offered by the university has a designated course prefix (or code) and number. The course prefix represents the discipline or field of study; the number indicates the level of the course content. MATH 108 thus indicates that the course is Mathematics and is appropriate for students in their first year; ECEN 492 indicates that the course is in Electronics and Communication Engineering and is intended for fourth year students.

Courses are offered at the discretion of the Senior Vice President for Academic Affairs and Student Affairs and the respective deans and department chairs. Not every course is offered every semester but are offered at a frequency that enables students to make satisfactory progress toward their degrees.

Course Credit

Each course has a credit value. A credit represents the in-class instruction and out-of-class study per week during the sixteen-week semester. Normally, one credit represents fifty minutes of class instruction per week per semester or 150 minutes of laboratory experience per week per semester.

Class Periods

Classes normally meet two times a week in one hour fifteen-minute periods or two. In some cases, usually advanced courses, classes may meet once per week for 150 minutes. The university traditionally holds classes four days a week from Sunday through Wednesday. If necessary, make up classes may be held on Thursdays or Saturdays. Graduate courses are usually held on Friday and Saturday. Though university administrative offices are usually closed on Friday and Saturday, appointments may be arranged to meet students during the weekends.

All classes are expected to meet on the days and times published in the course schedule. Changes may only be made with the approval of the Senior Vice President for Academic Affairs and Student Affairs.

Independent study or research, internships, other programs for experimental learning, and other study opportunities may follow a different, approved time frame and schedule.

Course Pre-requisites and Co-requisites

Course prerequisites or co-requisites state requirements for student entry into courses and reflect necessary preparation for enrolling in a course. It is the student's responsibility to be aware of these requirements as stated in the university catalog and to register for courses only if stated prerequisites have been completed. Similarly, if a course has a co-requisite, a student can only register for the course if the co-requisite is also selected for registration in the same semester. The Student Information System has been designed to prevent students from registering for courses for which prerequisites have not been completed or for which co-requisites have not been selected for simultaneous registration. However, if a student succeeds in enrolling in a course without having completed prerequisites or co-requisites for the course, the student will be dropped from the courses as soon as the registration error is detected. No waivers of prerequisites or co-requisites will be granted.

Course Selection and Registration

Typically, by mid-semester, the schedule of classes for the following semester is available through the Office of the Registrar and on the website. Students may select courses in consultation with their faculty/academic advisor, and they can then register online or submit a completed "Course Request Form" to the Office of the Registrar by the published deadlines.

Registration Criteria

Only registered students may participate in AURAK classes. Full-time status requires that an undergraduate student enroll in 12 or more semester credit hours in Fall and Spring semesters. Undergraduate students enrolling in fewer than 12 semester credit hours are considered to be enrolled on a part-time basis, but no undergraduate student may enroll for fewer than 6 semester credit hours, except in cases in which students need fewer than 6 credit hours to complete the degree. Undergraduate students whose semester credit hour loads are limited due to probation

requirements are considered to hold full-time status. All undergraduate students are accepted to the university on the assumption that they will be studying full time, and the timeframe for completing the program of study described in this catalog is based on full-time study. Registering as a part-time student results in a significantly higher per credit tuition fee, makes students ineligible for scholarships, and lengthens the program of study.

Independent Study

Occasionally, students may have the opportunity to benefit from an independent study experience that will significantly add to their preparation for a career or ensure timely completion of their program of study. Only one course (maximum of 4 credits) may be completed as an independent study in a student's program of study. The independent study must be approved by the instructor(s) involved. Students must have completed a minimum of sixty credits, must have a 3.0 (B) or better average, and seek approval through completion of the appropriate "Permission for Independent Study" Form. The instructor must hold a regular rank faculty appointment at AURAK within the department or program requesting the independent study and is responsible for submitting the final grade, and ensuring adherence to academic standards, policies, and procedures. In addition to the individual effort of the student, a student enrolled for independent study must meet weekly with the instructor for guidance in research, analysis, and composition of the project report or presentation. Independent study courses may not duplicate available course offerings in the semester in which the independent study is being taken.

The independent study must provide a rigorous academic experience equivalent to that of any other undergraduate course at the same level at AURAK. Grading of an independent study adheres to the established grading system.

Concurrent Enrollment at Other Institutions

Students who apply for admission to AURAK usually do not seek simultaneous enrollment with another institution. Should students seek concurrent enrollment, they must obtain written approval in advance from the VPAASS/Provost. Such approval enables a student to enroll elsewhere in a course unavailable at AURAK.

Catalog numbers and descriptions of courses to be taken elsewhere must be submitted with the request for approval. As for any transferred course, students must submit an official transcript to the Office of the Registrar to earn credit, and grades are not included in the GPA. Students who enroll elsewhere without advance written permission while enrolled at AURAK may not receive transfer credit for the course work taken.

Summer Term

During the summer term, students may register for one or two courses. All attendance, grading, and other academic policies apply to the summer term as well as to the two academic semesters.

Student Coursework Assessment Policy

Academic assessment of students must be robust, reliable, fair, accurate, aligned with learning outcomes and program goals, and undertaken at an appropriate level. As such, undergraduate students will be required to undergo a minimum of three (3) assessments distributed evenly throughout the semester. Acceptable assessment tools include:

- (a) Exam
- (b) Project
- (c) Paper
- (d) Presentation
- (e) Homework assignment
- (f) In class quiz
- (g) Case study/Analysis

'In class participation' will also be acceptable as an assessment tool, but will only be used to calculate the final grade where it does not exceed ten percent (10%) of the final grade.

Irrespective of the assessment tool used, it must be appropriate for the learning outcome(s) under assessment. Assessments may be adjusted to accommodate people of determination, however, such special adjustments must not invalidate the assessment.

Weighting of Assessment

- No single assessment tool can count for more than forty percent (40%) of the total grade.
- "In class participation" may be used in calculating final grade provided it does not exceed ten percent (10%) of the total grade.
- Attendance has no weight in the final grade as attendance requirements are explained under "attendance policy."

Midterm Exams

Midterm progress is reported for all semester-long courses so that students will have a clear idea of their standing in each course midway through the semester. The reporting period extends from the fifth through the ninth week of the semester, allowing flexibility as to when individual faculty provide reports for their classes. Midterm grades are reported as S (satisfactory - C+ and above), U (unsatisfactory - D - C) and F (failing). Students should check with their instructors as to when reports will be complete and available for viewing. These progress reports, which appear as 'Midterm Grades,' are not part of the student's official record. They are not calculated in the GPA, nor do they appear on any official or unofficial transcript. The midterm progress report is distributed to the student, his or her faculty advisor, and the Office of Student Affairs.

Final Exams

Final exams are usually given at the end of all undergraduate courses. Except for project submissions, laboratory courses, and studio courses, no exam or assessment tools of any kind may be given during the last week of classes. Exams may not exceed the scheduled length of two (2) hours for most courses. The final exam schedule is at least one week prior to the beginning of the semester by the

Office of the Registrar, and all faculty instructors include the day and time of the final exam on their syllabi. The final exam schedule is also posted on the campus website.

An instructor who is assigning a take-home exam or significant end-of-semester paper or project should so inform the students on the syllabus at the beginning of the semester. Take home exams should be distributed by the beginning of the last week of classes so that students can coordinate them with preparation for other exams. Students must not be required to submit exams before the date of the regularly scheduled exam for a course. Accommodations will be made for students with more than one final exam scheduled in the same time period. Retaking (or re-sitting) a final exam is not permitted. Students are not expected to take more than two final exams in one day.

Absence from Final Exams

Absences from final exams are not to be excused except for sickness on the day of the exam or for other causes approved by the dean of the respective school. The effect of an unexcused absence from an undergraduate final exam shall be determined by the weighted value of the exam as stated in the course syllabus provided by the instructor. If an absence from a final exam is unexcused, the grade for the course is entered as F.

Final Semester Grades

Final grades are available through the student portal on the university website and/or collected from the Office of the Registrar.. Official transcripts and official semester grade reports for tuition reimbursement are obtained through the Office of the Registrar.

Students may access and print an unofficial record of their semester grades and other academic information through the student portal on the university website by logging onto their Student Information System (SIS) account. Official transcripts are available upon request.

Change of Grade/Grade Appeal

Once a final grade has been recorded by the faculty member, it can be changed only through the Grade Appeal process. Where students believe their final course grades to be in error, students must request a meeting with the member of faculty to ask that the faculty member reconsider the grade and ensure that the appeal is submitted within the required timeline. Completed Grade Appeal Forms must be submitted within the following timeframe:

- For Fall Semester courses, completed Grade Appeal Forms must be submitted within the first three weeks of the following Spring Semester.
- For Spring Semester courses, completed Grade Appeal Forms must be submitted within three weeks of the "Grades Available" date listed in the Academic Calendar.
- For Summer Session courses, completed Grade Appeal Forms must be submitted within the first three weeks of the following Fall Semester.

The grade appeal will only be considered where:

- There was an error or unfairness in the assessment of the exam grade.
- There was an error or unfairness in the evaluation process for determining the course grade.

The faculty member will meet formally with the student and a record of the meeting will be recorded on the Grade Appeal Review Form. In the meeting with the student, the faculty member must explain how the grade in question was arrived at. If the faculty member decides, based on the

discussion with the student, that a change of grade is warranted, a Change of Grade Form will be completed by the faculty member, explaining the basis for the grade change and this will be forwarded to the Department Chair.

If the faculty member decides a change of grade is not warranted, the faculty member will complete the Grade Appeal Review Form, explaining why the original grade should not be changed and this will be forwarded to the Department Chair. The Change of Grade Form/Grade Appeal Review Form (as applicable) will then be reviewed in succession by the Department Chair, the Dean, and the AssAssociate Provost for Student Affairs and a decision will be made.

The Associate Provost for Student Affairs will render a decision only after the faculty member has responded to any requests for further consideration made by the Department Chair and the Dean. The decision by the Associate Provost for Student Affairs will be final and not subject to further appeal. The Registrar will not change a student's final grade in a course for any other reason or through any other procedure.

Where an appeal is successful, the Associate Provost for Student Affairs will forward the Change of Grade Form to the Registrar's Department so that the student's grade is changed on the Student Information System (SIS). A copy of the Grade Change Form will be filed in the student's record.

Pending Grade Appeal

A student may request the VPAASS/Provost to delay imposing an academic suspension because of a pending grade appeal that could change the student's status. An approved delay allows the student to register. If the grade appeal is successful, the official transcript is corrected, and the student continues in classes as a student in good academic standing. If the grade appeal is not successful, the student is required to stop attending all classes and settle the case with VPAASS/Provost appropriately.

Grading System

University course work is measured in terms of quantity and quality. A credit normally represents one hour per week of lecture or recitation or not fewer than two hours per week of independent or laboratory work throughout a semester. The number of credits is a measure of quantity. The grade is a measure of quality. The university system for undergraduate grading is as follows:

Grade	Quality Points	Percentage Scores
A	4.0	95-100
A-	3.7	90-94
B+	3.3	86-89
B	3.0	83-85
B-	2.7	80-82
C+	2.3	76-79
C	2.0	73-75
C-	1.7	70-72
D+	1.3	66-69
D	1.0	60-65
F	0	0-59

No credit toward graduation is received from a failing grade or additional credit gained from a grade that is replaced by a repeated course.

Grade Point Average (GPA)

Grade or quality point values are assigned to letter grades as indicated in the grading system table. A quality or grade point score is computed by multiplying the value of a letter grade by the number of credits for the course. For example, a student receiving an A in a three-credit course earns twelve quality points. The GPA is computed by dividing the quality points earned by the number of credits graded "A" through "F" (GPA hours).

For undergraduates, the GPA computed for the current semester gives the current GPA, which is the measure of academic performance in one semester and affects eligibility for the honor's list. The GPA computed for all institutional credit gives the CGPA, which is the basis for the university's retention policies, including good standing, warning, probation, suspension, and dismissal. The CGPA also determines students' eligibility to graduate and to have university honors posted to their record upon graduation.

Additional Grade Notations

Satisfactory/Unsatisfactory (S/U):

An "S" grade reflects satisfactory, or passing, work in a course; a "U" grade reflects unsatisfactory, or failing, work in a course. S and U will have no effect on GPA.

Incomplete (I):

This grade is given to students who are passing a course but are unable to complete all the course work or the final exam for a verified reason beyond their control or the instructor agrees to allow the student additional time to complete substantial aspects of the course's required work. The student must complete the final exam or the required work for the course by the end of the subsequent registered semester, not including summer terms, and the instructor must turn in the final grade by the end of that semester's grading deadline. Unless an explicit written extension is filed, the grade of "I" is changed to an "F" if the course requirements have not been fulfilled. Students who have filed their intention to graduate have six weeks from the date of degree conferral to resolve any incomplete grades. An "I" is not calculated in a student's GPA.

In Progress (IP):

The grade of "IP" indicates that the student is making progress in a course, writing a thesis or dissertation, or participating in an internship that extends beyond one semester or summer term. The grade of "IP" has no effect on the student's GPA.

Withdrawal (W):

The grade of "W" on a course indicates that the student has dropped the course. While it has no effect on the GPA, dropped courses are part of attempted course credits that serve as the basis for the student's credit level. A "W" for all courses in a given semester and the transcript notation "withdrawn" indicates that the student withdrew from AURAK.

Repeating a Course

Undergraduate students may repeat courses for which they seek a higher grade. A grade received in a repeated course replaces the previous grade in the same course in the calculation of the GPA, even if the more recent grade is lower. Every registration for a course and its grade remains as part of a student's transcript. Duplicate credit is not given. No adjustment to the CGPA will be made if the grade in the repeated course is a W. No more than three courses can be repeated at AURAK over the course of a given degree program.

A grade in an AURAK course will not be excluded from the CGPA based on the subsequent taking of an equivalent course at a transfer institution. Note that individual programs may disallow students from retaking certain high-demand courses simply for the purpose of improving their grade.

Transfer Credits

Courses taken at other institutions of higher education may be counted toward a degree but are not computed in the AURAK GPA.

Academic Progress

Academic Status

Only registered students may participate in AURAK classes. Fulltime status requires that an undergraduate student enroll in 12 or more semester credit hours in Fall and Spring semesters. Undergraduate students enrolling in less than 12 semester credit hours are considered to be enrolled on a part-time basis. Undergraduate students whose semester credit hour loads are limited due to probation requirements are considered to hold fulltime status. All undergraduate students are expected to enroll on a fulltime basis unless granted permission by the dean of the respective school to enroll on a part-time basis.

All students are required to demonstrate sufficient academic performance to maintain enrollment with the university. Accordingly, AURAK establishes the following to recognize high academic performance and establish parameters for improving academic performance falling below minimal expectations.

Academic Excellence

An undergraduate student will be named to the Dean's List for a fall or spring semester if during that semester the student has all of the following:

- Completed twelve or more semester credit hours in courses counted in the CGPA
- Received no grade below B
- Received no grade of I (Incomplete) in any course
- Attained a semester's GPA of 3.60 or above

The dean of the school will notify students who have been named to the Dean's List.

An undergraduate student who qualifies for the Dean's List in a given semester with a GPA of 3.90 or above will also be named to the President's List.

Please note that these lists are only compiled for fall and spring semesters.

Satisfactory Academic Progress

In order to be considered in good standings, an undergraduate student must maintain a CGPA of 2.0 or higher. To be considered in good standing, a graduate student must maintain a CGPA of 3.0 or higher. A student, undergraduate or graduate, must be in good standing to be eligible for graduation.

Every student, undergraduate or graduate, must evidence sufficient academic progress towards the completion of his or her program requirements. Academic progress is evaluated based on the student's maintenance of a satisfactory CGPA and satisfaction of a minimum course completion pace. All undergraduate students enrolled in mathematics/science intensive programs (e.g. Engineering, Biotechnology, Computer Science) are required to meet academic progression expectations relating to the required pace for completing mathematics and science courses. Any undergraduate student enrolled in a mathematics/science intensive program must earn a grade of C or higher in a required mathematics/science course before progressing to the subsequent course. A student enrolled in a mathematics/science intensive program will only be allowed to repeat a required mathematics/science course once after achieving a grade of C- or lower. Failure to maintain a sufficient level of academic progress in a mathematics/science course will result in removal from mathematics/science intensive programs, with the student required to enroll in an AURAK program that is not intensive in mathematics/science.

Academic Probation, Suspension, and Dismissal

If the CGPA of an undergraduate student falls below 2.0 in any semester, the student will be given an Academic Warning and will be required to enroll in an appropriate course in the subsequent semester. If the student's CGPA remains below 2.0 at the end of the subsequent semester, the student will be placed on Academic Probation 1. All students on academic probation will be required to sign the AURAK Probation Contract before being allowed to register for any coursework.

A student on Probation 1 will be allowed to take a maximum of 13 semester credit hours. Students earning a CGPA of 2.0 or higher in the subsequent semester will be removed from Probation 1.

Students on Probation 1 earning a CGPA of 1.5 and higher but less than 2.0 will be placed on Probation 2. A student on Probation 2 is allowed a maximum of 10 semester credit hours. Students earning a CGPA of 2.0 or higher in the subsequent semester will be removed from Probation 2. Students completing a semester of Probation 2 with a CGPA less than 2.0 will be academically suspended.

Suspended students will not be allowed to enroll in AURAK courses for the subsequent semester unless they change their program of study to a distinctly different academic discipline, utilizing the appropriate procedure. Suspended students who elect not to change their program of study to a different academic discipline can only request re-admission once they have been out for one full academic fall or spring semester and have completed during that time not-for-credit remedial coursework. Students readmitted following suspension will be dismissed from the university if they are suspended for a second time unless they change their program of study to a different academic discipline.

Academic dismissal from the university is permanent.

New Beginnings for Suspended Students

1. A student who has been academically suspended from the university may apply for re-admission through the Department of the Registrar as part of the New Beginnings Initiative immediately following suspension from the original program of study. The New Beginnings Initiative will only enable the student to undertake an alternative program that is distinctly different from the program from which the student was suspended. Students suspended from a math/science intensive program such as an Engineering program or Biotechnology cannot undertake an alternative program in another math/science intensive program.
2. The Department of the Registrar will:
 - Verify that the student has not previously been re-admitted under the New Beginnings Initiative.
 - Seek the approval of the Department Chair for the program in which the student will be enrolled.
 - Seek the approval of the Dean of School for the program in which the student will be enrolled.
 - Seek approval from the Provost or his or her designate.
3. Once a student is re-admitted to a different program, the Registrar will remove from the calculation of the CGPA:
 - Grades below C in the major/program for which the student was not suited.
 - Grades for courses that do not meet requirements in the new degree program and having a grade below C.
4. The following will not be removed from the calculation of the CGPA:
 - Courses in which a student was guilty of academic dishonesty i.e., cheating, plagiarism, etc.
 - General education courses that continue to meet requirements in the new degree program.
 - Courses with grades C and above, even if they are not a part of the student's new program.
5. The Registrar will recalculate both the semester GPA and CGPA based on the above and the adjusted courses will remain on the transcript and will receive an "E" notation, meaning the grades and the credit hours will be "Excluded from GPA calculations."

Intellectual Property (Student Work)

Intellectual property (IP) is property (an idea, invention, or process) that derives from the work of the mind or intellect. IP is also an application, right or registration of an idea, invention, or prose. Copyrightable works, including publications and patentable works developed in connection with course work by students who are not AURAK employees, are deemed to be intellectual property that belongs to the student. However, AURAK may claim copyright ownership of a work or ownership of a patentable invention when extraordinary use of university facilities, personnel, or resources is made in the development of the materials or invention, especially when unrelated to course work. Students are subject to the university's stated policies regarding patents and copyrights.

University Withdrawal

Official withdrawal removes students from any academic program and cancels student status at AURAK. In order to return to the university, students need to apply for readmission through the Department of Admissions. Students who wish to withdraw have to complete a withdrawal form available at the Office of the Registrar and get clearance from all university departments. If the withdrawal process is completed satisfactorily and all financial obligations to the university are cleared, the effective date of withdrawal is noted on the student's permanent academic record. The effective date is the date used for calculating billing or refunds. No grades other than a W are recorded for the current semester.

Leave of Absence

Occasionally, students are forced to leave the university for a semester or two because of circumstances beyond their control. Others find they simply need a break from studying. In such circumstances, taking a leave of absence might be wise. Students who have an approved leave of absence for a semester or a year may register for the semester in which they plan to return without applying for readmission. Unless there are extenuating circumstances such as illness, a leave of absence is not normally given to a student who leaves the university during a term.

Fields of Study - Degree Offerings

American University of Ras Al Khaimah has three schools that offer both undergraduate and graduate degree programs. Undergraduate programs are listed below:

- Bachelor of Architecture
- Bachelor of Arts in Interior Design
- Bachelor of Arts in Mass Communication
- Bachelor of Science in Artificial Intelligence
- Bachelor of Science in Biotechnology
- Bachelor of Science in Business Administration
- Bachelor of Science in Chemical Engineering
- Bachelor of Science in Civil and Infrastructure Engineering
- Bachelor of Science in Computer Engineering
- Bachelor of Science in Computer Science
- Bachelor of Science in Electrical and Electronics Engineering
- Bachelor of Science in Mechanical Engineering
- Bachelor of Science in Petroleum Engineering

Concentrations

Undergraduate and graduate degree programs may offer students one or more concentrations within the major from which to choose. A concentration is a set of related courses that focus on a specific sub-field within the general discipline. A concentration must consist of no fewer than fifteen (15) credit hours of study.

The concentration is specified on the student's diploma and transcript. To be recognized as completing an official concentration, a student must complete and submit the Request for Concentration Form before having completed 105 credit hours towards the degree.

Double Major

An undergraduate student may elect to complete a double major, which is a single degree with two majors.

A double major may be undertaken under the following conditions:

- A student must have a CGPA of 2.5 or above at the time of electing the second major.
- A student must have completed no more than 105 credit hours when electing to pursue a second major.
- A student must have completed two full years of study toward the first major (sixty credit hours) before applying for a double major.
- The second major may not be in the same academic department as the first.

To complete a double major, the following conditions must be met:

- A student must meet all the degree requirements for both majors.
- A student must complete at least twelve credits in addition to those required by the first major.
- No course substitutions will be available to a student in order to complete the requirements for the second major.

A student who completes a double major will receive one degree that indicates both majors on the transcript.

Change of Major

A student may change his or her major at any point during matriculation at AURAK. At the time of the major change, the student must indicate which major and the General Education requirements that will be met: those in the catalog stipulated at the time the student entered AURAK as a matriculating student or those stipulated at the time of the change of major. The student must choose one set of General Education requirements or the other; they may not split the General Education requirements listed in either catalog.

Unless the student invokes the GPA Adjustment Policy, all previous courses taken remain on the student's transcript and grades received are calculated in the student's CGPA.

To take effect, the student must complete and submit the Change of Major form with all signatures.

Course Substitution

All degree programs, concentrations, minors, and minors have published requirements for graduation, including GPA requirements applicable to all university students. All requirements are met by successful completion of specific courses.

The need occasionally arises for a course substitution in a student's program, which AURAK permits under certain circumstances:

- If a required course has not been offered in at least two different semesters during the student's period of full-time matriculation
- OR
- Under exceptional circumstances (e.g., certain disabilities make it impossible for a student to enroll in a required course)

In such cases, a course substitution is possible only if:

- The substitute course covers material similar in content and skills to the initial requirement.
- The substitute course is at the same level as, or more advanced than, the course for which it is being substituted.
- The replacement is approved by the student's academic advisor, a content specialist, the chair of the department of the program for which the substitution is being made, and the dean of the appropriate school, or, in the case of General Education, by the chair of the General Education Committee.

A course being transferred from another university may be used to substitute for an AURAK requirement if all other conditions are met.

Permission for a course substitution is granted only rarely, and a student may receive no more than two course substitutions in any program, including General Education. For a double major, no course substitutions are permissible in the second major.

Among others, the following will not be considered appropriate justifications for applying for a course substitution:

- A desire to complete the program of study quickly
- Failure to enroll and complete a required course that is offered regularly
- Dislike of a professor teaching a required course
- Substitution for a required course which the student has failed previously
- An independent choice to diversify courses taken for personal preference

Undergraduate Enrollment in Graduate Courses

AURAK recognizes that there are occasions where it is academically valuable and appropriate for an undergraduate student to enroll in and receive credit for a graduate level course. To be eligible to undertake such courses the undergraduate student must gain the approval of their Academic Advisor and Department Chair and must have:

- senior status.
- a CGPA of no less than 3.0.

Students granted permission to undertake graduate level courses must complete the same assignments as the graduate students enrolled on the course(s) and will only be permitted to enroll in **two** graduate level courses.

The credit achieved may count toward the student's Bachelor degree as determined by the Program's Specification and contribute to the relevant graduate program should the student later matriculate into the graduate program.

This policy is subject to approval by the Commission for Academic Accreditation (CAA).

Internship Policy

AURAK considers Internships to be an important education-related experience in the workplace and an opportunity for learning that integrates theory learned in the classroom with practical application. The Internship is also seen as an opportunity for skills development on the job that contribute to the development of personal and professional maturity as well as ethical decision-making.

Undergraduate students will only be permitted to register for the Internship if they:

- (a) have completed a minimum of two years of coursework (or higher if determined by the applicable School). *In exceptional circumstances a student who has not meet this condition, may be permitted to undertake the Internship on the recommendation of the Dean of School and with approval of the Provost.*
- (b) are in good standing (at least 2.0 CGPA). *In exceptional circumstances a student who has not meet this condition, may be permitted to undertake the Internship on the recommendation of the Dean of School and with the approval of the Provost.*
- (c) have the written approval of their Academic Advisor and the School's Internship Coordinator.

Internship Arrangements

In compliance with accrediting body requirements, AURAK will ensure that the Office/Department responsible for Internships:

- (a) Has an effective method for matching students to potential placement employers.
- (b) Prepares an Internship Manual which covers the following:
 - i. Policies and procedures related to Internships and other experiential learning opportunities.
 - ii. A delineation of the responsibilities of the faculty or academic supervisor, the provider and the student.
 - iii. A delineation of the placement of the internship in the curriculum or program of the student.

- (c) Prepares official agreements that clearly specify the responsibilities of AURAK, its student and the employer during the Internship, obtains the signature of all parties to the agreement in advance of the placement of students within an organization and retains a copy on file.
- (d) Organizes an Internship for each undergraduate student of a duration that is compliant with the relevant accrediting body(ies) for the program in which the student is enrolled.
- (e) Provides an orientation for students as to the skills, behaviors and knowledge which they need to succeed in the work environment.
- (f) Has an effective process in place to monitor the employment site and gather feedback from students and the provider.

Student Petitions and Appeals

Students' Responsibility

All official university communications are distributed through the AURAK-issued email address. These are considered official notifications. Students are responsible for checking their AURAK email accounts and for acting upon messages accordingly.

Students should keep their own records of all transactions with the university (e.g., registration schedules and forms, grade reports, payment records, etc.). It is also advisable to keep copies of all tests, digital files, papers, and so forth submitted in fulfillment of course work. Students should keep copies of all course syllabi.

Appeal of Academic-Related Issues

Academic grievances include grievances students may have against a faculty member, procedures implemented in class, or related academic issues. Note that grade appeals follow a different procedure and are described elsewhere in the Student Handbook and the AURAK catalog. For information regarding grievances please contact the Associate Provost of Student Affairs.

- The grievant must file a formal, written complaint with the VPAASS/Provost.
- The VPAASS/Provost will ascertain if the complaint falls within the scope of an action appropriate for a grievance and will attempt to resolve the complaint informally before instituting the official grievance process.
- If such action is deemed appropriate, the VPAASS/Provost will convene a committee of a minimum of three people including faculty, administration, and student representative(s) within three working days.
- The committee will review the complaint, interview the complainant and any witnesses cited in the complaint.
- The committee will recommend a cause of action to the VPAASS/Provost within seven working days.
- The VPAASS/Provost will forward to the student the recommendation of the Committee or a modification of its recommendation deemed appropriate after consultation with the committee within three working days.

- If the student does not agree with the decision, the student has the right to file an appeal with the President. The President will review the decision and render a final decision in the matter within three working days. The President is the final authority to whom appeal may be made.
- If a grievance is with the VPAASS/Provost, then the student may submit the grievance in writing to the President to initiate the process.

Graduation

Undergraduate Graduation Requirements

The graduation requirements for any individual student are normally determined by the catalog that was effective when the student began matriculated studies in the degree program. Every individual student is personally responsible for meeting all graduation requirements as detailed in his or her catalog year.

If a required course within an undergraduate program changes its number of credits, then the number of credits required by the program for graduation may, at the discretion of the school, change by the same amount provided the minimum total number of credits for graduation is 120 and the CGPA is at least 2.00. In case of substantial changes, in course offerings, equivalent graduation requirements are determined by the dean of the student's school.

Important: Course information, content, and prerequisites may be subject to change as a result of the university's commitment to a process of continual improvement in academic programs. Students must comply with the most up-to-date course requirements.

Academic Standing Requirement

A student must be in good academic standing to be eligible for graduation.

Application for Graduation

Candidates for undergraduate degrees must file an Application for Graduation form with the Office of the Registrar during the registration period of the last expected term of study. Only after an Application for Graduation form has been filed can the Office of the Registrar begin processing the necessary information for final certification for graduation.

Students who fail to complete all degree requirements by the end of the term for which they apply to graduate need not reapply for graduation. Their previous application will be automatically moved to the following semester.

Conferral of Degrees

Only students who have successfully completed degree requirements by the end of the term for which they have applied to graduate qualify for conferral of a degree. Degrees are conferred at the end of the semester in which requirements have been met. Conferral of the degree is noted on the academic transcript of the graduate with the date of graduation.

Names on Degrees

The names of AURAK students will be spelled in English exactly as they appear on the student's passport or identity card when printed on degrees. If a name on a passport or an identity card does not appear in English, then the spelling of the name will be printed according to the personal preference of the student.

Attestation of Degrees and Transcripts

The Office of the Registrar provides information relevant to the attestation of degrees and transcripts to the UAE Ministry of Education.

ACADEMIC PROGRAMS

A degree program—the major or field—is a program of study that normally requires at least thirty credits of coursework in a specific field. The degree and major, or field, appear on the diploma. The Bachelor of Science degree programs at AURAK are professionally oriented, and all require the completion of a minimum of 120 credits, approximately forty courses, including General Education courses. Some programs require additional coursework, and all require an internship.

Degrees normally take four years to complete with a minimum of three years, assuming summer enrollment. If a degree is not completed within six years, all coursework in the major is re-evaluated for its current relevance. The Bachelor of Architecture degree is a five-year program.

AURAK currently offers the following thirteen undergraduate majors:

- Bachelor of Architecture
- Bachelor of Arts in Interior Design
- Bachelor of Arts in Mass Communication
- Bachelor of Science in Artificial Intelligence
- Bachelor of Science in Biotechnology
- Bachelor of Science in Business Administration
 - Major in Business Administration
 - Major in Accounting
 - Major in Business Analytics
 - Major Finance
 - Major in Hospitality and Tourism Management
 - Major in Human Resource Management
 - Major in Marketing
- Bachelor of Science in Chemical Engineering
- Bachelor of Science in Civil and Infrastructure Engineering
- Bachelor of Science in Computer Engineering
- Bachelor of Science in Computer Science
- Bachelor of Science in Electrical and Electronics Engineering
- Bachelor of Science in Mechanical Engineering
- Bachelor of Science in Petroleum Engineering

AURAK Allocates Academic Credit According to the Following Criteria

AURAK follows the standard US calibration of academic credit, such that 1 semester credit hour implies 15 hours of class time per semester. Two (2) semester credit hours of credit implies 30 hours of class time. Three (3) semester credit hours implies 45 hours of class time. Four (4) semester credit hours implies 60 hours of class time. The class meetings are typically distributed evenly across a 15-week semester in fall and spring, excluding vacation days. Summer sessions and accelerated (short) sessions/courses vary in length.

Each course has a credit value. A credit represents the in-class instruction and out-of-class study per week during the 15-week semester. Normally one credit represents 50 minutes of class instruction per week per semester.

- Lectures carry one academic credit for 50 minutes of class instruction per week for 15 weeks.
- Laboratories are usually 3 contact hours per week and carry one credit.
- Courses in drawing and design, currently taught in the Bachelor of Architecture, carry one academic credit for three contact hours in class.
- Independent Study courses carry 1-4 credits, depending on the amount of work required for the specific project as determined by the supervisor of the project.
- Internships carry 3 credits for a full-time experience.

Degree Completion Requirements

- Successful completion of all credit hours as specified by the program curriculum with a minimum CGPA of 2.0
- Completion of the minimum enrollment period and not exceeding the maximum enrollment period
- Transfer or concurrently enrolled students are required to complete a minimum of fifty percent of the required credit hours at AURAK including the majority of the final thirty credit hours (please refer to this catalog for additional information on transfer credit policies)

Enrollment Periods

Degrees normally take four years to complete with a minimum of three years, assuming summer enrollment. At AURAK the degree must be completed within a specified period in order to remain current and the maxima for programs is as follows:

- Programs requiring up to 140 credits must be completed within seven (7) years from initial registration (inclusive of periods of non-enrollment).
- Programs requiring more than 140 credits must be completed within eight (8) years from initial registration (inclusive of periods of non-enrollment).

AURAK will permit an appeal for extenuating circumstances to extend the period by up to two additional semesters.

GENERAL EDUCATION COMPONENT

Introduction

The General Education courses required of each baccalaureate degree at AURAK are based on a coherent rationale. Courses offered to meet the General Education Component requirements are designed to provide students with the foundational knowledge necessary to engage in higher order thinking, analysis of information, and problem-solving. These courses are intended to cultivate an informed, capable, and progressive citizenry. All courses included in the General Education Component are university-level courses and emphasize experiential and activity-based learning. Specifically, the General Education Component examines the application of knowledge in concrete situations to foster critical thinking skills, communication, teamwork, empirical and quantitative skills, and personal and social responsibility. While the General Education Component is not a degree program, all undergraduate degree students must complete the University's General Education Component requirements as a part of their academic program of study.

Mission

The General Education Component is the fundamental constituent of all degree-granting programs. It provides students with fundamental knowledge that broadens their acquaintance with disciplines. As the component providing transferable skills of cortical thinking, numeracy, and synthesis, the General Education Component supports the University mission by preparing students to face the challenges of a globalized, multicultural, and rapidly changing world. The General Education Component complements the students' major program of study by examining achievements in areas of human inquiry and creative endeavor to cultivate those abilities so essential for the acquisition of knowledge.

Purpose

The purpose of the General Education Component is to explore broad disciplinary areas in order to expand students' perspectives and improve their skills in analytical and critical thinking. The General Education Component is intended to facilitate a well-rounded and informed educational experience established on a holistic awareness of a broad spectrum of knowledge areas to ensure that students are exposed to a wide range of disciplines and topics. The General Education Component is designed to provide a relevant and challenging educational experience.

Completion of General Education Requirements

AURAK's General Education Component embodies the vision, mission, and purpose of the University. The courses are required of all graduates as the intellectual foundation upon which AURAK's educational framework is built. They are an essential element in the University's educational process and lay the foundation for professional success, personal growth, and responsible citizenship. Students have the opportunity to acquire knowledge, learn communication skills, and examine their goals, values, and potentialities. Students are encouraged to complete their General Education Component requirements by the end of their second year of study.

Opportunities for Students to Develop and Demonstrate Competencies in Four Areas

The General Education Component provides opportunities for students to develop and demonstrate competencies in the following four areas:

Critical Thinking Skills: including reasoning, inquiry, reflection, creativity, and innovation, and the ability to analyze and synthesize information.

Communication and Teamwork: including the development, interpretation, and expression of ideas, the ability to consider and integrate multiple points of views and to work with others on shared objectives.

Empirical and Quantitative Skills: including the ability to derive knowledge from observable facts, the manipulation and analysis of data, resulting in informed, verifiable solutions and conclusions. Effective understanding of technological concepts and the appropriate use of technology as a problem-solving tool.

Personal and Social Responsibility: including the ability to consider ethical and moral aspects of actions and consequences, intercultural awareness, and knowledge of regional, national, and global responsibility.

General Education Component Learning Outcomes

Upon completion of the General Education Component requirements, completers will be able to:

Learning Outcome One: Solve abstract, familiar, and non-routine problems without assistance using appropriate information drawn from relevant fields of work or disciplines.

Learning Outcome Two: Present collaborative group analyses of topics that include complex ideas and integrate multiple point of view.

Learning Outcome Three: Analyze empirical data using appropriate information retrieval and analysis methods, tools and techniques to arrive at informed, verifiable solutions to abstract, familiar, and non-routine problems.

Learning Outcome Four: Articulate the ethical and moral aspects of possible courses of action in particular situations and explain the consequences that might follow with reference to intercultural issues at regional, national, and global levels.

Required Pathway to Completion of General Education Component Learning Outcomes:

AURAK ensures that graduates acquire a holistic awareness of a broad range of knowledge areas. The General Education Component addresses these needs in a diverse, connect, globalized environment through courses that are not only broad and deep in content but use up-to-date pedagogical methods. The diverse range of topics in the General Education Component, the interdisciplinary and globalized focus of the content, the pedagogical shift to experiential and activity-based learning, the focus on research, innovation and enhanced aesthetic sensitivity all combine to aid in the development of responsible, self-directed individuals who will engage in life-long learning.

The General Education Component at AURAK fulfills the requirements of a liberal arts and sciences education by providing a comprehensive foundation in the competencies that enable students to grow academically as well as professionally. To achieve the General Education Component learning outcomes, students must complete 31-34 semester credit hours consisting of:

- Orientation Courses (O) – 13-14 semester credit hours
- Humanities/Fine Arts (H) – minimum of 6 semester credit hours
- Social and Behavioral Sciences (S) – minimum of 6 semester credit hours
- Natural Sciences/Mathematics (N) – minimum of 6 semester credit hours

AURAK distinguishes between categories of **Orientation Courses (O)** that provide a coherent foundation and academic orientation, and **Knowledge Domains** that offer students opportunities to develop and pursue a variety of academic interests. The Knowledge Domains are delineated into **Humanities/Fine Arts (H)**, **Social and Behavioral Sciences (S)**, and **Natural Sciences/Mathematics (N)**. In addition, the General Education Component requires that of the courses taken to fulfil General Education Component requirements, five (5) courses must qualify as **Writing Intensive** courses. The codes “O”, “H”, “S”, and “N” identify the category to which a course belongs.

All AURAK undergraduate students must complete all General Education Component requirements to graduate. Completion of these requirements will amount to a total of 31-34 credit hours. All students must complete the Orientation Course requirements (amounting to 13-14 credits), and a minimum of 18 credits from the three remaining Knowledge Domains (amounting to a 18-20 credits). The courses for each of these are as follows.

ORIENTATION COURSES (13-14 semester credit hours required)

Students must complete from 13 to 14 semester credit hours from the following list of courses.

Orientation Courses		
Required Courses	Title	Semester Credit Hours
ENGL 101 (O)	Composition (Writing Intensive)	3
UNIV 100 (O)	University First-Year Transition	1
UNIV 200 (O)	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
ITEC 103 (O) or CSCI 112/113 (O)	Fundamentals of Information Technology Introduction to Computer Programming	3-4
ARAB 101 (O) or ARAB 110 (O)	Arabic Language and Culture for Non-Native Learners I or Arabic Language and Culture for Native Arabic Speakers I	3
Subtotal =		13 to 14 Credit Hours

KNOWLEDGE DOMAIN (18-20 semester credit hours required)

Students must complete a minimum of 18 semester credit hours in this area. The requirements are divided into the following three categories: 1) Humanities/Fine Arts (H), 2) Social and Behavioral Sciences (S), and 3) Natural Sciences/Mathematics (N). Expectations are outlined below.

Humanities/Fine Arts		
Required Courses	Title	Semester Credit Hours
PHIL 100 (H) OR ENGL 200 (H)	Critical Thinking and Reasoning (Writing Intensive) Advanced Composition (Writing Intensive)	3
MEST 100 (H)	Introduction to Islam in World Culture (Writing Intensive)	3
Subtotal =		6 Credit Hours
Social and Behavioral Sciences		
Required Courses	Title	Semester Credit Hours
UAES 200 (S)	Survey of United Arab Emirates Studies (Writing Intensive)	3
PSYC 100 (S) ECON 103 (S) POLI 100 (S) POLI 101 (S) GEOG 100 (S) or COMM 101 (S)	Introduction to Psychology Principles of Microeconomics Contemporary Global Issues Politics of Scarcity World Regional Geography Interpersonal Communication and Group Interaction	3

Subtotal =		6 Credit Hours
Natural Sciences/Mathematics		
Required Courses	Title	Credit Hours
MATH 101 (N) STAT 100 (N) MATH 108 (N) MATH 111 (N) MATH 113 (N)	Numbers and Data Interpretation Statistics Calculus with Business Applications Calculus with Life Sciences Applications Calculus I	3
BIOL 100 (N) CHEM 100/101 (N) CHEM 211/212 (N) ENVS 102 (N)	Humankind in a Biological World Chemistry in Everyday Life General Chemistry Sustainability and Human-Environment Relations	3-4
Subtotal =		6 to 8 Credit Hours
TOTAL = 31 to 34 Credit Hours		

Academic programs are encouraged to include additional courses focused on promoting creative and aesthetic understanding from *COMM 102 Reading Image and Film* and *ENGL 102 Public Speaking*. Academic programs are also encouraged to include additional courses focused on promoting cross-cultural and ethical understanding from *HIST 100 Contemporary Middle Eastern History*, *HIST 101 Ancient History of the Arabian Peninsula*, *PHIL 101 Ethics in Today's World*, and *PHIL 102 World Philosophies*.

Writing Intensive Courses

To promote students' achievement of the Written Communication competency, the University's General Education Component also requires that students complete five writing-intensive courses. One of these writing-intensive courses is ENGL 101. Of the remaining four required writing-intensive courses, three must be other writing-intensive General Education courses, and the fifth writing-intensive course must be a course in the major program of study. Programs are encouraged to require students to take more than the university minimum of five writing-intensive courses.

The criteria for writing-intensive courses are provided below:

- The course includes a sustained focus on individual student writing, as demonstrated in the syllabus through multiple writing assignments.
- Students receive actionable feedback on their writing from the course instructor.
- Students substantially revise and resubmit written drafts, incorporating feedback from the course instructor and using their own critical thought to improve their writing.
- The final versions of written assignments total to at least 14 double-spaced, printed pages.

To be designated as “Writing-Intensive,” courses must be approved by the General Education Committee as satisfying the criteria listed above. Courses that have been approved as Writing-Intensive are clearly designated as such in the catalog, on transcripts, and in study plans for all programs.

The General Education Component Requirements and Course Substitutions

The degree requirements for the individual undergraduate programs outline the course options from which a student may select courses to meet the General Education requirements. Other General Education courses that fulfill the requirements listed in the General Education Component may be substituted to complete respective General Education requirements in accordance with the established Course Substitution Policy and Procedure. Although students are encouraged to identify and select courses that meet their needs, faculty advisors must approve these course selections to ensure that they satisfy General Education Component requirements.



SCHOOL OF ARTS & SCIENCES

SCHOOL OF ARTS AND SCIENCES

Bachelor of Arts in Mass Communication

The Bachelor of Arts degree in Mass Communication has been developed to provide students with a high-quality education in line with emerging market trends in the media industry within the UAE, the Middle East, and the wider world. Rapid advancements in the field of communication and new media have created a demand for qualified professionals and leaders who possess the knowledge to address global issues and, by so doing, shape the future of the media industry.

The Bachelor of Arts in Mass Communication is overseen by the School of Arts and Sciences and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The Bachelor of Arts in Mass Communication with concentrations in Digital Media or Public Relations will produce competent graduates with the ability to apply both knowledge and skills of their area of concentration to engage in research to benefit the community and the nation, to ensure quality in the learners' professional lives.

Program Educational Objectives

The goals and objectives of the Bachelor of Arts in Mass Communication are simultaneously aligned with the mission statement of the university and seek to achieve the objectives embedded in the university mission.

The Goals of the Mass Communication Program are to:

1. Provide the student with a broad and deep knowledge about the study of communication, the similarities and differences between communication delivery modes, including their structure, processes, an ethical practice thereof, and the significance in social and professional uses
2. Foster an understanding of the interdisciplinary nature and the cultural, ethical, and historical contexts of human communication;
3. Enhance the student's proficiency in research skills, the use of technology and media, critical thinking, creativity, and analytical abilities to pursue knowledge independently in the pursuit of advanced higher education;
4. Produce graduates who possess advanced practical and interpersonal skills that are grounded in truth and ethically practiced who demonstrate the capacity to establish and evaluate communication relationships in order to lead and serve the community; and
5. Produce responsible and competent professionals with practical skills, social commitment, and ethical standards that satisfy the demands of employment in the UAE, the region, and the world.

Program Learning Outcomes

AURAK Mass Communication graduates should demonstrate the ability to:

1. Explain communication theory and demonstrate applications of relevant theoretical concepts for different mass communication fields;

2. Recognize key skills and employ contemporary tools for communicating effectively in realistic media environments;
3. Utilize appropriate messages and techniques for communicating effectively;
4. Apply research methodologies and models and implement these in communication work;
5. Identify factors characterizing the global nature of modern media systems;
6. Distinguish cultural issues in planning various forms of communication; and
7. Analyze principles of ethics, fairness, and regulation as they relate to media practices.

Goals for the Digital Media Concentration

The goals of the Digital Media Concentration are to:

1. Provide a strong foundation in the theoretical principles, approaches, techniques, and practices of communication through digital media;
2. Produce graduates that have proficiency in communication and communication delivery modes that is applicable in diverse contexts;
3. Enhance the student's proficiency in research skills by developing their critical thinking skills, creativity, analytical abilities, and their ability to work with communication technology and communication media; and
4. Produce responsible professionals competent in communication in digital media who satisfy the demands of employment and the needs of the community in the UAE and the wider region.

Program Learning Outcomes for the Digital Media Concentration

Upon completing the Bachelor of Arts in Mass Communication with a concentration in Digital Media, graduates will demonstrate the ability to:

1. Employ convergent technologies for production and expression of communication pieces for news, documentaries, entertainment, and persuasive communication; and
2. Apply latest trends in multimedia practices and uses for professional digital communication to a local or international context.

Goals for the Public Relations Concentration

The goals of the Public Relations Concentration are to:

1. Foster an understanding of the interdisciplinary nature and the historical and cultural contexts of human communication in English and in another modern language in all public relations roles;
2. Enhance the learner's proficiency in research skills, the use of technology and the media, critical thinking, creativity, and analytical abilities to pursue public relations communication knowledge independently in advanced higher education;
3. Demonstrate how contemporary public relation approaches and strategies may be used to improve communication with the audience in both public and private spheres;

4. Listen to and to speak, and to read and write public relations texts thoughtfully and to demonstrate an ability to support any communication with truthful and honest responses that increase the trust of the audience; and
5. Write effective public relations texts and to create public relations in media.

Program Learning Outcomes for the Public Relations Concentration

Upon completing the program in Mass Communication with a concentration in Public Relations, graduates will demonstrate the ability to:

1. Discuss the characteristics of unique publics and design strategies to best establish symmetrical relationships with them; and
2. Summarize modern principles of public relations practices and analyze how these can be used in the UAE and globally.

Degree Requirements

To earn a BA degree in Mass Communication, students must satisfactorily complete at least 120 approved credits, fulfill all the requirements for the BA in Mass Communication degree, and achieve a GPA of 2.00 or higher. Thirty-one credit hours from approved courses are required to fulfill the General Education requirements. Thirty-two credit hours from the program's core courses are similarly required. The BA in Mass Communication offers concentrations in Digital Media and Public Relations; these courses are offered in the second, third, and fourth year of study. Students complete eighteen credits in their concentration, an additional six credits through the completion of concentration electives, and then fifteen credits in free electives. After completing eighty-one to eighty-four credits, students are required to complete an internship (3 credit hours) which is required for their concentration i.e. Digital Media or Public Relations.

Students also fulfill a modern language and culture competency requirement. Involving a total of fifteen credit hours, this requirement consists of a block of four courses (12 credit hours) focusing on language and culture, with one further course (3 credit hours) focusing on wider cultural themes.

Students may enroll in one of the two sequences of Arabic courses. No native-level Arabic speaker, however, is permitted to take the Arabic language and culture sequence of course that is designed for non-Native learners.

The BA degree in Mass Communication is designed to be completed in four years, assuming students do not interrupt their study. Students who withdraw or take a leave of absence from the program must meet requirements for returning that are outlined in the AURAK Catalog. Students must meet specific standards to progress, as well as the maximum time allowed to complete the program, which are also detailed in the catalog. If a degree is not completed within a period of six years, all coursework in the major will be re-evaluated for its current relevance.

BA MASS COMMUNICATION DEGREE COMPLETION REQUIREMENTS

The Bachelor of Arts in Mass Communications degree requires the completion of the following requirements:

Requirement	Credit
I. University General Education Requirements	31
II. Program Core Course Requirements	32
III. Concentration Course Requirements: Digital Media or Public Relations	18
IV. Mass Communication Electives	6
I. Modern Language Competency Requirement	15
V. Free Electives	15
VI. Internship	3
TOTAL	120

University General Education Requirements

31 Credits

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. ***The fifth writing intensive course for the BA in Mass Communication is COMM 450 Selected Topics in Communication.***

Course Code	Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
ITEC 103	Fundamentals of Information Technology	3
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
UAES 200	Survey of United Arab Emirates Studies	3
Gen Ed Course	Social and Behavioral Science Course	3
Gen Ed Course	Mathematics Course	3
Gen Ed Course	Natural Science Course	3

PROGRAM CORE COURSE REQUIREMENTS**32 Credits**

Course No.	Title	No. of Credits
COMM 111	Introduction to Mass Communication	3
COMM 112	Introduction to Public Relations	3
COMM 113	Introduction to Digital Media	3
COMM 212	Media Writing	3
COMM 215	Feature Writing	3
COMM 222	Intercultural MassCommunication	3
COMM 223	Globalization and Media Culture	3
COMM 311	Media Law and Ethics	3
COMM 321	Theories of Media Communication	3
COMM 391	Communication Research Methodology	2
COMM 492	Mass Media and Public Opinion	3

CONCENTRATION COURSE REQUIREMENTS**Concentration in Digital Media****18 Credits**

Course No.	Title	No. of Credits
COMM 214	New Media and Digital Culture	3
COMM 224	Visual Storytelling	3
COMM 323	News Reporting	3
COMM 334	Digital Journalism	3
COMM 423	Multimedia Magazine	3
COMM 424	Capstone: Multimedia Storytelling	3

Concentration in Public Relations**18 Credits**

Course No.	Title	No. of Credits
COMM 213	Public Relations Writing	3
COMM 334	Digital Journalism	3
COMM 337	Public Relations Cases	3
COMM 344	Public Relations Theory	3
COMM 421	Social Media: Audiences and Messages	3
COMM 427	Capstone: Public Relations Campaigns	3

MASS COMMUNICATIONS ELECTIVES**6 Credits**

Course No.	Title	No. of Credits
COMM 221	Communication Analysis and Criticism	3
COMM 322	Digital Resources and Content	3
COMM 422	Streaming	3
COMM 450	Selected Topics in Communication (Writing Intensive) - Required	3

* Note: Students may choose one course from the other concentration as an elective course.

MODERN LANGUAGE COMPETENCY REQUIREMENT 15 Credits**Arabic for Non-Native Arabic Learners**

Course No.	Title	Pre-requisites and Co-requisites	No. of Credits
ARAB 101	Arabic Language and Culture for Non-Native Learners I		3
ARAB 102	Arabic Language and Culture for Non-Native Learners II	ARAB 101	3
ARAB 201	Intermediate Level Arabic Language and Culture I	ARAB 102	3
ARAB 202	Intermediate Level Arabic Language and Culture II	ARAB 201	3
	Special Topics in Arab Culture and Civilization (course to be designed)	ARAB 202	3

Arabic for Native Arabic Learners

Course No.	Title	Pre-requisites and Co-requisites	No. of Credits
ARAB 311	Upper Intermediate Level Arabic Language and Culture I		3
ARAB 312	Upper Intermediate Level Arabic Language and Culture II	ARAB 311	3
ARAB 411	Advanced Level Arabic Language and Culture I	ARAB 312	3
ARAB 412	Advanced Level Arabic Language and Culture II	ARAB 411	3
ARAB 420	Special Topics in Arab Culture and Civilization	ARAB 412	3

FREE ELECTIVES**15 Credits****Internship****3 Credits**

Course No.	Title	No. of Credits
UNIV 390	Internship	3

TOTAL PROGRAM COURSE REQUIREMENTS = 120 CREDIT HOURS

B.A. Degree in Mass Communication with Concentration in Digital Media Four-Year Schedule

Year 1 Semester 1			Year 1 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
COMM 111	Introduction to Mass Communication	3	COMM 112	Introduction to Public Relations	3
ENGL 101	Composition	3	COMM 113	Introduction to Digital Media	3
ITEC 103	Fundamentals of Information Technology	3	MEST 100	Introduction to Islam in World Culture	3
UNIV 100	University First-Year Transition	1	UNIV 200	Innovation, Entrepreneurship, and Sustainability	3
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3	PHIL 100 or ENGL 200	Critical Thinking & Reasoning or Advanced Composition	3
Gen Ed	Social and Behavioral Science	3			
Subtotal =		16	Subtotal =		15
Year 2 Semester 1			Year 2 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
COM 212	Media Writing	3	COMM 215	Feature Writing	3
COMM 222	Intercultural Mass Communication	3	COMM 223	Globalization and Media Culture	3
COMM 214	New Media and Digital Culture	3	COMM 224	Visual Storytelling	3
UAES 200	Survey of United Arab Emirates Studies	3		Free Elective	3
Gen Ed	Mathematics Course	3		Modern Language Requirement	3
Subtotal =		15	Subtotal =		15
Year 3 Semester 1			Year 3 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
COMM 311	Media Law and Ethics	3	COMM 391	Communication Research Methodology	2
COMM 321	Theories of Media Communication	3	COMM 334	Digital Journalism	3
COMM 323	News Reporting	3		Mass Communication Elective	3
Gen Ed	Natural Science Course	3		Modern Language Requirement	3
	Modern Language Requirement	3		Free Elective	3
Subtotal =		15	Subtotal =		14
Year Three, Summer Semester					
Course No.	Description	CR			
UNIV 390	Internship	3			
Year 4 Semester 1			Year 4 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
COMM 492	Mass Media and Public Opinion	3	COMM 424	Capstone: Multimedia Storytelling	3
COMM 423	Multimedia Magazine	3		Free Elective	3
	Mass Communication Elective	3		Free Elective	3
	Free Elective	3		Modern Language Requirement	3
	Modern Language Requirement	3			
Subtotal =		15	Subtotal =		12
TOTAL CREDIT HOURS 120					

B.A. Degree in Mass Communication with Concentration in Public Relations Four-Year Schedule

Year 1 Semester 1			Year 1 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
COMM 111	Introduction to Mass Communication	3	COMM 112	Introduction to Public Relations	3
ENGL 101	Composition	3	COMM 113	Introduction to Digital Media	3
ITEC 103	Fundamentals of Information Technology	3	MEST 100	Introduction to Islam in World Culture	3
UNIV 100	University First-Year Transition	1	UNIV 200	Innovation, Entrepreneurship, and Sustainability	3
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3	PHIL 100 or ENGL 200	Critical Thinking & Reasoning or Advanced Composition	3
Gen Ed	Social and Behavioral Science	3			
Subtotal =		16	Subtotal =		15
Year 2 Semester 1			Year 2 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
COM 212	Media Writing	3	COMM 215	Feature Writing	3
COMM 222	Intercultural Mass Communication	3	COMM 223	Globalization and Media Culture	3
	Mass Communication Elective	3	COMM 213	Public Relations Writing	3
UAES 200	Survey of United Arab Emirates Studies	3		Free Elective	3
Gen Ed	Mathematics Course	3		Modern Language Requirement	3
Subtotal =		15	Subtotal =		15
Year 3 Semester 1			Year 3 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
COMM 311	Media Law and Ethics	3	COMM 391	Communication Research Methodology	2
COMM 321	Theories of Media Communication	3	COMM 334	Digital Journalism	3
COMM 337	Public Relations Cases	3	COMM 344	Public Relations Theory	3
Gen Ed	Natural Science Course	3		Modern Language Requirement	3
	Modern Language Requirement	3		Free Elective	3
Subtotal =		15	Subtotal =		14
Year Three, Summer Semester					
Course No.	Description	CR			
UNIV 390	Internship	3			
Year 4 Semester 1			Year 4 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
COMM 492	Mass Media and Public Opinion	3	COMM 427	Capstone: Public Relations Campaigns	3
COMM 421	Social Media: Audiences and Message	3		Free Elective	3
	Mass Communication Elective	3		Free Elective	3
	Free Elective	3		Modern Language Requirement	3
	Modern Language Requirement	3			
Subtotal =		15	Subtotal =		12
TOTAL CREDIT HOURS 120					

Bachelor of Science in Biotechnology

Biotechnology is the modern face of the biological sciences. It applies technological advances, particularly those involving the sciences of molecular genetics, recombinant DNA techniques, and molecular diagnostics to solve biological problems through the production of materials, processes, and services to benefit our environment.

In medicine, Biotechnology aids in the manufacture of therapeutics, enzymes, antibiotics, vitamins, and vaccines. In agriculture, Biotechnology helps improve food quality, quantity, and processing. Additionally, Biotechnology has applications in green manufacturing technologies. It minimizes contamination and pollution and locates alternative energy sources. In the past decade, industry has increasingly turned to biotechnology to improve product quality and to make the production process easier, cleaner, and more cost-effective.

Biotechnology graduates work in fields such as molecular biology, genetics, forensics, plant and animal sciences, food production, therapeutics, agriculture, and environmental sciences.

The Bachelor of Science in Biotechnology is overseen by the School of Arts and Sciences and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The Bachelor of Science in Biotechnology Program, with separate concentrations in Cell and Molecular Biotechnology, Medical Biotechnology, and Forensic Biology, aims to produce skilled and knowledgeable biotechnologists who are capable of applying their acquired knowledge to scientific research and development in medicine, science, agriculture, forensics and the environment for the benefit of their societies, while maintaining the highest standards of professional ethics in their careers.

Program Educational Objectives

The goals of the Biotechnology Program are to:

1. Supply society with biologically literate citizens who are capable of advising and making decisions;
2. Produce ambitious, creative graduates who are interested in continuing their education in the biosciences;
3. Practice scientific inquiry and appreciate its role in the development of research, science, technology, and society;
4. Produce responsible biotechnology professionals to fulfill the employment and research needs in the biotechnology industry in the UAE and the region;
5. Contribute to the advancement of agriculture, medicine, and environmental sciences through the application of biotechnology theory and recombinant DNA technology; and
6. Enhance the students' ability to integrate their acquired math, computer, and bioscience knowledge and skills to investigate and solve biological problems.

Program Learning Outcomes

On completion of the program graduates will be able to:

1. **PLO1:** Identify, formulate, and solve broadly defined technical or scientific problems by applying knowledge of mathematics and science and/or technical topics to areas relevant to the discipline.
2. **PLO2:** Formulate or design a system, process, procedure or program to meet desired needs.
3. **PLO3:** Develop and conduct experiments or test hypotheses, analyze and interpret data and use scientific judgment to draw conclusions.
4. **PLO4:** Communicate effectively with a range of audiences.
5. **PLO5:** Discuss ethical and professional responsibilities and the impact of technical and/or scientific solutions in global, economic, environmental, and societal contexts.
6. **PLO6:** Function effectively on teams that establish goals, plan tasks, meet deadlines, and analyze risk and uncertainty.

Program Concentration Learning Outcomes – Cell and Molecular Biology

7. **PLO7:** Lead and implement appropriate research strategies to produce verifiable data in Cell and Molecular Biotechnology.
8. **PLO8:** Evaluate and manage complex and unpredictable work procedures to be able to function successfully within the Cell and Molecular Biotechnology graduate programs and employment sectors.

Program Concentration Learning Outcomes – Medical Biotechnology

7. **PLO9:** Lead and implement appropriate research strategies to produce verifiable data in Medical Biotechnology.
8. **PLO10:** Evaluate and manage complex and unpredictable work procedures to be able to function successfully within the Medical Biotechnology graduate programs and employment sectors

Program Concentration Learning Outcomes – Forensic Biology

7. **PLO11:** Lead and implement appropriate research strategies to produce verifiable data in Forensic Biology.
8. **PLO12:** Evaluate and manage complex and unpredictable work procedures to be able to function successfully within the Forensic Biology graduate programs and employment sectors.

Degree Requirements

The Bachelor of Science in Biotechnology degree is a four-year degree program with concentration in Cell and Molecular Biotechnology, Medical Biotechnology, and Forensic Biology. For the first three years of the Bachelor of Science in Biotechnology program, students follow a common sequence of courses, with courses specific to the respective concentration taken in the fourth year

of the program. To earn a bachelor's degree, students must satisfactorily complete at least 128 approved credits, fulfill all the requirements for the Bachelor of Science in Biotechnology degree, and achieve a GPA of 2.00 or higher. Students must apply for at least forty-five credits of upper-level courses (designated 300- level or higher) toward graduation requirements. Thirty-one to Thirty-four credit hours from approved courses are required to fulfill the General Education Program requirements. The degree is designed to be completed in four years, assuming students do not interrupt their study. Students who withdraw or take a leave of absence from the program must meet requirements for returning that are outlined in AURAK catalog. Students are required to meet specific standards to progress, as well as the maximum time allowed to complete the program, which are also detailed in the catalog. If a degree is not completed within a period of six years, all coursework in the major will be re-evaluated for its current relevance.

The Bachelor of Science in Biotechnology degree necessitates the completion of the following requirements:

Requirements	Credits
University General Education Requirements	31
Biotechnology Program Core Courses	63
Biotechnology Program Elective courses	6
Concentration (Cell and Molecular Biotechnology, Medical Biotechnology, or Forensic Biology) Course Requirements	21
Free Electives	6
TOTAL	127

University General Education Requirements

31 Credit Hours

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The 5th writing intensive course for the BS in Biotechnology is BIOL 450 Bioethics.**

Course Code	Title	Credits
ENGL 101	Composition (Writing Intensive)	3
UNIV 100	University First-Year Transition	1
ITEC 103	Fundamentals of Information Technology	3
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
UAES 200	Survey of United Arab Emirates Studies	3

MATH 111	Calculus for Life Sciences	3
Gen Ed	General Education Social Sciences Course	3
BIOL 100 or ENVS 102	Humankind in a Biological World or Sustainability and Human Environment Relations	3

Biotechnology Program Core Courses

63 Credit Hours

Course Code	Title	No. of Credits
BIOL 112	University Biology I	3
BIOL 113	University Biology I Lab	1
BIOL 114	University Biology II	3
BIOL 115	University Biology II Lab	1
BIOL 230	General Microbiology	3
BIOL 231	General Microbiology Lab	1
BIOL 270	General Genetics	3
BIOL 271	General Genetics Lab	1
BIOL 330	Applied and Industrial Microbiology	3
BIOL 331	Techniques in Applied and Industrial Microbiology	1
BIOL 250	Biochemistry I	3
BIOL 251	Biochemistry I Lab	1
BIOL 350	Biochemistry II	3
BIOL 356	Virology	3
BIOL 380	Biotechnology and Genetic Engineering	3
BIOL 381	Biotechnology Lab Methods and Techniques	1
BIOL 450	Bioethics (<i>Writing Intensive</i>)	3
BIOL 491	Senior Seminar	2
BIOL 493	Research Methodology in the Biological Sciences	2
CHEM 111	Principles of General Chemistry	3
CHEM 112	Principles of General Chemistry Lab	1
CHEM 120	Analytical Chemistry	3
CHEM 121	Analytical Chemistry Laboratory	1
CHEM 215	Organic Chemistry I	3

CHEM 216	General Organic Chemistry Lab I	1
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab	1
STAT 100	Introductory Probability and Statistics	3
UNIV 390	Internship	3

Biotechnology Program Elective Courses

6 Credit Hours

Course Code	Title	No. of Credits
BIOL 240	Ecology	3
BIOL 322	Microbial Genetics	3
BIOL 354	Biophysics	3
BIOL 355	Biophysics Lab	1
BIOL 360	Developmental Biology	3
BIOL 480	Food Biotechnology	3
BIOL 481	Bioprocessing Technology in the Pharmaceutical Industry	3
BIOL 492	Special Topics in Biotechnology	2
	Students may choose one course from the other concentration as an elective course	3- 4

Concentration Course Requirements

21 Credit Hours

Concentration in Cell and Molecular Biotechnology

21 Credit Hours

Course Code	Course Title	No. of Credits
BIOL 420	Molecular Biology and Genetics	3
BIOL 421	Molecular Biology and Recombinant DNA Lab	1
BIOL 423	Advanced Molecular Biology and Forensic Sciences	3
BIOL 430	Cell Culture Theory and Technology	3
BIOL 431	Cell Culture Techniques Lab	1
BIOL 434	Cell Communication and Signal Transduction	3
BIOL 473	Bioinformatics and Computational Biology	3
BIOL 494	Senior Project in Cell and Molecular Biotechnology	4

Concentration in Medical Biotechnology**21 Credit Hours**

Course Code	Course Title	No. of Credits
BIOL 411	Drug Discovery, Design and Development	3
BIOL 435	Stem Cell Biology and Regenerative Medicine	3
BIOL 436	Immunology	3
BIOL 440	Medical Microbiology	3
BIOL 441	Medical Microbiology Lab	1
BIOL 442	Molecular Diagnostics	3
BIOL 443	Molecular Diagnostics Lab	1
BIOL 495	Senior Project in Medical Biotechnology	4

Concentration in Forensic Biology**21 Credit Hours**

Course Code	Course Title	No. of Credits
CHEM 420	Forensic Chemistry	3
CHEM 421	Forensic Chemistry Lab	1
BIOL 423	Advanced Molecular Biology and Forensic Sciences	3
BIOL 424	DNA Fingerprinting and Serology Lab	1
BIOL 462	Crime Scene Investigation	3
BIOL 464	Criminal Justice	3
BIOL 473	Bioinformatics and Computational Biology	3
BIOL 496	Senior Project in Forensic Biology	4

Program Free Electives**6 Credits Hours**

Course Code	Course Title	No. of Credits
	Free Electives	6

B.S. Degree in Biotechnology with Concentrations

Proposed 4 Year Schedule

Year 1 Semester 1			Year 1 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
BIOL 112	University Biology I	3	BIOL 114	University Biology II	3
BIOL 113	University Biology I Lab	1	BIOL 115	University Biology II Lab	1
MATH 111	Calculus for the Life Sciences	3	MEST 100	Introduction to Islam in World Culture	3
ENGL 101	Composition	3	STAT 100	Introductory Probability and Statistics	3
CHEM 111	Principles of General Chemistry	3	CHEM 120	Analytical Chemistry	3
CHEM 112	Principles of General Chemistry Lab	1	CHEM 121	Analytical Chemistry Lab	1
UNIV 100	University First-Year Transition	1	Gen Ed	Social Sciences Course	3
Subtotal =		15	Subtotal =		17
Year 2 Semester 1			Year 2 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
BIOL 230	General Microbiology	3	BIOL 270	General Genetics	3
BIOL 231	General Microbiology Lab	1	BIOL 271	General Genetics Lab	1
UNIV 200	Innovation, Entrepreneurship, & Sustainability	3	BIOL 250	Biochemistry I	3
CHEM 215	Organic Chemistry I	3	BIOL 251	Biochemistry I Lab	1
CHEM 216	Organic Chemistry I Lab	1	PHYS 110	University Physics I	3
ITEC 103	Fundamentals of Information Technology	3	PHYS 111	University Physics I Lab	1
			PHIL 100 or ENGL 200	Critical Thinking and Reasoning or Advanced Composition	3
Subtotal =		14	Subtotal =		15
Year 3 Semester 1			Year 3 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
BIOL 350	Biochemistry II	3	BIOL 380	Biotechnology and Genetic Engineering	3
UAES 200	Survey of United Arab Emirates Studies	3	BIOL 381	Biotechnology Lab Methods and Techniques	1
BIOL 330	Applied and Industrial Microbiology	3	BIOL 356	Virology	3
BIOL 331	Techniques in Applied and Industrial Microbiology	1	Program Elective	Biotechnology Elective	3
Elective	Free Elective	3	BIOL 100 Or	Humankind in a Biological World or Sustainability and Human Environment Relations	3

			ENVS 101		
ARAB 101 or ARAB 110	Arabic Language and Culture for Non- Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3	Elective	Free Elective	3
Subtotal =		17	Subtotal =		16
Year 3, Summer Semester					
Course No.	Description	CR			
UNIV 390	Internship	3			
Year 4 with Concentration in Medical Biotechnology					
Year 4 Semester 1			Year 4, Semester 2		
Course No.	Description	CR	Course No.	Description	CR
BIOL 435	Stem Cell Biology and Regenerative Medicine	3	BIOL 440	Medical Microbiology	3
BIOL 442	Molecular Diagnostics	3	BIOL 441	Medical Microbiology Lab	1
BIOL 443	Molecular Diagnostics Lab	1	BIOL 436	Immunology	3
BIOL 493	Research Methodology in the Biological Sciences	2	BIOL 411	Drug Design and Drug Discovery	3
BIOL 450	Bioethics	3	BIOL 494	Senior Project Medical Biotechnology	4
BIOL 491	Senior Seminar	1			
Program Elective	Biotechnology Elective	3			
Subtotal =		16	Subtotal =		14
Year 4 with Concentration in Cell and Molecular Biotechnology					
BIOL 423	Advanced Molecular Biology and Forensic Sciences	3	BIOL 420	Molecular Biology and Genetics	3
BIOL 430	Cell Culture Theory and Technology	3	BIOL 421	Molecular Biology Recombinant DNA Lab	1
BIOL 431	Cell Culture Theory and Technology Lab	1	BIOL 434	Cell Communication and Signal Transduction	3
BIOL 493	Research Methodology in the Biological Sciences	2	BIOL 473	Bioinformatics and Computational Biology	3
BIOL 450	Bioethics	3	BIOL 495	Senior Project Cell and Molecular Biotechnology	4
BIOL 491	Senior Seminar	1			
Program Elective	Biotechnology Elective	3			
Subtotal =		16	Subtotal =		14

Year 4 with Concentration in Forensic Biology					
BIOL 423	Advanced Molecular Biology and Forensic Science	3	CHEM 420	Forensic Chemistry	3
BIOL 424	DNA Fingerprinting and Serology Lab	1	CHEM 421	Forensic Chemistry Lab	1
BIOL 462	Crime Scene Investigation	3	BIOL 464	Criminal Justice	3
BIOL 493	Research Methodology in the Biological Sciences	2	BIOL 473	Bioinformatics and Computational Biology	3
BIOL 450	Bioethics	3	BIOL 496	Senior Project Forensic Biology	4
BIOL 491	Senior Seminar	1			
Program Elective	Biotechnology Elective	3			
Subtotal =		16	Subtotal =		14
TOTAL CREDIT HOURS 127					



SCHOOL OF BUSINESS

SCHOOL OF BUSINESS

Bachelor of Science in Business Administration

Program Description

The Bachelor Science in Business Administration (BSBA) at AURAK aims to provide a broad background of study for students who wish to pursue a career in one of many business sectors - retail, hospitality, information technology, manufacturing, or non-profit organizations.

The BSBA program provides students with tools and knowledge valuable to any business and with practice in developing skills such as teamwork, analysis of numerical data, and communicating in writing and orally.

BSBA majors may choose to specialize in accounting, finance, human resource management, marketing, business analytics or hospitality and tourism management by taking seven major required courses and three major electives in one of these areas or choose to be a generalist, by taking a combination of courses from all the six specialist areas.

The Bachelor of Science in Business Administration is overseen by the School of Business and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The mission of the Business Administration Program is to prepare graduates who are knowledgeable, skilled, globally oriented, and ethical for entrance into business, government, other professions, and/or graduate programs.

Program Educational Objectives

The overall aims of the program are to:

1. Develop a breadth of knowledge across business functions
2. Create an understanding of ethical and social responsibility
3. Demonstrate clarity of written and oral communication using appropriate technology
4. Develop teamwork and leadership skills
5. Enable Professional reflection and self-development
6. Develop critical thinking, analytical and problem-solving skills
7. Develop proficiency in a chosen business discipline

Program Learning Outcomes

On completion of the program graduates will be able to demonstrate

PLO1: (CORE) Breadth of Business Knowledge

Critically expand on the basic principles of economics, accounting, finance, management, information systems, marketing and operations in the context of a global economy.

PLO2: (CORE) Ethical Reasoning

Apply concepts and theories of ethics and social responsibility to practical business dilemmas, recognizing the implications of management decisions for the interests of key internal and external stakeholders.

PLO3: (CORE) Communication Skills

Demonstrate competency in presentation and writing skills using the latest business communication tools.

PLO4: (CORE) Teamwork

Effectively work in teams and take the lead in team initiatives.

PLO5: (CORE) Professional Reflection

Engage in critical self-analysis and professional reflection about potential roles in a business context.

PLO6: (CORE) Critical Thinking and Problem Solving

Evaluate business situations and critique managerial decisions, using financial statements, statistical tools, and other appropriate methods to organize, analyze and present data.

Major Learning Outcomes (MLOs): Disciplinary Proficiency

Achievement of the learning outcomes specified for one of the BSBA majors.

Degree Requirements

The Bachelor of Science in Business Administration (BSBA) program includes forty-eight (48) credits of core courses in business administration covering a broad knowledge of business areas and functions, emphasizing global business environment and practices.

BSBA majors may choose to specialize in accounting, finance, human resource management, marketing, business analytics or hospitality and tourism management by taking 21 credits of major required courses and nine credits of major electives in one of these areas.

Additionally, students must fulfill the General Education requirements with thirty-one credits and complete 9 credits of general business electives and a three-credit general electives to complete the 121-credit program.

The Bachelor of Science in Business Administration degree requires the completion of the following degree requirements:

	BSBA Major in Accounting/ Business Analytics/ Finance/ HRM/ Marketing/ Hospitality & Tourism Management	BSBA – Generalist
Program Requirements	No. of Credits	No. of Credits
University General Education Requirements	31	31
School Requirements	48	48
Major Requirements	21	-
Major Elective Requirements	9	-

Business Electives	9	39
Free Electives	3	3
Total	121	121

University General Education Component Requirements

31 Credits

The General Education Component requires undergraduate students to take 31-34 credits in designated categories. To review the General Education Component requirements, please refer to the section of the catalog entitled, "General Education Component." ***The fifth writing intensive course for the BS in Business Administration is BUSN 301 Business Communication.***

Note that those seeking the BS of Business Administration are required to *take ECON 103 Principles of Economics* to satisfy the Social and Behavioral Sciences requirements *and STAT 100 Statistics* to satisfy the mathematics requirement for the Natural Sciences. For further guidance in selecting courses to satisfy the General Education Component requirements, please consult your academic advisor.

Course Code	Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
ITEC 103	Fundamentals of Information Technology	3
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
UAES 200	Survey of United Arab Emirates Studies	3
ECON 103	Principles of Microeconomics	3
STAT 100	Introduction to Probability & Statistics	3
Gen Ed Course	Natural Science Course	3

School of Business Core Course Requirements

48 Credits

Course Code	Title	Pre-Requisites	Credits
ACCT 204	Principles of Financial Accounting	ECON 103	3
ACCT 205	Principles of Managerial Accounting	ECON 103	3
BUAN 301	Analytics for Decision Making	OPMT 311	3
BUSN 301	Business Communication (A) (Writing Intensive)	ENGL 101	3
BUSN 304	Commercial Law	Junior Standing	3
BUSN 306	International Business (A)	ECON 104	3
BUSN 307	Business Research Methodology (A)	Junior Standing	3

BUSN 312	Public Policy and Business Ethics (A)	ECON 103	3
BUSN 498	Strategic Management (A)	Senior Standing	3
ECON 104	Macroeconomic Policy	ECON 103	3
FNAN 300	Principles of Business Finance	ACCT 204	3
MGMT 301	Principles of Management	Sophomore	3
MGMT 401	Organizational Behavior	Senior Standing	3
MIST 301	Introduction to Business Information Systems	Sophomore	3
MKTG 301	Principles of Marketing	ECON 103	3
OPMT 311	Methods and Models of Management Sciences	STAT 100	3

(A) – Applied Course

Major Requirements

21 credits

BSBA Major Specialization

To complete a major, graduates must complete thirty (30) credits in the chosen discipline, twenty-one (21) credits from the major required courses, and nine (9) credits from major elective courses

BSBA (Major in Accounting)

The BSBA (Major in Accounting) is a specialized major within the BSBA program that complements general education and core courses taken by all students within the BSBA by providing the student tools for business accounting used by financial managers. The degree is useful for students who wish to pursue careers in business, especially for company accountants, or auditors.

Accounting Major Program Learning Outcomes (AMLO)

On completion of a BSBA (Major in Accounting) students will be able to:

AMLO 1: Evaluate, aggregate, and synthesize knowledge relating to the accounting function from diverse sources for planning, controlling and performance evaluation;

AMLO 2: Assess how auditing and fiscal control measures can be implemented in manual and electronic accounting information systems: and

AMLO 3: Prepare, analyze, and interpret financial statements using International Financial Reporting Standards on accounting practice for business organizations and other corporate entities.

BSBA (Major in Accounting) Required Courses

21 Credits

Course	Course Title	Credits	Pre-requisites
ACCT 311	Managerial and Cost Accounting (A)	3	ACCT 205
ACCT 331	Intermediate Accounting I (A)	3	ACCT 204
ACCT 332	Intermediate Accounting II (A)	3	ACCT 331
ACCT 361	Accounting Information Systems (A)	3	ACCT 331
ACCT 390	Internship in Accounting (A)	3	Senior Standing
ACCT 412	Auditing (A)	3	ACCT 331
ACCT 444	Accounting Project Based Internship (A)	3	ACCT 331

(A) – Applied Course

BSBA (Major in Accounting) Elective courses (Any 3 courses to be taken)

9 Credits

Course Code	Course Title	Credits	Pre-requisites
ACCT 352	Taxation and Managerial Decisions (A)	3	ACCT 331
ACCT 472	International Accounting (A)	3	ACCT 331
ACCT 491	Special Topics in Accounting (A)	3	ACCT 331
FNAN 303	Financial Analysis, Forecasting, and Valuation (A)	3	ACCT 204

(A) – Applied Course

BSBA (Major in Finance)

The BSBA (Major in Finance) is a specialized major within the BSBA program that complements general education and core courses taken by all students within the BSBA by providing a sound base of knowledge of the field of Finance. The BSBA (Major in Finance) provides tools for analysis of business, and investment. The degree is useful for students who wish to pursue careers in government, services, and manufacturing in corporate finance.

Finance Major Program Learning Outcomes (FMLO)

On completion of a BSBA (Major in Finance) students will be able to:

FMLO 1: Assess the salience to specific decision scenarios of the structures and regulatory frameworks encompassing financial institutions and markets

FMLO 2: Evaluate, aggregate and synthesize knowledge relating to financial operations from diverse sources for the benefit of the organization; and

FMLO 3: Formulate and craft effective solutions to finance-related business problems, based on sound application of underlying financial principles.

BSBA (Major in Finance) Required Courses

21 Credits

Course Code	Title	Credits	Pre-requisite
FNAN 303	Financial Analysis, Forecasting, and Valuation	3	ACCT 204
FNAN 304	Risk Management and Insurance (A)	3	FNAN 303
FNAN 321	Financial Institutions (A)	3	FNAN 300
FNAN 390	Internship in Finance (A)	3	Senior Standing
FNAN 401	Advanced Financial Management (A)	3	FNAN 303
FNAN 411	Investment Analysis and Portfolio Management	3	STAT 100, FNAN
FNAN 444	Finance Project Based Internship (A)	3	FNAN 303

(A) – Applied Course

BSBA (Major in Finance) Elective courses (Any 3 courses to be taken)

9 Credits

Course	Title	Credits	Pre-requisites
ACCT 331	Intermediate Accounting I (A)	3	ACCT 204 & Sophomore Status
FNAN 412	Futures and Options Markets (A)	3	FNAN303, &FNAN411, OPMT311

FNAN 421	Money and Capital Markets (A)	3	FNAN 303
FNAN 431	International Financial Management (A)	3	FNAN 303

(A) – Applied Course

BSBA (Major in Human Resource Management)

The BSBA (Major in Human Resource Management) is a specialized major within the BSBA program that is intended to provide a broad background of study for students who wish to pursue a human resource career in one of the many business sectors: retail, hospitality, information technology, manufacturing, or non-profit organizations. The BSBA (Major in Human Resource Management) provides students with tools and knowledge valuable to any business and with practice in developing skills such as teamwork, analysis of numerical data, and communicating in writing and orally.

Human Resource Management Major Program Learning Outcomes (HMLO)

On completion of a BSBA (Major in Human Resource Management) students will be able to

HMLO 1: Assess sustainable paths to develop the human resource within an organization;

HMLO 2: Evaluate, aggregate, and synthesize knowledge relating to people management functions from diverse sources for the benefit of the organization; and

HMLO 3: Formulate and craft viable solutions to people management problems at both higher strategic and operational levels.

BSBA (Major in Human Resource Management) Required Courses

21 Credits

Course Code	Course Title	Credits	Pre-requisites
MGHR 301	Human Resource Management	3	MGMT 301
MGHR 302	Compensation Management	3	MGHR 301
MGHR 304	Human Resource Recruitment & Selection (A)	3	MGHR 301
MGHR 390	Internship in Human Resource Management (A)	3	Senior Standing
MGHR 403	Cross Cultural and International Human Resource Management (A)	3	MGHR 301
MGHR 431	Employee Relations	3	MGHR 301
MGHR 444	HRM Project Based Internship (A)	3	MGHR 301

(A) – Applied Course

BSBA (Major in Human Resource Management) Elective Courses

9 Credits

Course Code	Course Title	Credits	Pre-requisites
MGHR 461	Diversity in organization	3	MGMT 301
MGHR 462	Strategic Human Resource Management (A)	3	MGHR 301
MGHR 463	Negotiations in Organizations (A)	3	MGHR 301
MGHR 464	Training and Development (A)	3	MGMT 301

(A) – Applied Course

BSBA (Major in Marketing)

The BSBA (Major in Marketing) exposes students to the latest research and theory in global and multicultural marketing and examines research driven strategic advertising and promotion based on established analytics. Students are equipped with marketing knowledge and skills to develop specialized marketing professionals.

Marketing Major Program Learning Outcomes (MMLO)

On completion of a BSBA (Major in Marketing) students will be able to

MMLO 1: Apply salient marketing techniques to product/service positioning within market segments;

MMLO 2: Evaluate, aggregate, and synthesize knowledge relating to the marketing function from diverse sources for the benefit of the organization; and

MMLO 3: Formulate and craft integrated market plans and marketing strategies for various business domains.

BSBA (Major in Marketing) Required Courses

21 Credits

Course Code	Course Title	Credits	Pre-requisites
MKTG 312	Consumer behavior	3	MKTG 301
MKTG 313	Integrated Marketing Communications (A)	3	MKTG 301
MKTG 351	Marketing Research Techniques & Applications (A)	3	MKTG 301, OPMT 311
MKTG 390	Internship in Marketing (A)	3	Senior Standing
MKTG 412	Marketing of Services (A)	3	MKTG 301
MKTG 471	Marketing Strategy	3	Senior Standing
MKTG 444	Marketing Project Based Internship (A)	3	MKTG 301

(A) – Applied Course

BSBA (Major in Marketing) Elective Courses

9 Credits

Course Code	Course Title	Credits	Pre-requisites
MKTG 311	Sales Management	3	MKTG 301
MKTG 332	Retailing and E-Commerce (A)	3	MKTG 301
OPMT 405	Supply Chain Management	3	OPMT 311
MKTG 407	International Marketing (A)	3	MKTG 301

(A) – Applied Course

BSBA (Major in Business Analytics)

The BSBA (Major in Business Analytics) will equip the students to use the power of data science, big data, statistics, and machine learning to optimize business performance and to make the right decisions, organizations need people with expert skills and an analytical approach. The students will combine statistics, information technology, computer science, and business administration to come up with innovative solutions to problems in the business world.

Business Analytics Major Program Learning Outcomes (BMLO)

On completion of a BSBA (Major in Business Analytics) students will be able to

BMLO 1: Integrate information technologies with quantitative analysis and data science methods to extract business value from data sets, and to enable data-driven business decisions.

BMLO 2: Identify opportunities, needs and constraints for data analytics within specific organizational contexts.

BMLO 3: Demonstrate proficiency and creativity with the integrated use of analytics tools and techniques to formulate useful data analytics solutions.

BSBA (Major in Business Analytics) Required Courses

21 Credits

Course Code	Course Title	Credits	Pre-requisites
BUAN 311	Business Data Mining (A)	3	BUAN 301
BUAN 312	Visual Analytics (A)	3	BUAN 301
BUAN 390	Internship in Business Analytics (A)	3	Senior Standing
BUAN 401	Business Data Engineering (A)	3	MIST 301,
BUAN 411	Business Intelligence & Big Data Analytics	3	MIST 301,
MKTG 351	Marketing Research Techniques and Applications (A)	3	MKTG 301,
BUAN 444	Business Analytics Project Based Internship (A)	3	BUAN 301

(A) – Applied Course

BSBA (Major in Business Analytics) Elective Courses

9 Credits

Course Code	Course Title	Credits	Pre-requisites
ACCT 361	Accounting Information Systems (A)	3	ACCT 331
FNAN 303	Financial Analysis, Forecasting, and Valuation (A)	3	ACCT 204
OPMT 405	Supply Chain Management	3	OPMT 311
OPMT 301	Operations Management	3	Sophomore Standing

(A) – Applied Course

BSBA (Major in Hospitality and Tourism Management)

The BSBA (Major in Hospitality and Tourism Management) aims to provide a broad background of study for students who wish to pursue a career in hospitality and tourism management. The BSBA (Major in Hospitality and Tourism Management) program provides students with valuable tools and knowledge required in business. It also aims to provide students with skills such as teamwork, analysis of numerical data, and communicating in writing and orally. The BSBA (Major in Hospitality and Tourism Management) is offered in collaboration with Appalachian State University. Students will undertake one semester in a partner organization in the United States of America (USA).

Hospitality and Tourism Management Major Program Learning Outcomes (HTMLO)

On completion of a BSBA (Major in Hospitality and Tourism Management) students will be able to

HTMLO 1: Apply core hospitality and tourism management principles to an evaluation of hospitality and tourism systems, structures and processes;

HTMLO 2: Apply practical considerations to an evaluation of hospitality and tourism systems, structures and processes; and

HTMLO 3: Integrate an analysis of the external environment into an evaluation of management decisions in the hospitality and tourism sector.

BSBA (Major in Hospitality and Tourism Management) Required Courses **21 Credits**

Course Code	Course Title	Credits	Pre-requisites
HTMT 311	Survey of Hospitality and Tourism	3	None
HTMT 321	Hospitality Operations I – F&B	3	HTMT 311
HTMT 322	Hotel Operations (A)	3	HTMT 311
HTMT 390	Internship in Hospitality & Tourism Management (A)	3	Senior Standing
HTMT 411	Destination Management (A)	3	HTMT 311
HTMT 412	Meetings and Conventions (A) (C)	3	Completion of 84 Credits
HTMT 444	Hospitality and Tourism Management Project Based Internship (A)	3	HTMT 311

(A) – Applied Course, (C) Compulsory Course, (O) Optional Course

BSBA (Major in Hospitality and Tourism Management) Elective Courses **9 Credits**

Course Code	Course Title	Credits	Pre-requisites
HTMT 312	Sustainable Tourism	3	HTMT 311
BUSN 391	International Internship (A)	3	Completion of 90 Credits
MKTG 313	Integrated Marketing	3	MKTG 301
MKTG 412	Marketing of Services (A)	3	MKTG 301
MGHR 301	Human Resource Management (O)	3	MKTG 301

(A) – Applied Course, (C) Compulsory Course, (O) Optional Course

Business Elective Courses

9 Credits

Any three courses from the major requirements and major electives listed above other than the declared major required and major elective courses taken to complete a major.

Free Elective Courses

3 Credits

Any Course from other Programs and not already taken.

Other Paths to complete the 121 credit hours requirement for a BSBA degree

The Generalist Option

Instead of completing thirty (30) credits (21 + 9) of major elective courses in a specific major, students can complete thirteen (13) courses, 39 credit hours of business electives (including 2 internship courses in any of the majors) provided these comprise major required and major elective courses in any of the 6 majors.

TOTAL DEGREE REQUIREMENT

121 Credit Hours

Bachelor of Business Administration Four-Year Sample Schedule of Delivery

(Major in Accounting/Business Analytics/Finance/HRM/Marketing)

Year One, First Semester			Year One, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
ENGL 101	Composition	3	ECON 104	Principles of Macroeconomics	3
ECON 103	Principles of Microeconomics	3	UNIV 200	Innovation, Entrepreneurship, and Sustainability	3
ITEC 103	Introduction to Computing	3	ARAB 101 or ARAB110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
STAT 100	Introductory Probability & Statistics	3	PHIL 100	Critical Thinking & Reasoning	3
UNIV 100	University First-Year Transition	1	UAES 200	Survey of United Arab Emirates Studies	3
MEST 100	Introduction to Islam in World Culture	3			
Subtotal =		16	Subtotal =		15
Year Two, First Semester			Year Two, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
GEN ED	Any Natural Science Course	3	ACCT 205	Principles of Managerial Accounting	3
ACCT 204	Principles of Financial Accounting	3	MKTG 301	Principles of Marketing	3
MKTG 301	Principles of Marketing	3	BUAN 301	Analytics for Decision Making	3
FNAN 300	Principles of Business Finance	3	BUSN 301	Business Communication	3
OPMT 311	Methods and Models of Management Science	3	MIST 301	Introduction to Business Information Systems	3
Subtotal =		15	Subtotal =		15
Year Three, First Semester			Year Three, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
BUSN 304	Commercial Law	3	BUSN 307	Research Methodology	3
BUSN 312	Public Policy and Ethics	3	MAJOR	PROGRAM MAJOR 3	3
BUSN 306	International Business	3	MAJOR	PROGRAM MAJOR 4	3
MAJOR	PROGRAM MAJOR 1	3	Major Elective	MAJOR ELECTIVE 1	3
MAJOR	PROGRAM MAJOR 2	3	Elective	BUSINESS ELECTIVE 1	3
Subtotal =		15	Subtotal =		15
Year Three, Summer Session					
Major Internship 390	Major Specific Internship	3			

Year Four, First Semester			Year Four, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
BUSN 498	Strategic Management	3	Major Elective	MAJOR ELECTIVE 3	3
MGMT 401	Organizational Behavior	3	Elective	BUSINESS ELECTIVE 2	3
MAJOR	PROGRAM MAJOR 5	3	Elective	BUSINESS ELECTIVE 3	3
Major Elective	MAJOR ELECTIVE 2	3	Elective	FREE ELECTIVE	3
Major Specific 444	Major Specific Project Based Internship	3			3
Subtotal =		15	Subtotal =		12
121 Semester Credit Hours					

Bachelor of Business Administration Four-Year Sample Schedule of Delivery

(Major in Hospitality and Tourism Management)

Year One, First Semester			Year One, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
ENGL 101	Composition	3	ECON 104	Principles of Macroeconomics	3
ECON 103	Principles of Microeconomics	3	UNIV 200	Innovation, Entrepreneurship, and Sustainability	3
ITEC 103	Introduction to Computing	3	ARAB 101 or ARAB110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
STAT 100	Introductory Probability & Statistics	3	PHIL 100	Critical Thinking & Reasoning	3
UNIV 100	University First-Year Transition	1	UAES 200	Survey of United Arab Emirates Studies	3
MEST 100	Introduction to Islam in World Culture	3			
Subtotal =		16	Subtotal =		15
Year Two, First Semester			Year Two, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
GEN ED	Any Natural Science Course	3	ACCT 205	Principles of Managerial Accounting	3
ACCT 204	Principles of Financial Accounting	3	MKTG 301	Principles of Marketing	3
MKTG 301	Principles of Marketing	3	BUAN 301	Analytics for Decision Making	3
FNAN 300	Principles of Business Finance	3	BUSN 301	Business Communication	3
OPMT 311	Methods and Models of Management Science	3	MIST 301	Introduction to Business Information Systems	3
Subtotal =		15	Subtotal =		15
Year Three, First Semester			Year Three, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
BUSN 304	Commercial Law	3	BUSN 307	Research Methodology	3
BUSN 312	Public Policy and Ethics	3	Major Elective	MAJOR ELECTIVE 1	3
BUSN 306	International Business	3	Major Elective	MAJOR ELECTIVE 2	3
HTMT 321	Hospitality Operations I – F&B Management	3	Elective	BUSINESS ELECTIVE 2	3
Elective	BUSINESS ELECTIVE 1	3			
Subtotal =		15	Subtotal =		12
Year Three, Summer Session					

Major Internship 390	Major Specific Internship	3	
----------------------	---------------------------	---	--

Year Four, First Semester			Year Four, Second Semester		
BUSN 498	Strategic Management	3	HTMT 412	Meetings and Conventions	3
MGMT 401	Organizational Behavior	3	HTMT 411	Destination Management	3
HTMT 322	Hotel Operations	3	Major Elective	MAJOR ELECTIVE 3	3
HTMT 311	Survey of the Hospitality & Tourism Industry	3	Elective	BUSINESS ELECTIVE 3	3
Major Specific 444	Major Specific Project Based Internship	3	Elective	FREE ELECTIVE	3
Subtotal =		15	Subtotal =		15
121 Semester Credit Hours					

Bachelor of Business Administration Four-Year Sample Schedule of Delivery (Generalist)

Year One, First Semester			Year One, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
ENGL 101	Composition	3	ECON 104	Principles of Macroeconomics	3
ECON 103	Principles of Microeconomics	3	UNIV 200	Innovation, Entrepreneurship, and Sustainability	3
ITEC 103	Introduction to Computing	3	ARAB 101 or ARAB110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
STAT 100	Introductory Probability & Statistics	3	PHIL 100	Critical Thinking & Reasoning	3
UNIV 100	University First-Year Transition	1	UAES 200	Survey of United Arab Emirates Studies	3
MEST 100	Introduction to Islam in World Culture	3			
Subtotal =		16	Subtotal =		15
Year Two, First Semester			Year Two, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
GEN ED	Any Natural Science Course	3	ACCT 205	Principles of Managerial Accounting	3
ACCT 204	Principles of Financial Accounting	3	MKTG 301	Principles of Marketing	3
MKTG 301	Principles of Marketing	3	BUAN 301	Analytics for Decision Making	3
FNAN 300	Principles of Business Finance	3	BUSN 301	Business Communication	3
OPMT 311	Methods and Models of Management Science	3	MIST 301	Introduction to Business Information Systems	3
Subtotal =		15	Subtotal =		15
Year Three, First Semester			Year Three, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
BUSN 304	Commercial Law	3	BUSN 307	Research Methodology	3
BUSN 312	Public Policy and Ethics	3	Elective 3	Business Elective 3	3
BUSN 306	International Business	3	Elective 4	Business Elective 4	3
Elective 1	Business Elective 1	3	Elective 5	Business Elective 5	3
Elective 2	Business Elective 2	3	Elective 6	Business Elective 6	3
Subtotal =		15	Subtotal =		15
Year Three, Summer Session					
Major Internship 390 (Elective 7)	Major Specific Internship	3			

Year Four, First Semester			Year Four, Second Semester		
Course Code	Course Title	CR	Course Code	Course Title	CR
BUSN 498	Strategic Management	3	Elective 11	Business Elective 11	3
MGMT 401	Organizational Behavior	3	Elective 12	Business Elective 12	3
Elective 8	Business Elective 8	3	Elective 13	Business Elective 13	3
Elective 9	Business Elective 9	3	Free Elective	Free Elective (Any Gen Ed Course)	3
Major Specific 444 (Elective 10)	Major Specific Project Based Internship	3			3
Subtotal =		15	Subtotal =		12
121 Semester Credit Hours					



SCHOOL OF ENGINEERING

SCHOOL OF ENGINEERING

Bachelor of Architecture

Program Description

Bachelor of Architecture program is designed to prepare well educated, critical-thinkers and problem-solvers who possess and can effectively apply highly refined aesthetic, technical knowledge, are capable of facing professional and global challenges. Focused on sustainability challenges, which are as broadly cultural as they are technical, the program graduates are prepared to become credible stewards of the built and surrounding environment. The program leads students towards the application of principles of architecture as a science to analyze physical environments, components, and processes to identify problems and propose solutions through design and modeling. It navigates graduates to integrate into multidisciplinary teams as architects, to respect professional ethics, and through highly professional attitude contribute to the local, regional, and global sustainability.

The Bachelor of Architecture is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

Bachelor of Architecture program is focused on sustainability challenges and prepares graduates that are critical-thinkers, problem-solvers and capable of facing professional and global challenges. Equipped with highly refined aesthetic and technical knowledge, capable of respecting professional ethics and work in multidisciplinary teams, graduates can easily integrate into architectural practices for built environment problem solving.

Program Educational Objectives

The Bachelor of Architecture Program at AURAK produces graduates who are armed with knowledge and skills as articulated in its student outcomes to become:

1. Industrious architects who create healthy and safe spaces of different scale (buildings, cities) that are responsive to the environment and users.
2. Practitioners who value professional ethics and architecture business as influencing the built environment and human wellness.
3. Professionals capable of continuous learning to meet the requirements of altering built environment and comply with the regional/ national laws and regulations.
4. Designers experienced in evaluating solutions as it relates to building systems design, materials, technologies against project objectives, cost and building performance.
5. Team leaders capable of making decisions within architectural projects for universal design with minimal impact on environment.
6. Active contributors to the local community and the world through an integrative design process with a multidisciplinary team.

Program Learning Outcomes

Graduates of the Bachelor of Architecture program will have the ability to:

1. Employ critical thinking that takes into consideration alternative perspectives by analyzing, evaluating, and synthesizing ideas and information gathered through research grounded in acquired information.
2. Implement complex two and three-dimensional graphic representation techniques using a wide variety of manual and digital media as well as oral and written forms to reflect on and convey the architectural design process to a wide range of stakeholders.
3. Demonstrate a solid base of knowledge in history and theory grounded in the intellectual understanding of social, cultural, and economic aspects and through applied research by raising clear and precise questions to clarify and express information to reach logical conclusions and evaluate options against relevant design criteria, building standards, and program requirements.
4. Utilize and integrate technical skills and professional knowledge in comprehensive application of life safety, accessibility, and sustainability issues in integrating the principles of environmental stewardship and making appropriate design decisions of varying scales and levels of complexity through a technical design process based on appropriate documentation in a manner that is client-centered, sustainable, aesthetic, cost effective, and socially responsible.
5. Develop the capability to synthesize variable components into an integrated design solution through the employment of appropriate building materials, building systems, and construction practices which are environmentally friendly and based on research and design decisions for a variety of systems and levels of complexity.
6. Show the ability to lead teams of stakeholders collaboratively within the process of understanding, developing, and applying solutions to problems in the built as well as the natural environments, utilizing knowledge and parameters of professional practice associated with ethical, legal, financial and social responsibilities.
7. Apply reasoning skills including math, physics, and logic to investigate problems related to gravity and other forces in structural systems, thermal heat gain and loss in buildings, material quantity estimates, budget management, and life-cycle cost analysis.

Degree Requirements

Students complete the Bachelor of Architecture by completing twelve (12) courses (32 credit hours) of University General Education, five (5) course (13 credits) of Faculty Requirements, three (3) course (9 credits) of Department of Architecture requirements, thirty-one (31) courses (101 credits) of Architecture Major requirements and two (2) courses (6 credits) of Architecture Major Electives. Total number of credits – 161.

The required courses are:

The University General Education Component Requirements

32 Credit Hours

Course Code	Title	Credits
MATH 113	Calculus I	3
ENGL 101	Composition (Writing Intensive)	3
UNIV 100	University First-Year Transition	1
CSCI 112	Introduction to Computer Programming	3
CSCI 113	Introduction to Computer Programming Lab	1
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
Required Electives – The field of Art and Humanities (6 credits)		
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
Electives – The field of Social and Behavioral Sciences (6 credits)		
UAES 200	Survey of United Arab Emirates Studies (Required)	3
PSYC 100	Introduction to Psychology (Or another course from the same list)	3
Required Elective – The field of natural Sciences (3 credits)		
ENVS 102	Sustainability and Human Environment Relations	3

The fifth writing intensive course for the Bachelor of Architecture is ARCH 591 Graduation Project Thesis Research.

Faculty (Engineering) Requirements

13 Credit Hours

Course Code	Title	Credits
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab.	1
IENG 321	Engineering Economy	3
ENGR 390	Internship 1	3
ENGR 391	Internship 2	3

Department Requirements

9 Credit Hours

Course Code	Title	Credits
INDS 112	Design Communication 1: Sketching and Drawing for Interiors	3
INDS 121	Interior Design Process, Human Factors and Ergonomic	3
INDS 122	Design Communication 2: Digital Media for Interiors	3

Major Compulsory Requirements

101 Credit Hours

Course Code	Title	Credits
ARCH 121	Basic Design 1	4
ARCH 122	Basic Design 2	4
ARCH 223	Computer Architectural Drawing	3
ARCH 221	Architectural Design Studio 1	4

ARCH 222	Architectural Design Studio 2	4
ARCH 231	Building Construction 1	3
ARCH 232	Building Construction 2	3
ARCH 241	History and Theory of Architecture 1	3
ARCH 242	History and Theory of Architecture 2	3
ARCH 321	Architectural Design Studio 3	4
ARCH 322	Architectural Design Studio 4	4
ARCH 331	Building Construction 3	3
ARCH 336	Construction Drawing	3
ARCH 341	History and Theory of Architecture 3	3
ARCH 342	History and Theory of Contemporary Architecture	3
ARCH 372	Sustainable Architecture	3
ARCH 421	Architectural Design Studio 5	4
ARCH 422	Architectural Design Studio 6	4
ARCH 454	Urban Planning	3
ARCH 457	Urban Design	3
3ARCH 463	Landscape Architecture	3
ARCH 466	Building Utilities II: Illumination, Acoustics, and Electrical Building Services	3
ARCH 485	Professional Practice	3
ARCH 487	Project Management	3
ARCH 521	Architectural Design Studio 7	4
ARCH 591	Graduation Project Thesis Research (Writing Intensive)	2
ARCH 592	Graduation Project Design	5
CIEN 216	Structures for Architecture 1	3
CIEN 261	Surveying	1
CIEN 316	Structures for Architecture 2	3
MENG 468	Building Utilities: HVAC and Mechanical Building Services	3

Major Electives

6 Credit Hours

Course Code	Title	Credits
ARCH 449	Technology of the Built Environment	3
ARCH 456	Sustainable Housing	3
ARCH 459	Conservation of Historic Environment	3
ARCH 473	Environmental Control	3

Bachelor of Architecture Five-Year Sample Schedule

Year 1 Semester 1			Year 1 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
ARCH 121	Basic Design 1	4	ARCH 122	Basic Design 2	4
MATH 113	Calculus I	3	INDS 121	Interior Design Process, Human Factors and Ergonomics	3
INDS 112	Design Communication 1: Sketching and Drawing for Interiors	3	INDS 122	Design Communication 2: Digital Media for Interiors	3
ENGL 101	Composition	3	PHYS 110	University Physics I	3
UNIV 100	University First-Year transition	1	PHYS 111	University Physics I Lab.	1
			CSCI 112	Introduction to Computer Programming	3
			CSCI 113	Introduction to Computer Programming Lab.	1
Subtotal =		14	Subtotal =		18
Year 2 Semester 1			Year 2 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
ARCH 221	Architectural Design Studio 1	4	ARCH 222	Architectural Design Studio 2	4
ARCH 231	Building Construction 1	3	ARCH 232	Building Construction 2	3
ARCH 241	History & Theory of Architecture 1	3	ARCH 242	History & Theory of Architecture 2	3
CIEN 216	Structures for Architecture 1	3	CIEN 261	Surveying	1
ARCH 223	Computer Architectural Drawing	3	ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
			PSYC 100	Introduction to Psychology (or other from the same list)	3
Subtotal =		16	Subtotal =		17
Year 3 Semester 1			Year 3 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
ARCH 321	Architectural Design Studio 3	4	ARCH 322	Architectural Design Studio 4	4
ARCH 331	Building Construction 3	3	ARCH 336	Construction Drawings	3
ARCH 341	History and Theory of Architecture 3	3	ARCH 342	History and Theory of Contemporary Architecture	3
CIEN 316	Structures for Architecture 2	3	ARCH 372	Sustainable Architecture	3
ENVS 102	Sustainability and Human-Environment Relations	3	ARCH 000	Technical Elective 1	3
Subtotal =		16	Subtotal =		16
Summer Semester					
Code	Course Title	Credits			
ARCH 390	Internship I	3			
Year 4 Semester 1			Year 4 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
ARCH 421	Architectural Design Studio 5	4	ARCH 422	Architectural Design Studio 6	4
ARCH 463	Landscape Architecture	3	ARCH 466	Building Utilities II: Illumination, Acoustics, and Electrical Building Services	3
MENG 468	Building Utilities I: HVAC and Mechanical Building Services	3	ARCH 000	Technical Elective 2	3
MEST 100	Introduction to Islam in World Culture	3	MENG 422	Building Utilities II: Illumination, Acoustics, and Electrical Building Services	3
ARCH 457	Urban Design	3	ENGL 200 or PHIL 100	Advanced Composition or Critical Thinking & Reasoning	3
			UAES 200	Survey of United Arab Studies	3
Subtotal =		16	Subtotal =		16
Summer Semester					
Course No.	Description	CR			
ARCH 391	Internship 2	3			
Year 5 Semester 1			Year 4 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
ARCH 521	Architectural Design Studio 7	4	ARCH 592	Graduation Project Design	5
ARCH 591	Graduation Project Thesis Research	2	IENG 321	Engineering Economy	3
ARCH 454	Urban Planning	3	ARCH 485	Professional Practice	3
ARCH 487	Project Management	3	UNIV 200	Innovation and Entrepreneurship	3
Subtotal =		12	Subtotal =		14
TOTAL CREDIT HOURS 162					

Bachelor of Arts in Interior Design

Program Description

The Bachelor of Art in Interior Design Degree is a comprehensive experience that provides those who are creative excellent learning opportunities in Interior Design theory and practice. The study for the degree last nine semesters (four academic years) and equips graduates with substantial knowledge and skills to work in multidisciplinary teams for creating healthy indoor environments within diverse historical, social, cultural, geographical and economic contexts. Capable of critical thinking, continuous self-development and lifelong learning, graduates are well prepared to meet the challenges of the ever-changing world and building sector, evaluate design theories and solve the problems related to indoor spaces, human and environmental wellbeing.

The Bachelor of Arts in Interior Design is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

Bachelor of Art in Interior Design at AURAK prepares students for continuous self-development and lifelong learning and equips them with professional, competitive skills and capability of producing contextually appropriate design solutions that meet the requirements of the rapidly developing building sector and in accordance with the cultural, social, and political sensibility, human and environmental wellbeing, safety and aesthetic values.

Program Educational Objectives

The Bachelor of Arts in Interior Design at AURAK produces graduates who are armed with knowledge and skills as articulated in its student outcomes to become:

1. Assiduous designers who engage in lifelong learning, continuous self-development and value business ethics to meaningfully develop their careers and contribute to the human and environmental wellbeing.
2. Problems-solvers with advanced knowledge and specialized skills in interior design who are capable of working in multidisciplinary team and in diverse historical, social, cultural, geographical and economic contexts.
3. Designers capable of evaluating design theories and philosophies conserving indoor spaces, built and natural environment and make data-driven design decisions.
4. Wellness practitioners that are well equipped with knowledge in codes and regulations related to the built environment and can face the challenges of the interior design industry and positively contribute to it.
5. Professionals armed with skills in latest design technologies and communication media capable of producing interior design projects that satisfy the requirements of the clients and users.

Program Learning Outcomes

Graduates of the program will have the ability to:

1. Value business ethics and interior design practice as influenced by the regional and global market and integral part of multidisciplinary teamwork for building environment problem solving, space quality enhancement and human wellbeing.
2. Appraise global context and its impact on interior design practice and construction decisions, using knowledge and theories related to the history of interior design, decorative arts, material culture and architecture in relationship to the social, political and physical factors that affect the changes of the built environment.
3. Evaluate theories related to design elements and principles, space organization and produce design problems solutions throughout the phases of design process for their two- and three-dimensional representation via various media and communication techniques.
4. Apply philosophies concerning interrelations between building, nature, virtual and technological environments as influencing human experience, behavior, wellbeing, and performance, considering human factors, ergonomics and recommend design solutions for building systems to contribute human wellbeing and environmental safety.
5. Develop appropriate color and natural/artificial lighting schemes for supporting design ideas with minimal impact on the environment through use of color and light doctrines that are dependent on context and influencing space perception and human wellbeing.
6. Compare properties, safety, and life cycle cost of furnishings, objects, materials and finishes, fabrication, installation, and maintenance methods for supporting design project criteria while minimizing the impact on human and environmental wellbeing.
7. Assess the impact of construction on the environment and recommend best design solutions; develop interior construction drawings and documents for the given project scope; appraise legislations concerning built and natural environments human wellbeing and universal design.

Degree Requirements

Student complete Bachelor of Art in Interior design by completing eleven (11) courses (31 credit hours) of University General Education, two (2) course (6 credits) of Faculty Requirements, four (4) courses (14 credits) of Department of Architecture requirements, twenty-two (22) courses (72 credits) of Interior Design Major requirements and two (2) courses (6 credits) of Interior Design Major Electives. The total number of credits is 129.

University General Education Requirement

31 Credits

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BA in Interior Design is INDS 412 Graduation Final Thesis Research.**

Course Code	Title	Credits
Core – (16 credits)		
STAT 100	Introductory Probability and Statistics	3
ENGL 101	Composition (Writing Intensive)	3
UNIV 100	University First-Year Transition	1
ITEC 103	Fundamentals of Information Technology	3
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
Electives – The field of Arts and Humanities (6 credits)		
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
Electives – The field of Social and Behavioral Sciences (6 credits)		
UAES 200	Survey of United Arab Emirates Studies	3
PSYC 100	Introduction to Psychology	3
Electives – The field of Natural Sciences – (3 credits)		
ENVS 102	Sustainability and Human Environment Relations	3

Faculty (Engineering) Requirements

6 Credits

Course Code	Title	Credits
INDS 390	Internship 1	3
INDS 391	Internship 2	3

Department (Architecture) Requirements

14 Credits

Course Code	Title	Credits
ARCH 121	Basic Design 1	4
ARCH 122	Basic Design 2	4
ARCH 223	Computer Architectural Drawing	3
ARCH 341	History and Theory of Architecture 3	3

Major (Interior Design) Requirements

72 Credits

Course Code	Title	Credits
INDS 111	Color Theory and Rendering	3
INDS 112	Design Communication 1: Sketching and Drawing for Interiors	3
INDS 121	Interior Design Process: Human Factors and Ergonomics	3
INDS 122	Design Communication 2: Digital Media for Interiors	3
INDS 211	Interior Design Studio 1	4
INDS 212	History of Art and Interior Design 1	3
INDS 213	Interior Construction Methods, Materials, and Finishes 1	3
INDS 221	Interior Design Studio 2	4
INDS 222	History of Art and Interior Design 2	3

INDS 223	Interior Construction Methods, Materials, and Finishes 2	3
INDS 224	Sustainable Approaches to Interior Design	3
INDS 311	Interior Design Studio 3	4
INDS 312	Construction Drawings and Interior Detailing	3
INDS 313	Environmental Systems 1: Indoor Comfort and Human Wellbeing	3
INDS 321	Interior Design Studio 4	4
INDS 322	Furniture Design	3
INDS 323	Environmental Systems 2: Design with Light and Sound	3
INDS 411	Interior Design Studio 5	4
INDS 412	Graduation Final Thesis Research (<i>Writing Intensive</i>)	2
INDS 414	Professional Practice 1: Codes and Regulations	3
INDS 421	Graduation Project Design	5
INDS 422	Professional Practice II: Ethics and Project Management	3

Major (Interior Design) Electives

6 Credits

Course Code	Title	Credits
INDS 351	Parametric Design	3
INDS 352	Sketching and Painting	3
INDS 353	Landscape and Garden Design	3
INDS 451	Portfolio Design	3
INDS 452	Photography	3
INDS 453	Advanced BIM: Scheduling and Coordination	3
INDS 454	Research Methods for Interior Design	3

Bachelor of Arts in Interior Design Four-Year Study Plan

Year 1 Semester 1			Year 1 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
ARCH 121	Basic Design 1	4	ARCH 122	Basic Design 2	4
INDS 111	Color Theory and Rendering	3	INDS 121	Interior Design Process, Human Factors and Ergonomics	3
INDS 112	Design Communication 1: Sketching and Drawing for Interiors	3	INDS 122	Design Communication 2: Digital Media for Interiors	3
STAT 100	Introductory Probability and Statistics	3	ITEC 103	Fundamental of Information Technology	3
ENGL 101	Composition	3	ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
UNIV 100	University First-Year Transition	1			
Subtotal =		17	Subtotal =		16
Year 2 Semester 1			Year 2 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
INDS 211	Interior Design Studio 1	4	INDS 221	Interior Design Studio 2	4
INDS 212	History of Art and Interior Design 1	3	INDS 222	History of Art and Interior Design 2	3
INDS 213	Interior Construction Methods, Materials and Finishes	3	INDS 223	Interior Construction Methods, Materials and Finishes 2	3
ARCH 223	Computer Architectural Drawing	3	INDS 224	Sustainable Approaches in Interior Design	3
ENVS 102	Sustainability and Human-Environment Relations	3	MEST 100	Introduction to Islam in World Culture	3
Subtotal =		16	Subtotal =		16
Year 2 Summer Semester					
Code	Course Title	CR			
INDS 390	Internship 1	3			
Year 3 Semester 1			Year 3 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
INDS 311	Interior Design Studio 3	4	INDS 321	Interior Design Studio 4	4
INDS 312	Construction Drawing and Interior Detailing	3	INDS 322	Furniture Design	3
INDS 313	Environmental Systems 1: Indoor Comfort and Human Wellbeing	3	INDS 323	Environmental Systems 2: Design with Light and Sound	3
ARCH 341	History and Theory of Architecture	3	PHIL 100 Or ENGL 200	Critical Thinking and Reasoning or Advanced Composition	3
UAES 200	Survey of United Arab Emirates Studies	3	INDS	Elective 1	3
Subtotal =		16	Subtotal =		16
Summer Semester					
Course No.	Description	CR			
INDS 391	Internship 2	3			
Year 4 Semester 1			Year 4 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
INDS 411	Interior Design Studio 5	4	INDS 421	Graduation Project Design	5
INDS 412	Graduation Final Thesis Research	2	INDS 422	Professional Practice: Ethics and Project Management	3
INDS 414	Professional Practice 1: Codes and Regulations	3	UNIV 200	Innovation, Entrepreneurship, and Sustainability	3
PSYC100	Introduction of Psychology	3	INDS	Elective 2	3
Subtotal =		12	Subtotal =		14
TOTAL CREDIT HOURS 129					

Bachelor of Science in Artificial Intelligence

Program Description

The Bachelor of Science in Artificial Intelligence (BSAI) degree program at AURAK is designed to equip learners with an in-depth knowledge of computer science theory and a thorough understanding of artificial intelligence (AI) and data science technologies. Students are exposed to state-of-the-art AI-related methods relevant to knowledge representation and reasoning. Such techniques will enhance human capabilities through automated analysis of complex inputs and facilitate automated decision making. The program includes techniques and topics such as machine learning, computer vision, robotics, data analysis and visualization, natural language processing, and data mining.

The global demand for AI and data science professionals is on the rise due to the integration of the field into most industries. This need is further exacerbated by a worldwide shortage of AI talent.

The Bachelor of Science in Artificial Intelligence is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The mission of the Bachelor of Science in Artificial Intelligence program is to graduate students with the knowledge and skills to enable them to design and develop computer systems and data models using the latest advances in the field and hence become effective leaders capable of utilizing artificial intelligence and data science locally and globally across various domains.

Program Educational Objectives

A few years after earning their degree, our graduates will be:

1. Successful professionals and innovators in theoretical and practical areas of computer science, artificial intelligence, and data science;
2. Engaged in creating a positive technological impact with a sufficient awareness of the ethical, legal, and security issues related to computing, artificial intelligence, and data science; and
3. Equipped with the skills required for professional practice including functioning in teams and communicating effectively.

Program Learning Outcomes

Graduates of the program will have the ability to:

1. Analyze a complex computing problem and to apply principles of computing, artificial intelligence, statistics, and other relevant disciplines to identify solutions.
2. Design, implement, and evaluate computer science or artificial intelligence solutions to meet a given set of computing requirements in the context of the program's discipline.
3. Communicate effectively in a variety of professional contexts.
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.

6. Apply computer science and artificial intelligence theory and software development fundamentals to produce computing-based solutions.
7. Build, apply, and evaluate data science models to solve problems using theoretical fundamentals.

Degree Requirements

The BS in Artificial Intelligence requires the completion of 128 credits in the following areas:

Area	Credit
University General Education Requirements	32
School of Engineering Requirements	26
Artificial Intelligence Program Requirements	70 (61 compulsory & 9 technical electives)

General Education Requirements

32 Credits

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, “General Education Component.” You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS in Artificial Intelligence is CSAI 451 Machine Learning Lab.**

Course Code	Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
CSCI 112	Introduction to Computer Programming	3
CSCI 113	Introduction to Computer Programming Lab	1
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
UAES 200	Survey of United Arab Emirates Studies	3
MATH 113	Calculus I	3
Gen Ed Course	Social and Behavioral Sciences Course	3
Gen Ed Course	Natural Sciences Course	3

School of Engineering Requirements

26 Credits

Course Code	Title	Credits
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab	1
MATH 114	Calculus II	3
MATH 213	Calculus III	3

PHYS 220	University Physics II	3
PHYS 221	University Physics II Lab	1
MATH 203	Linear Algebra	3
ENGR 200	Engineering Statistics	3
CSAI 390	Internship 1	3
CSAI 391	Internship 2	3

Artificial Intelligence Program Requirements

70 Credits

Core Course

61 Credit Hours

Course Code	Title	Credits
MATH 225	Discrete Mathematics	3
CSCI 104	Introduction to Computing	3
CSCI 211	Object Oriented Programming	3
CSCI 215	Data Structures and Algorithms	3
CSCI 232	Computer Organization	3
CSAI 350	Introduction to Artificial Intelligence	3
CSCI 315	Design and Analysis of Algorithms	3
CSCI 326	Database Systems	3
CSCI 312	Operating System Fundamentals	3
CSAI 351	Data Science	3
EEEN 331	Digital Systems Design	3
EEEN 332	Digital Systems Design Lab	1
CENG 411	Software Engineering	3
CSCI 415	Introduction to Parallel Programming	3
CSAI 450	Machine Learning	3
CSAI 451	Machine Learning Lab (Writing Intensive)	1
CSCI 462	Data Communications and Computer Networks	3
CSAI 452	Natural Language Processing	3
CSAI 453	Data Visualization	3
CSAI 490	Professional Software Practice	2
CSCI 492	Senior Design Project I	2
CSCI 493	Senior Design Project II	4

Technical Electives

9 Credit Hours

Course Code	Title	Credits
CSCI 450	Information Security and Privacy	3
CSAI 480	Big Data	3
CSAI 481	Computer Vision	3
CSAI 482	Data Mining	3
CSAI 483	Information Retrieval	3

CSAI 484	Internet of Things System	3
CSAI 485	Introduction to Deep Learning	3
CSCI 416	Human Computer Interaction	3
CENG 431	Embedded Systems	3
CENG 432	Embedded Systems Lab	1
CSAI 486	Special Topics in Artificial Intelligence	3
CSAI 487	Introduction to Robotics	3
ENGR 399	Undergraduate Research Project	3
MEPM 511	Project Management Fundamentals	3
MEPM 533	Information Systems for Project Management	3

Bachelor of Science in Artificial Intelligence Four-Year Study Plan

Year 1 Semester 1			Year 1 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
CSCI 104	Introduction to Computing	3	ARAB 101 Or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition	3	CSCI 112	Introduction to Computer Programming	3
MATH 113	Calculus I	3	CSCI 113	Introduction to Computer Programming Lab	1
PHYS 110	University Physics I	3	MATH 114	Calculus II	3
PHYS 111	University Physics I Lab	1	PHYS 220	University Physics II	3
UNIV 100	University First-Year Transition	1	PHYS 221	University Physics II Lab	1
			MEST 100	Introduction to Islam in World Culture	3
Subtotal =		14	Subtotal =		17
Year 2 Semester 1			Year 2 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
CSCI 211	Object Oriented Programming	3	CSCI 215	Data Structures and Algorithms	3
MATH 213	Calculus III	3	EEEN 331	Digital Systems Design	3
MATH 225	Discrete Mathematics	3	EEEN 332	Digital Systems Design Lab	1
CSCI 232	Computer Organization	3	ENGR 200	Engineering Statistics	3
GEN ED	Social and Behavioral Sciences	3	PHIL 100 Or ENGL 200	Critical Thinking and Reasoning or Advanced Composition	3
			MATH 203	Linear Algebra	3
Subtotal=		15	Subtotal=		15
Year 2 Summer Session					
Course No.	Description	CR			
CSAI 390	Internship 1	3			
Year 3 Semester 1			Year 3 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
CSAI 350	Introduction to Artificial Intelligence	3	CSCI 315	Design and Analysis of Algorithms	3
GEN ED	Natural Sciences	3	CSAI 351	Data Science	3
CSCI 312	Operating System Fundamentals	3	CSCI 462	Data Comm. and Computer Networks	3
CSCI 326	Database Systems	3	CSCI 415	Introduction to Parallel Programming	3
UNIV 200	Innovation, Entrepreneurship, and Sustainability	3	UAES 200	Survey of United Arab Emirates Studies	3
Subtotal =		15	Subtotal =		15
Year 3 Summer Session					
Course No.	Description	CR			
CSAI 391	Internship 2	3			
Subtotal =		3			
Year 4 Semester 1			Year 4 Semester 2		
Course No.	Description	CR	Course No.	Description	CR
CSAI 450	Machine Learning	3	CSCI 493	Senior Design Project II	4

CSAI 451	Machine Learning Lab	1	CSAI 452	Natural Language Processing	3
CENG 411	Software Engineering	3	CSAI 453	Data Visualization	3
CSCI 492	Senior Design Project I	2	Elective	Technical Elective II	3
CSAI 490	Professional Software Practice	2	Elective	Technical Elective III	3
Elective	Technical Elective I	3			
Subtotal =		14	Subtotal =		16
TOTAL CREDIT HOURS 128					

Bachelor of Science in Chemical Engineering

Program Description

The Bachelor of Science in Chemical Engineering (CHEN) program was established in 2014 and started accepting students in Fall 2014. Over 31 students are currently enrolled in the program.

The CHEN program is a unique four-year undergraduate program. The program covers areas of sustainable engineering, global issues, and multidisciplinary problem solving and system analysis components across its curriculum to solve society and industrial most challenging problems. In addition, the program closely matches local professional opportunities and the direction of professions in chemical engineering.

The CHEN program has 133 credit hours over the duration of eight semesters including six credit hours of internship program. The curriculum of the program covers reaction engineering, process design and control, fluid and material transportation, renewable energy, mass and heat transfer and water and environmental engineering. The program was prepared taking into consideration criteria set by the Accreditation Board for Engineering and Technology (ABET) and the Commission for Academic Accreditation (CAA) at the Ministry of Higher Education and Scientific Research, United Arab of Emirates.

The Chemical Engineering faculty members have graduated from well-known universities in the world, and their expertise covers a broad range of technical areas, including reaction engineering, process design, renewable energy and management, environmental engineering, and wastewater treatment.

The Bachelor of Science in Chemical Engineering is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The Bachelor of Science in Chemical Engineering program at AURAK educates students to become qualified engineers who are capable of generating effective solutions by using engineering approaches in the field of Chemical Engineering. The graduates of the program will be well versed in technology and in social and environmental issues.

To fulfill this mission, the program provides the undergraduate student with a thorough foundation in the basic tenets of Chemical Engineering and technologies and a broad introduction into chemistry, engineering materials, thermal sciences, reaction and separation sciences, energy and environment protection, and chemical systems. The program provides a strong background for graduate study in the diverse areas branching out of the Chemical Engineering field. The technical focus is complemented with topics in general education leading to a well-rounded member of the global society.

Program Educational Objectives

The Program Educational Objectives (PEOs) describe the expected accomplishments of the graduates of the Bachelor of Science in Chemical Engineering. The PEOs have been derived from and support the mission statement of AURAK. The chemical engineering program has a strong focus on the integration of engineering, science, and process design with complementary areas of study in environmental engineering, fuels and energy, pharmaceutical, petrochemicals, materials, and renewable energy.

The graduates of the Chemical Engineering Program are expected to:

1. Pursue a successful professional career in engineering, scientific, or complementary disciplines through technical competence, effective communication, teamwork, and leadership;

2. Engaged in creating a positive impact with a sufficient awareness of the ethical, cultural, legal, and environmental issues; and
3. Maintain a lifelong interest in learning for personal and professional development.

Program Learning Outcomes

AURAK has adopted ABET's student outcomes for the Chemical Engineering Program. The outcomes for the program are for the student to acquire:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. An ability to communicate effectively with a range of audiences
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Degree Requirements

For students beginning their program of study at AURAK in Fall 2022, please be aware that the Ministry of Education is requiring all program to incorporate into their curriculum an internship experience of 16 weeks in duration. Therefore, the study plan for this program will be modified soon to accommodate this requirement. As soon as these changes are made, students enrolled in the program will be notified.

The Bachelor of Science in Chemical Engineering program requires the completion of 133 credits in the following areas:

Area	Credit
University General Education Requirements	32
School of Engineering Requirements	32
Chemical Engineering Program Requirements	69 (63 hrs compulsory and 6 hrs technical electives)

University General Education Requirements

32 Credit Hours

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS Chemical Engineering is CHEN 422 Chemical Engineering Laboratory II.**

Course Code	Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
CSCI 112	Introduction to Computer Programming	3
CSCI 113	Introduction to Computer Programming Lab	1
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
UAES 200	Survey of United Arab Emirates Studies	3
MATH 113	Calculus I	3
ECON 103	Principles of Microeconomics	3
CHEM 211	General Chemistry I	3

School of Engineering Requirements

32 Credit Hours

Faculty Requirements are (32) Credit Hours include the following courses:

Course Code	Course Title	Credits
ENGR 107	Introduction to Engineering	2
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab.	1
MATH 114	Calculus II	3
MATH 203	Linear Algebra	3
MATH 214	Elementary Differential Equations	3
PHYS 220	University Physics II	3
PHYS 221	University Physics II Lab.	1
ENGR 200	Engineering Statistics	3
ENGR 390	Internship 1	3
ENGR 391	Internship 2	3
IENG 321	Engineering Economy	3
CHEM 212	General Chemistry Lab. I	1

Chemical Engineering Program Requirements

69 Credit Hours

Program Requirements are (69) Credit Hours as follow:

Compulsory:

63 Credit Hours

Courses include the following:

Course Code	Course Title	Credits
CHEM 213	General Chemistry II	3
CHEM 214	General Chemistry II Lab.	1

BIOL 112	University Biology I	3
IENG 231	Engineering Materials	3
IENG 232	Materials Lab	1
CHEM 215	Organic Chemistry I	3
CHEM 216	General Organic Chemistry Lab. I	1
CHEM 315	Physical Chemistry	3
CHEM 316	Physical Chemistry Lab.	1
MENG 211	Thermodynamics I	3
CHEN 201	Principles of Chemical Engineering	3
CIEN 251	Fluid Mechanics	3
MENG 361	Heat Transfer	3
CHEN 302	Computer Applications in Chemical Engineering	1
CHEN 312	Chemical Engineering Thermodynamic	3
CHEN 321	Chemical Engineering Lab. I	2
CHEN 351	Chemical Reaction Engineering	3
CHEN 371	Mass Transfer	3
CHEN 404	Safety and Environmental Impact	3
CHEN 422	Chemical Engineering Lab. II (Writing Intensive)	2
CHEN 472	Separation Processes	3
CHEN 481	Process & Plant Design	3
CHEN 482	Chemical Process Dynamics & Control	3
CHEN 491	Senior Design Project I	3
CHEN 492	Senior Design Project II	3

Technical Electives:

6 Credit Hours

To be selected from the following list of courses:

Course Code	Course Title	Credits
CHEN 403	Fundamental of Biochemical Engineering	3
CHEN 462	Petroleum Refining Engineering	3
CHEN 463	Natural Gas Processing	3
ELEN 426	Renewable Energy Systems	3
CHEN 473	Water Desalination	3
CHEN 474	Industrial and Wastewater Treatment	3
CHEN 493	Special Topics in Chemical Engineering	3
ENGR 399	Undergraduate Research Project	3

Chemical Engineering Four-Year Sample Schedule

First Year, First Semester			First Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
PHYS 110	University Physics I	3	Arab 101 OR Arab 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
PHYS 111	University Physics I Lab.	1	MATH 114	Calculus II	3
ENGL 101	Composition	3	MEST 100	Introduction to Islam in World Culture	3
ENGR 107	Introduction to Engineering	2	PHYS 220	University Physics II	3
MATH 113	Calculus I	3	PHYS 221	University Physics II Lab.	1
UNIV 100	University First-Year Transition	1	CSCI 112	Introduction to Computer Programming	3
CHEM 211	General Chemistry	3	CSCI 113	Introduction to Computer Programming Lab.	1
CHEM 212	General Chemistry Lab.	1			
Subtotal =		17	Subtotal =		17
Second Year, First Semester			Second Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CHEM 213	General Chemistry II	3	IENG 231	Engineering Materials	3
CHEM 214	General Chemistry II Lab.	1	IENG 232	Materials Lab.	1
MENG 211	Thermodynamics I	3	CHEM 215	Organic Chemistry I	3
MATH 203	Linear Algebra	3	CHEM 216	General Organic Chemistry Lab. I	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability	3	MATH 214	Elementary Differential Equations	3
BIOL 112	University Biology I	3	CHEN 201	Principles of Chemical Engineering	3
			ECON 103	Principles of Microeconomics	3
Subtotal =		16	Subtotal =		17
Second Year, Summer Session					
Course Code	Course Title	Credits			
ENGR 390	Internship 1	3			
Third Year, First Semester			Third Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGR 200	Engineering Statistics	3	PHIL 100 OR ENGL 200	Critical Thinking and Reasoning OR Advanced Composition	3
CIEN 251	Fluid Mechanics	3	CHEN 312	Chemical Engineering Thermodynamics	3
CHEM 315	Physical Chemistry	3	MENG 361	Heat Transfer	3
CHEM 316	Physical Chemistry Lab.	1	CHEN 371	Mass Transfer	3
IENG 321	Engineering Economy	3	CHEN 302	Computer Applications in Chemical Engineering	1
UAES 200	Survey of United Arab Emirates Studies	3	CHEN 351	Chemical Reaction Engineering	3
Subtotal =		16	Subtotal =		16

Third Year, Summer Semester						
Course Code	Course Title	Credits	Course Code	Course Title	Credits	
ENGR 391	Internship 2	3				
Fourth Year, First Semester			Fourth Year, Second Semester			
Course Code	Course Title	Credits	Course Code	Course Title	Credits	
CHEN 321	Chemical Engineering Lab. I	2	CHEN 422	Chemical Engineering Lab. II	2	
CHEN 472	Separation Processes	3		CHEN Technical Elective 2	3	
	CHEN Technical Elective 1	3	CHEN 482	Chemical Process Dynamics and Control	3	
CHEN 481	Process & Plant Design	3	CHEN 404	Safety and Environmental Impact	3	
CHEN 491	Senior Design Project I	3	CHEN 492	Senior Design Project II	3	
Subtotal =		14	Subtotal =			14
Total 132 Semester Hour Credits						

Bachelor of Science in Civil and Infrastructure Engineering

Program Description

The Civil and Infrastructure Engineering (CIEN) program is a unique undergraduate program in the United Arab Emirates and in the region as it concentrates on infrastructure engineering in addition to traditional civil engineering areas. The CIEN program prepares students to excel in several sectors including building design and construction, highways and transportation, planning and management, and consulting.

The program has 13 credit hours over the duration of eight semesters including hours at internship programs. The curriculum of the program covers construction engineering and management, structural and materials engineering, highway and transportation engineering, geotechnical engineering, and water and environmental engineering. The program was prepared according to the International Standards of Accreditation Board for Engineering and Technology (ABET) and has passed all the conditions and criteria set by the Commission for Academic Accreditation (CAA) at the Ministry of Higher Education and Scientific Research, United Arab of Emirates.

The faculty members of the Civil and Infrastructure Engineering program have graduated from top universities in the world and their expertise covers a broad range of technical areas, including structural engineering, geotechnical engineering, construction engineering and management, environmental systems, hydraulic engineering and water resources, and transportation systems engineering.

The Bachelor of Science in Civil and Infrastructure Engineering is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The Bachelor of Science in Civil and Infrastructure program at AURAK educates students to become qualified engineers who are capable of generating effective solutions by using engineering approaches in the field of Civil and Infrastructure Engineering. The graduates of the program will be well versed in technology and social and environmental issues.

To fulfill this mission, the program provides the undergraduate student with a thorough foundation in the basic tenets of civil and infrastructure technologies and a broad introduction into structures, engineering materials, transport systems, soil engineering, and environment protection. The program provides a strong background for graduate study in the diverse areas branching out of the civil and infrastructure field. The technical focus is complemented with topics in general education leading to a well-rounded member of a global society.

Program Educational Objectives

The Bachelor of Science in Civil and infrastructure Engineering program at AURAK, based on student outcomes, produces graduates, who are armed with knowledge and skills to engage in lifelong learning and application of sustainable technologies to become:

1. Leaders in their profession who serve society in the construction industry, the transportation sector, project management, and in shaping public environmental and infrastructure policies;
2. Competent civil engineers who are engaged in the design of civil engineering and infrastructure systems; and
3. Contributors to the advancement of the civil engineering discipline at the national and global levels.

Program Learning Outcomes

AURAK has adopted ABET's student outcomes for the Civil and Infrastructure Engineering program. The outcomes for the program are for the student to acquire:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate effectively with a range of audiences
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Degree Requirements

The Bachelor of Science in Civil and Infrastructure Engineering requires the completion of 134 credits in the following areas:

Area	Credit
University General Education Requirements	32
School of Engineering Requirements	30
Civil and Infrastructure Engineering Department	72

University General Education Requirements

32 Credit Hours

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education

Component requirements are satisfied. **The fifth writing intensive course for the BS in Civil and Infrastructure Engineering is CIEN 334 Civil Engineering Testing and Materials.**

Course Code	Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
CSCI 112	Introduction to Computer Programming	3
CSCI 113	Introduction to Computer Programming Lab	1
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
UAES 200	Survey of United Arab Emirates Studies	3
MATH 113	Calculus I	3
ECON 103	Principles of Microeconomics	3
CHEM 211	General Chemistry I	3

School of Engineering Requirements

30 Credit Hours

Course Code	Course Title	Credits
ENGR 107	Introduction to Engineering	2
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab.	1
MATH 114	Calculus II	3
MATH 213	Calculus III	3
MATH 214	Elementary Differential Equations	3
PHYS 220	University Physics II	3
PHYS 221	University Physics II Lab.	1
ENGR 200	Engineering Statistics	3
ENGR 390	Internship 1	3
ENGR 391	Internship 2	3
ECEN 491	Engineering Seminar	1
CHEM 212	General Chemistry Lab	1

Civil and Infrastructure Engineering Department Requirements

72 Credit Hours

Core Courses**63 Credit Hours**

Course Code	Course Title	Credits
CIEN 201	Computer Aided Drawings	3
CIEN 211	Statics	3
CIEN 212	Mechanics of Materials	3
CIEN 241	Infrastructure Management	3
CIEN 250	Engineering in Global Environment	2
CIEN 251	Fluid Mechanics	3
CIEN 261	Surveying	1
CIEN 265	Geographical Information System	2
CIEN 301	Numerical Analysis	3
CIEN 311	Structural Analysis	3
CIEN 321	Reinforced Concrete Design	3
CIEN 331	Construction Materials	3
CIEN 332	Construction Materials Lab	1
CIEN 333	Geotechnical Engineering	3
CIEN 334	Civil Engineering Testing and Materials (Writing Intensive Course)	1
CIEN 351	Environmental Engineering	2
CIEN 361	Highway Engineering and Design	3
CIEN 362	Transportation Engineering	3
CIEN 421	Structural Steel Design	3
CIEN 440	Infrastructure Financing	3
CIEN 441	Construction Management	3
CIEN 451	Infrastructure Systems	3
CIEN 491	Senior Design Project (1)	2
CIEN 492	Senior Design Project (2)	4

Basic Science Electives**3 Credit Hours**

Course Code	Course Title	Credits
PENG 101	Physical Geology	3
BIOL 112	University Biology	3

Technical Electives**6 Credit Hours**

Course Code	Course Title	Credits
CIEN 422	Advanced Reinforced Concrete Design	3
CIEN 424	Bridge Design	3
CIEN 431	Foundation Engineering	3
CIEN 442	Construction Planning and Scheduling	3
CIEN 444	Construction Cost Analysis and Estimating	3
CIEN 454	Water and Wastewater Supply	3
CIEN 455	Solid Waste Management	3
CIEN 456	Sustainable Urban Building Sites	3
CIEN 462	Advanced Pavement Design	3
CIEN 464	Airport Planning and Design	3
CIEN 493	Special Topics in Civil Engineering	3
ENGR 399	Undergraduate Research Project	3

Civil and Infrastructure Engineering Four-Year Sample Schedule

First Year, First Semester			First Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGL 101	Composition	3	ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGR 107	Introduction to Engineering	2	MATH 114	Calculus II	3
MATH 113	Calculus I	3	MEST 100	Introduction to Islam in World Culture	3
UNIV 100	University First-Year Transition	1	CSCI 112	Introduction to Computer Programming	3
PHYS 110	University Physics I	3	CSCI 113	Introduction to Computer Programming Lab.	1
PHYS 111	University Physics I Lab.	1	PHYS 220	University Physics II	3
CHEM 211	General Chemistry I	3	PHYS 221	University Physics II Lab.	1
CHEM 212	General Chemistry I Lab.	1			
Subtotal =		17	Subtotal =		17
Second Year, First Semester			Second Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CIEN 201	Computer Aided Drawings	3	CIEN 212	Mechanics of Materials	3
CIEN 211	Statics	3	CIEN 251	Fluid Mechanics	3
CIEN 241	Infrastructure Management	3	CIEN 261	Surveying	1
CIEN 250	Engineering in Global Environment	2	CIEN 265	Geographical Information System	2
MATH 213	Calculus III	3	MATH 214	Elementary Differential Equations	3
ECON 103	Principles of Microeconomics	3	PENG 101 or BIOL 112	Physical Geology or University Biology	3
Subtotal =		17	Subtotal =		15
Second Year, Summer Session					
Course Code	Course Title	Credits			
ENGR 390	Internship 1	3			
Third Year, First Semester			Third Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CIEN 301	Numerical Analysis	3	CIEN 321	Reinforced Concrete Design	3
CIEN 311	Structural Analysis	3	CIEN 334	Civil Engineering Testing and Materials	1
CIEN 331	Construction Materials	3	CIEN 351	Environmental Engineering	2
CIEN 332	Construction Materials Lab.	1	CIEN 361	Highway Engineering and Design	3
CIEN 333	Geotechnical Engineering	3	CIEN 362	Transportation Engineering	3
UNIV 200	Innovation, Entrepreneurship, and Sustainability	3	ENGR 200	Engineering Statistics	3
Subtotal =		16	Subtotal =		15
Third Year, Summer Semester					
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGR 391	Internship 2	3			
Fourth Year, First Semester			Fourth Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CIEN 421	Structural Steel Design	3		Technical Elective	3
CIEN 440	Infrastructure Financing	3	CIEN 492	Senior Design Project (2)	4
CIEN 441	Construction Management	3	PHIL 100 or ENGL 200	Critical Thinking and Reasoning or Advanced Composition	3
ECEN 450	Engineering Seminar	1	UAES 200	Survey of United Arab Emirates Studies	3
CIEN 491	Senior Design Project (1)	2			
CIEN 451	Infrastructure Systems	3			
CIEN ---	Technical Elective	3			
Subtotal =		18	Subtotal =		13
Total 133 Semester Credit Hours					

Bachelor of Science in Computer Engineering

Program Description

The Computer Engineering program integrates both electronics engineering and computer science in the development of computer systems. Computer Engineers are involved in many hardware and software aspects of computing including design of microcontrollers, computers, and circuit design. These engineers focus not just on how computers work but on how to integrate them into larger systems.

Students learn the fundamentals of circuits, systems, and software associated with the design of programmable systems as used for general purpose computing, communications, control, or signal processing. Because of their broad skills in both hardware and software, students in computer engineering are in high demand for employment after graduation. Computer engineering students are also positioned to pursue graduate programs in either computer science or electronics engineering.

The Bachelor of Science in Computer Engineering is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The mission of the Computer Engineering program is to produce graduates who have the necessary theoretical background and the technical know-how and skills to create successful innovations and to solve problems in the field and to understand the social and environmental issues that may be involved. Graduates will have a thorough foundation in the basic principles of electrical engineering and information technology and a broad introduction to electronics, the design of digital computer systems, software development, data communications, and networking. Students will be prepared for employment in diverse areas of computer engineering or for further study in a graduate program.

Program Educational Objectives

A few years after earning their degree, our graduates will be:

1. Successful professionals in the theoretical and practical areas of Computer Engineering;
2. Ethical and responsible engineers with sufficient awareness of the impact of engineering solutions in various contexts; and
3. Life-long learners equipped with the skills required for professional practice including functioning in teams and communicating effectively.

Program Learning Outcomes

Graduates of the program will have the ability to:

1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;
2. Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors;
3. Communicate effectively with a range of audiences;

4. Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts;
5. Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives;
6. Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions; and
7. Acquire and apply new knowledge as needed, using appropriate learning strategies.

Degree Requirements

The BS in Computer Engineering (CE) requires the completion of 132 credits in the following areas:

Area	Credit
University General Education Requirements	32
School of Engineering Requirements	32
CE Program Requirements	68

University General Education Requirements

32 Credit Hours

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS Computer Engineering is CSCI 463 Data Communication and Computer Networks Laboratory.**

Course Code	Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
CSCI 112	Introduction to Computer Programming	3
CSCI 113	Introduction to Computer Programming Lab	1
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
UAES 200	Survey of United Arab Emirates Studies	3
MATH 113	Calculus I	3
Gen Ed	Social and Behavioral Sciences Course	3
Gen Ed	Natural Sciences Course	3

School of Engineering Requirements**32 Credit Hours**

Course Code	Course Title	Credits
ENGR 107	Introduction to Engineering	2
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab	1
MATH 114	Calculus II	3
MATH 213	Calculus III	3
MATH 214	Elementary Differential Equations	3
MATH 203	Linear Algebra	3
PHYS 220	University Physics II	3
PHYS 221	University Physics II Lab	1
ENGR 200	Engineering Statistics	3
ECEN 450	Engineering Seminar	1
ENGR 390	Internship 1	3
ENGR 391	Internship 2	3

CE Program Requirements**65 Credit Hours****Core Courses****62 Credit Hours**

Course Code	Course Title	Credits
CSCI 211	Object Oriented Programming	3
MATH 225	Discrete Mathematics	3
CSCI 215	Data Structures and Algorithms	3
CSCI 312	Operating System Fundamentals	3
CENG 315	Microprocessors	3
CENG 316	Microprocessors Lab	1
CENG 335	Computer Architecture	3
CENG 336	Computer Architecture Lab	1
CENG 401	Network Servers & Architecture	3
CENG 411	Software Engineering	3
CENG 431	Embedded Systems Design	3
CENG 432	Embedded Systems Design Lab	1
CSAI 350	Introduction to Artificial Intelligence	3
EEEN 220	Signal & Systems I	3
EEEN 221	Signal & Systems I Lab	1
EEEN 280	Electric Circuit Analysis I	3
EEEN 281	Electric Circuit Analysis I Lab	1
EEEN 331	Digital System Design	3
EEEN 332	Digital System Design Lab	1

Course Code	Course Title	Credits
EEEN 333	Linear Electronics I	3
EEEN 334	Linear Electronics I Lab	1
CSCI 326	Database Systems	3
CSCI 462	Data Communications and Computer Networks	3
CSCI 463	Data Communications and Computer Networks Lab (Writing Intensive)	1
CENG 492	Senior Design Project I	2
CENG 493	Senior Design Project II	4

Technical Electives

6 Credit Hours

Course Code	Course Title	Credits
CSAI 350	Introduction to Artificial Intelligence	3
CSAI 484	Internet of Things Systems	3
CSCI 415	Introduction to Parallel Programming	3
CSCI 450	Information Security and Privacy	3
CENG 435	Parallel Computer Architectures	3
CENG 437	Introduction to Robotics	3
CENG 461	Network Security	3
CSCI 499	Special Topics in Computing	3
ECEN 431	Digital Circuit Design	3
ECEN 481	Concepts of Multimedia Processing and Transmission	3
ENGR 399	Undergraduate Research Project	3
MEPM 511	Project Management Fundamentals	3
MEPM 533	Information Systems for Project Management	3

BS in Computer Engineering Four-Year Sample Schedule

First Year, First Semester			First Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
PHYS 110	University Physics I	3	ARAB 101 Or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
PHYS 111	University Physics I Lab	1	CSCI 112	Introduction to Computer Programming	3
ENGL 101	Composition	3	CSCI 113	Intro. to Computer Programming Lab	1
ENGR 107	Introduction to Engineering	2	MATH 114	Calculus II	3
MATH 113	Calculus I	3	PHYS 220	University Physics II	3
Gen Ed	Natural Sciences	3	PHYS 221	University Physics II Lab	1
UNIV 100	University First-Year Transition	1	MEST 100	Introduction to Islam in World Culture	3
Subtotal =		16	Subtotal =		17
Second Year, First Semester			Second Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MATH 225	Discrete Mathematics	3	CSCI 215	Data Structures and Algorithms	3
CSCI 211	Object-Oriented Programming	3	ECEN 220	Signal & Systems I	3
ECEN 280	Electric Circuit Analysis	3	ECEN 221	Signal & Systems I Lab	1
ECEN 281	Electric Circuit Analysis Lab	1	ECEN 331	Digital System Design	3
MATH 213	Calculus III	3	ECEN 332	Digital System Design Lab	1
ENGR 200	Engineering Statistics	3	MATH 203	Linear Algebra	3
			PHIL 100 Or ENGL 200	Critical Thinking and Reasoning or Advanced Composition	3
Subtotal =		16	Subtotal =		17
Second Year, Summer Semester					
Course Code	Course Title	Credits			
ENGR 390	Engineering Statistics	3			
Third Year, First Semester			Third Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CENG 335	Computer Architecture	3	CENG 315	Microprocessor	3
CENG 336	Computer Architecture Lab	1	CENG 316	Microprocessor Lab	1
CSCI 326	Database Systems	3	MATH 214	Elementary Differential Equations	3
EEEN 333	Linear Electronics I	3	CSCI 462	Data Comm. and Computer Networks	3
EEEN 334	Linear Electronics Lab I	1	CSCI 463	Data Comm. and Computer Networks Lab	1
CSCI 312	Operating System Fundamentals	3	UAES 200	Survey of United Arab Emirates Studies	3
			UNIV 200	Innovation, Entrepreneurship, and Sustainability	3
Subtotal =		14	Subtotal =		17
Third Year, Summer Semester					
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGR 390	Internship	3			

Senior Year, First Semester			Senior Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ECEN 450	Engineering Seminar	1	CENG 493	Senior Design Project II	4
CENG 401	Network Servers & Architecture	3	CENG 431	Embedded Systems Design	3
Gen Ed	Social and Behavioral Sciences	3	CENG 432	Embedded Systems Design Lab	1
CENG 411	Software Engineering	3		Technical Elective I	3
CENG 492	Senior Design Project I	2		Technical Elective II	3
CSAI 350	Introduction to Artificial Intelligence	3			
Subtotal =		15	Subtotal =		14
Total 132 Credits					

Bachelor of Science in Computer Science

Program Description

The Bachelor of Science in Computer Science (CS) degree program provides a solid foundation in computing and technology related sciences, along with exploration of the various disciplines of liberal arts and humanities. By examining the latest industry tools and techniques and developing valuable experience and professional skills in areas such as object-oriented programming, database systems, and network security, students are equipped with the in-depth understanding of their chosen professions.

The CS program aims to develop and prepare individuals for rewarding, cutting-edge careers in software engineering, system administration, and management. Furthermore, the CS graduates will be well trained to handle the problems faced in industry, academia, and daily life.

The Bachelor of Science in Computer Science is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The Bachelor of Science in Computer Science aims to provide students with high-quality education covering a broad and well-integrated knowledge in the concepts and methodologies underlying the analysis, design, and utilization of computer software, algorithms, and systems. The graduates of the program will be well versed in computer technology and its impact on the global society.

Program Educational Objectives

A few years after graduation, our alumni will be:

1. Successful professionals and innovators in theoretical and practical areas of computer science;
2. Engaged in creating a positive technological impact with a sufficient awareness of the ethical, legal, and security issues of computer science; and
3. Equipped with the skills required for professional practice including functioning in teams, communicating effectively, and life-long learning.

Program Learning Outcomes

Graduates of the program will have the ability to:

1. Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions;
2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline;
3. Communicate effectively in a variety of professional contexts;
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles;

5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline; and
6. Apply computer science theory and software development fundamentals to produce computing-based solutions.

Degree Requirements

The BS in Computer Science (CS) requires the completion of 127 credits in the following areas:

Area	Credit
University General Education Requirements	32
School of Engineering Requirements	26
CS Program Requirements	63
Free Electives	6

University General Education Requirements

32 Credit Hours

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS Computer Engineering is CSCI 463 Data Communication and Computer Networks Laboratory.**

C o u r s e	Title	C r e d i t s
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
CSCI 112	Introduction to Computer Programming	3
CSCI 113	Introduction to Computer Programming Lab	1
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3

UAES 200	Survey of United Arab Emirates Studies	3
MATH 113	Calculus I	3
Gen Ed	Social and Behavioral Sciences Course	3
Gen Ed	Natural Sciences Course	3

School of Engineering Requirements

26 Credit Hours

Course Code	Course Title	Credits
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab	1
MATH 114	Calculus II	3
MATH 213	Calculus III	3
PHYS 220	University Physics II	3
PHYS 221	University Physics II Lab	1
MATH 203	Linear Algebra	3
ENGR 200	Engineering Statistics	3
CSCI 390	Internship 1	3
CSCI 391	Internship 2	3

CS Program Requirements

63 Credit Hours

Core Courses

57 Credit Hours

Course Code	Course Title	Credits
CSCI 104	Introduction to Computing	3
CSCI 211	Object Oriented	3
MATH 225	Discrete Mathematics	3
CSCI 215	Data Structures and	3
CSCI 232	Computer Organization	3
CSAI 350	Introduction to Artificial	3
CSCI 315	Design and Analysis of	3
CSCI 326	Database Systems	3
CSCI 312	Operating System	3
CSCI 372	Compiler Design	3
CSCI 388	Programming Languages	3
CENG 335	Computer Architecture	3
CENG 336	Computer Architecture Lab	1
EEEN 331	Digital Systems Design	3
EEEN 332	Digital Systems Design Lab	1
CENG 411	Software Engineering	3

Course Code	Course Title	Credits
CSCI 462	Data Communications and	3
CSCI 463	Data Communications and Computer Networks Lab	1
CENG 461	Network Security	3
CSCI 492	Senior Design Project I	2
CSCI 493	Senior Design Project II	4

Technical Electives

6 Credit Hours

Course Code	Course Title	Credits
CENG 437	Introduction to Robotics	3
CSCI 411	Computer Graphics	3
CSCI 412	Computer Graphics Lab	1
CSCI 415	Introduction to Parallel	3
CSCI 416	Human Computer	3
CENG 435	Parallel Computer	3
ECEN 481	Concepts of Multimedia	3
CENG 401	Network Servers &	3
CSCI 450	Information Security and	3
CSCI 499	Special Topics in Computing	3
ENGR 399	Undergraduate Research	3
MEPM 533	Information Systems for	3
MEPM 511	Project Management	3

Free Electives

6 Credit Hours

Students must complete two courses (6 credits) of free electives

BS in Computer Science Four-Year Sample Schedule

First Year, First Semester			First Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CSCI 104	Introduction to Computing	3	ARAB 101 Or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition	3	CSCI 112	Introduction to Computer Programming	3
MATH 113	Calculus I	3	CSCI 113	Intro. to Computer Programming Lab	1
PHYS 110	University Physics I	3	MATH 114	Calculus II	3
PHYS 111	University Physics I Lab	1	PHYS 220	University Physics II	3
UNIV 100	University First-Year Transition	1	PHYS 221	University Physics II Lab	1
			MEST 100	Introduction to Islam in World Culture	3
Subtotal =		14	Subtotal =		1E
Second Year, First Semester			Second Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CSCI 211	Object Oriented Programming	3	CSCI 215	Data Structures and Algorithms	3
MATH 213	Calculus III	3	EEEN 331	Digital Systems Design	3
MATH 225	Discrete Mathematics	3	EEEN 332	Digital Systems Design Lab	1
CSCI 232	Computer Organization	3	MATH 203	Linear Algebra	3
UNIV 200	Innovation, Entrepreneurship, and Sustainability	3	ENGR 200	Engineering Statistics	3
			PHIL 100 Or ENGL 200	Critical Thinking and Reasoning or Advanced Composition	3
Subtotal =		15	Subtotal =		16
Second Year, Summer Session					
Course Code	Course Title	Credits			
CSCI 390	Internship 1	3			
Subtotal =		3			
Third Year, First Semester			Third Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CENG 335	Computer Architecture	3	Gen Ed	Social and Behavioral Sciences	3
CENG 336	Computer Architecture Lab	1	CSCI 372	Compiler Design	3
Gen Ed	Natural Sciences	3	CSCI 462	Data Comm. and Computer Networks	3
CSCI 326	Database Systems	3	CSCI 463	Data Comm. and Computer Networks Lab	1
UAES 200	Survey of United Arab Emirates Studies	3	CSCI 388	Programming Languages	3
CSCI 312	Operating System Fundamentals	3	CSCI 315	Design and Analysis of Algorithms	3
Subtotal =		16	Subtotal =		16
Third Year, Summer Session					
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGR 390	Internship	3			
Subtotal =		3			

Fourth Year, First Semester			Fourth Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
CSAI 350	Introduction to Artificial Intelligence	3	CSCI 493	Senior Design Project II	4
CENG 411	Software Engineering	3	CENG 461	Network Security	3
CSCI 492	Senior Design Project I	2		Free Elective II	3
	Free Elective I	3		Technical Elective II	3
	Technical Elective I	3			
Subtotal =		14	Subtotal =		
Total 127 Credits					

Bachelor of Science in in Electrical and Electronics Engineering

Program Description

The Bachelor of Science in Electrical and Electronics Engineering provides graduates with a sound understanding of fundamental engineering principles and applications in the field.

Graduates are prepared for post-graduate study or employment and are able to demonstrate their ability to apply the knowledge and methodologies of electrical and electronics engineering essential for a successful career. The American University of Ras Al Khaimah's (AURAK) Bachelor of Electrical and Electronics Engineering aims to produce effective, well educated, and globally competitive electrical and electronics engineers for the region and the wider world. The program aims to be amongst the top in the GCC region in the fields of electrical and electronics engineering with highly qualified and dedicated faculty and staff. It also aims to have a strong link with the local and regional government, public-sector organizations, and private sector enterprises. Moreover, the program aims to provide service to the community by sharing the program's technical expertise and human resources.

The Bachelor of Science in Electrical and Electronics Engineering is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The mission of the Electrical and Electronics Engineering program is to provide graduates with a sound understanding of fundamental engineering principles and applications in the field. Graduates will have a comprehensive understanding of subjects like signals and systems, digital logic, circuits and electronics, and the theory and applications of digital communications, particularly networks and wireless communications, in addition to power systems and electric machines. Graduates will be prepared for both graduate study and employment. They will be able to demonstrate their ability to apply the knowledge and methodologies in power systems, electronics, and communications engineering essential for a successful career.

Program Educational Objectives

The graduates of Electrical and Electronics Engineering Program, within a few years will be:

1. Become successful practitioners and innovators in the region and worldwide;
2. Continue their professional development throughout their careers; and
3. Work productively and ethically in teams, and function effectively in diverse cultural environments.

Program Learning Outcomes

AURAK Electrical and Electronics Engineering graduates should have:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors;
3. An ability to communicate effectively with a range of audiences;

4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts;
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives;
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions; and
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Degree Requirements

The BS in Electrical and Electronics Engineering (EE) requires the completion of 135 credits in the following areas:

Area	Credit
University General Education Requirements	32
School of Engineering Requirements	32
ECE Program Requirements	71 (62 CH compulsory and 9 CH technical electives)

University General Education Requirements

32 Credit Hours

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS in Electronics and Communication Engineering is ECEN 434 Linear Electronics Laboratory I.**

Course Code	Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
CSCI 112	Introduction to Computer Programming	3
CSCI 113	Introduction to Computer Programming Lab	1
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
Gen Ed	Social and Behavioral Science Course – ECON 103 is recommended.	3
UAES 200	Survey of United Arab Emirates Studies	3
MATH 113	Calculus I	3
CHEM 211	General Chemistry I	3

School of Engineering Requirements**32 Credit Hours**

Course Code	Course Title	Credits
ENGR 107	Introduction to Engineering	2
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab	1
MATH 114	Calculus II	3
MATH 203	Linear Algebra	3
MATH 213	Calculus III	3
MATH 214	Elementary Differential Equations	3
PHYS 220	University Physics II	3
PHYS 221	University Physics II Lab	1
ENGR 200	Engineering Statistics	3
ENGR 390	Internship 1	3
ENGR 391	Internship 2	3
ECEN 491	Engineering Seminar	1

EE Program Requirements**71 Credit Hours****Core Courses****62 Credit Hours**

Course Code	Course Title	Credits
EEEN 220	Signal & Systems I	3
EEEN 221	Signal & Systems I Lab	1
EEEN 280	Electric Circuit Analysis I	3
EEEN 281	Electric Circuit Analysis I Lab	1
EEEN 282	Electric Circuit Analysis II	3
EEEN 283	Electric Circuit Analysis II Lab	1
EEEN 305	Electromagnetic Theory	3
EEEN 331	Digital System Design	3
EEEN 332	Digital System Design Lab	1
EEEN 333	Linear Electronics I	3
EEEN 334	Linear Electronics I Lab	1
EEEN 350	Electric Machines	3
EEEN 351	Electric Machines Lab	1
EEEN 360	Random Signal & Noise	3
EEEN 412	Power Systems	3
EEEN 431	Digital Circuit Design	3
EEEN 433	Linear Electronics II	3
EEEN 434	Linear Electronics II Lab	1
EEEN 437	Power Electronics	3
EEEN 451	Control Theory	3
EEEN 460	Communication Systems	3
EEEN 461	Communication Engineering Lab	1
EEEN 464	Digital Communication Systems	3
EEEN 467	Mobile and Wireless Communications	3
EEEN 492	Senior Design Project I	2
EEEN 493	Senior Design Project II	4

Technical Electives**9 Credit Hours**

Course Code	Course Title	Credits
CENG 315	Microprocessor	3
EEEN 432	Nanotechnology Fundamentals & Applications	3
EEEN 435	Introduction to Optical Electronics *	3
EEEN 462	Data and Computer Communication	3
EEEN 466	Digital Signal Processing	3
EEEN 472	Antenna Theory and Design	3
EEEN 473	Radio Frequency and Microwave Engineering	3
EEEN 474	Advanced Information Theory and Coding	3
EEEN 481	Concepts of Multimedia Processing & Transmission	3
EEEN 499	Special Topics in EEEN	3
EEEN 421	Power System Protection	3
EEEN 422	High Voltage engineering	3
EEEN 423	Electrical energy systems and fault analysis	3
EEEN 425	Smart Power Grid Systems Theory and Implementation	3
EEEN 426	Renewable Energy Systems	3
MEPM 511	Project Management Fundamentals	3
MEPM 515	Project Scheduling	3
ENGR 399	Undergraduate Research	3

Electrical and Electronics Engineering Program EEE 4-Year Study Plan

First Year, First Semester			First Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
PHYS 110	Physics I	3	ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR	3
PHYS 111	Physics I Lab	1		Arabic Language and Culture for Native Arabic Speakers I	
ENGL 101	Composition	3	CSCI 112	Intro. to Computer Programming	3
ENGR 107	Introduction to Engineering	2	CSCI 113	Intro. to Computer Programming Lab	1
MATH 113	Calculus I	3	MATH 114	Calculus II	3
UNIV 100	University First-Year Transition	1	PHYS 220	Physics II	3
CHEM 211	General Chemistry I	3	PHYS 221	Physics II Lab	1
			MEST 100	Introduction to Islam in World Culture	3
Subtotal =		16	Subtotal =		17
Second Year, First Semester			Second Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
EEEN 280	Electric Circuit Analysis I	3	EEEN 220	Signals & Systems I	3
EEEN 281	Electric Circuit Analysis I Lab	1	EEEN 221	Signals & Systems I Lab	1
UAES 200	Survey of United Arab Emirates Studies	3	EEEN 282	Electric Circuit Analysis II	3
MATH 203	Linear Algebra	3	EEEN 283	Electric Circuit Analysis II Lab	1
MATH 213	Calculus III	3	ENGR 200	Engineering Statistics	3
EEEN 331	Digital System Design	3	MATH 214	Elementary Differential Equations	3
EEEN 332	Digital System Design Lab	1	PHIL 100 or ENGL 200	Critical Thinking and Reasoning Advanced Composition	3
Subtotal =		17	Subtotal =		17
Second Year, Summer Session					
Course	Course Title	Credits			
ENGR 390	Internship 1	3			
Third Year, First Semester			Third Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credit
EEEN 305	Electromagnetic Theory	3	EEEN 412	Power Systems	3
UNIV 200	Innovation, Entrepreneurship, and	3	EEEN 433	Linear Electronics II	3
EEEN 350	Electric Machines	3	EEEN 460	Communication Systems	3
EEEN 351	Electric Machines Lab	1	EEEN 431	Digital Circuit Design	3
EEEN 333	Linear Electronics I	3	EEEN 451	Control Theory	3
EEEN 334	Linear Electronics I Lab	1			
EEEN 360	Random Signal and Noise	3			
Subtotal =		17	Subtotal =		15
Third Year, Summer Semester					
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGR 391	Internship 2	3			
Fourth Year, First Semester			Fourth Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
GEN ED	Principles of Microeconomics (ECON 103)	3	EEEN 464	Digital Communication Systems	3
EEEN 434	Linear Electronics II Lab	1	EEEN 450	Engineering Seminar	1
EEEN 437	Power Electronics	3	EEEN 493	Senior Design Project II	4
EEEN 492	Senior Design Project I	2	EEEN 461	Communication Engineering Lab	1
EEEN 467	Mobile and Wireless Communications	3		Technical Elective II	3
	Technical Elective I	3		Technical Elective III	3
Subtotal =		15	Subtotal =		15
Total 135 Credits					

Bachelor of Science in Mechanical Engineering

Program Description

The mechanical engineering program is offered in the Mechanical and Industrial Engineering Department, the Bachelor of Science degree in mechanical engineering requires a minimum of 134 credit hours of coursework. Mechanical engineering is one of the largest, broadest, and oldest engineering disciplines. Mechanical engineers use the principles of energy, materials, and mechanics to design and manufacture machines and devices of all types. They create the processes and systems that drive technology and industry.

The key characteristics of the profession are its breadth, flexibility, and individuality. The career paths of mechanical engineers are largely determined by individual choices, a decided advantage in a changing world.

Mechanics, energy and heat, mathematics, engineering sciences, design and manufacturing form the foundation of mechanical engineering. Mechanics includes fluids, ranging from still water to hypersonic gases flowing around a space vehicle; it involves the motion of anything from a particle to a machine or complex structure.

The Bachelor of Science in Mechanical Engineering is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The Bachelor of Science in the mechanical engineering program at the American University of Ras Al Khaimah (AURAK) educates students to become qualified engineers who are capable of generating effective solutions by using engineering approaches in the field of mechanical engineering. The graduates of the program will be well versed in technology and in social and environmental issues.

To fulfill this mission, the program provides the undergraduate student with a thorough foundation in the basic tenets of mechanical engineering and technologies, and a broad introduction into machine design, engineering materials, thermal sciences, energy and environment protection and mechanical systems. The program provides a strong background for graduate study in the diverse areas branching out of the mechanical engineering field. The technical focus is complemented with topics in general education leading to a well-rounded member of the global society.

Program Educational Objectives

A few years after earning their degree, our graduates will:

1. Take leadership positions in energy and industrial sectors in response to regional and global market needs.
2. Pursue higher education, research and development, and/or other creative and innovative efforts in science, engineering and technology.
3. Create start-up companies that provide creative products and services for industry and society.

Program Learning Outcomes

Graduates of the program will have the ability to:

- 1 Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics

- 2 Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
- 3 Communicate effectively with a range of audiences
- 4 Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
- 5 Function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
- 6 Develop and conduct appropriate experimentation, analyze and interpretation of data, and use engineering judgment to draw conclusions
- 7 Acquire and apply new knowledge as needed, using appropriate learning strategies

Degree Requirements

The BS in Mechanical Engineering (MENG) requires the completion of 134 credits in the following areas:

Area	Credit
University General Education Requirements	32
School of Engineering Requirements	30
Mechanical Engineering Department Requirements	63
Technical Electives	9

University General Education Requirements

32 credit hours

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS in Mechanical Engineering is IENG 313 Manufacturing Processes Laboratory.**

Course Code	Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
CSCI 112	Introduction to Computer Programming	3
CSCI 113	Introduction to Computer Programming Lab	1
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
ECON 103	Principles of Microeconomics	3
UAES 200	Survey of United Arab Emirates Studies	3
MATH 113	Calculus I	3
CHEM 211	General Chemistry I	3

School of Engineering Requirements**30 Credit Hours**

Course Code	Course Title	Credits
ENGR 107	Introduction to Engineering	2
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab	1
MATH 114	Calculus II	3
MATH 213	Calculus III	3
MATH 214	Elementary Differential Equations	3
MATH 203	Linear Algebra	3
PHYS 220	University Physics II	3
PHYS 221	University Physics II Lab	1
ECEN 450	Engineering Seminar	1
ENGR 390	Internship 1	3
ENGR 391	Internship 2	3
CHEM 212	General Chemistry Lab	1

MENG Program Requirements**72 Credit Hours****Core Courses****63 Credit Hours**

Course Code	Course Title	Credits
CIEN 211	Statics	3
CIEN 212	Mechanics of Materials	3
CIEN 251	Fluid Mechanics	3
CIEN 301	Numerical Analysis	3
EEEN 280	Electric Circuits Analysis I	3
EEEN 281	Electric Circuits Analysis I Lab.	1
IENG 231	Engineering Materials	3
IENG 232	Engineering Materials Lab.	1
ENGR 200	Engineering Statistics	3
IENG 311	Manufacturing Processes	2
IENG 313	Manufacturing Processes Lab.	1
MENG 201	Mechanical Engineering Drawing	3
MENG 211	Thermodynamics I	3
MENG 212	Thermodynamics II	3
MENG 221	Dynamics	3
MENG 231	Engineering Measurements	2
MENG 321	Vibration and Control	3
MENG 342	Fluid Mechanics Lab.	1
MENG 361	Heat Transfer	3
MENG 362	Thermal Sciences Lab.	1
MENG 421	Theory of Machines	3
MENG 451	Mechanical Design I	3
MENG 452	Mechanical Design II	3
MENG 491	Senior Design Project I	2
MENG 492	Senior Design Project II	4

Technical Electives**9 Credit Hours**

Course Code	Course Title	Credits
MENG 311	Internal Combustion Engines	3
MENG 441	Turbo Machinery	3
MENG 453	Computer Aided Design	3
MENG 455	Finite Elements in Machine Design	3
MENG 461	HVAC and Refrigeration Systems	3
MENG 462	Design of Thermal System	3
MENG 463	Energy Conversion and Management	3
MENG 493	Special Topics in Mechanical Engineering	3
IENG 321	Engineering Economy	3
ENGR 399	Undergraduate Research	3
ELEN 426	Renewable Energy Systems	3
ELEN 451	Control Theory	3
MEPM 511	Project Management Fundamentals	3
MEPM 532	Engineering Management	3
MEPM 543	Risk Management for Project Managers	3
MSRE 513	Energy Economics	3
MSRE 515	Energy Efficiency and Management	3
MSRE 521	PhotoVoltaics	3
MSRE 526	Green Buildings	3

BS in Mechanical Engineering Four-Year Sample Schedule

First Year, First Semester			First Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
PHYS 110	University Physics I	3	ARAB 101 Or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
PHYS 111	University Physics I Lab	1	CSCI 112	Introduction to Computer Programming	3
ENGL 101	Composition	3	CSCI 113	Intro. to Computer Programming Lab	1
ENGR 107	Introduction to Engineering	2	MATH 114	Calculus II	3
MATH 113	Calculus I	3	PHYS 220	University Physics II	3
CHEM 211	General Chemistry	3	PHYS 221	University Physics II Lab	1
CHEM 212	General Chemistry Lab	1	MEST 100	Introduction to Islam in World Culture	3
UNIV 100	University First-Year Transition	1			
Subtotal =		17	Subtotal =		17
Second Year, First Semester			Second Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MENG 201	Mechanical Engineering Drawing	3	CIEN 212	Mechanics of Materials	3
CIEN 211	Statics	3	MENG 221	Dynamics	3
EEEN 280	Electric Circuit Analysis	3	MATH 203	Linear Algebra	3
EEEN 281	Electric Circuit Analysis Lab	1	MATH 214	Elementary Differential Equations	3
MATH 213	Calculus III	3	MENG 212	Thermodynamics II	3
MENG 211	Thermodynamics I	3	PHIL 100 Or ENGL 200	Critical Thinking and Reasoning Advanced Composition	3
MENG 231	Engineering Measurements	2			
Subtotal =		18	Subtotal =		18
Second Year, Summer Session					
Course Code	Course Title	Credit			
ENGR 390	Internship 1	3			
Third Year, First Semester			Third Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
IENG 231	Engineering Materials	3	MENG 361	Heat Transfer	3
IENG 232	Engineering Materials Lab.	1	MENG 362	Thermal Sciences Lab	1
CIEN 251	Fluid Mechanics	3	MENG 321	Vibration and Control	3
ECON 103	Principles of Microeconomics	3	ENGR 200	Engineering Statistics	3
MENG 342	Fluid Mechanics Lab	1	IENG 311	Manufacturing Processes	2
UAES 200	Survey of United Arab Emirates Studies	3	IENG 313	Manufacturing Processes Lab.	1
			CIEN 301	Numerical Analysis	3
Subtotal =		14	Subtotal =		16
Third Year, Summer Session					
Course Code	Course Title	Credits			
ENGR 391	Internship 2	3			
Fouthr Year, First Semester			Fourth Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MENG 451	Mechanical Design I	3	MENG 452	Mechanical Design II	3
MENG 421	Theory of Machines	3		ME Technical Elective 2	3
	ME Technical Elective I	3		ME Technical Elective 3	3
MENG 491	Senior Design Project I	2	MENG 492	Senior Design Project II	4
UNIV 200	Innovation, Entrepreneurship, & Sustainability	3	ECEN 450	Engineering Seminar	1
Subtotal =		14	Subtotal =		14
Total 134 Credits					

Bachelor of Science in Petroleum Engineering

Program Description

The Bachelor of Science in Petroleum Engineering program (PENG) was established in 2014 and started accepting students in Fall 2014. Over thirty students are currently enrolled in this program.

The PENG program is a unique four-year undergraduate program. The program covers areas of global issues, multidisciplinary problem solving and system analysis components across its curriculum to solve society and industrial most challenging problems. In addition, the program closely matches local professional opportunities and the direction of professions in petroleum engineering.

The Petroleum Engineering program has 133 credit hours over the duration of eight semesters including six credit hours of internship program. The curriculum of the program covers reservoir production and drilling engineering. The program was prepared by taking into consideration criteria set by the Accreditation Board for Engineering and Technology (ABET) and the Commission for Academic Accreditation (CAA) at the Ministry of Higher Education and Scientific Research, United Arab of Emirates.

The Petroleum Engineering program has very well-equipped laboratories. Some of these laboratories enable the students to visualize the various petroleum processes and how they are interrelated. Likewise, the program has advanced computation facilities, and the computation laboratories are equipped with a number of simulation software that is used by the students to enhance their understanding of the various petroleum engineering processes.

The Petroleum Engineering faculty members have graduated from top universities in the world and their expertise covers a broad range of technical areas, including reservoir engineering, drilling engineering, geothermal reservoir engineering, and reservoir economic evaluation.

The Bachelor of Science in Petroleum Engineering is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 7 requirements.

Program Mission

The Bachelor of Science in Petroleum Engineering program at AURAK educates students to become qualified engineers who are capable of generating effective solutions by using engineering approaches in the field of Petroleum Engineering. The graduates of the program will be well versed in technology and social and environmental issues.

The Bachelor of Science in Petroleum Engineering program at AURAK educates students to become qualified engineers who are capable of generating effective solutions by using engineering approaches in the field of Petroleum Engineering. The graduates of the program will be well versed in technology and social and environmental issues.

To fulfill this mission, the program provides the undergraduate student with a thorough foundation in the basic tenets of Petroleum Engineering and technologies and a broad introduction into geophysics, petroleum geology, fluid properties, formation evaluation, drilling, reservoir engineering and simulation, artificial lift systems, completions, oil and gas facilities engineering. The program provides a strong background for graduate study in the diverse areas branching out of the Petroleum Engineering field. The technical focus is complemented with topics in general education leading to a well-rounded member of a global society.

Program Educational Objectives

The Bachelor of Science in Petroleum Engineering program is developed to prepare students with a strong understanding of basic sciences, engineering, and a systematic study of the physical behavior of oil, water, and gas in porous media, flow lines and surface facilities in order to exploit petroleum resources with maximum economic benefits by natural and enhanced recovery methods.

The Program Educational Objectives (PEOs) of the Bachelor of Science in Petroleum Engineering program describe the career and professional activities that the program prepares graduates to:

1. Attain a broad knowledge in petroleum engineering applications in order to achieve critical thinking skills required to design and analyze petroleum engineering problems for maximum economic recovery of oil and gas resources, taking into account safety, environmental and social impacts;
2. Pursue life-long learning through training and possible post graduate education, engage in research, and contribute to scientific progressions through participation and leadership in professional organizations; and
3. Commit to professional ethics, environmental safety, and public welfare.

Program Learning Outcomes

AURAK has adopted ABET's a-k student outcomes for the Petroleum Engineering Program. The outcomes for the program are:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics;
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors;
3. An ability to communicate effectively with a range of audiences;
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts;
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives;
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions; and
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Degree Requirements

The Bachelor of Science in Petroleum Engineering program requires the completion of 132 credits in the following areas:

Area	Credit
University General Education Requirements	32
School of Engineering Requirements	32
Petroleum Engineering Program Requirements	69 (63 hrs. compulsory and 6 hrs. technical electives)

University General Education Requirements**32 Credit Hours**

The program requires completion of the General Education Component. For information relating directly to the General Education requirements, please review the catalog section entitled, "General Education Component." You must speak with your advisor to ensure that the General Education Component requirements are satisfied. **The fifth writing intensive course for the BS Petroleum Engineering program is PENG 322 Drilling Fluids Laboratory.**

Course Code	Title	Credits
ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
ENGL 101	Composition (Writing Intensive)	3
CSCI 112	Introduction to Computer Programming	3
CSCI 113	Introduction to Computer Programming Lab	1
UNIV 100	University First-Year Transition	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability (Writing Intensive)	3
PHIL 100 or ENGL 200	Critical Thinking and Reasoning (Writing Intensive) OR Advanced Composition (Writing Intensive)	3
MEST 100	Introduction to Islam in World Culture (Writing Intensive)	3
ECON 103	Principles of Microeconomics	3
UAES 200	Survey of United Arab Emirates Studies	3
MATH 113	Calculus I	3
CHEM 211	General Chemistry I	3

School of Engineering Requirements**32 Credit Hours**

Course Code	Course Title	Credits
ENGR 107	Introduction to Engineering	2
PHYS 110	University Physics I	3
PHYS 111	University Physics I Lab.	1
MATH 114	Calculus II	3
MATH 203	Linear Algebra	3
MATH 214	Elementary Differential Equations	3
PHYS 220	University Physics II	3
PHYS 221	University Physics II Lab.	1
ENGR 200	Engineering Statistics	3
ENGR 390	Internship 1	3
ENGR 391	Internship 2	3
IENG 321	Engineering Economy	3
CHEM 212	General Chemistry Lab. I	1

Petroleum Engineering Program Requirements**69 Credit Hours**

Program Requirements are 69 credits hours as follows:

Compulsory:

63 Credit Hours

Course Code	Course Title	Credits
PENG 101	Physical Geology	3
PENG 202	Introduction to Petroleum Engineering	1
PENG 231	Mechanics of Materials	3
CHEM 215	Organic Chemistry I	3
MENG 211	Thermodynamics I	3
CIEN 251	Fluid Mechanics	3
PENG 351	Reservoir Rock and Fluid Properties	3
PENG 321	Reservoir Rock and Fluid Properties Laboratory	1
PENG 322	Drilling Fluids Laboratory (Writing Intensive)	1
PENG 361	Drilling Engineering I	2
PENG 371	Petroleum Reservoir Engineering	3
PENG 381	Well Logging	3
PENG 382	Petroleum Production Engineering (Well Performance)	4
PENG 422	Fluid Flow in Porous Media Lab	1
PENG 403	Petroleum Property Evaluation	3
PENG 462	Drilling Engineering II	2
PENG 472	Applied Reservoir Geology	3
PENG 473	Reservoir Simulation	3
PENG 482	Natural Gas Engineering	3
PENG 483	Well Testing	3
PENG 484	Stimulation and Intervention in Petroleum Production Operations	3
PENG 485	Water Flooding	3
PENG 491	Senior Design Project I	3
PENG 492	Senior Design Project II	3

Technical Electives:

6 Credit Hours

Course Code	Course Title	Credits
PENG 486	Enhance Oil Recovery	3
PENG 442	Transportation and Storage of Petroleum	3
PENG 443	Separation and Treatment of Petroleum Fluids	3
PENG 474	Applied Reservoir Simulation	3
CHEN 404	Safety and Environmental Impact	3
PENG 493	Special Topics in Petroleum Engineering	3
ENGR 399	Undergraduate Research Project	3

Petroleum Engineering Four-Year Sample Schedule

First Year, First Semester			First Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
PHYS 110	University Physics I	3	ARAB 101 or ARAB 110	Arabic Language and Culture for Non-Native Learners I OR Arabic Language and Culture for Native Arabic Speakers I	3
PHYS 111	University Physics I Lab	1	MATH 114	Calculus II	3
ENGL 101	Composition	3	MEST 100	Introduction to Islam in World Culture	3
ENGR 107	Introduction to Engineering	2	PHYS 220	University Physics II	3
MATH 113	Calculus I	3	PHYS 221	University Physics II Lab	1
UNIV 100	University First-Year Transition	1	CSCI 112	Introduction to Computer Programming	3
CHEM 211	General Chemistry	3	CSCI 113	Introduction to Computer Programming Lab	1
CHEM 212	General Chemistry Lab	1			
Subtotal =		17	Subtotal =		17
Second Year, First Semester			Second Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ECON 103	Principles of Microeconomics	3	PHIL 100 OR NGL 200	Critical Thinking and Reasoning Advanced Composition	3
MENG 211	Thermodynamics I	3	PENG 231	Mechanics of Materials	3
MATH 203	Linear Algebra	3	CHEM 215	Organic Chemistry I	1
UNIV 200	Innovation, Entrepreneurship, and Sustainability	3	PENG 202	Introduction to Petroleum Engineering	3
PENG 101	Physical Geology	3	MATH 214	Elementary Differential Equations	3
			PENG 351	Reservoir Rock and Fluid Properties	3
			PENG 321	Reservoir Rock and Fluid Prop. Lab.	1
Subtotal =		15	Subtotal =		17
Second Year, Summer Session					
Course Code	Course Title	Credits			
ENGR 390	Internship 1	3			
Third Year, First Semester			Third Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGR 200	Engineering Statistics	3	PENG 382	Petroleum Production Engineering (Well Performance)	4
IENG 321	Engineering Economy	3	PENG 371	Petroleum Reservoir Engineering	3
CIEN 251	Fluid Mechanics	3	PENG 381	Well Logging	3
PENG 361	Drilling Engineering I	2	PENG 462	Drilling Engineering II	2
PENG 322	Drilling Fluids Laboratory	1	PENG 472	Applied Reservoir Geology	3
UAES 200	Survey of United Arab Emirates Studies	3			
Subtotal =		15	Subtotal =		15
Third Year, Summer Session					
Course Code	Course Title	Credits	Course Code	Course Title	Credits
ENGR 391	Internship 2	3			
Fourth Year, First Semester			Fourth Year, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
PENG 482	Natural Gas Engineering	3	PENG 483	Well Testing	3
PENG 403	Petroleum Property Evaluation	3	PENG 484	Stimulation and Intervention in Petroleum Production Operations	3
PENG 485	Water Flooding	3	PENG 473	Reservoir Simulation	3
	Technical Elective 1	3		Technical Elective II	3
PENG 491	Senior Design Project I	3	PENG 492	Senior Design Project II	3
PENG 423	Fluid Flow in Porous Media Lab.	1			
Subtotal =		16	Subtotal =		15

Total 133 Semester Hour Credits





GRADUATE PROGRAMS



ADMISSION TO GRADUATE STUDIES

AURAK offers a Post Graduate Diploma in Education and six Master Programs in response to the needs of Ras Al Khaimah, the region, and the international community. AURAK's programs meet the highest international standards, and our students will be well prepared for the professional challenges they will face in their careers. The programs engage relevant contemporary issues as well as crucial intellectual contexts. Students can expect to engage with one another and their professors in stimulating discussions and challenging assignments. They will study with experienced faculty from many of the best university in North America, Europe, Asia, and the Middle East.

Degree Offerings

AURAK currently offers one (1) post graduate diploma and five (5) programs of graduate study leading to a master's degree. These are:

- Post Graduate Diploma in Education
- Master of Education: Educational Leadership
- Executive Master of Business Administration
- Master of Business Administration
- Master of Science in Engineering Project Management
- Master of Science in Sustainable and Renewable Energy

Application Process

Matriculating Students

Applications for all AURAK graduate programs are processed through the Department of Admissions under the guidance of the appropriate school and program of study.

Prospective students can apply online or in person. The online application is available on <https://apply.aurak.ac.ae>.

The following documents would be required to submit:

- Attested transcripts for bachelor's degree(s) and certificate(s);
- English proficiency scores (EmSAT/TOEFL/IELTS or equivalent);
- Two (2) recommendation letters;
- Updated resume;
- Valid passport copy, Emirates ID, and residence visa (If Applicable);
- Health insurance card;
- Health history form; and
- Two (2) passport photos.

Incomplete applications are not processed.

Upon receiving a complete application, the Department of Admissions coordinates with the appropriate school and program to determine if the applicant meets the requirements for admission to the program. The relevant Graduate Program Coordinator makes all final admissions decisions. The Department of Admissions will notify the applicant of the university's final decision and provide all relevant information.

Application Deadlines

AURAK Admissions Application Deadline for Fall 2022 Semester

10th of September 2022

AURAK Admissions Final Application Deadline for Spring 2023 Semester

22nd of January 2023

Note: International students must apply two weeks in advance of the stated deadline to meet visa timeline requirements.

International applicants (i.e., graduates of universities located outside the UAE) are required to submit completed application forms and all supporting documents to the Department of Admissions.

Graduates of universities outside the UAE are required to present an equivalency of their degree from the UAE Ministry of Education located in Abu Dhabi, UAE. The procedure for obtaining an equivalency can be obtained from the Ministry by calling +971 2 695 1300 or +971 2 642 8000.

Admitted international students who need visas for the UAE should submit the visa application form, which is available from the Department of Admissions, at least three weeks prior to the first day of class.

Admission is valid only for the semester immediately following the admission decision. If an admitted student would like to have the admission date delayed, he or she must submit a formal request in writing to the Department of Admissions. After consultation with the Graduate Program Coordinator and the Dean of the school in which the admitted student's degree program is located, the student will be informed of the university's decision concerning delay of admission.

University Requirements for Admission to Graduate Programs

To be considered for admission, all applicants must meet the general university requirements for graduate admission. There are also program-specific requirements, information about which applicants will find in the section of the catalog in which the specific program is described.

For admission to a graduate degree program at AURAK, an applicant must:

- Hold a four-year bachelor's degree from an independently accredited university recognized by the UAE Ministry of Education (applicants with a bachelor's degree obtained outside the UAE must submit an equivalency of their degree from the UAE Ministry of Education); and
- Have attained the minimum score of one of the specified exams as indicated in the Graduate English Language Proficiency requirement table below. Note that the EmSAT is the preferred assessment of English proficiency.

Type of English Proficiency Exam	Minimum Score
EmSAT	1400
IELTS	6.0
TOEFL	575
TOEFL (Internet-based)	88

Table: Graduate English Proficiency Requirement

Program-Specific Requirements

- Post Graduate Diploma in Education
 - A minimum of two (2) years of experience of teaching or equivalent educational work experience is required.
 - Students granted entry to the education programs with less than a 3.0 CGPA on a 4.0 scale in their baccalaureate work must enter the PGDE rather than the MEd.
 - The minimum entry requirement is a CGPA of 2.5 on a 4.0 scale.
- Master of Education in Educational Leadership:
 - A minimum of two (2) years of experience of teaching or equivalent work experience is required.
 - Students transitioning from the Post Graduate Diploma in Education must have earned a CGPA of 3.0 or higher on a 4.0 scale in their program of study for admission into the Master of Education.
 - The minimum entry requirement for the Master of Education is a CGPA of 3.0 on a 4.0 scale. Student having an entry CGPA less than 3.0 on a 4.0 scale should enter the Post Graduate Diploma in Education.
- Executive Master of Business Administration:
 - A baccalaureate degree in Business Administration, with a minimum cumulative GPA of 3.00 on a 4.0 scale, or its established equivalent in the applicant's undergraduate degree program;

- A baccalaureate degree, or its equivalent, with a major or minor in Business Studies, Marketing, Finance, Accounting, Economics, or Management/Business Information Systems with a minimum CGPA of 3.00 on a 4.0 scale or its recognized equivalent in the applicant's undergraduate degree program, or
- A baccalaureate degree, or its equivalent, in a non-business field with a minimum of CGPA of 3.00 on a 4.0 scale, followed by a postgraduate certificate or diploma in a business-related field.
- The students must also submit an experience certificate, which proves at least five years of work experience in a managerial position to be eligible.
- Successful Personal Interview with School of Business Faculty.
- Master of Business Administration:
 - A baccalaureate degree in Business Administration, with a minimum cumulative GPA of 3.00 on a 4.0 scale, or its established equivalent in the applicant's undergraduate degree program;
 - A baccalaureate degree, or its equivalent, with a major or minor in Business Studies, Marketing, Finance, Accounting, Economics, or Management/Business Information Systems with a minimum CGPA of 3.00 on a 4.0 scale or its recognized equivalent in the applicant's undergraduate degree program, or
 - A baccalaureate degree, or its equivalent, in a non-business field with a minimum of CGPA of 3.00 on a 4.0 scale, followed by a postgraduate certificate or diploma in a business-related field.
 - This MBA program is geared toward applicants with or without work experience. Leveling courses may be required by the School of Business.
 - Personal interview.
- Master of Science in Engineering Project Management:
 - A baccalaureate degree in Engineering, Computer Science, or Information Technology, with a minimum CGPA of 3.0 on a 4.0 scale.
 - Qualified applicants requiring prerequisite courses may be required to take such courses in addition to their regular graduate program courses.
 - Students with no prior background in probability and statistics will be required to take IENG 241 Engineering Statistics.
 - Students with no prior background in engineering economy will be required to take IENG 321 (Engineering Economy).
- Master of Science in Sustainable and Renewable Energy
 - A baccalaureate degree in engineering or a science discipline.
 - Completion of specific courses in the areas of thermodynamics, fluid mechanics and heat transfer.

Conditional Admission

AURAK offers conditional admission under specific circumstances and to those who meet the following requirements:

- Hold a four-year bachelor's degree from an independently accredited university recognized by AURAK; and
- Have attained a minimum English Proficiency Exam score (as outlined in the table above).

To receive full admission into a graduate program, a conditional admission student must achieve the required minimum English Proficiency Score prior to the beginning of the second semester.

If the English proficiency requirement is not met by the beginning of the second semester of study, the student will not be allowed to continue his or her studies in the admitted graduate program at AURAK.

There may also be program-specific prerequisite courses for conditional admission students. Credits from these courses do not satisfy credit requirements for the degree and are not used to calculate the graduate CGPA. Conditionally admitted students must satisfy CGPA requirements in prerequisite courses at 3.0 or higher.

Non-Degree Seeking Graduate Student Admission

A non-degree seeking graduate student is one who would like to take AURAK courses for academic credit but who does not seek a master's degree. Students are admitted to AURAK with non-degree status if they meet the normal requirements for graduate admission or receive permission from the Dean of the appropriate school of study. Complete applications should be submitted to the Department of Admissions.

A non-degree seeking graduate student may take a maximum of six credit hours at the graduate level.

Visiting Students Admission

A visiting student is one who is not formally admitted to AURAK, and who is not participating in a formal exchange agreement between AURAK and a partner university but is typically enrolled in a degree program at another university. Such a student may take courses at AURAK for transfer to the student's home institution. It is wise for visiting students to check with their home institutions. Checking for the transferability of credit hours from a course from AURAK to another institution is the responsibility of the student.

To be admitted as a visiting graduate student, a student must be enrolled in a graduate program at an accredited institution and be in good academic standing in his/her current institution.

Students must also have attained the same minimum score on the English Proficiency Exam as students gaining admission to the Graduate Degree Programs at AURAK. Applicants seeking visiting graduate student status must submit to AURAK's Department of Admissions a complete application form and evidence of current enrollment in another university's graduate program, typically an official transcript. The application will be considered for admission. Visiting students

will be admitted as space allows. Registration in courses is subject to approval by the respective graduate program chair/coordinator.

Applicants seeking visiting student status must submit to AURAK's Department of Admissions the following documents:

- Transcripts for bachelor's degree and certificate;
- Evidence of enrollment in another university;
- An official letter from the university allowing the student to enroll for the course(s) at AURAK;
- English proficiency scores (TOEFL/IELTS or equivalent) – expired will also be accepted;
- Two (2) recommendation letters;
- Updated resume;
- Valid passport copy, Emirates ID, and residence visa (If Applicable); and
- Two (2) passport photos.

A complete application form is required with evidence of current enrollment in another university's graduate program, typically an official transcript. The application will be considered and visiting students will be admitted as space allows. Registration in courses is subject to approval by the relevant graduate program director.

Change of Status

A student may apply at any time for a change of status from non-degree to degree status, or from visiting to degree status. To change status, a student should submit a complete application through the Department of Admissions. All admissions and program requirements in place at the time of the change of status request must be met to be admitted and to graduate. Grades earned in courses taken as a visiting student in AURAK will be counted towards the CGPA (CGPA).

Courses already taken at previous institution may be accepted with the approval of the **relevant school and program**. University rules and regulations governing transfer courses and credits will apply.

Transfer Credit Policy for Graduate Studies

Applicants may be allowed to transfer up to 25% of the total program credit hours subject to review and recommendation of the program faculty. Partial transfer credit will not be awarded. Transfer credit may not be awarded for graduation projects or theses courses.

Grades earned in transferred courses do not count in the student's CGPA, though transferred credits count toward the cumulative earned hours and may apply towards meeting graduation requirements. Applicants must request that credit transfers be reviewed at the time of application.

GRADUATE ACADEMIC POLICIES

Each student is responsible for knowing AURAK's rules, regulations, requirements, and academic policies. Both the graduate information in the catalog and the institutional website are repositories of policy statements. Corrections, changes, or interpretations may be communicated by other means including electronically. Any student in doubt about an academic matter should consult the Associate Provost of Academic Affairs, the Associate Provost of Research, Department of the Registrar, Dean of the appropriate school, or their assigned faculty advisor. Students are subject to the university's stated policies regarding patents and copyrights.

Academic Advising

Academic Advisement is a critical dimension of the AURAK student experience, one that involves both students and faculty. For students it helps them to make acceptable progress toward their degree, and for faculty it is an important aspect of their teaching responsibilities.

Each student is responsible for his or her progress towards his or her degree completion and for meeting all requirements for graduation. The advisement process is available to assist the student in completing his or her studies, but responsibility for doing so lies ultimately with each student.

Academic advising is the responsibility of full-time faculty members. A full-time faculty member will be assigned to serve as a student's academic advisor from each department where a degree or minor program is housed. Typically, advisees are distributed equally among the full-time faculty in the relevant academic departments. The advisor's responsibilities include:

- Meeting with each advisee in the two-week period before early registration each semester to guide the advisee in course selection;
- Following up on the student's course requests during early and regular registration to facilitate the student's successful registration;
- Meeting with each advisee early in the semester to ensure that the student has registered for the appropriate courses;
- Meeting with each advisee roughly at mid-semester to gauge the student's progress and advise accordingly; and
- Serving as a mentor to advisees to facilitate their academic success and to provide guidance when appropriate on their future studies and careers.

Academic advisors are expected:

- To be familiar with all academic requirements for graduationcccccccccc; and
- To collaborate with university colleagues to provide the strongest academic advisement possible.

Attendance Policy

Please see the Student Attendance Policy section of this catalog.

Courses and Course Schedules

Each course offered by the university has a designated course prefix (or code) and number. The course prefix represents the discipline or field of study; the number indicates the level of the course content.

Courses are offered at the discretion of the VPAASS/Provost, the Associate Provost of Academic Affairs, the Registrar, the respective deans, and department chairs. Not every course is offered every semester. The Registrar has primary responsibility for approving the course schedule.

Course Credit

Each course has a credit value. A credit represents the in-class instruction and out-of-class study per week during the sixteen-week semester. Normally one credit represents fifty minutes of class instruction per week per semester.

Course Prerequisites and Co-requisites

Course prerequisites or co-requisites state requirements for student entry into courses and reflect necessary preparation for attempting a course. It is the student's responsibility to be aware of these requirements as stated in the catalog, and to have taken prerequisites recently enough to be of value. The instructor of the course may drop students who have enrolled in a course for which they have not met the prerequisites.

Course Selection and Registration

By mid-semester, the class schedule for the following semester is available through the Department of Registration and on the website. Students may select courses in consultation with their faculty/academic advisor, and they can then register online or submit a completed "Course Request Form" to the Department of the Registrar by the published deadlines.

Registration Criteria

Only registered students may participate in AURAK classes. Full-time status requires that a graduate student enroll in 9 semester credit hours in Fall and Spring semesters. This is the expected credit hour load for graduate students. Graduate students enrolling in fewer than 9 semester credit hours in Fall and Spring semesters are considered to be enrolled on a part-time basis. However, please note the following:

- A student with a CGPA of less than 3.0 is limited to registering for six credit hours the following semester subject to the approval of the academic advisor.
- No more than six (6) credit hours are allowed during summer terms.
- A student with a CGPA of 3.50 or higher can enroll in 12 semester credits in the subsequent semester subject to the approval of the academic advisor and the appropriate school dean.

Independent Study

Independent study courses are not allowed for graduate programs.

Summer Courses Outside AURAK

Requirements

An enrolled student is eligible to apply to take courses at another college/university during the summer with the aim of transferring credits to AURAK provided the following conditions are met:

- The student must be in good academic standing at AURAK.
- The summer courses at the host university must not be taken as attempts to repeat AURAK courses in which F grades were previously earned.
- The host university must be recognized by the UAE Ministry of Education.
- The host university must provide learning experiences similar to those offered by AURAK.
- Contact/credit hours for the course at the host university must be equivalent to, or greater than, the contact/credit hours required for its equivalent course at AURAK.
- The language of instruction of the course(s) taken at the host university must be English (except for language courses conducted in other languages) unless otherwise approved by AURAK.
- The student must obtain approval from AURAK prior to registering the summer course(s).
- Some programs may reserve the right not to allow any courses to be taken at another college/university.

Final Semester Grades

Final grades are made available on-line through the Student Portal. Official transcripts and official semester grade reports are obtained through the Department of the Registrar.

Students may access and print an unofficial record of their semester grades and other academic information from the Student Portal on the university website by logging onto their account. Official transcripts are available on request.

Change of Grade

Once a final grade has been recorded, it can be changed only in cases of computational error, recording error, or pursuant of a successful appeal of grade. Additional work of any type submitted to improve a grade after the final grade has been assigned will not be accepted. All changes of final grades must be initiated, approved, and recorded by the second week of classes of the next regular semester (spring for fall grades and fall for spring grades).

Grade Appeal

Although faculty members are generally the best judges of student performance, sometimes a student believes a grade is unfair. This procedure is therefore designed to specify the process through which students can appeal final course grades and the process through which faculty members can initiate a change in a final course grade following the official posting of students' grades in a course.

Where a student believes their final course grades to be in error the student must request a meeting with the member of faculty to ask that they reconsider the grade and ensure that the appeal is

submitted within the required timeline. Completed Grade Appeal Forms must be submitted within the following timeframe:

- For Fall Semester courses, completed Grade Appeal Forms must be submitted within the first three weeks of the following Spring Semester.
- For Spring Semester courses, completed Grade Appeal Forms must be submitted within three weeks of the “Grades Available” date listed in the Academic Calendar.

For Summer Session courses, completed Grade Appeal Forms must be submitted within the first three weeks of the following Fall Semester.

The grade appeal will only be considered where:

- There was an error or unfairness in the assessment of the exam grade.
- There was an error or unfairness in the evaluation process for determining the course grade.

The faculty member will meet formally with the student and a record of the meeting will be recorded on the Grade Appeal Review Form. In the meeting with the student, the faculty member must explain how the grade in question was arrived at. If the faculty member decides, based on the discussion with the student, that a change of grade is warranted, a Change of Grade Form will be completed by the faculty member, explaining the basis for the grade change and this will be forwarded to the Department Chair.

If the faculty member decides a change of grade is not warranted, the faculty member will complete the Grade Appeal Review Form, explaining why the original grade should not be changed and this will be forwarded to the Department Chair. The Change of Grade Form/Grade Appeal Review Form (as applicable) will then be reviewed in succession by the Department Chair, the Dean, and the Associate Provost for Student Affairs and a decision will be made.

The Associate Provost for Student Affairs will render a decision only after the faculty member has responded to any requests for further consideration made by the Department Chair and the Dean. The decision by the Associate Provost for Student Affairs will be final and not subject to further appeal. The Registrar will not change a student's final grade in a course for any other reason or through any other procedure.

Where an appeal is successful, the Associate Provost for Student Affairs will forward the Change of Grade Form to the Registrar's Department so that the student's grade is changed on the Student Information System (SIS). A copy of the Grade Change Form will be filed in the student's record.

Grading System

University course work is measured in terms of quantity and quality. One semester credit hour normally represents fifty minutes per week of lecture throughout a semester. The number of credits is a measure of quantity. The grade is a measure of quality. The university system for graduate grading is as follows:

Grade	GPA Points	Percentage Scores
A	4.0	95-100
A-	3.7	90-94

B+	3.3	86-89
B	3.0	83-85
B-	2.7	80-82
C+	2.3	76-79
C	2.0	70-75
F	0	0-69

Grade Point Average (GPA)

Grade or quality point values are assigned to letter grades as indicated in the grading system table. A quality or grade point score is computed by multiplying the value of a letter grade by the number of credits for the course. For example, a student receiving an A in a three-credit course earns twelve quality points. The GPA is computed by dividing the quality points earned by the number of credits graded A through F (GPA hours).

Additional Grade Notations

Satisfactory/Unsatisfactory (S/U):

An “S” grade reflects satisfactory work and is considered passing in a course; a “U” grade reflects unsatisfactory work and is considered a failure. Courses evaluated with the “S” and “U” assessment have no effect on GPA.

Incomplete (I):

This grade is given to students who are passing a course but are unable to complete all the course work or the final exam for a verified reason beyond their control. The student must then complete all the requirements by the end of the next semester, not including any summer term, and the instructor must turn in the final grade by the end of the that semester’s grading deadline. Unless an explicit written extension is filed, the grade of “I” is changed to an “F” if the course requirements have not been fulfilled. Students who have filed their intention to graduate have only six weeks from the date of degree conferral to resolve any incomplete grades. An “I” grade is not calculated in a student’s GPA.

In Progress (IP):

The grade of “IP” indicates that the student is making progress in a course, writing a thesis or dissertation, or participating in an internship that extends beyond one semester or summer term. The grade of “IP” has no effect on the student’s GPA.

Withdrawal (W):

The grade of “W” on a course indicates that the student has dropped the course. While it has no effect on the GPA, dropped courses are part of attempted course credits that serve as the basis for the student’s credit level. A “W” for all courses in a given semester and the transcript notation “withdrawn” indicates that the student withdrew from AURAK.

Repeating a Course

Normally, graduate courses cannot be repeated. With the recommendation of the dean of the student’s school and with the approval of the Associate Provost of Research, a graduate student may

be allowed to repeat any course in which a grade of “C+,” “C,” or “C-” is received. The original grade and the new grade will appear on the transcript, but only the new grade will be calculated into the GPA.

No course may be taken more than twice.

Academic Progress / GPA Retention Levels

Satisfactory Progress

To make satisfactory progress toward degree completion, the student must attain a CGPA of 3.0 (“B”) or better.

Academic Probation and Dismissal

Graduate students receiving a grade of F in any given term will be subject to academic dismissal from the university. If the CGPA of a graduate student falls below 3.0 in any semester and no grade of F is earned, the student will be placed on Probation. Graduate students on probation will be allowed a maximum of 6 semester credit hours in the subsequent semester. Graduate students failing to earn a CGPA of 3.0 or higher in the subsequent semester will be subject to academic dismissal from the university. Probation will be removed at the end of any semester in which the student attains a CGPA of 3.00.

Academic Dismissal

A graduate student on probation who does not achieve good academic standing by the end of the regular semester following the term in which the CGPA fell below 3.00 will be dismissed from the university.

University Withdrawal

Official withdrawal removes students from any academic program and cancels student status at AURAK. In order to return to the university, students need to apply for readmission through the Department of Admissions. Students who wish to withdraw have to complete a “Withdrawal Form” available at the Department of the Registrar and get clearance from all university departments. If the withdrawal process is completed satisfactorily and all financial obligations to the university are cleared, the effective date of withdrawal is noted on the student’s permanent academic record. The effective date is the date used for calculating billing or refunds. No grades other than a “W” for the current semester are recorded.

Leave of Absence

Occasionally, students are forced to leave the university for a semester or two because of circumstances beyond their control. Others find they simply need a break from studying. In such circumstances, taking a leave of absence might be wise. Students who have an approved leave of absence for a semester or a year may register for the semester in which they plan to return without applying for readmission. Unless there are extenuating circumstances such as illness, a leave of absence is not normally given to a student who leaves the university during a term.

Fields of Study

Degree Offerings

The American University of Ras Al Khaimah has three schools that offer both undergraduate and graduate degree programs. Graduate programs are listed below:

- Post Graduate Diploma in Education
- Master of Education: Educational Leadership
- Executive Master of Business Administration
- Master of Business Administration
- Master of Science in Engineering Project Management
- Master of Science in Sustainable and Renewable Energy

Transferring from Non-Degree or Visiting to Degree Status

Students may request a change of status from non-degree to degree status or from visiting to degree status by submitting a complete application through the Department of Admissions. All admissions requirements in place at the time of the change-of-status request must be met. Schools may choose to take into consideration the student's academic performance in the 500-level courses taken with AURAK.

Courses taken while under the non-degree or the visiting status may apply towards the degree program given the approval of the dean of the student's school. Grades earned in courses that are accepted will count in the CGPA. The university rules and regulations governing transfer courses and credits will apply.

The graduation requirements will be determined by the catalog that is effective when the student joins a degree program. Additional leveling courses may apply if appropriate.

Change of Program

Students seeking to change their graduate degree program must complete the Change of Major Form available from the Department of the Registrar. Requests for a change of program should be submitted to the dean of the school of the program of the student's choice by the last day of the 12th week of classes of the fall or spring semester. The dean will forward the approved forms to the Department of Registration. Forms received by the Department of the Registrar by the end of the Add and Drop period will be effective as of the following semester.

To be eligible for a change of program, the graduate student must meet the requirements for admission to the new program. Please refer to the relevant program's catalog section for information on admission requirements.

Student Petitions and Appeals

Students' Responsibility

All official university communications are distributed through the AURAK-issued email address. These are considered official notifications. Students are responsible for checking their AURAK email accounts and for responding to or acting upon messages accordingly.

Students should keep their own records of all transactions with the university (e.g., registration schedules and forms, grade reports, payment records, etc.). It is also advisable to keep copies of all tests, digital files, papers and so forth submitted in fulfillment of course work. Students should keep copies of all course syllabi.

Appeal of Academic-Related Issues

If a student wishes to discuss an issue pertaining to a course, instructor, or other academic-related issues, the student may direct his or her concern to the involved faculty member. If a satisfactory resolution of the situation is not reached, the student may appeal to the Department Chair, then to the Dean of the School, and finally to the VPAASS/Provost (or his/her designee), who has the final authority to decide the merit of the appeal.

If the instructor is no longer associated with the university, the VPAASS/Provost will appoint a faculty surrogate, who will assume the authority of the instructor. If the matter is not resolved at this point, the VPAASS/Provost will hear the case and make a decision that is not subject to further appeal. Academic appeal requests must be submitted no later than the end of the first day of orientation week of the following semester.

Degree Completion

AURAK requires, for a Postgraduate Diploma, at least one year of study and at least 24 credit hours of study beyond the Bachelor's level. Master's degree programs will typically require at least one year of full-time study, or a minimum of 30 semester credits of course work (or equivalent) beyond the Bachelor's degree. These minimum credits shall not be inclusive of any non-credit bridge courses which may be required. Master's degrees offered by AURAK must include no less than 15 credit hours of courses (excluding thesis/dissertation credits) that are distinctive to that particular Master's program. AURAK also requires that all graduate students hold a CGPA of 3.0 in order to graduate and that Masters students complete a thesis.

AURAK reserves the right to change program or course requirements, grading procedures, or the requirements for acceptance into particular programs, academic standing, or graduation. These changes generally apply to all students enrolled at the time of implementation of the change and thereafter. When changes are made to degree requirements, however, students may elect to complete either the requirements in effect at the time of their admission or the new set of requirements, as detailed in the program requirements section of the catalog. Degree or program requirements do not include modality of program delivery (e.g., online or in-person delivery). With a change in modality of program delivery (e.g., shifting from in-person classes to online delivery of classes), the new modality of program delivery will typically apply only to students admitted following the change in modality.

For all degrees, requirements must be met as stated in the University Catalog. Candidates for graduate degrees who have been continuously enrolled may choose to graduate under the terms of any catalog in effect at or after their admission.

Master's Thesis

In AURAK, a Master's degree will require a substantial thesis of at least six and no more than nine credit hours. A master's thesis documents research conducted by AURAK graduate students undertaking a master's degree program. Preparation of the thesis includes training in research methods, the development of the thesis is supervised by appropriately qualified and experienced, full-time faculty. The thesis is the culmination of the students' programs of study and is expected to reflect appropriate scholarly depth and rigor.

Students who do not complete their thesis in the final semester of the program of study will be required to maintain continuous enrollment in the thesis in succeeding semesters, with an In-progress (IP) grade assigned in the course designed to be the final thesis course until subsequent thesis coursework leads to the successful completion of the thesis. At that time, all IP grades originally assigned will be converted by the thesis supervisor to standard letter grades.

Where a student wishes to pursue an additional Master's degree from AURAK, the student must meet the admission requirements for the second degree and complete requirements for the additional degree. The requirements of the second degree must include at least 15 credit hours of courses (excluding thesis/dissertation credits) that are distinctive and not taken to meet requirements of the first degree.

It is AURAK policy to maintain master's theses in AURAK's Saqr Library so that they are available to other students and scholars. Students completing their thesis will be required to submit a bound copy of their theses to the Library.

Maximum Graduate Degree Completion Periods

In order to graduate successfully from a graduate degree program after full admission to the program, the following periods for completion will be applied:

- Postgraduate Degree program (Diploma/Certificate) – four regular semesters.
- Master's Degree program
 - One year degree program must be completed within four regular semesters.
 - Two year degree program must be completed within six regular semesters.

Academic Standing Requirement

A student must be in good academic standing to be eligible for graduation.

Application for Graduation

Candidates for graduate degrees must file an "Application for Graduation Form" with the Department of the Registrar during the registration period of the last expected term of study. Only after an "Application for Graduation Form" has been filed can the Department of the Registrar begin processing the necessary information for final certification for graduation.

Students who fail to complete all degree requirements by the end of the term for which they apply to graduate need not reapply for graduation. Their previous application will be automatically moved to the following semester.

Conferral of Degrees

Only students who have successfully completed degree requirements and all thesis requirements, including corrections and final submission of the completed thesis to the library, by the end of the term for which they have applied to graduate are certified for conferral of a degree. Degrees are conferred at the end of the semester in which requirements have been met. Conferral of the degree is noted on the academic transcript of the graduate with the date of graduation.

Names on Degrees

The names of AURAK students will be spelled in English exactly as they appear on their passports or identity cards when printed on degrees. If a name on a passport or an identity card does not appear in English, then the spelling of the name will be printed according to the personal preference of the student.

Attestation of Degrees and Transcripts

The Department of the Registrar provides information relevant to the attestation of degrees and transcripts with the UAE Ministry of Education.

ACADEMIC PROGRAMS

Executive Master of Business Administration

Program Description

The Executive Master of Business Administration (EMBA) program mission is to disseminate advanced and contextualized business management knowledge, suitable for leading and managing organizations at senior level. In the process, produce high-quality business leaders who are able to craft creative and effective solutions to a variety of complex business problems; and also provide a learning environment that ensures that graduates lead and manage organizations ethically in a variety of local and international business settings.

The Executive Master of Business Administration is overseen by the School of Business and is designed to satisfy QF Emirates Level 9 requirements.

Vision Statement

Our vision is to be students' preferred choice for internationally recognized business education in the Northern Emirates enriched by the impactful research and outreach of our dedicated faculty.

School Mission Statement

The mission of the School of Business is to work with the AURAK 'family' to prepare tomorrow's business leaders to be thoughtful, creative, collaborative, and responsible with an ability to think critically, communicate their ideas clearly and take initiative in applying what they have learned. Backed by a diverse faculty committed to enrich their teaching and enhance their community impact by engaging in applied, cross-disciplinary, and pedagogical scholarship, the School of Business offers programs based on the North American model and the cultural characteristics of the Gulf region in a student-centered teaching environment.

Program Educational Goals

The overall goals of the program are:

1. To impart contemporary knowledge, and provide an integrated background in advanced concepts, frameworks and methodologies underpinning the high-level management of organizations.
2. To afford candidates the opportunity to concentrate on specific areas and issues within the business field. It also seeks to encourage life-long learning beyond the duration of the program.
3. To enhance professional communications skills and peer-based interaction, through its curriculum and associated group-based coursework.

Program Learning Outcomes

The program learning outcomes for the Executive MBA are as follows:

- PLO1** Evaluate advanced business knowledge for the effective functioning of local and international organizations across business sectors.
- PLO2** Enhance strategic leadership role through application of program knowledge for analyses of corporate strategies.
- PLO3** Assess effectiveness of strategic leadership in corporate governance, ethical functioning, and socially responsible organizational behavior.
- PLO4** Develop skills of collaborative decision making through teamwork and leadership to communicate complex business requirements to organizational personnel and stakeholders.
- PLO5** Produce critical and entrepreneurial thinking for effective decision-making towards business innovation and sustainability.
- PLO6** Integrate research principles and techniques with prior work experience to execute a major business project.
- PLO7** Combine personal reflection with program learning for professional development to executive level.

EMBA Remedial Courses

If an EMBA applicant holds a baccalaureate degree, or its equivalent, in a non-business field with a minimum CGPA of 3.00, followed by a postgraduate certificate or diploma in a business-related field, he or she can qualify for full admission to the program, assuming all other admission requirements (such as English language proficiency) are met.

However, for those prospective students who have a CGPA of 2.0-2.4 and/or hold a degree from a non-business field the student will have to complete remedial courses designed to prepare them for graduate-level coursework. The remedial courses consist of one compulsory course plus two courses chosen from a restricted list as detailed in the table below.

Compulsory Course:	EMFN 500 EMBA Foundation (3 CH)
Restricted List (any two)	EDUC 525 Theories of Leadership and Administration (3 CH) EDUC 640 Inclusive Leading of Diverse Groups (3 CH) ENGR 500 Engineering Foundation for Graduate Students (3 CH)

The student must achieve a minimum CGPA of 3.0 on a 4.0 scale, or its established equivalent, in these nine credits of remedial courses in order to progress to the graduate program or be subject to dismissal.

Only after passing the three remedial courses within two semesters will students be allowed to register for the EMBA Program. An unsuccessful student who has not completed the courses within two semesters will not be able to continue the EMBA program and will have to leave the institution.

Where feasible, the foundational business course and the other remedial courses, which do not count toward the required credits for the degree, may be run in a condensed format (e.g., as condensed multi-weekend course modules) provided the learning outcomes remain unchanged and achievable.

AURAK will also accept candidates who have taken and passed the EMBA preparatory courses from other UAE Ministry or Higher Education-accredited universities, provided the courses taken have compatible syllabi and learning outcomes.

Degree Requirements

The Executive Master of Business Administration requires the completion of 36 credits in the following areas:

Area	Credits
I. Core Course Requirements	30
II. Elective Course Requirements	6
Total	36

CORE COURSES

30 Credits

EMBA Core Courses			
Course Code	Course Title	Pre-requisite/ Co-requisite	Credits
EMMG 510	Leadership and Managing People	Graduate Standing	3
EMAC 512	Managerial and Financial Accounting	Graduate Standing	3
EMFN 514	Managerial Finance	Graduate Standing	3
EMMG 515	Strategic Management	Graduate Standing	3
EMMK 518	Marketing Management	Graduate Standing	3
EMMI 520	Managing Information Systems Performance	Graduate Standing	3
EMIB 524	International Business	Graduate Standing	3
EMGN 525	Research Methods for Business	Graduate Standing	3
EMGN 555	EMBA Investigative Thesis or Project I (Thesis Research Course)	EMGN 525	3
EMGN 556	EMBA investigative Thesis or Project II (Thesis Research Course)	EMGN 555	3

ELECTIVE COURSES

6 Credits

EMBA Elective Course Requirements

Course Code	Course Title	Pre-requisite/ Co-requisite	Credits
Elective 1	Any course from the EMBA Electives List	-	3
Elective 2	Any course from the EMBA Electives List	-	3

EMBA Elective Courses – Any Two Courses			
Course Code	Course Title	Pre-requisite/ Co-requisite	Credits
EMFN 560	International Corporate Finance	EMFN 514	3
EMFN 562	Investment Planning & Management	EMFN 514	3
EMFN 564	Financial Markets	EMFN 514	3
EMHR 590	Applied and Strategic Human Resources Management	EMMG 510	3
EMHR 592	Managing Change and Innovation	EMMG 510	3
EMHR 594	Employee Relations and Compensation Management	EMMG 510	3

For Further Detailed information, Please Visit: <https://aurak.ac.ae/en/academics/school-business/academic-programs/executive-master-of-business-administration/>

Executive Master of Business Administration - Two Year Plan

Year 1, First Semester			Year 1, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
EMMG 510	Leadership and Managing People	3	EMGN 525	Research Methods for Business	3
EMFN 514	Managerial Finance	3	EMAC 512	Managerial and Financial Accounting	3
EMMG 515	Strategic Management	3	EMMK 518	Marketing Management	3
Subtotal =		9	Subtotal =		9
Year 2, First Semester			Year 2, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
EMGN 555	EMBA Investigative Thesis or Project I	3	EMGN 556	EMBA Investigative Thesis or Project II	3
EMIB 524	International Business	3	EMMI 520	Managing Information Systems Performance	3
Elective 1	Any course from the EMBA Elective List	3	Elective 2	Any course from the EMBA Elective List	3
Subtotal =		9	Subtotal =		9
Total 36 Semester Credit Hours					

Master of Business Administration

Program Description

The Master of Business Administration (MBA) program prepares potential students to become effective business managers and leaders of the future. The program provides basic and advanced knowledge and skills in various aspects of business management. The program provides an opportunity for students to specialize in an area of Finance or Human Resources Management.

The Master of Business Administration is overseen by the School of Business and is designed to satisfy QF Emirates Level 9 requirements.

Vision Statement

Our vision is to be students' preferred choice for internationally recognized business education in the Northern Emirates enriched by the impactful research and outreach of our dedicated faculty.

School Mission Statement

The mission of the School of Business is to work with the AURAK 'family' to prepare tomorrow's business leaders to be thoughtful, creative, collaborative and responsible with an ability to think critically, communicate their ideas clearly and take initiative in applying what they have learned. Backed by a diverse faculty committed to enrich their teaching and enhance their community impact by engaging in applied, cross-disciplinary, and pedagogical scholarship, the School offers programs based on the North American model and the cultural characteristics of the Gulf region in a student-centered teaching environment.

Program Educational Goals

The overall goals of the program are:

1. To impart contemporary knowledge, and provide an integrated background in advanced concepts, frameworks and methodologies underpinning the high-level management of organizations.
2. To afford candidates the opportunity to concentrate on specific areas and issues within the business field. It also seeks to encourage life-long learning beyond the duration of the program.
3. To enhance professional communications skills and peer-based interaction, through its curriculum and associated group-based coursework.

Program Learning Outcomes

On successful completion of this program the graduate will be able to:

- PLO1** Evaluate core business knowledge for the effective functioning of local and international organizations across business sectors.
- PLO2** Enhance strategic management role through analysis of the corporate strategy, constructing viable and effective solutions in the process.
- PLO3** Assess effectiveness of corporate governance, ethical functioning, and socially responsible organizational behavior.

- PLO4** Develop teamwork, interpersonal, communication and leadership skills for communicating business requirements to organizational personnel and stakeholders.
- PLO5** Produce critical and entrepreneurial thinking to support decision-making for business innovation and sustainability.
- PLO6** Integrate research principles and techniques to address business issues.
- PLO7** Combine personal reflection with further learning for professional development of own career.

MBA Remedial Courses

If an MBA applicant holds a baccalaureate degree, or its equivalent, in a non-business field with a minimum CGPA of 3.00, followed by a postgraduate certificate or diploma in a business-related field, he or she can qualify for full admission to the program, assuming all other admission requirements (such as English language proficiency) are met.

However, for those prospective students who have a CGPA of 2.0 – 2.4 and/or hold a degree from a non-business field the student will have to complete remedial courses designed to prepare them for graduate-level coursework. The remedial courses consist of one compulsory course plus two courses chosen from a restricted list as detailed in the table below.

Compulsory Course:	MBFN 500 MBA Foundation (3 CH)
Restricted List (any two)	EDUC 525 Theories of Leadership and Administration (3 CH) EDUC 640 Inclusive Leading of Diverse Groups (3 CH) ENGR 500 Engineering Foundation for Graduate Students (3 CH)

The student must achieve a minimum CGPA of 3.0 on a 4.0 scale, or its established equivalent, in these nine credits of remedial courses in order to progress to the graduate program or be subject to dismissal.

Only after passing the three remedial courses within two semesters will students be allowed to register for the MBA. An unsuccessful student who has not completed the courses within two semesters will not be able to continue to the MBA program and will have to leave the institution.

Where feasible, the foundational business course and the other remedial courses, which do not count toward the required credits for the degree, may be run in a condensed format (e.g. as condensed multi-weekend course modules) provided the learning outcomes remain unchanged and achievable.

AURAK will also accept candidates who have taken and passed the MBA preparatory courses from other UAE Ministry of Higher Education-accredited universities, provided the courses taken have compatible syllabi and learning outcomes.

Program Requirements

The Master of Business Administration requires the completion of 36 credits in the following areas:

MBA DEGREE REQUIREMENTS	
Area	Credit
Core Courses	30
Elective Courses	6

Total	36
--------------	-----------

Degree Requirements

Core Course Requirements (30 Credits)			
Course Code	Course Title	Pre-requisite/ Co-requisite	Credits
MBMG 510	Leadership and Managing People	Graduate Standing	3
MBAC 512	Managerial and Financial Accounting	Graduate Standing	3
MBFN 514	Managerial Finance	Graduate Standing	3
MBMG 515	Strategic Management	Graduate Standing	3
MBMK 518	Marketing Management	Graduate Standing	3
MBMI 520	Managing Information Systems Performance	Graduate Standing	3
MBIB 524	International Business	Graduate Standing	3
MBGN 525	Research Methods for Business	Graduate Standing	3
MBGN 555	MBA Investigative Thesis or Project I (Thesis Research Course)	MBGN 525	3
MBGN 556	MBA Investigative Thesis or Project II (Thesis Research Course)	MBGN 525	3
MBA Elective Course Requirements (6 Credits)			
Elective 1	Any MBA Elective Course	-	3
Elective 2	Any MBA Elective Course	-	3

Further Detailed information, please visit: <https://aurak.ac.ae/en/academics/school-business/academic-programs/master-business-administration-mba/>

MBA Elective Course List – Any 2 Courses			
Course Code	Course Title	Pre-requisite/ Co-requisite	Credits
MBFN 560	International Corporate Finance	MBFN 514	3
MNFN 562	Investment Planning & Management	MBFN 514	3
MBFN 564	Financial Markets	MBFN 514	3
MBHR 590	Applied and Strategic Human Resources Management	MBMG 510	3
MBHR 592	Managing Change and Innovation	MBMG 510	3

MBHR 594	Employee Relations and Compensation Management	MBMG 510	3
----------	--	----------	---

Further Detailed Information, please visit: <https://aurak.ac.ae/academics/school-business/academic-programs/master-business-administration-mba/>

Master of Business Administration - Two Year Plan

Year 1, First Semester			Year 1, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MBMG 510	Leadership and Managing People	3	MBGM 525	Research Methods for Business	3
MBFN 514	Managerial Finance	3	MBAC 512	Managerial and Financial Accounting	3
MBMG 515	Strategic Management	3	MBMK 518	Marketing Management	3
Subtotal =		9	Subtotal =		9
Year 2, First Semester			Year 2, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MBGN 555	MBA Investigative Thesis or Project	3	MBGN 556	MBA Investigative Thesis or Project II	3
MBIB 524	International Business	3	MBMI 520	Managing Information Systems Performance	3
Elective 1	Any MBA Elective Course	3	Elective 2	Any MBA Elective Course	3
Subtotal =		9	Subtotal =		9
Total 36 Semester Credit Hours					

Master of Education: Educational Leadership & Post Graduate Diploma in Education

Program Description

The Master of Education Degree is a thorough scholarly experience that provides educators with excellent learning opportunities in leadership theory and practice. Students will acquire practical knowledge in research, data-driven decision-making, and real-world leadership. Upon completing this three-semester program, Graduates will have the necessary skills to become positive change agents capable of leading the continuous improvement of schools and student learning.

The program is designed to provide educational professionals with the opportunity to develop the necessary skills to become effective educational leaders, school administrators, cluster managers, educational researchers, employees of governmental agencies, and/or higher education professionals.

The program provides two exit points. The first is the Post Graduate Diploma in Education (PGDE) with twenty-four (24) credit hours, and the second is the Master of Education with an additional nine (9) credit hours, for a total of thirty-three (33) credit hours.

The Master of Education consists of a total of thirty-three (33) credit hours. Students may choose to complete the Post Graduate Diploma only (in two semesters) or continue for a third semester to complete the Master of Education Degree. All students will take four courses (12 credit hours) the first two semesters, completing the PGDE in one year. Those choosing to obtain the Master of Education Degree will continue for an additional semester and complete three additional courses (9 credit hours). PGDE completers must evidence a CGPA of 3.3 or higher to progress to the Master of Education program.

During the first and second semesters, student will take two courses over 16 weeks, with an additional two classes offered in an intensive mode: one in Fall A (during the first eight weeks) and one in Fall B (during the last eight weeks). The same applies for Spring semesters: students take two 16-week courses and two additional, intensive mode classes: one in Spring A and one in Spring B. This approach facilitates a more timely completion of the program, while also providing important time to master course content and skills. Again, students must maintain a 3.3 CGPA to continue into the third semester. During the third semester students will take three courses (9 credit hours) over 16 weeks to complete the Master of Education pending final approval of their research project.

The Cohort Model

The PGDE and Med program codifies, in an intentional way, the value of learning communities. This is accomplished through the implementation of a Graduate Cohort. In a cohort model, all students take all the same courses at the same times, altogether. Cohort models are common in Graduate education programs across the United States, as this approach yields immensely valuable benefits. Notable, a cohort model allows the program to achieve the following goals: (1) to implement a meaningful progression of knowledge and skills while promoting rigor, and (2) to build an authentic learning community.

All courses are scheduled according to a pre-set timetable, and the language of instruction is English. Courses are offered in a face-to-face format with designated out-of-classroom experiences/activities conducted collaboratively or independently to complement course content and interactions. The

program is collaborative with teaching and learning occurring through applied interactive lectures, discussions, case studies, and projects. A student-centered, active learning, and inquiry-based approach is consistently utilized, and the entire program incorporates reflective practice.

The Post Graduate Diploma in Education is designed to satisfy QF Emirates Level 8 requirements. The Master of Education: Educational Leadership is designed to satisfy QF Emirates Level 9 requirements. Both awards are overseen by the School of Arts and Sciences.

Program Mission

AURAK's Master of Education and Post Graduate Diploma in Education offer a transformational experience to graduate candidates as they actively apply theory, critical reflection and analysis, and best practice. Graduate students in these programs actively engage in practical diverse discourse on topics related to leadership, change agency, and governance within the educational systems of the UAE and surrounding region. Candidates obtain the requisite knowledge, skills, and tools to meaningfully contribute to positive change in addressing the dynamic challenges that occur when leading educational institutions, communities, faculty, and staff, in both their daily activities and in overall systemic change.

Program Educational Objectives

Graduates of the Master of Education and the Post Graduate Diploma in Education are prepared to actively contribute to K-12 schools, higher education institutions, and governmental entities in leadership roles such as instructional, administration and community building. The program is designed to prepare educational leaders who can substantially impact the continuous improvement of student learning based on data informed decision-making, real-world research skills, meaningful assessment, and the implementation of best practice.

The specific program goals of the Master of Education and Post Graduate Diploma in Education are to:

1. Prepare scholar-practitioners who engage in lifelong learning to meaningfully develop their careers in education;
2. Produce educators who apply in reflective practice to promote inclusion and diverse discourse grounded in the culture of the region;
3. Prepare scholar-practitioners who engage in data-driven decision making, to effect meaningful, research-based positive change in academic settings; and
4. Develop problem-solvers with advanced knowledge of education and specialized skills in education who contribute to overcoming challenges in education, academic settings, and at the intersection of education and industry.

Post Graduate Diploma in Education (PDGE) Program Learning Outcomes

Graduates of the Post Graduate Diploma in Education are expected to:

1. Assess advanced knowledge of leadership theories and/or methodologies as the basis for original thinking, encompassing reflective practice and ethical processes of knowledge production;
2. Evaluate the advanced skills required in educational leadership research, analysis, and the innovation of complex ideas in academic settings and where education meets industry;

3. Construct a comprehensive internalized, personal worldview, while accepting responsibility to society, and the sociocultural norms and relationships;
4. Take responsibility leading in the strategic performance and development of professional teams and self within the ethical boundaries of the discipline;
5. Critically reflect and actively choose to maintain and enhance leadership skills in the field of education; and
6. Participate collectively in developing academic settings and the public sector where education meets industry using best practices in leadership.

Master of Education (MEd) Program Learning Outcomes

Graduates of the Master of Education degree are expected to:

1. Assess highly specialized knowledge of leadership theories and/or methodologies as the basis for original thinking, encompassing reflective practice and ethical processes of a detailed body of knowledge of recent developments in the field of educational leadership;
2. Evaluate the specialized and advanced problem-solving skills required in educational leadership, research and analysis, while utilizing specialist communication and technology skills to present, explain and/or critique highly complex matters;
3. Construct a comprehensive, internalized, personal worldview, while accepting responsibility to society, and the sociocultural norms and relationships;
4. Take responsibility for leading strategic performance and development of professional teams, and self, in learning contexts that are complex and may require new strategic approaches within the ethical boundaries of educational leadership;
5. Critically reflect, and actively choose to enhance, leadership skills in the field of Education to contribute to, and implement, ethical leadership standards that advances professional knowledge and practice; and
6. Lead and participate in developing academic settings, and where education meets industry, demonstrating best practice in leadership.

Diploma/Degree Requirements

The following program requirements must be met for degree completion.

1. Successful complete the appropriate 24 (PGDE) or 33 (MEd) credit hours;
2. Maintain a Grade Point Average of 3.00 or higher throughout the program;
3. Successful participation in Research Presentations designed to disseminate knowledge of best practice within local educational environments; and
4. Successful construction, implementation, analysis, and presentation of an original research project (MEd).

Assuming students do not interrupt their study, the MEd degree is designed to be completed in three semesters (generally, this equates to 18 months). Students who withdraw or take a leave of absence from the program must meet requirements for returning that are outlined in AURAK's catalog. Students must meet specific standards to progress and do so within the maximum time allowed to complete the program; these standards and time allotment are also detailed in the catalog. If a degree is not

completed within a period of six years, all coursework in the degree may be re-evaluated for its current relevance.

The PGDE program requires twenty-four credit hours detailed as presented below.

Program Course Requirements for the Post Graduate Diploma in Education		
Course No.	Title	Credit Hours
EDUC 511	Introduction to Applied Research (Thesis Research Course I)	3
EDUC 515	Theories of Teaching and Learning	3
EDUC 525	Theories of Leadership and Administration	3
EDUC 545	Coaching, Mentoring, and Developing School Staff	3
EDUC 620	Supervision and Evaluation of Instruction	3
EDUC 640	Inclusive Leading of Diverse Groups	3
EDUC 660	Implementation and Evaluation of Comprehensive School Reforms	3
EDUC 680	Applied Research I (Thesis Research Course II)	3
TOTAL =		24

The M.Ed. program requires thirty-three credit hours detailed as presented below.

Program Course Requirements for the Master of Education		
Course No.	Title	No. of Credit Hours
EDUC 511	Introduction to Applied Research (Thesis Research Course I)	3
EDUC 515	Theories of Teaching and Learning	3
EDUC 525	Theories of Leadership and Administration	3
EDUC 545	Coaching, Mentoring, and Developing School Staff	3
EDUC 620	Supervision and Evaluation of Instruction	3
EDUC 640	Inclusive Leading of Diverse Groups	3
EDUC 660	Implementation and Evaluation of Comprehensive School Reforms	3
EDUC 675	School Governance and Community Service	3
EDUC 680	Applied Research I (Thesis Research Course II)	3
EDUC 685	Continuous Improvement Through Data-Informed Leadership	3
EDUC 695	Applied Research II	3
TOTAL =		33

Post Graduate Diploma in Education Two-Semester Plan

Schedule of Delivery PGDE in Educational Leadership					
Year One, First Semester					
Course No	Course Title	CR	Course No		CR
EDUC 511	Introduction to Applied Research (Thesis Course I) - 16 weeks				3
EDUC 525	Theories of Leadership and Administration – 16 weeks				3
Intensive Course: Fall A (8 weeks)			Intensive Course Fall B (8 weeks)		
EDUC 515	Theories of Teaching and Learning	3	EDUC 545	Coaching, Mentoring, and Developing School Staff	3
Semester Subtotal =					12
Year One, Second Semester					
EDUC 660	Implementation and Evaluation of Comprehensive School Reform (16-weeks)				3
EDUC 680	Applied Research I (Thesis Course II) (16-weeks)				3
Intensive Course: Spring A (8 weeks)			Intensive Course Spring B (8 weeks)		
EDUC 620	Supervision and Evaluation of Instruction	3	EDUC 640	Inclusive Leading of Diverse Groups	3
Semester Subtotal =					12
Total Program Credit Hours = 24					

Master of Education: Educational Leadership Three-Semester Plan

Schedule of Delivery M. Ed in Educational Leadership					
Year One, First Semester					
Course No	Course Title	CR	Course No		CR
EDUC 511	Introduction to Applied Research (Thesis Course I) - 16 weeks				3
EDUC 525	Theories of Leadership and Administration – 16 weeks				3
Intensive Course: Fall A (8 weeks)			Intensive Course Fall B (8 weeks)		
EDUC 515	Theories of Teaching and Learning	3	EDUC 545	Coaching, Mentoring, and Developing School Staff	3
Semester Total =					12
Year One, Second Semester					
EDUC 660	Implementation and Evaluation of Comprehensive School Reform (16-weeks)				3
EDUC 680	Applied Research I (Thesis Course II) (16-weeks)				3
Intensive Course: Spring A (8 weeks)			Intensive Course Spring B (8 weeks)		
EDUC 620	Supervision and Evaluation of Instruction	3	EDUC 640	Inclusive Leading of Diverse Groups	3
Semester Total =					12
Year Two, First Semester					
EDUC 675	School Governance and Community Service (16 – weeks)				3
EDUC 685	Continuous Improvement Through Data-Informed Leadership (16-weeks)				3
EDUC 695	Applied Research II (Thesis Course III) (16-weeks)				3
Semester Total =					9
Total Program Credit Hours = 33					

Master of Science in Engineering Project Management

Program Description

The curriculum consists of thirty-six credit hours that provide the students with balanced coverage of project management: nine core courses and three elective courses. The core courses include a field application project, a capstone course in which the graduate student evidences the ability to apply the program learning outcomes to an applied project that integrates all basic elements of project management; planning; organizing; securing; managing; leading; and controlling resources to result in achievable specific goals project. The student will work with an instructing advisor to achieve project approval.

The Master of Science in Engineering Project Management is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 9 requirements.

Program Mission

The Master of Science in Engineering Project Management (MSEPM) program at AURAK mission is to meet the needs of graduates in disciplines requiring an advanced understanding of the theoretical and practical principles of the project management function.

The MSEPM program requires students to apply principles of engineering and management science to contribute to the growing body of knowledge in project management. The program will prepare project managers from diverse engineering disciplines to efficiently manage complex projects in the UAE and the Gulf Region countries. The program provides students with excellent technical and managerial skills: Change Control, Cost Management, Documentation Management, Information / Communication Management, Procurement, Project Context / Environment, Quality Management, Reporting, Resource Management, Risk Management, Time Management / Scheduling / Planning, Stakeholder / Relationship Management, Team Building / Development / Teamwork, Work Content, and Scope Management to better satisfy customer needs. After completing the requirements for the degree, students will be able to apply engineering and management knowledge to successfully lead and execute engineering projects. Students will be trained to design, model, manage and control complex projects, and prepare to play leadership roles in industry and government.

Program Educational Objectives

The Program Educational Objectives describe the expected accomplishments of graduates during their first few years after graduation. The program objectives have been derived from and support the mission statement of American University of Ras Al Khaimah. On successful completion of the Master of Science in Engineering Project Management program, a graduate should have:

1. Acquired skills in the initiation of projects, methods and techniques to control time cost and quality, resource management and long-term stewardship of assets;
2. Acquired a comprehensive understanding of the whole process of project management;
3. Gained project team leadership skills; and
4. Understood the establishment of project management methodologies.

Program Learning Outcomes

The program learning outcomes of the MSEPM program are aligned with those of the Project Management Institute (PMI), Global Accreditation Center (GAC), and with level nine descriptors of the QF Emirates. Students who successfully complete this program will be able to:

- PLO 1:** Apply the principles of project management knowledge throughout the project life cycle within a broad range of scope and complexity;
- PLO 2:** Utilize the full range of project management tools and techniques to satisfy industry and stakeholders' requirements best;
- PLO 3:** Demonstrate professional responsibilities concerning people, resources, and the environment adherence to the project management code of ethics and professional conduct.
- PLO 4:** Integrate scope, schedule, budget, resource, quality, procurement, and risk processes in projects to meet stakeholders' requirements.
- PLO 5:** Direct projects with a clear understanding of project manager responsibilities of professional teams within each domain area of project management to motivate and influence team members and stakeholders.
- PLO 6:** Create (Design) effective project management communications and problem-solving techniques, project status reporting, conflict management, negotiation, and stakeholder engagement.
- PLO 7:** Conduct research using highly developed cognitive and creative skills and intellectual independence that significantly contributes to expanding project management knowledge.
- PLO 8:** Develop a project management framework by tailoring the processes based on project complexity and the competing constraints of scope, schedule, cost, resources, quality, and risk.

Degree Requirements

The requirements of the MSEPM program are thirty-six (36) credit hours as follows: twenty-seven (27) credit hours are compulsory and nine (9) are electives.

Compulsory

27 Credit Hours

Course Code	Course Title	Credits
MEPM 511	Project Management Fundamentals	3
MEPM 512	Engineering Contracts and Procurement	3
MEPM 515	Project Scheduling	3
MEPM 521	Project Cost Accounting and Finance	3
MEPM 532	Engineering Management	3
MEPM 541	Organizational Behavior and Project Team Management	3
MEPM 543	Risk Management for Project Managers	3
MEPM 590	Thesis I (Thesis Research Course)	3
MEPM 591	Thesis II (Thesis Research Course)	3

Course Code	Course Title	Credits
MEPM 513	Project Implementation and Performance	3
MEPM 514	Global Projects Management	3
MEPM 516	Project Quality Management	3
MEPM 522	Financial Analysis and Decision Making	3
MEPM 531	Operation Research	3
MEPM 533	Information Systems for Project Management	3
MEPM 542	Supply Chain Management for Project Managers	3

Master of Science in Engineering Project Management

Two-Year Study Plan

Year 1, First Semester			Year 1, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MEPM 511	Project Management Fundamentals	3	MEPM 512	Engineering Contracts and Procurement	3
MEPM 532	Engineering Management	3	MEPM 515	Project Scheduling	3
MEPM 521	Project Cost Accounting and Finance	3	MEPM 543	Risk Management for Project Managers	3
Subtotal =		9	Subtotal =		9
Year 2, First Semester			Year 2, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MEPM 590	Thesis I	3		MEPM Elective (2)	3
MEPM 541	Organizational Behavior and Project Team Management	3		MEPM Elective (3)	3
	MEPM Elective (1)	3	MEPM 591	Thesis II	3
Subtotal =		9	Subtotal =		9
Total 36 Semester Credit Hours					

Master of Science in Sustainable and Renewable Energy

Program Description

The Master of Science in Sustainable and Renewable Energy program is designed to provide the students with balanced coverage of advanced renewable energy technologies. Students will be exposed to the status of energy resources, their interaction with the environment, different renewable energy sources technologies, different techniques and technologies for energy management and energy conservation along with the economic aspects of renewable energy-based power generation. The curriculum developed for this Master's degree brings together the necessary tools to learn how to identify energy needs and propose the application of techniques for the saving and efficient use of energy as well as integrate the knowledge acquired in the field of sustainability and energy efficiency. The curriculum consists of 36 credit hours composed of six core courses, three elective courses and nine credits for the master thesis. The program can be completed in as little as two years. The program also targets full-time working students.

The Master of Science in Sustainable and Renewable Energy is overseen by the School of Engineering and is designed to satisfy QF Emirates Level 9 requirements.

Program Mission

The mission of the Master of Science in Sustainable and Renewable Energy program at the American University of Ras Al Khaimah (AURAK) is to provide a high-quality education for United Arab Emirates students in the key aspects of Renewable and Sustainable Energy. Thus, it will enable the students to take responsible, creative, challenging and stimulating posts in research and industry in this exciting field.

The Master of Science in Sustainable and Renewable Energy program is designed to close the strategic gap between the technical aspects of renewable energy and the policy drivers for sustainable development. The program aims also to address the social, economic, and environmental issues involved with renewable energy systems. In this way, students will not only learn about technical and operational details, but also about the policy context, market mechanisms, and business models for renewable energy technologies and installations.

After completing the requirements for the degree, students will be in an excellent position to start or continue a career in renewable and sustainable energy technologies. Students will be trained to design, model, manage and control complex projects and prepared to play leadership roles in industry and government.

Program Learning Objectives (PLOs)

The Program goals describe the expected accomplishments of graduates during their first few years after graduation. The program goals have been derived from and support the mission statement of the American University of Ras Al Khaimah. On successful completion of the Master of Science in Sustainable and Renewable Energy, a graduate should have:

1. Acquired skills and competency in using renewable energy solutions in place of fossil fuel energy sources.
2. Acquired a comprehensive understanding of renewable and sustainable energy systems.
3. Teamwork skills in multidisciplinary projects and the identification of problems and proposal of solutions, including rigorous analysis and design of a process, component, or system.
4. Appreciation of the ethical and societal responsibilities entailed in the environmental engineering profession and the need for continuous education in the field and commitment to life-long learning.

Program Learning Outcomes

The outcomes of the Master of Science in Sustainable and Renewable Energy program have developed the Student Outcomes (SOs) that resulted from an extensive outreach effort and analysis. The outcomes are based on program educational objectives and are in line with level 9 descriptors of the QF Emirates. Students who successfully complete this program will develop:

1. An ability to acquire and apply knowledge in the field of renewable and sustainable energy systems, their components and interactions between the components;
2. An ability to identify and critically evaluate current developments and emerging trends within the renewable energy sector;
3. An ability to contextualize renewable technology projects within a local, national and international framework, and consider social, ethical and environmental impacts and consequences;
4. An ability to analyze, formulate, and handle technical problems from a system perspective, with an overview on their life-cycle, from idea/need to specification, development, maintenance and termination, and the ability to set conditions, decide necessary resource consumption and manage procedures to solve and realize problems;
5. An ability to communicate effectively with peers, other engineering experts and the community using a variety of media and instruments of communication;
6. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts; AND
7. An ability to work effectively and proactively within cross-cultural, multi-disciplinary teams, to accomplish high-level engineering results.

Degree Requirements

The requirements of the Master of Science in Sustainable and Renewable Energy are thirty-six (36) credit hours as follows: eighteen (18) credit hours are compulsory, nine (9) are electives, and nine (9) are for the thesis.

Degree Requirements	Credits
Core Courses	18
Elective Courses	9
Thesis	9
Total	36

Core Requirements

18 Credit Hours

Code	Course Title	Credit
MEPM 511	Project Management Fundamentals	3
MSRE 511	Energy Conversion	3
MSRE 512	Energy Systems Modeling and Optimization	3
MSRE 513	Energy Economics	3
MSRE 514	Direct Research	3
MSRE 515	Energy Efficiency and Management	3

Electives from the Following List

9 Credit Hours

Code	Course Title	Credit
MSRE 521	Arabic Language and Culture for Native Arabic Speakers I	3
MSRE 522	Introduction to Visual Arts	3
MSRE 523	Biofuels	3
MSRE 524	Sustainable Desalination	3
MSRE 525	Concentrated Solar Power	3
MSRE 526	Green Buildings	3

Master Thesis

9 Credit Hours

Code	Course Title	Credit
MSRE 590	Thesis I (Thesis Research Course)	3
MSRE 591	Thesis II (Thesis Research Course)	6
Program Total 36 Credit Hours		

Master of Science in Sustainable and Renewable Energy Two-Year Study Plan

Year 1, First Semester			Year 1, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
MEPM 511	Project Management Fundamentals	3	MSRE 513	Energy Economics	3
MSRE 511	Energy Conversion	3	MSRE 515	Energy Efficiency and Management	3
MSRE 512	Energy Systems Modeling and Optimization	3	MSRE 514	Direct Research	3
Subtotal =		9	Subtotal =		9
Year 2, First Semester			Year 2, Second Semester		
Course Code	Course Title	Credits	Course Code	Course Title	Credits
	MSRE Elective (1)	3		MSRE Elective (3)	3
	MSRE Elective (2)	3	MSRE 591	Thesis II	6
MSRE 590	Thesis I	3			
Subtotal =		9	Subtotal =		9
Total 36 Semester Credit Hours					



FACULTY LIST



FACULTY LIST

Abu Hantash, Tawfiq

Associate Professor, MS, Architecture Studies, Massachusetts Institute of Technology, USA

Abu-Raed, Abeer

Instructor, M.Sc., Architectural Engineering, Anhalt University of Applied Sciences, Germany

Ahmed, Irshad

Professor, PhD, Chemistry, Bhavnagar University, India

Al Alkim, Hassan

Professor and President, PhD, Political Science, Exeter University, UK

Al Hazza, Muataz

Associate Professor, PhD, Manufacturing Engineering, International Islamic University, Malaysia

Al Mansoori, Khaledah

Assistant Professor, PhD, Education (TEFL), University of Exeter, UK

Al Matroushi, Eisa

Associate Professor, PhD, Chemical Engineering, Pennsylvania State University, PA, USA

Al Shidrawi, Shadi

Instructor, M.A., English Literature, American University of Beirut, Lebanon

Al Zarooni, Mohammad

Associate Professor and Associate Provost, PhD, Chemical Engineering, University of Tennessee, USA

Al-Allaf, Mashhad

Associate Professor, PhD, Modern Philosophy, Science and Metaphysics, University of Tennessee, USA

Alamir, Islam

Lab Engineer, BS, Water Resources Engineering, Sudan University of Science and Technology, Sudan

Alamurin, Babatunde

Lab Engineer, BS Science Laboratory technology, University of Jos, Nigeria

Alassi, Eman

Associate Professor, PhD, Urban Design, University of Edinburgh, UK

Al-Hameedi, AboTaleb

Assistant Professor, PhD, Petroleum Engineering, Missouri University of Science & Technology, USA

Alkhanbouli, Ayman

Assistant Professor, PhD, Business Model Innovation, University of Hull, UK

Alnahhal, Mohammed

Assistant Professor, PhD, Industrial Engineering/Project Management, Duisburg-Essen University, Germany

Alnoman, Ali

Assistant Professor, PhD, Electrical and Computer Engineering, Ryerson University, Canada

Altawil, Shadi

Assistant Professor, PhD, Construction Engineering and Management, University of Alberta, Canada

Alzioud, Mohmoud

Assistant Professor, PhD, Civil Engineering, University of Akron, USA

AlZouabi, Hussain

Lab Technician, BS, Biotechnology, American University of Ras Al Khaimah AURAK, UAE.

Amirova, Gulbahor
Instructor, M.Ed., Educational Leadership, American University of Ras Al Khaimah AURAK, UAE.

Arman, Abdelfatah
Assistant Professor, PhD, Human Resource Management, Pennsylvania State University, USA

Arshi, Tahseen
Associate Professor, PhD, Entrepreneurship and Innovation, University of Bedfordshire, UK

Assaf, Hamed
Associate Professor, Associate Dean, PhD, Civil Engineering (Water Resources), University of British Columbia, Canada

Attia, Hussain
Senior Instructor, PhD, Electronics, University of Malaya, Malaysia

Awad, Mohammed
Associate Professor, PhD, Computer Science, University of Houston, USA

Awadallah, Ibrahim
Instructor, MS, Mathematics and Statistics, Mu'tah University, Jordan

Balhasan, Saad
Assistant Professor, PhD, Petroleum Engineering, University of Wyoming, USA

Batti, Tariq
Associate Professor, PhD, Business and Industry Operation

Begum, Vazeerjan
Associate Professor, PhD, Management & Marketing, Mother Teresa Women's University, India

Berriche, Hamid
Professor, PhD, Quantum Physics, Paul Sabatier University, France

Butt, Atif
Associate Professor, PhD, Supply Chain Management, Monash University, Australia

Cazacova, Liudmila
Associate Professor, PhD, Architecture, European University of Lefke, Cyprus

Dorstewitz, Philipp
Associate Professor, PhD, Philosophy and Planning Theory, London School of Economics, UK

El-Jammal, Walid
Senior Instructor, Ed.D., Innovation & Leadership in Education, Wilmington University, USA

Elmenghaw, Fathia
Assistant Professor, PhD, Urban Systems, New Jersey Institute of Technology – Newark, USA

Elsayed, Hanan
Instructor, MA in Media and Communication, University of Wollongong, UAE

Faiz, Sara
Assistant Professor, PhD, Chemical Engineering, Universiti Teknologi Petronas, Malaysia

Farhat, Maissa
Associate Professor, PhD, Electrical Engineering, University of Gabès, Tunisia

Getu, Beza
Associate Professor, PhD, Electrical and Communication Engineering, Aalborg University, Denmark

Ghani, Arfan
Associate Professor, PhD, Computer Systems Engineering, University of Ulster, UK

Haque, Sabir
Assistant Professor, PhD, Communications, Manipal Academy of Higher Education, India

Harrison, Gbubemi
Associate Professor, PhD, Petroleum Engineering (Reservoir & Production), Heriot-Watt University, UK

Hazirbaba, Kenan
Professor, PhD, Civil Engineering, University of Texas at Austin, USA

Hodeify, Rawad
Assistant Professor, PhD, Biomedical Sciences, University of Arkansas, USA

Holmes, Bryn
Associate Professor, PhD, Education, University of Cambridge, UK

Hossin, Khaled
Assistant Professor, PhD, Mechanical (Thermal) Engineering, Northumbria University, UK

Hussain, Khalid
Professor and Dean, PhD, Mathematical Sciences, Loughborough University, UK

Jaber, Nasser
Assistant Professor, PhD, Mechanical and Industrial Engineering, University of Toronto, Canada

Kalyebara, Baliira
Assistant Professor, PhD, Finance, Victoria University, Australia

Kemp, Linzi
Professor, PhD, Management, Manchester Metropolitan University, UK

Kharseh, Mohamad
Assistant Professor, PhD, Renewable Energy, Lulea University of Technology, Sweden

Khodr, Majed
Associate Professor, PhD, Electrical and Computer Engineering, University of Oklahoma, USA

Kilic, Kutbettin
Assistant Professor, PhD, Near Eastern Languages and Cultures, Indiana University, USA

Kim, Bong Sik
Assistant Professor, PhD, Mathematics, Arizona State University, USA

Kumar, Uday
Assistant Professor, PhD, Energy Technology, KTH Royal Institute of Technology, Sweden

Mahdi, Mohammad
Associate Professor, PhD, Management, University of Aberdeen, Scotland, U.K.

Marton, John
Senior Instructor, MS, Biochemistry, Bharathidasan University, India

Matar, Rachel
Associate Professor, PhD, Structural and Functional Biochemistry, University Claude Bernard, France

Merabet, Mustapha
Associate Professor, Associate Dean, PhD, Energy Sciences, University of Quebec, Canada

Merheb, Maxime
Associate Professor, PhD, Forensic Biology and Biophysics, Ecole Normale Supérieure, France

Mhairat, Walaa
Lab Instructor, MS, Civil Engineering (Transportation), University of Jordan, Jordan.

Miqdadi, Issam
Assistant Professor, PhD, Civil Engineering (Environmental Engineering), University of Newcastle, U.K.

Moutassem, Fayez
Assistant Professor, PhD, Civil Engineering, McMaster University, Canada

Nassar, Roz-Ud-Din
Associate Professor, PhD, Civil Engineering, Michigan State University, USA

Omar, Amjad
Professor, PhD, Electrical Engineering, University of Waterloo, Canada

Omar, Khaled
Assistant Professor, PhD, Computer Science, La Trobe University, Australia

Omari, Mohammed
Professor, PhD, Computer Science, New Mexico Institute of Mining and Technology, USA

Osman, Saoussene
Instructor, MS, Applied Mathematics, Université Ferhat Abbas, Sétif 1, Algeria

Proff, Alexandria
Assistant Professor, EdD, Organizational Change and Leadership, University of Southern California, USA

Sakhrieh, Ahmad
Professor, PhD, Mechanical Engineering (Combustion), Lehrstuhl Für Technische Thermodynamik Friedrich-Alexander University, Germany

Salameh, Khouloud
Assistant Professor, PhD, Computer Science, University of Pau and Pays de l'Adour, France; PhD, Control Engineering, Automation and Robotics, the University of the Basque Country, Spain.

Schoch, Daniel
Associate Professor, PhD, Philosophy of Science (Economics), Frankfurt University, Germany

Sediadi, Eka
Associate Professor, PhD, Architecture, Hanover University, Germany

Semlali, Anis
Associate Professor, PhD, Planning, University of Montreal, Canada

Shublaq, Inshirah
Teaching Assistant, BS, Architectural Engineering, American University of Sharjah, UAE

Sohaib, Osama
Associate Professor, PhD, Information Systems, University of Technology, Sydney, Australia

Tadele, Haileslasie
Assistant Professor, PhD, Finance, University of Otago, New Zealand

Takruri, Maen
Associate Professor, PhD, Electrical Engineering, University of Technology, Australia.

Thomas, Rinku
Assistant Professor, PhD, Polymer Chemistry, the Cochin University of Science and Technology, India

Uaje, Arnold
Lab Engineer, BS, Computer Science, Republic Central Colleges, Philippines

Ulusoy, Suleyman
Professor, PhD, Mathematics, Georgia Institute of Technology, USA.

Vazhappily, Cijo
Assistant Professor, PhD, Biology, VIT University, India

Waller, Sharon
Assistant Professor, PhD, Special Education, Texas Women's University, USA

Wallis, Joseph
Professor and Dean, PhD, Economics, Rhodes University, South Africa

Waseem, Shagufta
Associate Professor, PhD, Chemistry, Lucknow University, India

Wilhite, Stephen
Professor and Provost, DPhil, Human Experimental Psychology, University of Oxford, UK

Yousef, Basem
Associate Professor, PhD, Engineering Science (Robotics), University of Western Ontario, Canada

Zaid, Mohammed
Instructor, MS, Industrial Engineering, University of Jordan, Jordan

Zdravkovska, Suzana
Associate Professor, PhD, Communication Science, University Ss. Cyril and Methodius, Macedonia

Zengos, Hariclea
Professor and Dean, PhD, English, Tufts University, USA



COURSE DESCRIPTIONS



COURSE DESCRIPTIONS

Program Title	Program Code
Accounting	ACCT
Architecture	ARCH
Arabic	ARAB
Artificial Intelligence	CSAI
Biology, Bioinformatics, Biotechnology	BIOL
Business, Business Analytics (Undergraduate)	BUAN/BUSN/MGHR/MGMT/ MIST/MKTG/OPMT
Business (MBA/EMBA Graduate)	EMAC/EMFN/EMGN/EMHR/EMIB/ EMMG/EMMI/EMMK/ MBAC/MBFN/MBGN/MBHR/MBIB/ MBMG/MBMI/MBMK/ MBOM
Chemical Engineering	CHEN
Chemistry	CHEM
Civil & Infrastructure Engineering	CIEN
Communication	COMM
Computer Engineering	CENG
Computer Sciences	CSCI
Economics	ECON
Education	EDUC
Electrical and Electronics Engineering	EEEN
Electrical Engineering	ELEN
Electronics and Communication Engineering	ECEN
Engineering	ENGR
Engineering Project Management	MEPM
English	ENGL
Environmental Science	ENVS
Finance	FNAN
Geography	GEOG
History	HIST
Hospitality and Tourism	HTMT
Industrial Engineering	IENG
Information Technology	ITEC
Interior Design	INDS
Mathematics	MATH
Mechanical Engineering	MENG
Middle Eastern Studies	MEST
Petroleum Engineering	PENG
Philosophy	PHIL
Physics	PHYS
Political Sciences	POLS
Psychology	PSYC
Sociology	SOCI

Statistics	STAT
Sustainable and Renewable Energy (Graduate)	MSRE
University Studies	UNIV

Course Descriptions

Each course provides a course number, name, semester credit hours, prerequisites and corequisites if any, and a detailed course description. Course titles are followed by values in parentheses (0:0:0) separated by colons. The first number refers to the number of lecture hours per week; the second number refers to the number of laboratory hours per week; and the third number refers to the number of practicum or internship hours per week. For independent study, readings, topics, or similar courses individual instructors set hours.

Course Level Designations

AURAK offers degree programs that provide a coherent course of study. The coursework in all undergraduate programs follows a logical pattern from the 100 through the 200, 300, and 400 levels. The distinctions among these levels are generally understood as follows:

000-level course designation

Prerequisite courses serving to prepare students for successful performance in college level courses.

100-level course designation

Courses with no prerequisites, or survey courses, or courses defining basic concepts or presenting the terminology of a discipline.

200-level course designation

Courses of intermediate university-level difficulty; courses with 100-level course(s) as prerequisite(s); or survey courses devoted to specific areas within a discipline.

300-level course designation

Courses of advanced university-level difficulty taken by majors and upper division students who have a clear interest and who are qualified in the subject.

400-level course designation

Advanced courses and/or seminars, tutorials and special courses for majors and upper-division students.

For Masters Programs courses are offered at the 500 and 600 levels:

500-level course designation

Courses that assume bachelor's level familiarity with the subject, typically serve as the core for graduate programs, and are the foundation for more advanced and focused study.

600-level course designation

Courses at a more advanced graduate level through which students attend to focused and specialized topics in the discipline.

The number designations of the course descriptions in this chapter have the following significance:

- The single number (BIOL 100) indicates the course is complete within a single semester, and the semester course may be taken separately with credit toward a degree.
- The double number separated by a comma indicates that the subject matter or content of the course extends through two semesters, but that either semester may be taken by itself.
- Unless otherwise specified, the first semester is not a pre-requisite to the second semester.

COURSE DESCRIPTIONS

ACCT 204 Principles of Financial Accounting

(3:0:0)

3 Semester Credit Hours

Prerequisite(s): ECON 103

This is an introductory course in financial accounting which incorporates International Financial Reporting Standards (IFRS)/GAAP and emphasizes the use of accounting information in an ethical manner. Topics covered include: Basic accounting concepts and processes with particular emphasis on double entries; the accounting cycle; transaction analysis; accounting for assets, liabilities, owner's equity, expenses and revenues recognition, preparation of cash flow statements and ethical decision-making.

ACCT 205 Principles of Managerial Accounting

(3:0:0)

3 Semester Credit Hours

Prerequisite(s): ECON 103

This is the first course in cost/managerial accounting. It discusses a range of introductory managerial accounting topics such as cost measurement and terminology, costs systems; analyzing cost-volume-profit relationships, job order and process costing, operating budgets, standard costing, introduction to capital budgeting and ethical managerial accounting-related decision-making.

ACCT 311 Managerial and Cost Accounting

(3:0:0)

3 Semester Credit Hours

Prerequisite(s): ACCT 205

This course extends topics studied in Principles of Managerial Accounting (ACCT 205), as well as covers topics; pricing decisions, and joint product costing. The course examines the application of Managerial Management Accounting concepts and skills in decision making, of contemporary management philosophies and techniques in relation to JIT, quality, transfer pricing, performance measurement and questions the relevance of various aspects of traditional management accounting for internal management decision.

ACCT 331 Intermediate Accounting I

(3:0:0)

3 Semester Credit Hours

Prerequisite(s): ACCT 204

The aim of this course is to develop the learner's understanding of financial accounting that comprises complex topics related to measuring and reporting of cash, inventories, investments, property, receivables, revenue recognition principles, plant, and intangible assets, and equipment. Emphasis is placed on both accounting theory and practice and on the development of professional opinion and significant thoughts.

ACCT 332 Intermediate Accounting II

(3:0:0)

3 Semester Credit Hours

Prerequisite(s): ACCT 331

In this course, students examine the liability and equity sections of the balance sheet. Topics include current and other liabilities; applications to long-term liabilities; dilute securities and earnings per share (EPS); income taxes; pensions and other employee future benefits; leases; accounting changes and error analysis; statement of cash flows; and full disclosure in financial reporting.

ACCT 352 Taxation and Managerial Decisions

(3:0:0)

3 Semester Credit Hours

Prerequisite(s): ACCT 331

The objective of this course is to develop a sound conceptual and technical foundation for the study of federal tax system and thus provide students with the tools necessary to stay up to date with ever-changing tax laws, where these apply. The course will imbue students with sufficient understanding of the tax environment to identify important issues when evaluating business transactions.

ACCT 361	Accounting Information Systems	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): ACCT 331		
This course discovers the study and development of accounting information systems. The focus is on business processes covering many industries with an emphasis on accounting data modeling and internal control. Special topics including Extensible Business Reporting Language (XBRL) and commercial systems are incorporated throughout the course.		
ACCT 390	Internship in Accounting	(0:0:21)
3 Semester Credit Hours		
Prerequisite(s): Senior Standing		
The Internship is designed for candidates on the undergraduate accounting major programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation takes place at the beginning of the internship, and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.		
ACCT 412	Auditing	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): ACCT 331		
This course will cover: [1] Study and evaluation of internal control; [2] audit planning and assessment of risk; [3] audit evidence and documentation; [4] professional ethics; [5] Application of auditing and [6] audit report.		
ACCT 444	Accounting Project Based Internship	(0:0:21)
3 Semester Credit Hours		
Prerequisite(s): ACCT 331		
The accounting project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem- solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the accounting specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.		
ACCT 472	International Auditing	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): ACCT 331		
This course discovers a broad overview of the global financial and accounting environment and addresses three areas: financial reporting, financial analysis, and planning and control of multinational enterprises. In addition therefore the course analyses how different businesses operating in global environments influence accounting practices.		
ACCT 491	Special Topics in Accounting	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): ACCT 332		
This is an advanced subject and seminar course which covers major applications of Classical/current issues in Accounting. The topics to be covered will vary and will be announced by the end of the first week of commencing the course.		
ARAB 100	Arabic Language and Culture for Visiting Students	(3:0:0)
3 Semester Credit Hours		
The course provides an introduction to the basics of the Arabic language. It aims to develop students' communicative competence in spoken as well as written Arabic. Intended to resonate with the experiences of non-native learners of Arabic who are visiting the UAE, the course's linguistic content is based within the contemporary culture of the region. The course is intended to stand-alone and not to serve as a prerequisite for any other university courses.		

Arabic for speakers of other languages helps develop students' communicative competence in the skills of speaking, listening, reading, and writing. The course provides for the acquisition of vocabulary, the study of grammar, and structures of spoken and written Arabic. Non-native speakers of Arabic gain insights into target language countries.

Prerequisite(s): ARAB 101

Prerequisite(s): ENGL 101

Prerequisite(s): ARAB 102

Prerequisite(s): ARAB 201

Prerequisite: Arabic Placement Test

Prerequisite: ARAB 311

Prerequisite(s): ARAB 312

Catalog 2022 - 2023

- ARAB 412 Advanced Level Arabic Language and Culture for Native Arabic Speakers II (3:0:0)**
3 Semester Credit Hours
Prerequisite: ARAB 411
 Advance -Level Arabic Language and Culture for Native Arabic Speakers II is a direct continuation of ARAB 411 and represents the fourth course in the Arabic Language and Culture for native speakers' sequence. This course provides a survey of themes and genres of Arabic literature from the mid- 19th century to the present. While focus will be on content, students will continue their acquisition of MSA through written and oral assignments designed for their advanced level of competency.
- ARAB 420 Special Topics in Arab Culture and Civilization for Native Arabic Speakers (3:0:0)**
3 Semester Credit Hours
Prerequisite: ARAB 412
 Supplementing the four-course sequence in Arabic Language and Culture for native speakers, ARAB 420 will be taught in MSA. The course critically examines issues, values, and institutions of the contemporary Arab world primarily through analysis and discussion of current events. While focus will be on content, students will continue their acquisition of Modern Standard Arabic through written and oral assignments designed for their level of competency.
- ARCH 121 Basic Design I (0:12:0)**
4 Semester Credit Hours
 An introduction to the basic design components: subject, form, content, with a focus on their dynamic interrelation. Through observation, analysis, discussion of traditional and contemporary approaches in fine arts, architecture and interior design, students learn to apply the elements and principles of visual organization to create projects using various methods.
- ARCH 122 Basic Design II (0:12:0)**
4 Semester Credit Hours
Pre-requisite(s): ARCH 121
 The course introduces basics of design process and theory as it relates to the 3-dimensional world. Students are involved in hand-on exploration of simple structures and spaces using diverse materials and textures. Space function, volume, mass are discovered through designing a small-scale residential space(s), where design elements/principles are applied.
- ARCH 221 Architectural Design Studio 1 (0:12:0)**
4 Semester Credit Hours
Pre-requisite(s): ARCH 122
 Introduction to the application of design principles in architecture and to the technical conventions and expressions. Focus on elementary architectural design as it relates to function, structure, site, land plot and the environment.
- ARCH 222 Architectural Design Studio 2 (0:12:0)**
4 Semester Credit Hours
Pre-requisite(s): ARCH 221
 Design of medium scale buildings with an integrated program and moderate site conditions. Emphasis on human centered design process, impact of human and climatic factors on design solutions. Exploration of sustainability and particularly natural ventilation, daylighting, building materials in regards to the design intent and environment.
- ARCH 223 Computer Architectural Drawing (1:3:0)**
3 Semester Credit Hours
Pre-requisite(s): ARCH 115
 Concepts, features and applications related to Building Information Modeling (BIM) are introduced. Students explore Autodesk® Revit Architecture structure, features, modeling and editing techniques, sheet creation and organization. Emphasis on modelling BIM models and rendering 2D and 3D interiors and exteriors (including site), furnishings, accessories and details for architectural projects.
- ARCH 231 Building Construction 1 (2:3:0)**
3 Semester Credit Hours
Pre-requisite(s): ARCH 114
 This course aims to introduce building materials, their properties, the impact of building materials and construction on environment and building codes. It focuses on sustainable methods of building technology as it relates to the site and construction and explores building systems, structure's types and elements (including foundations).

(2:3:0)

This course explores materials and buildings technologies employed in building envelope (including windows, doors, other openings) and interior construction (including fixtures, fitments, built-ins) and systems, thermal and acoustic insulation, waterproofing and damp proofing. Accents on selection of materials and building technologies with minimal negative impact on environment.

(3:0:0)

Introduction to history of art and architecture, and their development through the ancient civilizations of Egypt, Mesopotamia, Greece, and Rome. Analysis of historical monuments and buildings and their architectural characteristics. Emphasis on the Architectural principles, theories, and impact on the development of buildings throughout ancient times. Examining ancient historical patterns.

(3:0:0)

Introduction to Early Christianity Architecture during the Middle Ages in Europe. The course begins with the origins of Christian architecture of the Early Christianity, Byzantine Architecture, Romanesque Architecture Gothic, Gothic Style in France and England, Gothic Style in Italy. The course concludes with the Architecture of the Baroque and Renaissance age.

(0:12:0)

Design of a building or a group of buildings with moderate complexity through different spatial, formal, and functional strategies. Consideration of psychological, environmental, and technical requirements influencing architecture and design. Experiment with green buildings design and explore passive design solutions.

(0:12:0)

Create sustainable multifunctional buildings or a group of buildings design, within an urban context, that responds to the nature of the site and intricate topography. Solve design problems related to building structure, building materials and technologies as influenced by the building type, function, urban context and environmental protection.

(2:3:0)

This course explores advanced building technologies and complex building structural configurations for long span buildings such as structural frames, space structure, suspended roof structure and air-supported structures. Offsite constructions, prefabricated and modular housing, envelopes to above listed types of building structures in relation to impact on environment are discussed.

(2:3:0)

This course will lead the students to explore the principles of BIM used to visualize, document and communicate professional architectural working drawings respecting national, regional and international building codes and regulations.

- ARCH 341 History and Theory of Architecture 3 (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): ARCH 242
 The study of the development of Islamic art and architecture and its characteristics since the dawn of Islam and the impact of social, cultural, environmental and functional factors on architecture in the Moslem world. Analysis of selected examples of historical monuments from Islamic countries and interpretations of contemporary Islamic Architecture.
- ARCH 342 History and Theory of Contemporary Architecture (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): ARCH 341
 The formation of modern theories and trends in contemporary art and architecture since the industrial revolution until today. Analysis of works by the pioneers of modern movements in architecture. Regional and vernacular architecture with special regards to environmental, social and technological issues. Selected examples of characteristic contemporary architecture in the UAE.
- ARCH 372 Sustainable Architecture (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): ENVS 102
 The course addresses the relationship of the built environment to natural environment through a whole system approach, with a focus on sustainable design of buildings. The emphasis is on local green design approaches, energy efficiency, renewable energy, and the appropriate use and conservation of resources, including materials, water, and land.
- ARCH 390 Internship I (0:3:12-20)**
3 Semester Credit Hours
Pre-requisite(s): Completion of 90 credits and a CGPA of 2.0 or higher
 This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.
- ARCH 391 Internship II (0:3:12-20)**
3 Semester Credit Hours
Pre-requisite(s): Completion of 112 credits and a CGPA of 2.0 or higher
 This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.
- ARCH 421 Architectural Design Studio 5 (0:12:0)**
4 Semester Credit Hours
Pre-requisite(s): ARCH 322
 Introduce students to the integrated design process with multiple stakeholders and multidisciplinary approaches (structural, mechanical, environmental) through buildings with shared functions. Via independent study and creative solutions students explore approaches and processes of integrated design process and produce environmentally comprehensive and human-centered design solutions.
- ARCH 422 Architectural Design Studio 6 (0:12:0)**
4 Semester Credit Hours
Pre-requisite(s): ARCH 421
 Design of a complex site and buildings through an integrated process taking in consideration the boundaries and exploring their mutual reciprocities. Emphasis on professional design thinking approaches that reflect student's personal values and ethical attitude that address issues of sustainability and technology in response to the project's functional requirements and regional context.

(3:0:0)

Pre-requisite(s): ARCH 336

(1:6:0)

Pre-requisite(s): ARCH 322

(3:0:0)

Pre-requisite(s): ARCH 372

(1:6:0)

Pre-requisite(s): ARCH 463

(3:0:0)

Pre-requisite(s): ARCH 341

(1:6:0)

Pre-requisite(s): ARCH 322

(2:3:0)

Pre-requisite(s): MENG 468

227 | Page

(3:0:0)

ARCH 485 Professional Practice
3 Semester Credit Hours

(3:0:0)

ARCH 487 Project Management
3 Semester Credit Hours
Pre-requisite(s): ARCH 336

(3:0:0)

ARCH 521 Architectural Design Studio 7
4 Semester Credit Hours
Pre-requisite(s): ARCH 422

(0:12:0)

ARCH 591 Graduation Project Thesis Research
2 Semester Credit Hours
Pre-requisite(s): ARCH 422

(1:3:0)

ARCH 592 Graduation Project Design
6 Semester Credit Hours
Pre-requisite(s): ARCH 521 and ARCH 591

(0:18:0)

BIOL 095 **Introductory Biology**
3 Semester Credit Hours

(3:0:0)

BIOL 100 Humankind in a Biological World
3 Semester Credit Hours

(3:0:0)

228 | Page

BIOL 112	University Biology I	(3:0:0)
3 Semester Credit Hours		
Co-requisite: BIOL 113 University Biology I Laboratory		
Introduction to cell chemistry, metabolism, and genetics.		
BIOL 113	University Biology I Lab.	(0:3:0)
1 Semester Credit Hour		
Corequisite: BIOL 112		
Introduction to Cell Structure & Function and Basics on plant biology.		
BIOL 114	University Biology II	(3:0:0)
3 Semester Credit Hours		
Corequisite: BIOL 115		
This course is intended for science majors and pre-professionals in life sciences and provides an introduction to Genetics, Microbiology and animal form and function.		
BIOL 115	University Biology II Lab.	(0:3:0)
1 Semester Credit Hour		
Corequisite: BIOL 114		
This course encompasses the basic techniques in Biology for science majors and pre-professionals in life sciences that accompany BIOL 114.		
BIOL 230	General Microbiology	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 114		
Corequisite: BIOL 231		
This course covers the classification of microorganisms, microbial metabolism, microbial growth, microbial genetics, structure of eubacteria, archaea, bacteria, fungi, and viruses, specific defenses, and diseases.		
BIOL 231	General Microbiology Lab	(0:3:0)
1 Semester Credit Hour		
Corequisite: BIOL 230		
This course focuses on laboratory techniques involved in culturing, staining, and identifying microorganisms.		
BIOL 240	Ecology	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 114		
This course introduces students to concepts and issues related to physical environment, energy flow, structure and function of populations, dynamics of communities, and succession.		
BIOL 250	Biochemistry I	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): CHEM 111 or CHEM 120		
Corequisite: BIOL 251		
Structure and chemical behavior of biochemical compounds, levels of protein structure, steady state enzyme kinetics and activities, protein purification, protein functions. Structure and functions of Carbohydrates, Nucleotides and Nucleic Acids, DNA-Based Information Technologies. Lipids and Biological Membranes and Transport.		
BIOL 251	Biochemistry I Lab	(0:3:0)
1 Semester Credit Hour		
Corequisite: BIOL 250		
This experimental course will introduce students with hand practice of some major biochemistry techniques. This course has two parts, part A is wet lab where students learn and practice separation, identification, and quantization techniques that exploit properties of biological molecules. To analyze results, data and write reports to develop analytical reasoning and problem-solving skills. The second part B is dry/computer lab where students will use Geneious to study the 3D structures of biological macromolecules from PDB files (Protein Data Bank).		

BIOL 270	General Genetics	(3:0:0)
3 Semester Credit Hours		
Prerequisite: BIOL 112 or BIOL 114		
This course focuses on the basic concepts of heredity and modern developments in Genetics.		
BIOL 271	General Genetics Lab.	(0:3:0)
1 Semester Credit Hour		
Corequisite: BIOL 270		
This course covers basic laboratory techniques in Genetics that accompany the course BIOL 270.		
BIOL 322	Microbial Genetics	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 270		
This course covers the basic concepts of microbial genetics including gene structure, gene expression and its control, and evolution of bacterial genetic diversity. The emphasis will be on topics such as mechanisms of genetic exchange, recombination, plasmids, analysis of bacteria and phage genomes, mutagenesis, and DNA repair. The course will also discuss the common methods using bacteria in gene cloning and identification and isolation of mutants		
BIOL 330	Applied and Industrial Microbiology	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 230		
Corequisite: BIOL 331		
This course covers the biology of microorganisms of ecological and industrial significance and includes topics such as food production, spoilage and preservation, fermentation technology, waste disposal, water purification, biodeterioration, and decomposition.		
BIOL 331	Techniques in Applied and Industrial Microbiology Lab.	(0:3:0)
1 Semester Credit Hour		
Corequisite: BIOL 330		
This course includes laboratory exercises that illustrate applied methodologies in microbiology, including isolation of commercially useful strains, and discusses the production and purification of industrial products.		
BIOL 350	Biochemistry II	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): CHEM 250		
This course covers the bioenergetics, Glycolysis, Gluconeogenesis, the Citric Acid Cycle and the oxidative phosphorylation and photophosphorylation. Lipid, amino acids and nucleotides Biosynthesis. DNA and RNA metabolism and gene expression regulation.		
BIOL 354	Biophysics	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 114		
Corequisite: BIOL 355		
This course introduces students to the use of physical methods in the study of biological systems including macromolecules, membranes, nerves, muscle, photosynthetic systems and visual systems.		
BIOL 355	Biophysics Lab.	(0:3:0)
1 Semester Credit Hours		
Corequisite: BIOL 354		
This course introduces students to common laboratory techniques used in Biophysics.		
BIOL 356	Virology	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 230		
This course provides an overview of the characteristics of major families of viruses; the intrinsic properties of viruses that cause disease, their development, and life cycles; interaction with host cells; genetics; and tumor-inducing properties and epidemiology.		

BIOL 360	Developmental Biology	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 270		
This course exposes students to the principles of embryonic development and differentiation in animal species at the cellular, molecular, tissue, and whole organism levels.		
BIOL 380	Biotechnology and Genetic Engineering	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 270		
Corequisite: BIOL 381		
This course focuses on the theory and applications of Biotechnology to real-world problems, emphasizing significance and societal applications.		
BIOL 381	Biotechnology Lab Methods and Techniques	(0:3:0)
1 Semester Credit Hour		
Corequisite: BIOL 380		
This course focuses on the fundamental practical techniques used in Biotechnology research and industry.		
BIOL 411	Drug Discovery, Design and Development	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 350		
This course will explore the process of drug development, from target identification to final drug registration. It will present drug development as a process involving target selection, lead discovery using computer-based methods and combinatorial chemistry/high-throughput screening. Safety evaluation, bioavailability, clinical trials, and the essentials of patent law will also be discussed. Along the way, you will learn about molecular recognition, computer aided drug design, and toxicology as applied to the development of new medicines.		
BIOL 420	Molecular Biology	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 270		
Corequisite: BIOL 421		
This course focuses on the basic concepts of structure and function of genetic material at the molecular level. Topics such as DNA replication, chromosome structure, gene expression and mutation, DNA repair, and transposable elements are covered in molecular detail.		
BIOL 421	Molecular Biology and Recombinant DNA Lab.	(0:3:0)
1 Semester Credit Hour		
Corequisite: BIOL 420		
This course focuses on the techniques in modern molecular biology techniques such as restriction mapping, isolation of DNA and RNA from eukaryotic cells, Southern hybridization, reverse transcription, and polymerase chain reaction (PCR and RT-PCR).		
BIOL 423	Advanced Molecular Biology and Forensic Sciences	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 270		
This is for forensic DNA practitioners, those interested in a career in forensic DNA and all students interested in broadening their understanding of the genetic principles that underlie the forensic application of DNA typing methods.		
BIOL 424	DNA Fingerprinting and Serology Lab	(0:3:0)
1 Semester Credit Hour		
Prerequisite(s): BIOL 270		
Laboratory course to accompany BIOL 423 (Advanced Molecular Biology and Forensic Sciences). Focusing on the basic techniques used in forensic DNA laboratories, this course will introduce the students to the most recent DNA fingerprinting techniques and serological analysis.		

- BIOL 425 Advanced Genetics** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): BIOL 270
 This course discusses the principles of genetics and the role of genetic variations on the expressed phenotype in unicellular and multicellular organisms. The topics include genome structure and function, gene mutations, recombination, transposable elements, experimental genetics analysis, population genetics and diseases.
- BIOL 430 Cell Culture Theory and Technology** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): BIOL 270
Corequisite: BIOL 431
 This course aims at providing a strong foundation in the structure, growth, and function of plant and animal cells as well as the technology involved in cell and tissue culture, cell preservation, protoplast culture and fusion, cell cloning and fusion, monoclonal antibody production, breeding and genetic engineering. The applications of stem cells to therapy will also be emphasized in this course.
- BIOL 431 Cell Culture Techniques Lab.** **(0:3:0)**
1 Semester Credit Hour
Corequisite: BIOL 430
 This course deals with the principles and practical applications of animal tissue culture. Molecular techniques or manipulation and maintenance of cells in culture using animal systems.
- BIOL 434 Cell Communication and Signal Transduction** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): BIOL 270
 This course focuses on how cells communicate with each other and their microenvironment, the signaling events that result from these interactions, and their translation into cellular responses.
- BIOL 435 Stem Cell Biology and Regenerative Medicine** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): BIOL 270
 This course provides a broad overview of the biological principles influencing stem cell populations and emphasizes the functional role of stem cells in regulating normal development and contributing to disease, as well as examines the therapeutic potential of stem cells in regenerative medicine.
- BIOL 436 Immunology** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): BIOL 270
 This course focuses on the structure and function of immunoglobulins, the role of cell-mediated immunity, the protective role of the immune system, and disease and injury related to malfunctions of the immune system.
- BIOL 440 Medical Microbiology** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): BIOL 230
Corequisite: BIOL 441
 This course focuses on mechanisms of microbial pathogenesis, the host response, and the scientific approaches that are used to investigate these processes. It provides students with: (1) in depth-understanding of the pathogenic microorganisms and particularly address the fundamental mechanisms of their pathogenicity. (2) the necessary foundation to identify and characterize microorganisms as well as a foundation in the methods to prevent, diagnose and treat infectious diseases. (3) opportunities for a student to develop diagnostic skills in microbiology, including the identification of and interpretation of laboratory tests for the diagnosis of infectious diseases.

(0:3:0)

Corequisite: BIOL 440

The course covers a collection of techniques used for culture, isolation, and identification of common microorganisms. The topics aim to provide students with practical training on standardized methods for microbial identification including staining and microscopy, antibiotic susceptibility, and biochemical tests. In addition, the course provides in depth emphasis on some methods of disinfection, sterilization, and antimicrobial therapy.

(3:0:0)

3 Semester Credit Hours

Prerequisite(s): BIOL 380

Corequisite: BIOL 443

This course covers a collection of techniques used to analyze biological markers in the genome and proteome to diagnose human disease. The course will examine and understand modern tools employed to identify variations in the genetic material or proteins associated with specific health conditions or disease. Topics include the most commonly used molecular tests to diagnose disease such as DNA and RNA isolation, amplification, hybridization, PCR, DNA, and RNA sequencing, ELISA, and flow cytometry. In addition, the course provides in depth emphasis on specimen handling, advantages, and disadvantages of molecular diagnostics, and quality assurance in the molecular lab.

(0:3:0)

1 Semester Credit Hour

Prerequisite(s): BIOL 380 and BIOL 381

Corequisite: BIOL 442

This course covers the basic concepts and techniques that are used in the diagnosis of disease. The students will be trained to have hands-on experience on several assays such as nucleic acid isolation, PCR, gel electrophoresis, restriction fragment length polymorphisms analysis, and ELISA. In addition, the course will prepare students to evaluate the controls for validating results, standardize tests, and discuss quality control for diagnostic laboratory.

(3:0:0)

3 Semester Credit Hours

Prerequisite(s): 90 credits

This course examines selected ethical issues arising from the application of modern biotechnology to microorganisms, plants, animals, and humans. It also discusses the ethics and misconduct in research, authorship, and peer-review. **(Writing Intensive Course)**

(3:0:0)

3 Semester Credit Hours

Prerequisite(s): BIOL 270

This course will introduce students to crime scene investigation procedures and techniques. With a focus on crime scene processing, this course discusses the most recent topics on database technologies and DNA analyses.

(3:0:0)

3 Semester Credit Hours

Prerequisite(s): Completion of 90 credit hours

This course will introduce students to the real world of criminal justice with a focus on topics related to understanding crimes, roles of the police, and the court system.

(3:0:0)

3 Semester Credit Hours

Prerequisite(s): BIOL 270

This course is designed to provide instruction in the scientific concepts and computer skills currently used in searching biological databases, comparing sequences, looking at protein and DNA characteristics and answering biological and biomedical questions using databases.

BIOL 480	Food Biotechnology	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 270		
This course introduces students to the impact of Biotechnology on food production, the genetic tools applied in Food Biotechnology, the improvement of microbes used in food production by modern biotechnological approaches, and also discusses related ethical, legal, and regulatory issues.		
BIOL 481	Bioprocessing Technology in Pharmaceuticals Industry	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): BIOL 380		
This course provides an overview of the main elements, theory and application of biotechnology procedures, and bioprocess design settings and operations related to the development of biopharmaceutical products including good manufacturing practices and bioprocessing equipment standards. The course lays emphasis on fermentation, harvest, primary and final purification, media and buffer preparation, equipment cleaning and sterilization, and critical process utilities.		
BIOL 491	Senior Seminar	(2:0:0)
2 Semester Credit Hours		
Prerequisite(s): 90 credits		
This is a capstone course and involves student participation in seminars on various Biotechnology topics.		
BIOL 492	Special Topics in Biotechnology	(2:0:0)
2 Semester Credit Hours		
Prerequisite(s): Minimum 90 credits and Minimum CGPA 2.0		
This course focuses on the major advances in genomics, proteomics, and metabolomics and their significance in biotechnological applications.		
BIOL 493	Research Methodology in the Biological Sciences	(2:0:0)
2 Semester Credit Hours		
Prerequisite(s): Senior Status/Standing		
This course provides in-depth knowledge of research design and methodology and to train the student in writing a research proposal and critically reviewing scientific literature. The emphasis will be on interpreting and evaluating recent studies from the scientific literature and communicating biological findings in a written and oral format to a range of audiences. The course focus is to train students to critically review literature, identify a research question, lay out a detailed experimental plan including the statistical analysis, and discuss anticipated results.		
BIOL 494	Senior Project in Cell and Molecular Biotechnology	(0:12:0)
4 Semester Credit Hours		
Prerequisite(s): Senior Status/Standing BIOL 493		
Under the guidance of Biotechnology faculty mentors, students will conduct a research project in the field of Cell and Molecular Biotechnology and present their research findings in both oral and written formats.		
BIOL 495	Senior Project in Medical Biotechnology	(0:12:0)
4 Semester Credit Hours		
Prerequisite(s): Senior Status/Standing BIOL 493		
Under the guidance of Biotechnology faculty mentors, students will conduct a research project in the field of Medical Biotechnology and present their research findings in both oral and written formats.		
BIOL 496	Senior Project in Forensic Biology	(0:12:0)
4 Semester Credit Hours		
Prerequisite(s): Senior Status/Standing BIOL 493		
Under the guidance of Biotechnology faculty mentors, students will conduct a research project in the field of Forensic Biology and present their research findings in both oral and written formats.		

BUAN 301 Analytics for Decision Making**(3:0:0)****3 Semester Credit Hours****Prerequisite(s): OPMT 311**

This course is an introductory survey course in business analytics and is geared towards providing a broad coverage of descriptive, predictive, and prescriptive analytics techniques and applications within a business context. The course aims to introduce students to foundational concepts, methodologies, and tools required to understand the emerging role of business analytics in organizations, with a focus on decision making for efficiency, profitability, and sustainability. Through hands-on practice tutorials and assignments, students will also be introduced to computer-based modeling approaches to facilitate formulation, resolution, and interpretation of the results, and the process of transforming data into actions.

BUAN 311 Business Data Mining**(3:0:0)****3 Semester Credit Hours****Prerequisite(s): BUAN 301**

Data Mining (DM) is the non-trivial process of extracting implicit, novel, and useful information from large volume of data. It has emerged as a unique combination of several fields of science and technology including statistics, database systems, computer programming, machine learning, and artificial intelligence. This course is designed to provide undergraduate students with an understanding of data mining techniques, including data pre-processing, classification, clustering and association rules, as well as their applications in business intelligence, customer relationship management, marketing, credit scoring, churn and survival analysis. The students will also be introduced to Big data and Big data analytics. Students gain hands-on experience with the Rapid Miner software application. They perform pre-processing and apply data mining techniques to analyze the selected data to discover useful and actionable patterns and information.

BUAN 312 Visual Analytics**(3:0:0)****3 Semester Credit Hours****Prerequisite(s): BUAN 301**

This course provides an in-depth view of working with data to extract and present valuable information. Students will learn to collect, clean, manipulate, analyze, and visualize data from various sources correctly and efficiently. Through hands-on application, students will gain an understanding of what problems can occur when dealing with a variety of data and what solutions exist. This course introduces data visualization as an analytical tool, a medium of communication, and the basis for interactive information dashboards. Students will learn best practices in data visualization, sharpen analytical skills, and learn how to design dashboards for use by stakeholders. Computing is a major component of this course, and students will learn a number of in-demand technical skills.

BUAN 390 Internship in Business Analytics**(0:0:21)****3 Semester Credit Hours****Prerequisite(s): Senior Standing**

The Internship is designed for candidates on the undergraduate business analytics major programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation take place at the beginning of the internship and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.

BUAN 401 Business Data Engineering**(3:0:0)****3 Semester Credit Hours****Prerequisite(s): MIST 301, BUAN 311**

This course introduces students to the foundations of database management systems (DBMS), data warehousing fundamentals, practices, and technologies, and the challenges faced by businesses in managing large amounts of data and making meaningful use of this data for informed decision making. The course focuses on the basics of data pipelines, data pipeline flows and associated business use cases, and how organizations derive value from data and data engineering. Throughout the course, students will be introduced to various computing platforms for developing, deploying, configuring a wide range of data science applications for different business use-cases.

- BUAN 411 Business Intelligence & Big Data Analytics (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): MIST 301, BUAN 311
 As an intermediate-level course in business analytics, this course covers foundational theories and practices concerned with business intelligence and big data analytics. The course focuses on business applications of big data and provides an introduction to supporting technologies for big data analytics including MapReduce, Hadoop, Spark, and their ecosystems. The course also provides an introduction to the data analytics lifecycle to address business challenges that leverage big data. Examples of various types of BI use-cases for business decision-making will be discussed to allow students to identify and apply appropriate tools and methodologies to real-world business challenges and to facilitate data-driven decision-making.
- BUAN 444 Business Analytics Project Based Internship (0:0:21)**
3 Semester Credit Hours
Prerequisite(s): BUAN 301
 The business analytics project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem-solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the business analytics specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.
- BUSN 301 Business Communications (3:0:0)**
3 Semester Credit Hours
Prerequisite ENGL 101
 This course introduces fundamentals of business models and writing as a learning tool. The interrelationships between accounting, finance, information systems, marketing, and operations are the subject of several “learning by writing” deliverables. The course teaches business students effective methods for communicating business-related issues to clients, management, and fellow employees. **(Writing Intensive Course)**
- BUSN 304 Commercial Law (3:0:0)**
3 Semester Credit Hours
Prerequisite Junior Standing
 This course provides exposure to general legal environment of business, emphasizing government regulatory process, business ethics, and social responsibility. Regulatory topics include torts and crimes, product liability, intellectual property and cyber law, contracts, and issues related to employment and competition.
- BUSN 306 International Business (3:0:0)**
3 Semester Credit Hours
Prerequisite ECON 104
 This course offers a multi-disciplinary approach to global economic issues from the viewpoint of managing international business. It introduces salient aspects of managing in a globalized environment, including consideration of theoretical concepts. It provides a practical treatment of political-economic aspects of international trade. The course discusses foreign direct investment, global monetary systems, and strategy formulation for international business.
- BUSN 307 Research Methodology (3:0:0)**
3 Semester Credit Hours
Prerequisites: Junior Standing
 This course provides essential underpinning support for research elements of higher-level business courses. It allows candidates to make informed decisions and appropriate choices pertaining to research methodology. A range of business research tools, approaches and analytical techniques are discussed, and guidance is given on optimal structuring of research work in the business field, and the generation of research proposals.

(3:0:0)

This course introduces the policy and ethical dimensions of business as they relate to the various stakeholders inside and outside the organization. It includes topics such as actors, institutions and ideas in the policy process, business ethical theory, ethical decision making, ethical leadership, ethical and policy dilemmas, corporate social responsibility, and environmental sustainability.

(0:0:21)

The Internship is designed for candidates on the undergraduate finance, marketing, human resource management and accounting programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation take place at the beginning of the internship and will serve to introduce the student to real-world business practices, culture, and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.

(0:0:21)

This course is designed for candidates on the Bachelor of Science in Business Administration to provide a field experience in which students are able to apply knowledge and skills gained through course work, and to demonstrate their capacity to function successfully in a professional setting. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues in international settings. Through this internship, the students will learn, and apply real-world management skills on an international level, and throughout the process, the student will complete the internship report to fulfil the course requirements. The vast majority of the work and value of this course, however, will come in the form of the international, intercultural, and interpersonal experiences gained throughout the period in the international workplace.

(3:0:0)

This course is designed to help students apply functional knowledge gained from earlier courses and integrate these with more practical information and real-life experiences. The course takes the general management point of view, emphasizing the creation, implementation, and evaluation of strategy in organizations. Students adopt the viewpoint of upper management, strategize, and make essential "Big Picture" decisions, using standard strategic analysis frameworks. Students develop expertise in the analysis of complex business situations and in clearly presenting strategic findings both orally and in writing.

(3:0:0)

Microprocessors as components in a computer system; programmer's view of a microprocessor's architecture; microprocessor instruction set; assembly language programming; interrupts; input and output; interfacing a microprocessor to memory and I/O devices from the programmer's view. At the end of the course, the students should be able to program a modern microprocessor in assembly or C language and perform hardware I/O interfacing.

(0:3:0)

This course introduces microprocessors in a laboratory environment. Topics include: assembly language programming; hardware interface and design with common microprocessor peripherals, uVision Simulation, Software Setup, GPIO: LEDs and Joystick, LCD Display, Keypad, LED in Assembly, Stepper Motor in C and Assembly, System timer, PWM, ADC/Infrared Sensor and DAC / Digital Music.

(3:0:0)

3 Semester Credit Hours

Pre-requisite(s): EEEN 331

Co-requisite(s): CENG 336

This course covers details of microprocessor design including the instruction set architecture, memory design, and data path and control design. The course also emphasizes memory performance related concepts such as associativity and multi-level caching. Additional topics include virtual memory and performance speed-up techniques using pipelining, multithreading, and multiprocessing.

(0:3:0)

1 Semester Credit Hour

Pre-requisite(s): EEEN 331

Co-requisite(s): CENG 335

This course covers modern computer system architecture and computer design principles. A Hardware Description Language is used to design basic components of a microprocessor datapath and control. Additional topics covered include Adders, MUX, Counters, ALU, registers/shift registers, RAM, pipelining, and cache memory.

(3:0:0)

3 Semester Credit Hours

Pre-requisite(s): CSCI 462

This course discusses socket programming, client-server applications, peer-to-peer networks and applications, Web servers, datacenters and load balancing, CDNs, IPV6, NAT, anycasting, DHCP, multimedia networking, network operation and management and basic Internet measurements.

(3:0:0)

3 Semester Credit Hours

Prerequisite(s): CSCI 215

This course examines in detail the software development process. Topics include concepts such as software processes, software lifecycle models, requirement engineering, system modeling, architectural design, software design implementation, software testing, software evolution, and software reuse.

(3:0:0)

3 Semester Credit Hours

Pre-requisite(s):CSCI 232 or CENG 315

Co-requisite(s): CENG 432

This course introduces the design and analysis of embedded computer systems that interact with the physical world. The course provides an integrated approach to the development of low-power systems with hardware, software, sensors, actuators, processors and networks. Topics include hardware and software architectures, assembly and C programming, real-time design, interrupts, multitasking, embedded software tools, and embedded systems performance. Students will work on a comprehensive project to design, implement, and evaluate a prototype embedded system.

(0:3:0)

1 Semester Credit Hour

Co-requisite(s): CENG 431

In this lab course, students will design microcontroller-based embedded systems for various applications. Topics covered include interfacing to sensors and actuators from both hardware and software perspectives, assembly and C programming for embedded systems I/O, real-time design, and interrupts. Additionally, students will implement and evaluate a complete embedded system.

(3:0:0)

3 Semester Credit Hours

Pre-requisite(s): CENG 335

This course provides students with a deep understanding of parallel computing architectures including multicore CPUs and GPUs. The course emphasizes trade-offs involved in design of modern parallel computing systems, and techniques necessary to effectively utilize these machines. Topics covered include data-level parallelism, thread-level parallelism, GPU architectures, cache coherence, memory consistency, interconnection networks, reconfigurable computing and domain-specific parallel programming systems.

CENG 461	Network Security	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): MATH 225 and CSCI 462 or EEEN 462		
This course examines information security services and mechanisms in a network context. Topics include Cryptographic tools, User Authentication, Authorization and Access Control, Database Security, Malicious Software, Denial of Service Attacks, Firewall and Intrusion Prevention, Software Security, Operating System, Internet Security Protocols and Standards. Stack and Buffer Overflow, Security Auditing, Forensics, Ethical and Legal aspects, and Wireless Network Security.		
CENG 492	Senior Design Project I	(0:6:0)
2 Semester Credit Hours		
Pre-requisite(s): Senior Standing		
Co-requisite: ECEN 491		
The course requires seniors to work in small teams to solve significant problems. Over the duration of CENG 492 and CENG 493, students design, implement, and evaluate a solution to the problem in conjunction with a faculty advisor. The course reinforces gained design principles and serves as a capstone for computing knowledge obtained in the BSCE curriculum. The recognition of the ethical and legal principles are also aspects of the course.		
CENG 493	Senior Design Project II	(0:12:0)
4 Semester Credit Hours		
Pre-requisite(s): CENG 492		
Implementation of the project for which preliminary work was done in CENG 492. Project includes designing and constructing software and/or hardware, conducting experiments or studies, and testing and validating a complete system. At the end of the term, each team presents to a committee information related to its project in both written and oral formats.		
CHEM 095	Introductory Chemistry	(3:0:0)
3 Semester Credit Hours		
This course is an introduction to the basic chemistry notions such as atomic structure, molecules and mole concept, periodic classification of elements, naming of compounds, chemical formulas and equations and its balancing, the basic understanding of simple compounds of carbon and its applications, the influence of chemistry in the environment, and its scope for application in other fields of study. This course is not for degree credit.		
CHEM 100	Chemistry in Everyday Life	(3:0:0)
3 Semester Credit Hours		
This course focuses on chemistry and its impact in everyday life. Basic chemical principles, the impact, and practical everyday applications are examined. Environmental concerns including energy, resource consumption and pollution as well as the crucial role involved in making informed opinions and better consumer choices in our society is addressed.		
CHEM 101	Chemistry in Everyday Life Lab	(0:3:0)
1 Semester Credit Hour		
A chemistry appreciation course for non-science majors. Covers basic chemistry concepts with technical/fundamental aspects. Emphasizes the practical chemical world of human beings and the chemical nature of everyday products. Hands-on laboratory sessions provide the opportunity to put the scientific method into action allowing further exploration and reinforcement of chemical principles.		
CHEM 111	Principles of General Chemistry	(3:0:0)
3 Semester Credit Hours		
Corequisite: CHEM 112		
This course introduces students to the fundamental principles and laws of chemistry. This course covers coordination compounds, acid-base equilibrium, chemical equilibrium, solubility and complex-ion equilibria, thermochemistry, gas laws, and solutions. Introduction to organic chemistry, electrochemistry and nuclear chemistry.		
CHEM 112	Principles of General Chemistry Lab	(0:3:0)
1 Semester Credit Hour		
Corequisite: CHEM 111		
This course introduces chemistry lab techniques and provides hands-on experience in conducting chemical reactions that accompany the course CHEM111. It provides experience in basic techniques for the synthesis of coordination complexes and physical chemistry-based experiments.		

- CHEM 120 Analytical Chemistry** **(3:0:0)**
3 Semester Credit Hours
Corequisite: CHEM 121
 This course emphasizes quantitative chemical analysis and underlying principles. Specific analytical techniques or concepts covered are: basic understanding of analytical chemistry including experimental error, statistical analysis of data, quality assurance, calibration methods, introduction of titrations, and polyprotic acid base equilibria. This course will introduce spectroscopic, separation, gravimetric, and combustion methods for quantitative chemical analysis.
- CHEM 121 Analytical Chemistry Lab.** **(0:3:0)**
3 Semester Credit Hours
Corequisite: CHEM 120
 Laboratory techniques and experiments arranged to accompany course Analytical Chemistry (CHEM 120). The aim of this course is to teach students the basic principles of analytical chemistry. The course covers the preparation of reagent, statistical evaluation of replicated data measurements and studies the validity of results. In addition, the analysis of unknowns using different analytical qualitative and quantitative methods is also enclosed.
- CHEM 211 General Chemistry I** **(3:0:0)**
3 Semester Credit Hours
 This course forms the foundation of chemical concepts: basic facts and principles of chemistry, including atoms, molecules, ions, chemical reactions, gas theory, thermochemistry, electrochemistry, chemical kinetics and equilibrium, molecular geometry, and states of matter
- CHEM 212 General Chemistry I Lab** **(0:3:0)**
1 Semester Credit Hour
Corequisite: CHEM 211
 This is a Laboratory class that is a companion class to CHEM 211 with laboratory techniques and reactions arranged to reinforce concepts. This laboratory course provides experience in physical-chemical measurement, synthesis-based experiments, qualitative, and quantitative analysis.
- CHEM 213 General Chemistry II** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): CHEM 211
Corequisite: CHEM 214
 This course covers the foundations of chemical concepts including: solutions, chemical reactions, acid base equilibrium, acids and bases, quantum theory of atoms, solubility and complex-ion equilibrium, chemistry of main group elements, polymer chemistry, and introduction of organic and nuclear chemistry, transition elements and coordination complexes.
- CHEM 214 General Chemistry II Lab** **(0:3:0)**
1 Semester Credit Hour
Corequisite: CHEM 213
 This course introduces Laboratory techniques and reactions arranged to accompany CHEM-213. This laboratory course provides experience in physical /chemical measurement, synthesis-based experiments, qualitative, and quantitative analysis.
- CHEM 215 Organic Chemistry I** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): CHEM 111 or CHEM 120
Corequisite: CHEM 216
 This course provides a foundation in Organic Chemistry and includes the theoretical and synthetic aspects of the chemistry of carbon compounds.
- CHEM 216 General Organic Chemistry Lab I** **(0:3:0)**
1 Semester Credit Hour
Corequisite: CHEM 215
 This course provides hands-on laboratory experience in techniques and chemical reactions in organic chemistry that accompany CHEM 215. The course includes basic techniques for the preparation, isolation, purification, inter-conversion, and spectroscopic study of common classes of organic compounds.

CHEM 217	Organic Chemistry II	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): CHEM 215		
Co-requisite: CHEM 218		
This course focuses on aromatic organic chemistry, stereochemistry, organic chemistry reaction mechanisms, bio-molecules and the methods used for the synthesis of organic compounds.		
CHEM 218	General Organic Chemistry Lab. II	(0:3:0)
1 Semester Credit Hour		
Corequisite: CHEM 217		
This course provides practical experience in common techniques for synthesis, separation, purification, and identification of organic molecules such as NMR, Mass Spectrometry, and Fourier Transform Infrared spectroscopy, as well as the qualitative analysis of functional groups.		
CHEM 315	Physical Chemistry	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): CHEM 215		
Corequisite(s): CHEM 316		
This course will introduce the concepts of physical chemistry, properties of gases, first and second law of thermodynamics, physical transformation, phase diagram, chemical equilibrium, atomic structure and spectra, molecular structure, spectroscopy, chemical reactions, statistical thermodynamics, kinetics and dynamics of chemical reactions.		
CHEM 316	Physical Chemistry Lab.	(0:3:0)
1 Semester Credit Hour		
Corequisite(s): CHEM 315		
Introductions to the techniques and reasoning of experimental chemistry. Lab techniques and reactions arranged to accompany Physical Chemistry.		
CHEM 411	Medicinal Chemistry	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): CHEM 217		
This course examines principles of structural, physical, and physical-organic chemistry, including mechanistic considerations involved in synthetic organic chemistry, bioorganic chemistry, and the design of chemotherapeutic agents. This course includes mechanisms of action and factors that influence drug action within specific drug classes of pharmacodynamic and chemotherapeutic nature, drug structures, chiral drugs, and structure-activity relationships of drugs.		
CHEM 420	Forensic Chemistry	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): CHEM 120		
Corequisite: CHEM 421		
Foundation of forensic chemistry: applications of the principles, methods, and instrumentation of chemistry to forensic problems. Covers topics most commonly applicable to the functions of a crime laboratory professional.		
CHEM 421	Forensic Chemistry Lab	(0:3:0)
1 Semester Credit Hour		
Prerequisite(s): CHEM 121		
Corequisite: CHEM 420		
This course introduces laboratory techniques to the forensic analysis of trace evidences according to established forensic procedures. The focus of the course will be on identifying and understanding the nature of the samples, sample preparation methods, chemical and instrumental methods for analysis.		
CHEN 201	Principles of Chemical Engineering	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CHEM 211		
This course introduces the students to chemical engineering profession and basic calculations in mass and energy balance; phase equilibrium; and process flow sheeting. It includes applications on reactive and non-reactive chemical processes. Computer programs are used to implement these topics.		

CHEN 302	Computer Applications in Chemical Engineering	(0:3:0)
1 Semester Credit Hour		
Pre-requisite(s): CHEN 201 and CIEN 251		
This course will focus on computer applications in chemical engineering including available software packages. Students will be introduced to the applications of software packages such as E-Z solve, Polymath, MATLAB, COMSOL, MATHCAD, Hysys, Aspen Plus for solving systems of algebraic and differential equations related to chemical engineering processes.		
CHEN 312	Chemical Engineering Thermodynamics	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CHEM 315, CHEM 316, MENG 211		
This course presents: fundamentals of thermodynamics, such as thermodynamic properties (energy, entropy, enthalpy, heat capacity, etc.), the first and second law of thermodynamics (energy and entropy balance), heat of reactions, etc., thermodynamics of ideal and non-ideal gases and liquids, vapor-liquid equilibrium and thermodynamics of chemical processes.		
CHEN 321	Chemical Engineering Laboratory I	(0:6:0)
2 Semester Credit Hours		
Pre-requisite(s): MENG 211, CIEN 251, and MENG 361		
In this laboratory course, students conduct selected experiments to illustrate and use material covered in transport phenomena courses: fluid, heat and mass transfer. Students are evaluated on submitted reports covering experimental design, theory tested, and analyses of experimental data.		
CHEN 351	Chemical Reaction Engineering	(3:0:0)
2 Semester Credit Hours		
Pre-requisite(s): CHEN 201 and CHEM 315		
This course covers kinetics of homogeneous and heterogeneous reactions, design of isothermal reactors such as Batch, Continuous Stirred Tank Reactor (CSTR) and Plug Flow Reactor (PFR). Other topics include data collection and handling, catalytic reactions, non- isothermal reactor design and multiple reactions.		
CHEN 371	Mass Transfer	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 251		
This course covers molecular and convective steady- and unsteady-state mass transfer, interfacial mass transfer, and continuous and stage-wise contact operations, with applications in absorption, stripping, and humidification.		
CHEN 403	Fundamentals of Biochemical Engineering	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): BIOL 112 and CHEN 351 / Co-requisite(s): CHEN 371		
This course aims to apply the principles of Chemical Engineering in bioprocess, in which biocatalysts, i.e. cells and enzymes, play key roles in producing new bioproducts and chemicals. The purpose of this course is to introduce the fundamental principles of biochemical engineering and present a wide spectrum of potential technological applications. This course includes introduction to biochemical and microbiological applications to commercial and engineering processes, including industrial fermentation, ultrafiltration and resulting waste treatment. In addition to introduction to enzyme kinetics, cell growth, energetics and mass transfer.		
CHEN 404	Safety and Environmental Impact	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): Senior Standing		
Sources of hazard in chemical and petroleum industry, management of safety and loss prevention, safety programs and safety rules, typical and sources of pollutant in chemical and petroleum operations, environmental protection, rules and regulation.		
CHEN 422	Chemical Engineering Laboratory II	(0:6:0)
2 Semester Credit Hours		
Pre-requisite(s): CHEN 371 and CHEN 351 / Co-requisite: CHEN 472		
In this course the students conduct selected experiments to apply material presented in kinetics and reactor design, and separation processes. Students are evaluated based on their participation and submitted reports. (Writing Intensive Course)		

(3:0:0)

Pre-requisite(s): CHEN 351 and CHEN 371

CHEN 463 Natural Gas Processing

(3:0:0)

Pre-requisite(s): CHEN 371

CHEN 472 Separation Processes

(3:0:0)

Pre-requisite(s): CHEN 371

CHEN 473 Water Desalination

(3:0:0)

Pre-requisite(s): CHEN 371

CHEN 474 Industrial and Wastewater Treatment

(3:0:0)

3 Semester Credit Hours

Pre-requisite(s): Senior Standing

CHEN 481 Process and Plant Design

(3:0:0)

3 Semester Credit Hours

Pre-requisite(s): IENG 231 and IENG 321

Co-requisite: CHEN 472

CHEN 482 Chemical Process Dynamics and Control

(3:0:0)

3 Semester Credit Hours

Pre-requisite(s): CHEN 302 and CHEN 351

243 | Page

CHEN 491	Senior Design Project I	(0:9:0)
3 Semester Credit Hours		
Pre-requisite(s): Senior Standing		
The Senior Design Project entails team work involving development of a two-semester long workflow that includes preparing and presenting a proposal for a major design work. The design work may involve physicochemical process design and testing, or simulation workflow design related to one or more chemical engineering activities, such as, reaction, separation, mass or heat transfer and dynamic and control– with due consideration of economic, environmental and societal impacts. The SDP-I entails finalizing the proposal by interacting with the Faculty Supervisor. A progress report is due that includes the following: objective, review of literature, statement of the problem, scope of work, description of tasks to be executed and task plan. The recognition of the ethical and legal principles are also aspects of the course. Most of the tasks are planned to be pursued and finalized in Senior Design Project-II for the following semester.		
CHEN 492	Senior Design Project II	(0:9:0)
3 Semester Credit Hours		
Pre-requisite(s): CHEN 491		
The Senior Design Project entails team work involving development of a two-semester long workflow that includes designing, implementing and evaluate the propose solution to a certain engineering problem. The design work may involve process design and testing, or simulation workflow design related to one or more chemical engineering activities, such as, reaction, separation, mass or heat transfer and dynamic and control– with due consideration of economic, environmental and societal impacts. The SDP- II entails finalizing the tasks planned in SDP- I. Students must present their work in a detailed technical report and defend their project in oral presentation.		
CHEN 493	Special Topics in Petroleum Engineering	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): Department Consent		
A specific topic in Petroleum Engineering that is not covered in other program courses is presented in a course format.		
CIEN 201	Computer Aided Drawings	(1:6:0)
3 Semester Credit Hours		
Pre-requisite(s): None		
Introduction to computer graphics. Includes the following topics: geometric construction; line convention; orthographic projections, isometric projections; oblique projections; Descriptive projections; dimensioning, and sectional views. Computer-aided design and problem-solving techniques. Uses computer software AutoCAD in data analysis, data display and visualization techniques. Prepare drawings in civil and infrastructure engineering.		
CIEN 211	Statics	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): MATH 113 and PHYS 110		
Vectors, force systems (2D and 3D), equilibrium of particles and rigid bodies (2D and 3D), structures (trusses, cables, frames and machines), distributed forces (centroids and centers of mass), internal forces (shearing force and bending moment diagrams), friction, and moment of inertia.		
CIEN 212	Mechanics of Materials	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 211		
Types of loads, axial stress and strain of determinate and indeterminate system, normal and bending moment diagrams, torsion of determinate and indeterminate system, bending of beams, combined stresses, shearing stress and strain, Mohr's circle of stress and strain, thin walled pressure vessels.		
CIEN 216	Structures for architecture 1	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 211		
Equilibrium of a rigid body, force systems, geometric properties of various shapes, internal forces in beams, trusses, and frames, analysis of three-hinged and tied arches, construction of shear force and bending moment diagrams for beams and frames, mechanical properties of materials, axil, shear, and bending stresses in members, stability, determinacy and indeterminacy of beams and trusses.		

CIEN 241	Infrastructure Management	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): ECON 103		
An introduction to infrastructure management systems, including management process, data collection technologies, interdependence, benchmarking and best practices. Other related issues, such as, resilience security of infrastructure systems are addressed.		
CIEN 250	Engineering in Global Environment	(2:0:0)
2 Semester Credit Hours		
Introduction to environmental engineering as practiced in different societies around the world. Defining environmental engineering as organizational and physical infrastructure to manage natural resources. Focuses on how different societies respond to environmental challenges related to engineering opportunities. Identify and explain the global, economic, environmental, and societal impacts of specified civil and infrastructure engineering systems. Issues for sustainability in engineering practice. Include case studies of sustainable engineering in developing and developed world.		
CIEN 251	Fluid Mechanics	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): MATH 213 & CIEN 211 (Civil Engineering) and MATH 114 and PHYS 110 (Mechanical Engineering)		
Properties of fluids, flow regimes, pressure and force calculations under hydrostatic conditions, manometers, buoyancy and stability of floating and submerged bodies, elementary fluid dynamics, conservation equations: mass, energy and momentum, continuity and Bernoulli equations. Boundary layer theory, fluid flow in conduits and determination of head loss in pipes.		
CIEN 261	Surveying	(0:3:0)
1 Semester Credit Hour		
Pre-requisite(s): CIEN 201 (for Civil and Infrastructure Engineering); INDS 122 (for Architecture Students)		
Principles of surveying by conducting tests on distance measurements, levels and theodolites, directions and angular measurements, topographic surveys, areas and volumes; traverse surveys; setting out horizontal and vertical curves, training on Total Station.		
CIEN 265	Geographic Information System (GIS)	(1:3:0)
2 Semester Credit Hours		
Pre-requisite(s): CSCI 112		
This course exposes students to the principles of GIS (hardware, software, people, data, and methods) and its environmental and infrastructure management applications. Subjects also include the acquisition and compilation of data from maps, field surveys, air photographs and satellite images. The course includes hands-on GIS state-of-the-art software.		
CIEN 301	Numerical Analysis	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): MATH 214 and CSCI 112		
Matrix Algebra: Matrix operations, inverse, determinant, Eigen vectors, Eigen Values and solutions of systems of linear equations. Application of computers to solve engineering problems using various numerical methods, mathematical modeling and error analysis, solution of algebraic and differential equations, numerical differentiation and integration and curve fittings. Presents application examples from civil and infrastructure engineering, industrial engineering and mechanical engineering.		
CIEN 311	Structural Analysis	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 212		
Structural forms, reactions, determinate structures, degree of determinacy, shear and moment diagrams for beams and frames, influence lines for beams, deflections (principle of moment area theorems, virtual work and conjugate-beam methods), Analysis of indeterminate structures by moment force method, slope-deflection method and distribution method.		

CIEN 316	Structures for Architecture 2	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 212		
Introduction to concrete, structural steel, and masonry as construction materials. Design of rectangular reinforced concrete beams reinforced concrete one-way slabs, and simple design of reinforced concrete short columns according to ACI 318 design code. Design of simple masonry walls. Design of structural steel beams and structural steel columns according to LRFFD design method.		
CIEN 321	Reinforced Concrete Design	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 311 and CIEN 331		
Properties of concrete and steel, cracked and uncracked section analysis, strength design, stress block, design for bending and shear, singly, doubly reinforced sections, rectangular sections, and T-sections, load cases and moment envelopes, bond requirements, development length and bar cutoffs, one-way solid and one-way ribbed slabs, design of short columns.		
CIEN 331	Construction Materials	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 212 and CHEM 211		
Cement (types, manufacture, properties and hydration), aggregates, fresh concrete, hardened concrete (strength, strength development, shrinkage, creep), concrete in severe environment (hot and cold), durability, mix design by ACI and DoE methods, use of masonry, fiber reinforcement and metal form decking and structural steel joists.		
CIEN 332	Construction Materials Lab.	(0:3:0)
1 Semester Credit Hour		
Pre-requisite(s): CIEN 212, and CHEM 211		
Co-requisite: CIEN 331		
Introduction to testing & specifications, concrete and mortar tests, aggregate testing, fresh and hardened concrete testing, non-destructive tests, design & testing of concrete mixes, brick testing, tests on steel reinforcing bars.		
CIEN 333	Geotechnical Engineering	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 212 and CHEM 211		
Index and classification of soils, water flow in soils (one- and two-dimensional water flow), soil stresses, soil compaction, distribution of stresses in soil due to external loads, consolidation and consolidation settlement, shear strength of soils, slope stability.		
CIEN 334	Civil Engineering Testing and Materials	(0:3:0)
1 Semester Credit Hour		
Pre-requisite (s): CIEN 331, CIEN 333		
Co-requisite: CIEN 361		
Tests on soils: specific gravity, grain size distribution, consistency limits, coefficient of permeability (constant and falling head), consolidation test, direct shear and tri-axial and CBR. Tests on fluids: Center of pressure, orifice and jet flow, Pressure variation in flowing fluid, momentum principle, flow through pipelines and closed conduits. Tests on asphalt properties, and marshal mix design. (Writing Intensive Course)		
CIEN 351	Environmental Engineering	(2:0:0)
2 Semester Credit Hours		
Pre-requisite(s): CIEN 251, CHEM 211		
Introduces principles of environmental engineering management and design pertaining to water supply and treatment, wastewater treatment, solid waste management, air pollution control, noise pollution measurement and control, and environmental impact assessment. Includes case studies from UAE.		

CIEN 361	Highway Engineering and Design	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 331		
Pavement types; Materials used in flexible pavement layers (soil, aggregate, and asphalt); Calculation of Equivalent Standard Axle Loads (ESAL); flexible pavement thickness design, Hot Mix Asphalt (HMA) design; Highway construction operations; highway performance.		
CIEN 362	Transportation Engineering	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 261		
Role of transportation engineering; Transportation system issues and challenges; Modes of transportation, Main components of highway, mode of transportation (driver, pedestrian, traffic, road); Geometric design of highways and highway facilities; Highway functional classification and special facilities; Intersection design and control. Introduction to rail, air, and water transportation.		
CIEN 421	Structural Steel Design	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 311		
Introduction to steel structures and practical design methods, steel sections, load factors and load combinations, design of various steel elements using LRFD-method, design of tension and compression members, beam design: Compact section criterion, lateral-torsional buckling, lateral supports, and various design aspects of beams. Beam-column elements, design of column base plates, design of simple bolted (or welded) steel connections.		
CIEN 422	Advanced Reinforced Concrete Design	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 321		
Structural layout, estimation of dead and live loads, serviceability, deflections and crack control, design for torsion, design of frames, moment redistribution, slender columns, approximate methods for two-way slabs, detailing of reinforcement.		
CIEN 424	Bridge Design	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 321		
Covers design of highway bridges; history, classification and aesthetics of bridge structures; design philosophy; loading, girder distribution factors; and load combinations; design of concrete deck slab; design of reinforced concrete T-beam and box girders bridges; and design of piers, bearings and abutments. Introduces pre-stressed concrete bridges.		
CIEN 431	Foundation Engineering	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 333		
Site investigation, bearing capacity of shallow foundation, distribution of stresses in soils, settlement of shallow foundation, factors to be considered in foundation design, introduction to deep foundation, lateral earth pressure and retaining walls.		
CIEN 440	Infrastructure Financing	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 241		
Fundamentals of infrastructure project financing, Public Private Partnerships (PPP), Build Own Transfer (BOT), variation models, financing structures, framework, bonds, life cycle analysis, sponsors and risk management.		
CIEN 441	Construction Management	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 331		
Basic concepts of construction project management, Construction planning, project time Management, project cost management, project quality management, value engineering and project life cycle, construction process optimization, construction contracts, contracting methods, project specifications, bidding, procurement methods and contractor applications for payment procedures.		

CIE 442	Construction Planning and Scheduling	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIE 441		
Principles of planning, monitoring, and controlling construction projects. Developing schedules using bar charts, precedence diagrams, program evaluation and review techniques (PERT), and linear schedules. Resource histograms and s-curves. Resource allocation and resource leveling. Schedule constraints. Earned value concept. Includes MS project and Primavera Project Planner software.		
CIE 444	Construction Cost Analysis and Estimating	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIE 441		
Perceptions of construction cost, engineering economic analysis, risk and uncertainty, range estimating, cost fundamentals, types of cost estimating, estimating construction materials cost, estimating construction labor cost, direct versus indirect costs, estimating construction equipment cost, cost of concrete structures, estimating project cost, time/cost trade-off analysis and contractor general requirements.		
CIE 451	Infrastructure Systems	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIE 351 and CIE 362		
Role of planning, system demands and networking in infrastructure systems, energy systems, water and waste water infrastructure, transportation systems, waste disposal and resources conservation, smart growth and effects of infrastructure on the environment, models of creating sustainable future development, planning, design and architecture in sustainable communities, and integrated infrastructure system models.		
CIE 454	Water and Wastewater Supply	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIE 351		
Sources of water, requirements for water supply projects, population's studies, rates of water consumption, variation in water demand, collection and distribution of water, distribution networks. Sources of wastewater, quantities and quality, sewage collection works, sewage purification works and disposal, primary treatment, secondary treatment, activated sludge system, design of sewer systems.		
CIE 455	Solid Waste Management	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIE 351		
Evolution of solid waste management, integrated approach to solid waste management, sources, composition, and properties of solid waste, physical, chemical and biological properties of MSW and household hazardous wastes. Waste handling, separation, storage and collection. Building a sustainable future, application of life-cycle analysis to waste management systems, reuse technologies, energy recovery from liquid and solid wastes and product recovery from oily wastes.		
CIE 456	Sustainable Urban Building Sites	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIE 351		
Concept of sustainability, how sustainability applies to infrastructure projects and programs, measuring sustainability, identification and design of sustainable technologies associated with water and energy management for infrastructure projects, green buildings and sustainable housing, sustainable transportation, energy, use of materials and waste management and water use. Includes case studies in sustainability.		
CIE 462	Advanced Pavement Design	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIE 361		
Pavement types, Pavement materials; subgrade stabilization methods; Principles of mix design using SUPERPAVE; Analysis of stresses in flexible and rigid pavement, Design methods of highway flexible and rigid pavements; Overlay design, Computer applications.		

CIEN 464	Airport Planning and Design	(3:0:0)
3 Semester Credit Hours		
Co-requisite(s): CIEN 362		
Examines airport master planning, forecasting air travel demand and sustainable design of airports, including lighting, terminal facilities, noise-level control, aircraft control, airspace utilization and automobile parking.		
CIEN 491	Senior Design Project (1)	(0:6:0)
2 Semester Credit Hours		
Pre-requisite(s): Senior Standing		
Preparation and starting of project in civil and infrastructure engineering. The project is a multidisciplinary interaction for infrastructure design and management that includes system analysis and inculcates sustainable engineering principles. Includes use of engineering software especially project management such as MS project, Primavera Project Planner and CYCLONE.		
CIEN 492	Senior Design Project (2)	(0:12:0)
4 Semester Credit Hours		
Pre-requisite(s): CIEN 491		
Continuation of the senior design project I. The project is a multidisciplinary interaction for infrastructure design and management that includes system analysis and inculcates sustainable engineering principles. Includes use of engineering software's especially project management software such as MS project, Primavera Project Planner and CYCLONE, writing a technical report and developing project drawings, specifications and details.		
CIEN 493	Special Topics in Civil Engineering	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): Department Consent		
Special up-to-date topic in the civil and infrastructure engineering.		
COMM 101	Interpersonal Communication and Group Interaction	(3:0:0)
3 Semester Credit Hours		
The course presents the principles to develop appropriate and effective communication strategies in one-to-one and small group communication settings. It emphasizes analyzing and assessing communication skills to create and sustain effective communication in personal and professional relationships.		
COMM 102	Reading Image and Film	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): ENGL 101		
This course allows learners to study aesthetic effects and strategies of persuasion in contemporary visual images. Learning engagement includes images in electronic media, film, photographs, cartoons, advertisements, and public performances and events.		
COMM 111	Introduction to Mass Communication	(3:0:0)
3 Semester Credit Hours		
This course surveys the history and characteristics of mass communication as a field and set of intertwining industries and professions. Learners explore the role of mass media in modern society by considering the impact of technology, culture, government, and economics. Trends are considered in historical context.		
COMM 112	Introduction to Public Relations	(3:0:0)
3 Semester Credit Hours		
This course provides learners with basic knowledge about the theories, concepts, and best practices in communication within the public relations field.		
COMM 113	Introduction to Digital Media	(3:0:0)
3 Semester Credit Hours		
Introduction to Digital Media, allows students to learn and understand the basics of Digital Media. The course content discusses various technologies that will aid students to learn the language of visual imagery. The applications that will be focused on industry standard applications for many graphic design positions. In this course, students will build a blog as a		

means of communicating and presenting their work as producing print and digital layouts to a wide audience.

COMM 212 Media Writing (3:0:0)

3 Semester Credit Hours

Prerequisite(s): COMM 111 or COMM 112 or COMM 113

This course introduces learners to the basic norms, values, standards and practices for writing for the mass media.

COMM 213 Public Relations Writing (3:0:0)

3 Semester Credit Hours

Prerequisite(s): COMM 111 or COMM 112 or COMM 113

Learners improve their media-related writing skills with a focus on standard, professional presentation of information and messages in the public relations setting.

COMM 214 New Media and Digital Culture (3:0:0)

3 Semester Credit Hours

Prerequisite(s): COMM 111 or COMM 112 or COMM 113

This course examines the cultural impact of new digital technologies, such as the Internet and new telephone and audiovisual media. Students survey the origins of digital communication and the Internet; and they are introduced to contemporary scholarship on digital technologies, the Internet, and the institutions that control these technologies.

COMM 215 Feature Writing (3:0:0)

3 Semester Credit Hours

Prerequisite(s): COMM 111 or COMM 112 or COMM 113

This course is designed to give students experience in recognizing and producing high-quality feature articles. Course materials and lectures will cover the basics of writing the newspaper and magazine feature story. Students will be exposed to, and write in, a diverse variety of approaches and techniques.

COMM 221 Communication Analysis and Criticism (3:0:0)

3 Semester Credit Hours

This course investigates media through a variety of theories and methods. The focus will be on some of the dominant critical perspectives that have contributed to our understanding of media and its roles in society. The course provides the basic vocabulary and concepts used to analyze different types of media.

COMM 222 Intercultural Mass Communication (3:0:0)

3 Semester Credit Hours

This course explores the communication strategies and techniques within a specific cultural milieu and how those strategies and techniques differ among various cultural milieu, and learners practice in a variety of communication modes how best to accommodate their rhetorical strategies in communicating to both intracultural and intercultural audiences.

COMM 223 Globalization and Media Cultures (3:0:0)

3 Semester Credit Hours

Prerequisite(s): COMM 111 or COMM 112 or COMM 113

Learners in this course apply communication strategies to and investigate real-world case study challenges with a global worldview of various issues of global media cultures. Students learn how to develop a *Weltanschauung* from which to best communicate in modern technologies with world audience.

COMM 224 Visual Storytelling (3:0:0)

3 Semester Credit Hours

Prerequisite(s): COMM 111 or COMM 112 or COMM 113

This course combines storytelling arts with social, mobile, and digital media technologies. In this course, students apply digital storytelling theory and techniques to write, produce and publish digital stories. They integrate images, text, video and audio to create digital stories, and acquire competency in the use of digital media applications.

- COMM 311 Media Law and Ethics** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 111 or COMM 112 or COMM 113
 Learners engage the philosophical underpinnings of ethics and the core principles of journalism and mass communication to develop an understanding and appreciation of the field's normative ethical values. Students will learn how to apply an ethical decision-making framework to a variety of challenges.
- COMM 321 Theories of Media Communication** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 111 or COMM 112 or COMM 113
 Learners identify the major concepts, issues, and theories of media communication, and learners identify and use communication theories in a variety of best practices to demonstrate effective use of the theories learned in the course.
- COMM 322 Digital Resources and Content** **(3:0:0)**
3 Semester Credit Hours
Corequisite(s): COMM 212
 This course is designed to present fundamentals of open /data collecting, gathering, cleaning, analyzing, and create data visualizations. Students will learn how database journalism turns to data journalism; and how data became a strong role in transferring more efficient growing online resources, tools, and techniques.
- COMM 323 Advanced Journalism** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 212 or COMM 215
 This course will engage learners in the values, norms and professional practices newsgathering and writing. Emphasis is on traditional reporting methods, including interviewing and observation, and on the ethical, reliable presentation of news in print/digital formats.
- COMM 334 Digital Journalism** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 212 or COMM 213 or COMM 222
 This course introduces multimedia production skills to students. Learners will integrate text, audio, photos and video to produce online media packages that are attractive, accessible, easy to navigate, and appropriate for the platform and the audience.
- COMM 337 Public Relations Cases** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 212 or COMM 213 or COMM 222
 This course gives learners the context, knowledge, and skills to examine and critically analyze real-world public relations problems and cases. Students will study and apply communication and public relations theories to case studies and problems. Learners examine the four steps included in the design of public relations programs that include research, planning, implementation, and evaluation.
- COMM 344 Public Relations Theory** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 212 or COMM 213 or COMM 222
 This course provides students with an overall perspective of public relations theories as they are applied in various public relations activities and discover the underlying socio-cultural impact on public relations.
- COMM 391 Communication Research Methodology** **(2:0:0)**
2 Semester Credit Hours
Prerequisite(s): COMM 212 or COMM 222 or COMM 223
 This course provides an introduction to research methods and the philosophical underpinnings of research inquiry in the field of communication. It includes the topic adherence, overviews of quantitative, qualitative, and mixed-method research methodologies, a range of alternative research methods, including observation, archival research, questionnaire surveys, case studies, and experimentation, research design, data collection, and data analysis, the ethical implications of research with human and non-human subjects, and appropriate connections between research questions and methodologies.

- COMM 421 Social Media: Audiences and Messages (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 311 or COMM 321 or COMM 334
 Learners explore social media as a unique platform for interaction with multiple audiences and learn how to plan strategies to engage with key stakeholders and disseminate persuasive, effective messages.
- COMM 422 Streaming (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 334
 The course covers basic desktop digital video and audio applications, video streaming, and basic design for web and mobile products with a focus on aesthetics, functionality, and access.
- COMM 423 Multimedia Magazine (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 334
 This course allows students to explore magazine-style writing, editing, and presentation in the digital environment. Students collectively produce an online, magazine-style, publication.
- COMM 424 Capstone: Multimedia Storytelling (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 423
Prerequisite(s): COMM 321 or COMM 323 or COMM 334 or COMM 391
 this course, learners apply skills in newsgathering, writing, and multimedia to produce an individual and a group project of publication quality that demonstrates their proficiency in multimedia storytelling.
- COMM 427 Capstone: Media Production in Public Relations (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 321 or COMM 323 or COMM 334 or COMM 391
 The course provides learners with practical instruction and experience in a variety of media modes of communication. Students focus on producing clear, persuasive messages in attractive formats.
- COMM 450 Selected Topics in Communication (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): COMM 212 or COMM 222 or COMM 223
 This course explores contemporary issues in communication theory, research, and practice and deal with specific media and PR issues or trends. Topics, ideas, and issues not taught in the standard courses are investigated. **(Writing Intensive Course)**
- COMM 492 Mass Media and Public Opinion (3:0:0)**
3 Semester Credit Hours
Corequisite(s): COMM 321 and COMM 391
 This course is designed to prepare students to be informed and critical consumers of polls and media coverage of them, to introduce students to basic theories and findings regarding the influence of mass media on public opinion, to provide students with firsthand experience in conducting and writing about public opinion research.
- CSAI 350 Introduction to Artificial Intelligence (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): CSCI 215 and ENGR 200
 This course provides an introduction to the different sub-areas of Artificial Intelligence (AI). In addition, students learn basic concepts, methods, and algorithms of AI and how they can be used to solve practical AI problems. The topics include classical and adversarial search & heuristic, knowledge representation, probabilistic reasoning, convex optimization methods, Bayesian methods, reinforcement learning, and supervised and unsupervised learning techniques. Particular focus will be placed on real-world applications of the material.

CSAI 351	Data Science	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSAI 350 and MATH 203		
This course provides an introduction to data science and highlights its importance in real world context. Topics include data science concepts, project lifecycle, tools & programming environment, fundamentals of Python programming, numerical processing, data visualization, exploratory data analysis, data preprocessing, parameter optimization, model performance evaluation, and applications of machine learning algorithms in Python (i.e., Naïve Bayes, k-Nearest Neighbors, Linear/Multiple/Logistic Regressions, Decision Trees, and Clustering Applications), natural language processing, and real-world data science case studies.		
CSAI 390	Internship I	(0:3:12-20)
3 Semester Credit Hours		
Pre-requisite(s): Completion of 60 credits and a CGPA of 2.0 or higher		
This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.		
CSCI 391	Internship II	(0:3:12-20)
3 Semester Credit Hours		
Pre-requisite(s): Completion of 90 credits and a CGPA of 2.0 or higher		
This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.		
CSAI 450	Machine Learning	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSAI 350 and CSAI 351		
Co-requisite(s): CSAI 451		
This course introduces fundamental concepts of machine learning, and provides students with knowledge and understanding of the methods, mathematics, and algorithms used in machine learning. Topics include statistical learning concepts, linear & quadratic discriminant analysis, resampling methods, model selection and regularization, regression & smoothing splines, generalized additive models, regression trees, bagging and boosting, support vector machines, principal components analysis, k-means clustering, hierarchical clustering, and neural networks.		
CSAI 451	Machine Learning Lab	(0:3:0)
1 Semester Credit Hour		
Co-requisite(s): CSAI 450		
This course, which is conducted within a laboratory environment, aims to familiarize students with several techniques used in machine learning. The topics covered include Linear Regression, Classification, Resampling, Linear Model Selection, Tree-Based Methods, Support Vector Machines, and Neural Networks. (Writing Intensive Course)		
CSAI 452	Natural Language Processing	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSAI 450		
This course introduces the fundamental concepts and techniques of natural language processing (NLP). Topics include text corpora and conditional frequency distributions, lexical resources and WordNet, raw text processing and regular expressions, text normalization and lemmatization, structured natural language processing (NLP) programs, part-of-speech tagging, automatic tagging, n-gram, & transformation-based tagging, document and sequence classification, maximum entropy classifiers and modeling linguistic patterns, information extraction, linguistic structure, named entity recognition, & relation extraction, grammatical structure & context free grammar, context free grammar parsers & dependency grammar, and feature based grammars.		

CSAI 453	Data Visualization	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): CSAI 350		
Data visualization is an essential skill required in today's data-driven world. This course presents principles and techniques to design and create data visualization based on gathered data and the goals of the task at hand. Topics include the value of visualization, data, tasks, validation, marks and channels, design guidelines, tables, networks and trees, spatial, temporal and textual data, interaction and navigation, and data reduction.		
CSAI 480	Big Data	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSCI 326		
This course provides an in-depth coverage of various topics in big data from data generation, storage, management, transfer, to analytics. Topics include Hadoop ecosystem, Hadoop architecture, YARN & HDFS, Sqoop, Apache Hive & Impala, Apache Flume, Apache Spark, resilient distributed datasets (RDDs) in Spark, map reduce in Spark, Spark applications, Spark parallel processing, SparkSQL, workflow management with Apache Oozie, machine learning in Spark, and big data applications.		
CSAI 481	Computer Vision	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSAI 450		
This course provides introduction to fundamental topics in computer vision and the application of statistical estimation techniques in this field. It is intended to give students a good basis for work in this important field. Topics include image representation, image processing, image analysis, image segmentation, object tracking, 3D shape reconstruction, feature detection and tracking, object detection, and convolutional neural networks.		
CSAI 482	Data Mining	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSAI 350		
Data Mining studies algorithms and computational paradigms that allow computers to find patterns and regularities in datasets, then perform prediction/forecasting and generally improve their performance through interaction with data. The course introduces the fundamental concepts of data mining techniques. Topics include data preparation, data classification, performance measures, association rule mining, clustering, outlier detection, collaborative filtering and data mining case studies.		
CSAI 483	Information Retrieval	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSCI 326		
The course covers basic and advanced techniques necessary to define, design, and implement Information Retrieval (IR) systems, including the following topics: basic IR models, basic tokenizing, indexing, and implementation of vector-space retrieval, experimental evaluation of IR, query operations and languages, text representation, web search, text categorization, language-model based retrieval, recommender systems, information extraction, and integration.		
CSAI 484	Internet of Things System	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): (CSCI 232 or CENG 315) and CSCI 462		
The Internet of Things is an emerging technology in which interconnected devices are embedded within everyday objects enabling them to send and receive data. This course introduces core concepts and networking protocols for IoT applications. Topics covered include: IoT architectures, smart objects, sensing and actuation, displaying information, WPANs, WBANs, IoT clouds, security and privacy for IoT systems, IoT robustness and reliability, and IoT applications.		
CSAI 485	Introduction to Deep Learning	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSAI 450		
The course provides an introduction to neural networks and deep learning. The topics to be covered include basic conceptual understanding of neural networks, deep learning programming environments, shallow neural networks, radial basis function networks, recurrent neural networks, convolutional neural networks, and deep reinforcement learning.		

- CSAI 486 Special Topics in Artificial Intelligence (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): CSAI 350
This course gives instructors the opportunity to cover the latest developments and contemporary issues in Artificial Intelligence. Instructors will provide a detailed course outline at the beginning of the semester.
- CSAI 487 Introduction to Robotics (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): EEEN 331 and CSAI 350
This course introduces fundamentals of robot modelling and control. Topics include forward and inverse kinematics, Jacobians, trajectory design, motion planning, in addition to relevant applications and technologies such as mobile robotics, industrial robotics and computer vision.
- CSAI 490 Professional Software Practice (2:0:0)**
2 Semester Credit Hours
Pre-requisite(s): Senior Standing
Co-requisite(s): CSCI 492
The course develops student understanding about various non-technical issues related to the field of computing. The course covers the field from economic, legal, ethical, and professional perspectives. Covered topics include code of conduct, ethics, intellectual property, cyber security, privacy, risk management, and teamwork.
- CSCI 104 Introduction to Computing (3:0:0)**
3 Semester Credit Hours
This course serves as an introduction to the field of computer science and the computer's various layers. The course provides exposure to the following layers: information, hardware, programming, operating systems, applications, and communications. Additional topics include ethics, security, privacy, the impact of computing, and widely used software applications.
- CSCI 112 Introduction to Computer Programming (3:0:0)**
3 Semester Credit Hours
Co-requisite(s): CSCI 113
This course introduces students to computers and programming languages and more specifically the C++ language. Additionally, students learn to use computer programming as a problem-solving tool. The topics covered include basic operations, data types, input/output, selection statements, control structures, arrays, functions, and strings, knowledge representation, neural networks and natural language, data summarization and visualization.
- CSCI 113 Introduction to Computer Programming Lab (0:3:0)**
1 Semester Credit Hour
Co-requisite(s): CSCI 112
This course introduces the use of computer programming as a problem-solving tool in laboratory environment. Topics in procedural programming include, simple data types, input/output selection statements, control loops, testing, debugging, and programming environments.
- CSCI 211 Object Oriented Programming (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): CSCI 112
This course is an introduction to object-oriented programming principles and techniques using Java. Topics include Java elementary programming, and Java object-oriented features such as methods, objects, classes, access modifiers, constructors, immutable objects & classes, abstraction, encapsulation, inheritance, polymorphism, dynamic binding, object castings, abstract and interface classes, and exception handling.
- CSCI 215 Data Structures and Algorithms (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): CSCI 211 and MATH 225
This course introduces data structures and various fundamental computer science algorithms. The course covers abstract data-type concepts, stacks, queues, lists, and trees. Several sorting and searching algorithms are covered. Additional topics include an introduction to graphs and their implementation and running time and time complexity measurement.

CSCI 232	Computer Organization	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSCI 112		
This course provides a programmer's view of the execution of programs in computer systems. Topics covered include instruction sets, machine-level code, assembly language, performance evaluation and optimization, memory organization and management, address translation, and virtual memory.		
CSCI 312	Operating System Fundamentals	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSCI 215		
This course covers the principles, components, and design of modern operating systems, focusing on the UNIX platform. Topics include system structure, process concept, multithreaded programming, process scheduling, synchronization, atomic transaction, deadlocks, memory management, and file system.		
CSCI 315	Design and Analysis of Algorithms	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSCI 215		
This course introduces the design and analysis principles for various algorithms. The topics covered include searching algorithms, dynamic programming, greedy algorithms, Huffman coding, graph traversing algorithms, shortest path algorithms, linear programming, and NP-completeness.		
CSCI 326	Database Systems	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSCI 211		
This course is an introductory course on database management systems. The goal of the course is to present a comprehensive introduction to the use of data management systems. Some of the topics covered are the following: The Entity-Relationship Model, the Relational Data Model, the SQL language, the database design, and the database integrity and security.		
CSCI 372	Compiler Design	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSCI 232		
The course introduces the theory and practice of compilation. Topics include compiler architecture, components, phases, software tools, lexical roles and specifications, regular expressions, syntax roles and specifications, context-free grammars, top-down parsing, bottom-up parsing, LR parsers & parse trees, syntax directed translation, syntax tree, abstract syntax tree, and finite automata.		
CSCI 388	Programming Languages	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSCI 215		
This course is an introduction to basic concepts in the design of programming languages. The course focuses on programming languages within the functional and logic programming paradigms such as Scheme and Prolog. Topics include history of programming languages, language design criteria, functional programming, syntax, logic programming, semantics, and object-oriented principles.		
CSCI 390	Internship I	(0:3:12-20)
3 Semester Credit Hours		
Pre-requisite(s): Completion of 60 credits and a CGPA of 2.0 or higher		
This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.		

CSCI 391	Internship II	(0:3:12-20)
3 Semester Credit Hours		
Pre-requisite(s): Completion of 90 credits and a CGPA of 2.0 or higher		
This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.		
CSCI 411	Computer Graphics	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSCI 215		
Co-requisite(s): CSCI 412		
This course introduces interactive computer graphics and drawing algorithms. The topics covered include graphics output primitives and their implementations, two-dimensional and three-dimensional geometric transformations and viewing, hierarchical modeling, computer animation, spline representations, visible-surface detection methods, illumination models, and surface-rendering methods, and texturing and surface-detail methods.		
CSCI 412	Computer Graphics Lab	(0:3:0)
1 Semester Credit Hour		
Co-requisite(s): CSCI 411		
This course introduces computer graphics and drawing algorithms in a laboratory environment. The topics covered include graphics output primitives and their implementations, two-dimensional and three-dimensional geometric transformations and viewing, hierarchical modeling, computer animation, spline representations, visible-surface detection methods, illumination models, and surface-rendering methods, and texturing and surface-detail methods.		
CSCI 415	Introduction to Parallel Programing	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSCI 215		
This course is an introduction to parallel programming principles and techniques. Topics include parallel computing memory architecture, memory organization, parallel programming models, parallel program design, performance evaluation, thread-based parallelism, process-based parallelism, message passing, asynchronous programming, and heterogeneous programming.		
CSCI 416	Human Computer Interaction	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSCI 215		
This course provides an introduction to and overview of the field of human-computer interaction (HCI). The topics include usability principles, predictive evaluation, design management processes, graphic design, understanding users' requirements gathering, task analysis, handling errors & help, prototyping & UI software, interaction styles, user models, evaluation, and universal design.		
CSCI 450	Information Security and Privacy	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSCI 215 or Instructor permission		
This course is a survey of information security considerations as they apply to information systems analysis, design, and operations. Topics include information security vulnerabilities, threats, and risk management. Furthermore, the course introduces several cryptographic algorithms in addition to the privacy and secrecy of statistical databases and e-government applications.		
CSCI 462	Data Communications and Computer Networks	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSCI 112		
This course introduces computer networks. Topics include layering approach, functions of different layers, Internet applications (HTTP, DNS), reliable and unreliable transport (TCP and UDP), routing and IP addressing, data link layer services and protocols, and Ethernet.		

CSCI 463	Data Communications and Computer Networks Lab	(0:3:0)
1 Semester Credit Hour		
Pre-requisite(s): CSCI 112		
Co-requisite(s): CSCI 462		
This course provides students with hands on training on design, troubleshooting, modeling and evaluating of computer networks. Topics include network addressing, Address Resolution Protocol (ARP), basic troubleshooting tools, IP routing, and route discovery. Additionally, student will perform network modeling, simulation, and analysis using Packet tracer and WireShark analyzer. (Writing Intensive Course)		
CSCI 492	Senior Design Project I	(0:6:0)
2 Semester Credit Hours		
Pre-requisite(s): Senior standing		
The course requires seniors to work in small teams to solve significant problems. Over the duration of CSCI 492 and CSCI 493, students design, implement, and evaluate a solution to the problem in conjunction with a faculty advisor. The course reinforces programming principles and serves as a capstone for computing knowledge obtained in the BSCS and BSAI curricula. The recognition of the ethical and legal principles are also aspects of the course.		
CSCI 493	Senior Design Project II	(0:12:0)
4 Semester Credit Hours		
Pre-requisite(s): CSCI 492		
Implementation of the project for which preliminary work was done in CSCI 492. Project includes designing and constructing software and/or hardware, conducting experiments or studies, and testing and validating a complete system. At the end of the term, each team presents to a committee information related to its project in both written and oral formats.		
CSCI 499	Special Topics in Computing	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CSCI 215		
This course gives instructors the opportunity to cover the latest developments and contemporary issues in computing. Instructors will provide a detailed course outline at the beginning of the semester.		
ECON 103	Principles of Microeconomics	(3:0:0)
3 Semester Credit Hours		
This course introduces microeconomics in the context of current problems. Explores how market mechanism allocates scarce resources among competing uses; uses supply, demand, production, and distribution theory to analyze problems.		
ECON 104	Macroeconomic Economics Policy	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): ECON 103		
This course introduces students to the applications of macroeconomics in the context of current problems to business decision making and economic policy. Students will be introduced to the economic way of thinking and how key concepts; theories and methods of modern economic analysis can be applied to everyday economic issues and problems. Key areas covered include the national economy; national income accounting; business cycles; inflation; economic growth and stability; unemployment; and monetary and fiscal policy. Particular emphasis is placed on reviewing contemporary economic issues and how economics permeates almost every aspect of business, highlighting economics as the fundamental discipline underpinning the study of most business specializations.		
EDUC 511	Introduction to Applied Research	(3:0:0)
3 Semester Credit Hours		
This course provides students and introduction to quantitative, qualitative, and mixed-methods research design in the educational sciences. This is a practical course designed to guide students through the development of their applied research proposals into a gap in practice. Students will defend this research proposal at the conclusion of the semester at a graduate students' colloquium. Thesis Research Course I		
EDUC 515	Theories of Teaching and Learning	(3:0:0)
3 Semester Credit Hours		

This course will examine major theories regarding teaching and learning such as behaviorism, constructivism, social cognitivism, and humanism. An overview of the major theorists, contemporary learning theories, strategies, models, and approaches will provide the framing for the development of a personal philosophy of teaching and learning.

EDUC 525 Theories and Principles of Leadership

(3:0:0)

3 Semester Credit Hours

This course provides students with conceptual and practical knowledge in leadership theories, with an emphasis on application in real world education settings. Students will acquire skills grounded in various leadership theories and use this conceptual knowledge to create an individual, metacognitive Leadership Profile.

EDUC 545 Coaching, Mentoring, and Developing School Staff

(3:0:0)

3 Semester Credit Hours

This course is designed to prepare leaders to coach and mentor school faculty and staff for professional development. Students will acquire the requisite skills to coach and/or mentor school staff and faculty to facilitate positive change within schools and improve student performance.

EDUC 620	Supervision and Evaluation of Instruction	(3:0:0)
3 Semester Credit Hours		
This course will examine theory, research and practice of supervision; approaches to staff development; assessment and evaluation; clinical supervision techniques.		
EDUC 640	Inclusive Leading of Diverse Groups	(3:0:0)
3 Semester Credit Hours		
This leadership course is designed to prepare students to lead diverse groups and/or teams of individuals in schools and where education meets industry. Throughout the duration of this course, students will engage in meaningful dialogue centered on these authentic, real-world scenarios.		
EDUC 660	Implementation and Evaluation of Comprehensive School Reform	(3:0:0)
3 Semester Credit Hours		
This course provides students with the knowledge and skills to carry out comprehensive school reform, and it includes the major steps required for determining needs, planning the reforms, understanding challenges and roadblocks to reforms, and implementing and evaluating the reforms.		
EDUC 675	School Governance and Community Engagement	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): Completion of the PGDE with CGPA of 3.3 or Higher		
In this course students explore the governance of schools and the building of constructive relationships between schools and communities from socio-economical and sociological perspectives grounded in educational research.		
EDUC 680	Applied Research I	(1:6:0)
3 Semester Credit Hours		
Prerequisite(s): EDUC 511		
This course, designed using a metacognitive approach, enables students to apply critical reflective practice while completing their research fieldwork. This course will draw upon specified skills acquired in previous research and leadership courses to facilitate the necessary fieldwork for students' individual research projects. Thesis Research Course II		
EDUC 685	Continuous Improvement Through Data Driven Leadership	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): Completion of the PGDE with a CGPA of 3.3 or higher		
This course focuses on the role of research in the work of school leaders; ways leaders use research to identify opportunities for improvement and change; relationship between theory, research, and practice; role of research and benchmarking in school improvement; and strategies for change in education.		
EDUC 695	Applied Research II	(1:6:0)
3 Semester Credit Hours		
Prerequisite(s): Completion of the PGDE		
This research capstone prepares students who have transitioned into the Master of Education program to finalize their original research for submission to an academic journal. This course is designed to enable students with requisite skills in disseminating research.		
EEEN 220	Signal and Systems I	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): MATH 113		
Co-requisite(s): EEEN 221		
Introduction to methods of representing continuous-time signals and systems, and interaction between signals and systems. Analysis of signals and systems through differential equations and transform methods; Laplace and Fourier transforms; frequency response of systems; stability in time and frequency domains. Presents application examples from communications, circuits, control, and signal processing.		

EEEN 221	Signal and Systems I Lab	(0:3:0)
1 Semester Credit Hour		
Co-requisite(s): EEEN 220		
Laboratory course to accompany EEEN 220. In this course, the student will acquire hands-on experience with programming in MATLAB. Topics include representation of different signals, system linearity and time invariance, analysis of a first - order system, implementing matched filter for Barker codes, response of second - order systems and damping ratio, synthesis periodic signals		
EEEN 280	Electric Circuit Analysis I	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): PHYS 220		
Basic circuit concepts and DC analysis, circuit analysis techniques, circuit theories, fundamental operation of operational amplifiers and their applications, transient and steady state analysis of RL, RC, and RLC circuits and basic AC analysis.		
EEEN 281	Electric Circuit Analysis I Lab	(0:3:0)
3 Semester Credit Hours		
Co-requisite(s): EEEN 280 or MENG 231		
Laboratory course to accompany EEEN 280. In this course, students will experimentally verify circuit analysis concepts under DC excitation and transient response. They will use different measurement instruments and build DC electric circuits.		
EEEN 282	Electric Circuit Analysis II	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): EEEN 280		
Review of AC sinusoidal circuit analysis with active and reactive power. Covers magnetically coupled inductors and ideal transformers, three phase circuits, Laplace transform, application of Laplace transform in circuit analysis, passive and active filter analysis and design, two port networks.		
EEEN 283	Electric Circuit Analysis II Lab	(0:3:0)
1 Semester Credit Hour		
Pre-requisite(s): EEEN 281		
Co-requisite(s): EEEN 282		
Laboratory course to accompany EEEN 282. In this course, students will experimentally verify circuit analysis concepts under steady state AC excitation. Students will use different measurement instruments and build AC electric circuits.		
EEEN 305	Electromagnetic Theory	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): PHYS 220 and MATH 214		
Course uses vector algebra and vector calculus. Covers topics related to electrostatic and magnetostatic fields, electric and magnetic properties of media, electric boundary value problems, Maxwell's equations, electromagnetic waves and plane wave propagation, Poynting theorem and transmission line theory.		
EEEN 320	Signal and Systems II	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): EEEN 220 and MATH 203		
Co-requisite(s): EEEN 321		
Methods of representing and analyzing discrete-time signals and systems. Studies effects of converting from continuous-time to discrete time and presents Z-transform as convenient analysis tool. Emphasizes powerful concept of frequency response of systems developed in first semester. Studies random signals in continuous and discrete time. Presents application examples from communications, circuits, control, and signal processing.		
EEEN 321	Signal and Systems II Lab	(0:3:0)
1 Semester Credit Hour		
Co-requisite(s): EEEN 320		
Laboratory course to accompany EEEN 320. In this course, the student will acquire hands-on experience with programming in MATLAB. The experiments cover sound signals processing using MATLAB, smoothing data, difference equations, and discrete time Fourier transformation.		

(3:0:0)3

Pre-requisite(s): PHYS 220

This course covers principles of digital logic and digital system design. Topics include number systems; Boolean algebra; analysis, design, and minimization of combinational logic circuits; analysis and design of synchronous and asynchronous finite state machines; and an introduction to VHDL and behavioral modeling of combinational and sequential circuits.

(0:3:0)

Pre-requisite(s):

Laboratory course to accompany EEEN 331. In this course, the student will acquire hands-on experience with basic logic components, combinational and sequential logic circuits, and the use of VHDL.

(3:0:0)

Pre-requisite(s): EEEN 280

Principles of operation and application of electron devices and linear circuits. Topics include semiconductor properties, diodes, bipolar and field effect transistors, biasing, amplifiers, frequency response, operational amplifiers, and analog design.

(0:3:0)

Pre-requisite(s): EEEN 281

Laboratory course to accompany EEEN 333. In this course, the student will acquire hands-on experience with basic electronic components and circuits. Topics covered include: Semiconductor diodes, rectification, Zener diodes, BJT and FET transistors and Amplifiers.

(3:0:0)

Pre-requisite(s): EEEN 282

The general theory of electro-mechanical motion devices relating to electric variables and electromagnetic forces. Basic concepts and operational behavior of DC motors, induction and brushless DC Motors, and stepper motors used in control applications.

(0:3:0)

Co-requisite(s): EEEN 350

Laboratory course to accompany EEEN 350. In this course, students will acquire hands-on experience with the characteristics of dc motors and dc generators (separate, series, shunt and compound). They will learn to find the parameters of transformers and evaluate their performance characteristics. The starting, speed control and performance of 3-phase induction motors are also studied. **(Writing Intensive Course)**

(3:0:0)

Pre-requisite(s): ENGR 200 and EEEN 220.

Introduction to set theory, probability, random variables, and random processes. Modelling various types of noise encountered in communication systems as random processes. Analyzing the system response in the presence of random noise processes and calculating the power spectral density.

EEEN 412	Power Systems	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): EEEN 350		
Electric Power Systems, Elements of a Power Systems; The analysis of power systems starting with the calculation of line resistance, line inductance, and line capacitance of power transmission lines; Analysis of power systems in terms of current, voltage, and active/reactive power; Per-Unit Quantities; Load Flow Study; Economic Dispatch; Symmetrical Components; Fault Study; System Protection; System transient and Stability issues.		
EEEN 421	Power System Protection	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): EEEN 412		
Power systems protection schemes for transmission and distribution networks, connection and standards of current and voltage instrument transformers for protection and metering applications, Protective relays, Protection of generators; Differential protection - Problems with differential protection - Biased differential protection - Biased differential protection of generator - Over current and earth fault protection, P; Buchholz relay - Biased differential protection of transformers - Harmonic restraint – Harmonic blocking - Other transformer protections, Protection of transformers, Protection of transmission lines.		
EEEN 422	High Voltage Engineering	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): EEEN 282		
The components of power system and their characteristics. Fundamental electric field calculations (Laplacian fields) in insulation systems of simple geometries, introduction to gas discharge physics, Townsends theory of electric breakdown in air and Paschens law and its implications on gas insulation strength. Experimental techniques applied in high voltage engineering.		
EEEN 423	Electrical Energy Systems & Fault Analysis	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): EEEN 412		
Energy and power; forms of energy; energy conversion from energy sources including wind , solar, tidal, bio-fuel, wave, hydro, nuclear and fossil fuel. Structure of a modern power system: operating charts, voltage control, and matrix representation of transmission lines. Two port network representation of transmission lines, per unit system, fault analysis: symmetrical components, transformers: construction, operation, connections, and relevant calculations. Load flow analysis: network matrix representation, Gauss-Seidel and Newton-Raphson solution techniques. AC/DC conversion: converter types, dc transmission, advantages compared to AC transmission. Over-voltages: switching and fault over-voltages, Bewley Lattice diagrams, switchgear principles, current chopping, insulation coordination. Modal component theory: wave propagation.		
EEEN 425	Smart Power Grid Systems Theory & Implementation	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): EEEN 412		
This course explores a set of emerging concepts, technologies, applications and business models, and the related trade-off decisions involved in transforming the traditional centralized power grid into a climate and renewable energy-friendly “Smart Grid.” A cross-disciplinary approach intended to deepen individual areas of expertise in the context of multidisciplinary teamwork. Basic Smart Grid literacy, applications of this knowledge base to specific “real world” case studies.		
EEEN 426	Renewable Energy Systems	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): Senior standing		
The course aims to introduce a general engineering/science audience to the basic concepts of renewable energy. In the interest of time some mathematical criteria will be covered, e.g. Betz limit for wind, limit of efficiency of WEC point absorber. Each lecture contains several examples from real world applications and in-progress industrial developments.		
EEEN 431	Digital Circuit Design	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): EEEN 331 and EEEN 333		

Analysis and design of discrete and integrated switching circuits. Topics include transient characteristics of diodes, bipolar, and field-effect transistors; MOS and bipolar inverters; non regenerative and regenerative circuits; TTL, ECL, IIL, NMOS, and CMOS technologies; semiconductor memories; VLSI design principles; and SPICE circuit analysis.

EEEN 432 Nanotechnology Fundamentals & Applications (3:0:0)

3 Semester Credit Hours

Pre-requisite(s): Senior Standing

Students will be provided an introduction to the basic principles of the subject of nanotechnology with minimum mathematics involved. The applications of nanotechnology to various fields such as environment, health, alternative energy and electronic/optics will be addressed and explained. Students will be required to do research projects and presentations in class.

EEEN 433 Linear Electronics II (3:3:0)

3 Semester Credit Hours

Pre-requisite(s): EEEN 333.

Differential amplifiers, feedback circuits, power amplifiers, feedback amplifier frequency response, analog integrated circuits, operational amplifier systems, oscillators, wide band and microwave amplifiers, and computer-aided design.

EEEN 434 Linear Electronics II Lab (0:3:0)

1 Semester Credit Hour

Pre-requisite(s): EEEN 334.

Co-requisite: EEEN 433

Laboratory course to accompany EEEN 433. In this course, the student will acquire hands-on experience with Electronic Amplifiers, active filters and oscillators. Topics covered include: Cascade amplifiers, differential amplifier, active filters, oscillators, and feedback amplifier concepts. **(Writing Intensive Course)**

EEEN 435 Introduction to Optical Electronics (3:0:0)

3 Semester Credit Hours

Pre-requisite(s): EEEN 305

Co-requisite(s): EEEN 333

Introduces optoelectronic devices for generation, detection, and modulation of light. Topics include electro-optic modulators, gas, solid state and semiconductor lasers, photodetectors, and detector arrays.

EEEN 437 Power Electronics (3:0:0)

3 Semester Credit Hours

Pre-requisite(s): EEEN 333

Course examines the application of electronics to energy conversion and control. The subject covers modern power semiconductor devices e.g., diodes, thyristors, MOSFETS, and other insulated gate devices; Static and switching characteristics, gate drive and protection techniques; Various DC-DC, AC-DC and DC-AC converter circuit topologies, their characteristics and control techniques; Analysis of input and output waveforms of these circuits; and their applications. Utility interference and Harmonic issues for power electronics circuits.

EEEN 451 Control Theory (3:0:0)

3 Semester Credit Hours

Pre-requisite(s): EEEN 220 or MATH 214

Introduction to feedback control systems; Block diagram and signal flow Graph representation; Mathematical modeling of physical systems; Stability of linear control systems; Time-domain and frequency-domain analysis tools and performance assessment; Lead and lag compensator design; Multi input multi output systems; Routh, Nyquist; Bode and root locus diagrams; Introduction to state variable techniques; state transmission matrix and state variable feedback.

EEEN 460 Communication Systems (3:0:0)

3 Semester Credit Hours

Pre-requisite(s): EEEN 220 and ENGR 200

Introduction to analog and digital communications. Topics include review of important concepts from signals and systems theory and probability theory; Gaussian processes and power spectral density; digital transmission through additive white Gaussian channels; sampling and pulse code modulation; analog signal transmission and reception using amplitude, frequency and phase modulation; and effects of noise on analog communication systems.

(0:3:0)

Pre-requisite(s): EEEN 460

Laboratory course to follow EEEN 460 and accompany EEEN 464. In this course, the student will acquire hands-on experience with fundamental blocks of Analog and Digital communication systems. Topics covered include: Amplitude and Angle Modulation and demodulation, sampling and reconstruction, PCM Encoding & PCM Decoding and digital modulation and demodulation.

(3:0:0)

Introduction to modern data communications and computer networks. Topics include point-to-point communication links and transmission of digital information, modems, and codecs; packet switching, multiplexing, and concentrator design; multi-access and broadcasting; local area and wide area networks; ISDN; architectures and protocols for computer networks; OSI reference model and seven layers; physical interfaces and protocols; and data link control layer and network layer.

(3:0:0)

Co-requisite(s): EEEN 461
Introduces digital transmission systems. Topics include quantization, digital coding of analog waveforms, PCM, DPCM, DM, base band transmission, digital modulation schemes, ASK, FSK, PSK, MSK, QAM, pulse shaping, inter symbol interference, partial response, voice band and wideband modems, digital cable systems, regenerative repeaters, clock recovery and jitter, multi path fading, digital radio design, optimal receiver design, MAP receiver, and probability of error.

(3:0:0)

This course provides a thorough treatment of digital signal processing including the fundamental theorems and properties of discrete-time linear systems, filtering, sampling, and discrete-time Fourier Analysis.

(3:0:0)

Cellular systems design fundamentals, fading and multipath channels, Modulation techniques for mobile radio systems, Diversity and combining techniques for mobile radio systems, multiple access techniques for mobile systems, Mobile systems and standards.

(3:0:0)

Course provides the fundamental knowledge in the theory and design of antennas. The theory of electromagnetic radiation is introduced and the fundamental antenna properties and parameters are explained. Standard antenna characterization parameters such as impedance, far-field radiation pattern, gain, directivity, bandwidth, beam width, polarization, efficiency, antenna temperatures are studied. The electromagnetic theory behind antenna operation and an overview of different antenna systems such as monopoles, dipoles, wire antennas and loop antennas etc... are discussed. The principles of analysis and design of antenna arrays are discussed.

(3:0:0)

This course covers a broad range of topics in the field of radio frequency (RF) and microwave engineering. This includes transmission lines, waveguides, impedance matching, microwave resonators, RF filters, RF amplifiers, and passive RF and microwave devices (mixers, diplexers, etc.). Furthermore, RF/microwave communications link design will be provided.

- EEEN 474 Advanced Information theory and coding (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): EEEN 460
 Advanced topics in information theory and coding. The course is divided into two main parts, namely, Source coding and data compression, and channel coding and error detection/correction codes. The first part covers, entropy, amount of information source coding techniques, Shannon Fano, Huffman, and Lempel-Ziv codes. The second part covers binary symmetric channels, Z-channels, and E-channels, channel capacity, mutual information, linear block codes and convolutional codes, Viterbi decoders and cyclic redundancy check codes.
- EEEN 481 Concepts of Multimedia Processing and Transmission (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): EEEN 320 or CSCI 462
 The course introduces the fundamentals of signal processing and communications for multimedia applications. It covers various topics relating to audio, image and video processing, storage and transmission. It discusses the human visual and hearing systems and relates them to image and sound digitization processes. The course also covers various lossless and lossy methods for audio, image and video compression. In addition, it gives the student hands on experience on applying the presented processing techniques using suitable software packages.
- EEEN 491 Engineering Seminar (1:0:0)**
1 Semester Credit Hour
Pre-requisite(s): Senior Standing
 Engineering ethics, professionalism, the role of engineers in society, current topics, and employment opportunities.
- EEEN 492 Senior Design Project I (0:6:0)**
2 Semester Credit Hours
Pre-requisite(s): Senior Standing
 Conception of senior design project and determine of feasibility of proposed project. Includes development of a preliminary design and implementation plan.
- EEEN 493 Senior Design Project II (0:12:0)**
4 Semester Credit Hours
Pre-requisite(s): EEEN 492
 Implementation of project from EEEN 492. Project includes designing and constructing hardware, writing required software, conducting experiments or studies, and testing complete system. Requires oral and written reports during project and at completion.
- EEEN 499 Special Topics in EEEN (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): Senior Standing
 Advanced and emerging topics in electronics and communication engineering. Topics are announced through the Schedule of Classes.
- EMAC 512 Managerial and Financial Accounting (3:0:0)**
3 Semester Credit Hours
Prerequisite: Graduate Standing
 The course deals with managerial and financial accounting techniques and practices. The course examines the impact of cost and cost allocation on business performance analysis, using a variety of costing schemes. It presents accounting as an information system construct, and looks at methods for recording, presenting and analyzing accounting information. The course also looks at budgeting, ratio analysis and other fiscal management measures. The course discusses the impact, on Management and Financial Accounting, of contemporary management philosophies and techniques including JIT practices, quality assurance and performance measurement.

- EMFN 514 Managerial Finance** **(3:0:0)**
3 Semester Credit Hours
Prerequisite: Graduate Standing
The course provides detailed critical treatment of the theory and practice of financial management within organizations. Topics covered include financial modeling, breakeven analysis, investment decision-making (NPV, IRR, etc.), capital budgeting and structure, working capital management, analysis of financial statements, and risk analysis. There is additional treatment of managerial economics.
- EMFN 560 International Corporate Finance** **(3:0:0)**
3 Semester Credit Hours
Prerequisites: EMFN 514
This course focuses on corporate finance at a global environment context. It explores advanced concepts and practices applicable to multinational company's financing and investment decision- makings. Topics covered include financial analysis for investment decisions in an international setting, international money operations and capital markets, management of foreign exchange risk, direct foreign investment, funding of international projects, political risk analysis, currency derivatives and swap markets..
- EMFN 562 Investment Planning and Management** **(3:0:0)**
3 Semester Credit Hours
Prerequisite: EMFN 514
The course deals with the advanced analysis of equity securities and investment portfolios, predicated on relevant market hypothesis and capital market theory. The course looks at investment risk-return trade-off, asset pricing models, and stock price behavior. The course adds emphasis on stocks, bonds, and financial futures and options.
- EMFN 564 Financial Markets and Institutions** **(3:0:0)**
3 Semester Credit Hours
Prerequisite: EMFN 514
The course deals with financial markets, primarily global equity markets. It looks at markets for handling government debt instruments, and exchange-traded and over-the-counter financial derivative instruments such as futures, options, swaps, and asset-backed securities. There is detailed treatment of derivatives theories, derivatives exchanges, and valuation of derivatives, including standard and other non-standard options on a variety of underlying assets, in relation to relevant financial markets. The course also discusses emerging financial markets, and the effective management of risks emanating from these markets.
- EMGN 525 Research Methods for Business** **(3:0:0)**
3 Semester Credit Hours
Prerequisite: Graduate Standing
This course provides the necessary underpinning support for the EMBA dissertation and general business domain research. It allows candidates to make informed decisions and appropriate choices pertaining to research methodology. A range of business research tools, approaches and analytical techniques are discussed, and guidance is given on optimal structuring of business research documents and EMBA dissertations. **Thesis Research Course**
- EMGN 555 EMBA investigative Thesis or Project I** **(0:9:0)**
3 Semester Credit Hours
Pre-requisite(s): EMGN 525
EMGN 555 is the first part of the EMBA investigative Thesis or Project course, serving to integrate the various topics studied in the School of Business EMBA program. Students undertake a work-based study that focuses on a business /management topic. The outcome for EMGN 555 is to submit a research introduction, literature review, and research methodology that will be used for the completion of this course. Satisfactory performance of the students in this course will move the student into EMBA 556.

- EMGN 556 EMBA Investigative Thesis or Project II** **(0:9:0)**
3 Semester Credit Hours
Pre-requisite(s): EMGN 555
 EMGN 556 is the second part of the three-credit hour research requirement, serving to integrate the various topics studied in the School of Business EMBA program. Students undertake a work-based study that focuses on a chosen business/management topic. The outcome is a thesis, an extended piece of work necessitating investigation of secondary data and empirical fieldwork, through collection of primary data, pertaining to an organization and/ or a business domain problem. Analysis of collected data is expected to generate critical commentary and recommendations.
- EMHR 590 Applied and Strategic Human Resource Management** **(3:0:0)**
3 Semester Credit Hours
Prerequisite: EMMG 510
 The course deals with key areas of applied and strategic human resource management, essential for providing supervisors and human resource specialists with a thorough understanding of the strategic role of human resource management and the techniques available to management for making effective use of the human resources of an organization. The material is delivered from the perspective of strategic and applied HRM practice in both the UAE/GCC and internationally.
- EMHR 592 Managing Change and Innovation** **(3:0:0)**
3 Semester Credit Hours
Prerequisite: EMMG 510
 This course discusses issues related to the management of change in organizations. The different types of change typically encountered in organizations are characterized using well- defined analysis frameworks. The effective management of corporate innovation is also considered, looking in details at the ideas generation, development and diffusion phases
- EMHR 594 Employee Relations and Compensation Management** **(3:0:0)**
3 Semester Credit Hours
Prerequisite: EMMG 510
 This course deals with legal and regulatory issues pertaining to employee relations from a number of perspectives. The compensation management aspect focuses on the theories underlying direct compensation and reward systems in organizations, and the practical administrative practices used to implement such systems, with particular reference to the UAE/GCC sub-region. Compensation management practices, including the analysis and evaluation of jobs, criteria and procedures for determining wage levels, individual wage determination, forms of pay, and incentive systems will be covered.
- EMIB 524 International Business** **(3:0:0)**
3 Semester Credit Hours
Prerequisite: Graduate Standing
 The course looks in-depth at salient aspects of managing in a globalized environment, including consideration of theoretical concepts. It offers a practical treatment of political-economic aspects of international trade. The course discusses foreign direct investment, global monetary systems, and strategy formulation for international business practice. The course additionally places stress on international business case analysis, within the UAE/GCC context.
- EMMG 510 Leadership and Managing People** **(3:0:0)**
3 Semester Credit Hours
Prerequisite: Graduate Standing
 This course provides an in-depth study of concepts relating to leadership and people management within organizations. The work integrates theory, research and applications, with an emphasis on context-driven learning. Students apply principles of leadership and people management to their own occupational situations and will investigate topical case studies. The material is delivered from the perspective of people-management practice in both the UAE/GCC and internationally.
- EMMG 515 Strategic Management** **(3:0:0)**
3 Semester Credit Hours
Prerequisite: Graduate Standing
 The course deals with the formulation, implementation and evaluation of strategies designed to give organizations a competitive edge. Corporate, business and operational – level strategy types are treated, and the use of various strategy analysis frameworks and models are discussed. Additional emphasis is placed on strategy formulation in the context of business practice in the UAE/GCC.

(3:0:0)

Prerequisite: Graduate Standing

EMMK 518 Marketing Management

(3:0:0)

Prerequisite: Graduate Standing

ENGL 099 **Foundations of College English**

(3:0:0)

Prerequisite(s): Requisite score on the English Placement Test, IETLS, or TOEFL

ENGL 101 Composition

(3:0:0)

Prerequisite(s): ENGL 099, Requisite score on the English Placement Test, IETLS, or TOEFL

ENGL 102 **Public Speaking**

(3:0:0)

Prerequisite(s): ENGL 101

ENGL 200 Advanced Composition

(3:0:0)

Prerequisite(s): Completion of a minimum of 45 credit hours and ENGL 101

ENGR 107 Introduction to Engineering

(2:0:0)

Pre-requisite(s): Math Placement Test score qualifying student for MATH 113

269 | Page

ENGR 200	Engineering Statistics	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): Math 114		
The course introduces principles of statistics and probability for undergraduate students in Engineering. The course covers the basic concepts of probability, discrete and continuous random variables, probability distributions, expected values, joint probability distributions, and independence. The course also covers statistical methods and topics including data summary and description techniques, sampling distributions, hypothesis testing, and regression analysis.		
ENGR 390	Internship I	(0:3:12-20)
3 Semester Credit Hours		
Prerequisite(s): Completion of 60 credits and a CGPA of 2.0 or higher		
This is one of two supervised field experiences of professional-level duties where each is for 240 to 320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.		
ENGR 391	Internship II	(0:3:12-20)
3 Semester Credit Hours		
Prerequisite(s): Completion of 90 credits (97 credits for Interior Design students; 112 credits for Architecture students) and a CGPA of 2.0 or higher. This is one of two supervised field experience of professional-level duties where each is for 240 to 320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designate site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.		
ENGR 399	Undergraduate Research Project	(0:9:0)
3 Semester Credit Hours		
Prerequisite(s): A minimum of 90 C.H., 3.0 CGPA or Department Chair approval		
Student in this course will be engaged in a creative research project under the guidance of the faculty member from school of engineering. The student will prepare a paper suitable for publications in peer review journal, conference or scientific gathering. Fixed credit hours; 3 credits are assigned.		
ENGR 450	Engineering Seminar	(1:0:0)
1 Semester Credit Hours		
Pre-requisite(s): Senior Standing		
Engineering ethics, professionalism, the role of engineers in society, current topics, and employment opportunities.		
ENGR 500	Engineering Foundation for Graduate Students	(3:0:0)
3 Semester Credit Hours		
This course is designed to provide a foundation in engineering principles, including strategic and systems thinking, statistical analysis of scientific data and sustainability. It is required for students who do not meet the CGPA for enrollment in graduate programs. The course aims to provide students with the skills and education required to pursue a Master's program.		
ENVS 101	Energy and Environmental Science	(3:3:0)
4 Semester Credit Hours		
The course is an inter-disciplinary study of environmental disruption and management, natural environmental systems, and the human impact on them. Other topics include energy procurement and use, waste management, water resources and water pollution, acid rain, global warming and ozone depletion.		
ENVS 102	Sustainability and Human-Environment Relations	(3:0:0)
3 Semester Credit Hours		
This course is designed to present the student with an overview of environmental issues. Humans are closely related to their environment and depend upon it to supply all the necessary resources to maintain life and to provide inputs for industrial activities.		
FNAN 300	Principle of Business Finance	(3:0:0)
3 Semester Credit Hours		
Prerequisite ACCT 204		

This course teaches students the fundamental principles of finance. It covers the financial decisions that financial managers should make to maximize shareholders' wealth or a firm's value. These financial decisions include investment, financing, dividend, and working capital decisions. Students are taught how to apply financial theories to real life financial situations, financial ratio analysis, time value of money, stock valuation, bond valuation, capital budgeting and risk and return. It also discusses the difference between profit maximization and shareholders' wealth maximization. The course entails calculations and arriving at conclusions that help financial managers make financial decisions and measure financial performance.

FNAN 303	Financial Analysis, Forecasting, and Valuation	(3:0:0)
3 Semester Credit Hours		

Prerequisites: ACCT 204

This course teaches students a range of techniques for examining, analyzing, and interpreting financial statements, and applying the information for both forecasting and making operational or a strategic financial decision that maximize a firm's value. The course aims at providing comprehensive understanding of the role of financial statements play and their impact on financial decision-making. Topics covered in this course include financial statement analysis, analysis of risk and return, development of financial models, financial forecasting, and firm valuation.

FNAN 304	Principles of Risk Management and Insurance	(3:0:0)
3 Semester Credit Hours		

Prerequisites: FNAN 303

This course introduces students to the techniques of risk management and insurance. The course also discusses the theory and application of risk and insurance to real world financial decisions. Different disciplines define risk differently. However, in finance, we define risk as the possibility or chances of not achieving the firm's set objective of maximizing shareholder wealth or firm value. Students will study how financial managers identify, measure, control, and monitor risk in order to eliminate, reduce, transfer or avoid it in their operational and strategic financial decision-making.

FNAN 321	Financial Institutions	(3:0:0)
3 Semester Credit Hours		

Prerequisites: FNAN 300

The course examines the functions of financial institutions with particular references to industry structure and regulatory environment. It analyses decision variables that management should concentrate on to achieve objectives. Topics include types of financial institutions, the structure and functioning of financial institutions, regulation of financial institutions, measuring and managing the risk financial institutions are exposed to, and liquidity and liability management.

FNAN 390	Internship in Finance	(0:0:21)
3 Semester Credit Hours		

Prerequisites: Senior Standing

The Internship is designed for candidates on the undergraduate finance major programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation take place at the beginning of the internship and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.

FNAN 401	Advanced Financial Management	(3:0:0)
3 Semester Credit Hours		

Prerequisites: FNAN 303

The course analyzes financial decision-making in firms. It emphasizes the conceptual structure of problems and the use of advanced analytic techniques. The topics covered include investment decisions, financing decisions, dividend decisions, working capital management, current asset management, capital structure, and corporate planning models.

FNAN 411	Investment Analysis and Portfolio Management	(3:0:0)
3 Semester Credit Hours		

Prerequisites: STAT 100, FNAN 303

This course focuses on analysis and valuation of investment alternatives and strategies. It will mainly focus on classifying investment alternatives and examining how investors allocate their financial assets and manage their portfolio. The course analyzes modern techniques of portfolio management including evaluating standards for selecting individual securities to

include or remove from portfolio. The course also discovers risk-return analysis for portfolio, and portfolio performance measures. The course is practically oriented, but also aims to equip students with conceptual and theoretical background of investment analysis and portfolio management.

- FNAN 412 Futures and Options Markets (3:0:0)**
3 Semester Credit Hours
Prerequisite OPMT 311, FNAN 303, FNAN 411
 This course demonstrates the theoretical and practical aspects of derivative securities and their applications in financial risk management. Over the past few decades, the use of derivatives to manage financial risk has been increasing tremendously. The main goal of this course is to enhance students' knowledge and skills on the use of derivative securities and application of derivative trading strategies in financial risk management. The course emphasizes on the understanding of the features and functions of different derivative securities, and the operation of the derivative market. Topics covered include an overview of the derivative securities, the mechanics of forwards, futures, swaps and options, hedging, option trading strategies, valuation of derivative securities, option Greeks and Volatility smiles.
- FNAN 421 Money and Capital Markets (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): FNAN 303
 This course illustrates the conceptual and real-world applications of the functions and operations of the money and capital markets. The course starts with an introduction of the financial system and proceeds with a detailed analysis of the functions of the central bank, demonstrates the interest rates behaviors and term structure, emphasizes the practical applications related to the function, operation and performance of money and capital markets and the instruments traded within. The course also develops an analytical framework to understand the factors that affect the performance of the financial market with a special emphasis on financial regulations and financial crisis. It also discusses the importance of the mortgage and foreign exchange markets in the financial system. These issues are presented within a unified framework intended to encourage critical thinking and problem-solving skills of students.
- FNAN 431 International Financial Management (3:0:0)**
3 Semester Credit Hours
Prerequisite FNAN 303
 The course focuses on the management of contemporary financial operations of international corporations. The topics include foreign exchange risk, political risk, returns and risks of international projects, international money and capital markets, financial accounting, capital structure, and cost of capital.
- FNAN 444 Finance Project Based Internship (0:0:21)**
3 Semester Credit Hours
Prerequisite: FNAN 303
 The finance project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem- solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the finance specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.
- GEOG 100 World Regional Geography (3:0:0)**
3 Semester Credit Hours
 This course explores the major regions of the world through a critical examination of key geographic concepts. Physical and environmental factors of each region, grounded in each region's history, culture, and society, are discussed. This frames a larger exploration of globalization, which also takes into account political and economic systems.
- GEOG 101 Introduction to Urban Social Geography (3:0:0)**
3 Semester Credit Hours
 This course provides an overview of urban social geography. The student will learn about the history of cities, how they effect the lives of urban dwellers, and the importance of cities in a globalized world. Environmental concerns will also be addressed.
- HIST 100 Contemporary Middle Eastern History (3:0:0)**
3 Semester Credit Hours
 This course examines the major issues affecting the Middle East in the 21st century while reviewing the origins and development of the modern Middle East. Students will develop an understanding of the social, economic, and political foundations that set the stage for the region this century.

- HIST 101 Ancient History of the Arabian Peninsula (3:0:0)**
3 Semester Credit Hours
 This course concentrates on the geographical background of the Arabian-peninsula including location, descriptions of its provinces, routes, flora and fauna. It also focuses on the commercial importance and its political situation during the period from third Millennium B.C. to the rise of Islam or to the seventh century A.D.
- HIST 111 Introduction to World History (3:0:0)**
3 Semester Credit Hours
 Analytical approach to an overview of world history that surveys the major features of principal existing civilizations of the world, as originally formed and as altered by key global processes, including forces of modernity.
- HIST 281 Survey of Middle Eastern Civilization I (3:0:0)**
3 Semester Credit Hours
 Survey of Middle Eastern history from rise of Islam to present, emphasizing processes that led to emergence of economic, cultural, social, and political institutions that characterize region today. HIST 210 surveys the period from rise of Islam in 570 to medieval period (ca. 1258).
- HIST 282 Survey of Middle Eastern Civilization II (3:0:0)**
3 Semester Credit Hours
 Second half of the survey of Middle Eastern history. HIST 282 surveys the period from about 1258 to the present.
- HTMT 311 Survey of the Hospitality & Tourism Industry (3:0:0)**
3 Semester Credit Hours
 A survey of the history, trends, organizational structure, and economic impact of the hospitality and tourism industry on UAE economy. Some study of the problems originating in the operation and management of various segments of the hospitality industry in UAE will be introduced.
- HTMT 312 Sustainable Tourism (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): HTMT 311
 The course will introduce the issues associated with the sustainable management of destinations and tourist attractions. Students will: learn the variety of ways that sustainability can manifest in the hospitality and tourism industry in UAE; understand the roles of international stakeholders involved in sustainable tourism; analyze case studies in key areas related to social, economic, and environmental impacts and consequences; and learn broad perspectives on how tourism relates to ecosystems, poverty, health, education, and social equity.
- HTMT 321 Hospitality Operations I - F&B Management (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): HTMT 311
 This course involves the concepts and applications of management in food and beverage operations in various types of hospitality firms across UAE. Topics such as menu development, beverage management, catering, service, sanitation, foodservice design, and cost controls will be presented. Management approaches will be developed to provide quality products and services.
- HTMT 322 Hotel Operations (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): HTMT 311
 This course will offer a thorough overview of the management of various lodging properties, taking a generalist view of lodging management, moving from department to department in a typical hotel. While this course will be general in nature, we will take a much more strategic perspective, identifying and considering issues of concern to general managers of all types of lodging properties, with a particular focus on profit maximization (yield/revenue management) and distribution channel management. As such, we will consider the broader political, economic, social, and technological environments and trends and their impact on lodging operations. For example, the recent global economic downturn has had a devastating effect on the lodging industry. Thus, the recession will form a basic issue that we will deal with throughout the semester.

- HTMT 390 Internship in Hospitality and Tourism Management (0:0:21)**
3 Semester Credit Hours
Prerequisite(s): Senior Standing
 The Internship is designed for candidates on the undergraduate hospitality and tourism management major programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation take place at the beginning of the internship and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.
- HTMT 411 Destination Management (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): HTMT 311
 The course will introduce the various issues associated with the management of a convention visitor's bureau (CVB). The course will analyze the mission, structure, and business activities of organizations that develop and promote a complex tourism destination in UAE. This includes the management functions of membership services, visitor services, financial and marketing concerns, research activities, and the strategic planning and evaluation of these efforts. Topics covered will include developing the following consumer segments: meetings/conventions, pleasure/tours, and festivals/special events.
- HTMT 412 Meetings and Conventions (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): Prior Completion of 84 Credit Hours
 A course dealing with the many issues impacting the management of large convention and exposition centers. The course is taught from an organizational marketing base. Topics include meeting site selection, program planning and budgeting, legal issues and insurance problems, housing, food and beverage arrangements, transportation, exposition management, and audio-visual services."
- HTMT 444 Hospitality & Tourism Management Project Based Internship (0:0:27)**
3 Semester Credit Hours
Prerequisite(s): HTMT 311
 The hospitality and tourism management project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem-solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the hospitality and tourism management specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.
- INDS 111 Color Theory and Rendering (2:3:0)**
3 Semester Credit Hours
Pre-requisite(s): None
 The aim of this course is to introduce color theory as a component of the design process that influences human perception of the interior space, behavior, performance and wellbeing and therefore influences design solutions. It offers skills in color selection and rendering for diverse spaces function and design concept communication.
- INDS 112 Design Communication 1: Sketching and Drawing for Interiors (2:3:0)**
3 Semester Credit Hours
 This course introduces students to the principles and techniques of interior design's form and space representation and communication. The course makes the students familiar with manual drawing tools and modes of documenting to develop students' sketching, technical drawing and ideation skills.
- INDS 121 Interior Design Process, Human Factors and Ergonomics (3:0:0)**
3 Semester Credit Hours
 The aim of this course is to provide knowledge in design process steps, components, human factors, anthropometrics and ergonomics and how they influence design solutions and contribute to the human wellbeing. It offers skills in gathering, analyzing human-centered evidence, application of ergonomics and other human factors in inclusive, universal design.

- INDS 122 Design Communication 2: Digital Media for Interiors (2:3:0)**
3 Semester Credit Hours
Pre-requisite(s): INDS 112
This course introduces students to the most common 2D and 3D software used in the field of interior design to create and develop graphical and technical presentations. Emphasis will be on giving the students tools to digitize a sketches, drawings and concepts to examine design idea in terms of accuracy and workability of schematic design.
- INDS 211 Interior Design Studio 1 (0:12:0)**
4 Semester Credit Hours
Pre-requisite(s): INDS 122
Co-requisite(s): INDS 213 and ARCH 223
This course introduces a small-scale residential environment interior design project. A scenario-based design brief focuses on relationships between the users' needs and space's features leads students through the early phases of the design process (pre-design, schematic design, design development) with an aim of meeting space functional requirements and user(s) needs.
- INDS 212 History of Art and Interior Design 1 (3:0:0)**
3 Semester Credit Hours
The course aims to provide knowledge in history as it relates to decorative arts, architecture, interior design, furniture, material culture and understanding of how the design of the built environment changes along with historical periods that are influenced by the social, political, physical changes in design of the built environment.
- INDS 213 Interior Construction Methods, Materials and Finishes 1 (2:3:0)**
3 Semester Credit Hours
Pre-requisite(s): INDS 121 and INDS 122
Co-requisite(s): ARCH 223
The course aims to provide students with understanding of building material, finishes, and furnishings, their fabrication process, installation methods and maintenance requirements and skills in selecting materials, finishes and furnishings to support the design intent, project criteria and human and environmental wellbeing. The courses focuses on classic materials.
- INDS 221 Interior Design Studio 2 (0:12:0)**
4 Semester Credit Hours
Pre-requisite(s): INDS 211
Co-requisite(s): INDS 223
This studio introduces medium size interior design of office spaces. The studio encourages the students to investigate social interaction in multiusers environments and to use elements that reflect the nature and identity of work activities through the phases of data collection and analyzes, space programming, materials, furnishings and finishes selection.
- INDS 222 History of Art and Interior Design 2 (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): INDS 212
The course aims to provide knowledge in history as it relates to decorative arts, architecture, interior design, furniture, material culture and understanding of how the design of the built environment changes along with historical periods that are influenced by the social, political, physical changes in design of the built environment.
- INDS 223 Interior Construction Methods, Materials and Finishes 2 (2:3:0)**
3 Semester Credit Hours
Pre-requisite(s): INDS 213
Co-requisite(s): INDS 224
The course aims to provide students with understanding of building material, finishes, and furnishings, their fabrication process, installation methods and maintenance requirements and skills in selecting materials, finishes and furnishings to support the design intent, project criteria and human and environmental wellbeing. The course focuses on modern materials.

(3:0:0)

Pre-requisite(s): ENVS 102

INDS 311 **Interior Design Studio 3**

(0:12:0)

Pre-requisite(s): INDS 221

This design studio increases the complexity of the interior design projects by introducing design solutions for commercial spaces in urban, suburban, and shopping mall settings. Students explore space utilization, fixtures, and display as well as the role interior designer plays in supporting the business and branding of the commercial space.

(2:3:0)

Pre-requisite(s): INDS 223

This course will lead the students to continue exploring more advanced software application and the principles of Building Information Modelling used to visualize, document and communicate interior design working drawings.

(3:0:0)

Pre-requisite(s): INDS 223

The course aims to explore environmental control systems, types, size, their integration into interior design project, and awareness of these systems in relation to indoor thermal comfort, indoor air quality, environmental and human wellbeing. The course focuses on water, waste, HVAC, electrical, fire safety, conveyance, and communication and security systems.

(0:12:0)

Pre-requisite(s): INDS 311

This design studio deals with contemporary complex interior design project within the hospitality industry and will enable the students to explore the combination of wellness, aesthetics, functionally and operational aspects in designing city restaurants, cafes, and hotels.

(2:3:0)

Pre-requisite(s): INDS 121 and INDS 223

The aim of this course is to provide knowledge and skills in materials, construction methods and details for furniture design and making. The course analyzes materials for furniture design in relation to their impact on environment, healthy and safe living and efficient work within the interior space.

(2:3:0)

Pre-requisite(s): INDS 313

The course aim to provide knowledge in space's lighting and acoustical design for space quality enhancement in relation to safety, comfort, performance, and wellbeing. The course suggests skills in light and sound integration into design of residences, stages, exhibitions, theaters, halls, etc. with the application of high-tech materials and technologies.

(2:3:0)

3 Semester Credit Hours

Pre-requisite(s) INDS 312

This course is an advanced level course that will allow students to discover the fundamentals of Parametric modeling for design applications in Interior Design and Architecture. Students will investigate rule-based and parametric design concepts and techniques in the context of a generative modeling environment: Dynamo for Revit.

(2:3:0)

3 Semester Credit hours

This course aims to improve students' skills in sketching and painting and focuses on two-dimensional media. Students continue to explore and experiment various techniques in art making. Two-dimensional media art works such as still life, compositions, collage are practiced with the application of various two-dimensional media.

(2:3:0)

3 Semester Credit Hours

This course aims to introduce principles of landscape design throughout all its phases. Students study types of landscape, materials and its relationship with the context, environmental and human wellbeing and apply various representation techniques in landscape design projects.

(0:3:12-20)

3 Semester Credit Hours

Pre-requisite(s): Completion of 60 credits and a CGPA of 2.0 or higher

This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

(0:3:12-20)

3 Semester Credit Hours

Pre-requisite(s): Completion of 97 credits and a CGPA of 2.0 or higher

This is one of two supervised field experiences of professional-level duties where each is for 240-320 hours (8 weeks) of full-time training at approved internship sites. The internship takes place under the guidance of a designated site supervisor in coordination with a faculty supervisor. In addition to the regular reports during the internship, students must present their activities and learning experiences at the end of the internship.

(0:12:0)

4 Semester Credit Hours

Pre-requisite(s): INDS 321

Co-requisite(s): INDS 413 and INDS 414

This course extends the scope of interior design within a medium scale project of a healthcare environment. Students explore design approaches related to this complex and sensitive environment with respect to safety, efficiency, codes and regulations and that will satisfy all end users (including patients, staff, and visitors)' needs and wellness.

(1:3:0)

2 Semester Credit Hours

Pre-requisite(s): INDS 321

Co-requisite(s): INDS 411

The course aims to prepare students for Graduation Project Design that follows next semester. It offers students opportunity to consider their evolving interest, targeted career plan and select a project theme, gather, analyze, systemize relevant to the theme data, and develop a design brief, project statement, design concept and program. **(Writing Intensive Course)**

(3:0:0)

3 Semester Credit Hours

Pre-requisite(s): INDS 323

The aim of this course is to equip students with knowledge of regional and international laws, codes, standards and guidelines that influence the encounter of the inhabitants with interior spaces and skills in application of laws, codes, standards and guidelines concerning built environment, sustainability, wellness, inclusive and universal design.

INDS 421	Graduation Project Design	(0:15:0)
5 Semester Credit Hours		
Pre-requisite(s): INDS 412		
Co-requisite(s): INDS 422		
The graduation project is a capstone self-directed course under the supervision of one the program's senior faculty member where the project identified during the Graduation Final Thesis Research course is implemented. Students apply gained knowledge and skills in developing a holistic design approach to a design project.		
INDS 422	Professional Practice II: Ethics and Project Management	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): INDS 414		
The aim of the course is to provide awareness of the context of interior design practice, the influence of local and global context on design practices, understanding of the interior design profession principles, processes, responsibilities and its value to the society, interior design management, business formation, practices, and ethics.		
INDS 451	Portfolio Design	(2:3:0)
3 Semester Credit Hours		
Pre-requisite(s): None		
This course aims to assist and mentor students in assembling a comprehensive portfolio of their academic work, enhancing students' graphic techniques, and provides skills in organizing the layouts of their projects in both book and digital formats.		
INDS 452	Photography	(2:3:0)
3 Semester Credit Hours		
This course aims to provide knowledge in types of cameras and skills in camera operating, taking pictures in different contexts, enhancing the pictures, processing and organizing pictures and application of photographs in interior design presentations.		
INDS 453	Advanced BIM: Scheduling and Coordination	(2:3:0)
3 Semester Credit Hours		
Pre-requisite(s): INDS 312		
The course aims to introduce technologically based collaboration of interior design with multiple disciplines and provides knowledge and training on collaboration tools and methods using Building Information Modelling applications. It focuses on developing, reading, exploring, and coordinating working drawings and executive documents to avoid conflicts and ensure construction documents consistency.		
INDS 454	Research Methods for Interior Design	(3:0:0)
3 Semester Credit Hours		
This course aims to introduce research processes in design related tasks within the framework of Evidence-Based Design Approach. It offers a review of design research methods such as literature review, sampling procedures, questionnaires interviews, and observations.		
ITEC 103	Fundamentals of Information Technology	(3:0:0)
3 Semester Credit Hours		
The course focuses on the nature and uses of computers with an introduction to word processing, spreadsheets, presentation software and effective presentation information, in addition to related lab projects. Additional topics include computer systems organizations, communications and networking, legal and ethical issues, computer security, the Internet, and trending technologies.		

- MATH 091 Pre-algebra** **(3:0:0)**
3 Semester Credit Hours
 This course prepares students to take MATH 101 (Numbers and Data interpretation). It enables them to gain a command in operations with integers, fractions, decimals and percent, geometric figures and their measures, and pre-algebra topics including properties of rational numbers, operations of rational numbers, simplification of polynomials and equation-solving techniques. This course is not for degree credit.
- MATH 093 Intermediate Algebra** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): Math Placement Test
 This course covers basic algebraic skills including factoring, solving basic linear and quadratic equations, the rules of exponents and radicals, and basic concepts of functions with specific examples, such as polynomial and rational functions. The course serves as a prerequisite to MATH 108. This course is not for degree credit.
- MATH 095 Pre-calculus** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): Math Placement Test
 This course is a prerequisite to MATH 113. It reviews mathematical skills essential to studying calculus. Topics include inequalities, absolute values, graphs, functions, exponential and logarithmic functions, and trigonometry. This course is not for degree credit.
- MATH 101 Numbers and Data Interpretation** **(3:0:0)**
3 Semester Credit Hours
 This course meets the quantitative reasoning requirement, one of the Foundation requirements of the University General Education program. Topics include Logic Mathematics, Sets and Counting, probability and descriptive statistics. This course is broad in scope, emphasizing applications.
- MATH 102 Mathematics in Civilization** **(3:0:0)**
3 Semester Credit Hours
 Throughout the history of mankind, the mathematical spirit has been a driving force in the development of the civilized world. This course creates an understanding of the impact of mathematical ways of thinking on the arts and sciences, on the development of technologies, and on the study of history. The goal is to illuminate the role of mathematics in the development of Eastern and Western civilizations and, thereby, to shed on human civilization from a mathematical point of view.
- MATH 108 Calculus with Business Applications** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): MATH 093/103, or appropriate score on Math Placement test
 The course covers standard topics such as functions, limits, derivative, and integral calculus as well as applications of differentiation and integration. Simple introductory examples and applications are drawn from, but do not require, advanced knowledge of business applications.
- MATH 111 Calculus with Life Sciences Applications** **(3:1:0)**
3 Semester Credit Hours
Pre-requisite(s): MATH 095/105 or appropriate score on Math Placement test
 The course covers standard topics such as functions, limits, derivative, and integral calculus as well as applications of differentiation and integration. Simple introductory examples and applications are drawn from, but do not require advanced knowledge of the life sciences.
- MATH 113 Calculus I** **(3:1:0)**
3 Semester Credit Hours
Prerequisite(s): MATH 095 or appropriate score on Math Placement test
 This course covers the concepts of limits and continuity, derivatives of functions, extreme values, curve sketching, L'Hopital's rule, definite and indefinite integrals and integration by substitution.

- MATH 114 Calculus II** **(3:1::0)**
3 Semester Credit Hours
Prerequisite(s): MATH 113
 This course covers techniques and applications of integration, transcendental functions, infinite sequences and series and parametric equations.
- MATH 203 Linear Algebra** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): MATH 113
 This course covers systems of linear equations, linear independence, linear transformations, inverse of a matrix, determinants, vector spaces, eigenvalues, eigenvectors, and diagonalization.
- MATH 213 Calculus III** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): MATH 114
 This course covers partial differentiation, multiple integrals, line and surface integrals, and three-dimensional analytic geometry.
- MATH 214 Elementary Differential Equations** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): MATH 114
 This course covers first-order ODEs, higher-order ODEs, Laplace transforms, linear systems, nonlinear systems, numerical approximations, and modeling.
- MATH 225 Discrete Mathematics** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): MATH 113
 This course covers elementary discrete mathematics. Topics include propositional logic, tautology and equivalence rules, logical inference rules, logical consistency, predicate logic and quantifiers, graphs, planar graphs, binary and decision trees, and complexity of algorithms.
- MBAC 512 Managerial and Financial Accounting** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): Graduate Standing
 The course deals with managerial and financial accounting techniques and practices. The course examines the impact of cost and cost allocation on business performance analysis, using a variety of costing schemes. It presents accounting as an information system construct, and looks at methods for recording, presenting and analyzing accounting information. The course also looks at budgeting, ratio analysis and other fiscal management measures. The course discusses the impact, on Management and Financial Accounting, of contemporary management philosophies and techniques including JIT practices, quality assurance and performance measurement.
- MBFN 500 MBA Foundations** **(3:0:0)**
3 Semester Credit Hours
 The course is designed to provide a strong foundation in business for Graduate students who graduated from a non-business discipline. This course offers a comprehensive introduction to aspects of business. This course aims to equip students with business education and skills to successfully complete the business graduate program. **This course does not count as degree credit.**
- MBFN 514 Managerial Finance** **(3:0:0)**
3 Semester Credit Hours
Prerequisite(s): Graduate Standing
 The course provides detailed critical treatment of the theory and practice of financial management within organizations. Topics covered include financial modeling, breakeven analysis, investment decision-making (NPV, IRR, etc.), capital budgeting and structure, working capital management, analysis of financial statements, and risk analysis. There is additional treatment of managerial economics.

MBFN 560 International Corporate Finance**(3:0:0)****3 Semester Credit Hours****Prerequisite(s): MBFN 514**

This course focuses on corporate finance at a global environment context. It explores advanced concepts and practices applicable to multinational investing, financing and dividend decision making that maximize firm value. Topics covered include financial analysis for investment decisions in an international setting, international money operations and capital markets, management of foreign exchange risk, direct foreign investment, funding of international projects, political risk analysis, currency derivatives and swap markets.

MBFN 562 Investment Planning and Management**(3:0:0)****3 Semester Credit Hours****Prerequisite(s): MBFN 514**

The course deals with the advanced analysis of equity securities and investment portfolios, predicated on relevant market hypothesis and capital market theory. The course looks at investment risk-return trade-off, asset pricing models, and stock price behavior. The course adds emphasis on stocks, bonds, and financial futures and options.

MBFN 564 Financial Markets**(3:0:0)****3 Semester Credit Hours****Prerequisite MBFN 514**

The course deals with financial markets, primarily global equity markets. It looks at markets for handling government debt instruments, and exchange-traded and over-the-counter financial derivative instruments such as futures, options, swaps, and asset-backed securities. There is detailed treatment of derivatives theories, derivatives exchanges, and valuation of derivatives, including standard and other non-standard options on a variety of underlying assets, in relation to relevant financial markets. The course also discusses emerging financial markets, and the effective management of risks emanating from these markets.

MBGN 525 Research Methods for Business**(3:0:0)****3 Semester Credit Hours****Prerequisite(s): Prior completion of 6 MBA program credits**

This course provides the necessary underpinning support for the MBA dissertation and general business domain research. It allows candidates to make informed decisions and appropriate choices pertaining to research methodology. A range of business research tools, approaches and analytical techniques are discussed, and guidance is given on optimal structuring of business research documents and MBA dissertations.

MBGN 555 MBA Investigative Thesis or Project I**(0:9:0)****3 Semester Credit Hours****Pre-requisite(s): MBGN 525**

MBGN 555 is the first part of the MBA Investigative Thesis or Project course, serving to integrate the various topics studied in the School of Business MBA program. Students undertake a work-based study that focuses on a chosen business / management topic. The outcome of MBGN 555 is to submit a research introduction, literature review, and methodology

MBGN 556 MBA Investigative Thesis or Project II**(0:9:0)****3 Semester Credit Hours****Pre-requisite MBGN 555**

EBGN 556 is the second of the three-credit hour research requirement; serving to integrate the various topics studied in the School of Business EMBA program. Students undertake a work-based study that focuses on a chosen business/ management topic. The outcome is a thesis, an extended piece of work necessitating investigation of secondary data and empirical fieldwork, through collection of primary data, pertaining to an organization and/or a business domain problem. Analysis of collected data is expected to generate critical commentary and recommendations.

- MBHR 590 Applied and Strategic Human Resource Management (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): MBMG 510
 The course deals with key areas of applied and strategic human resource management, essential for providing supervisors and human resource specialists with a thorough understanding of the strategic role of human resource management and the techniques available to management for making effective use of the human resources of an organization. The material is delivered from the perspective of strategic and applied HRM practice in both the UAE/GCC and internationally.
- MBHR 592 Managing Change and Innovation (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): MBMG 510
 This course discusses issues related to the management of change in organizations. The different types of change typically encountered in organizations are characterized using well- defined analysis frameworks. The effective management of corporate innovation is also considered, looking in details at the ideas generation, development and diffusion phases.
- MBHR 594 Employee Relations and Compensation Management (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): MBMG 510
 This course deals with legal and regulatory issues pertaining to employee relations from a number of perspectives. The compensation management aspect focuses on the theories underlying direct compensation and reward systems in organizations, and the practical administrative practices used to implement such systems, with particular reference to the UAE/GCC sub-region. Compensation management practices, including the analysis and evaluation of jobs, criteria and procedures for determining wage levels, individual wage determination, forms of pay, and incentive systems will be covered.
- MBIB 524 International Business (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): Graduate Standing
 The course looks in-depth at salient aspects of managing in a globalized environment, including consideration of theoretical concepts. It offers a practical treatment of political-economic aspects of international trade. The course discusses foreign direct investment, global monetary systems, and strategy formulation for international business practice. The course additionally places stress on international business case analysis, within the UAE/GCC context.
- MBMG 510 Leadership and Managing People (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): Graduate Standing
 This course provides an in-depth study of concepts relating to leadership and people management within organizations. The work integrates theory, research, and applications, with an emphasis on context-driven learning. Students apply principles of leadership and people management to their own occupational situations and will investigate topical case studies. The material is delivered from the perspective of people-management practice in both the UAE/GCC and internationally.
- MBMG 515 Strategic Management (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): Graduate Standing
 The course deals with the formulation, implementation and evaluation of strategies designed to give organizations a competitive edge. Corporate, business and operational – level strategy types are treated, and the use of various strategy analysis frameworks and models are discussed. Additional emphasis is placed on strategy formulation in the context of business practice in the UAE/GCC.
- MBMI 520 Managing Information Systems Performance (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): Graduate Standing
 The course studies techniques and practices for analyzing business information systems performance with emphasis on support for business processes. The course gives a strategic and integrated view of the exploitation of information and communication technology, with particular emphasis on e-business for improving business performance and business efficiency. The course discusses IT-driven business strategy, and reviews IT-related factors that are necessary for enhancing the performance of organizations in the current competitive global business environment.

MBMK 518 Marketing Management
3 Semester Credit Hours

(3:0:0)

Prerequisite(s): Graduate Standing

This course looks in-depth at techniques for crafting effective marketing strategies, for a variety of market environments. Customer-focused marketing strategy design implementation and management are discussed using contemporary frameworks within a market-driven setting. The course emphasizes managerial aspects of marketing, including integrating specific elements of the marketing process. The course additionally places stress on case analysis, within the UAE/GCC context.

MENG 201 Mechanical Engineering Drawing
3 Semester Credit Hours

(1:6:0)

The course introduces the fundamental rules of an international language that enables ideas to be expressed and communicate in an easy and clear way through visual illustration. Includes the following topics: geometric construction; line convention; orthographic projections, Position student feet on the first step of the engineering design based on the visual representation. The course covers the following topics: isometric projections; oblique projections; Perspective projections; dimensioning, and sectional views. Computer software design, prepare parts drawing and assembly it using the software CREO.

MENG 211 Thermodynamics I
3 Semester Credit Hours

(3:0:0)

Pre-requisite(s): CHEM 211

This course will consider the fundamental science of classical thermodynamics and its practical applications. Problem solving will be emphasized, including problem formulation, analytic, and computational solutions. Topics include the first law of thermodynamics, work, heat, properties of substances and state equations, the second law of thermodynamics and applications to engineering systems.

MENG 212 Thermodynamics II
3 Semester Credit Hours

(3:0:0)

Pre-requisite(s): MENG 211

This course is a continuation of MENG 211 - Thermodynamics I. It provides more depth in the study of cycles, with applications to gas power and refrigeration cycles, and vapor and combined power cycles; mixtures of gases and vapors, psychrometrics, chemical reactions, and energy analysis.

MENG 221 Dynamics
3 Semester Credit Hours

(3:0:0)

Pre-requisite(s): CIEN 211

Kinematics of a Particle, Kinetics of a Particle: Force and Acceleration, Kinetics of a Particle: Work and Energy, Kinetics of a Particle: Impulse and Momentum, Planar Kinematics of a Rigid Body, Planar Kinetics of a Rigid Bodies.

MENG 231 Engineering Management
2 Semester Credit Hours

(2:0:0)

This course introduces the theory behind engineering measurements. Basic definitions, error analysis, characteristics of mechanical systems, system response and signal analysis. Students will apply the knowledge learned in this course in experiments preformed in the BSME program's required laboratory courses.

MENG 311 Internal Combustion Engines
3 Semester Credit Hours

(3:0:0)

Pre-requisite(s): MENG 212

Engine classifications and terminology. Engine operating characteristics and performance parameters. Air standard engine cycles including: Otto, Diesel, Dual and two-stroke cycles. Common fuels used in IC engines, combustion reactions and the associated thermochemical calculations. Engine emissions and their control technologies and strategies. Air and fuel induction methods and technologies, the physics of the combustion phenomena. Friction losses, lubricants and lubrication systems.

- MENG 321 Vibration and Control** **(3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): MATH 214, MENG 221
 Introduction to Vibration, Oscillatory Motion, Free Vibration, Forced Vibration, Rotating Unbalance, Multiple-Degree-of-Freedom Systems, Introduction to control, block diagrams, modeling of systems, state space representation, Laplace transform, solution of linear systems, stability, input/output description, PID controllers, transfer function methods, tracking.
- MENG 342 Fluid Mechanics Lab.** **(0:3:0)**
1 Semester Credit Hour
Pre-requisite(s): MENG 231
Co-requisite(s): CIEN 251
 This course is composed of a set of selected experiments about general fluid mechanics. The experiments will be either performed in groups by the students or demonstrated by the instructor. Individual class work will be strongly encouraged as well as teamwork. The lab also includes an open-ended design of experiment.
- MENG 361 Heat Transfer** **(3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): MENG 212 and MATH 214
 Introduction to heat transfer mechanisms, heat conduction equation, steady heat conduction including the thermal resistance networks, transient heat conduction, lumped systems, fundamental of convection and thermal boundary layers, external and internal forced convection, natural convection, boiling and condensation, thermal radiation, and heat exchangers.
- MENG 362 Thermal Sciences Lab.** **(0:3:0)**
1 Semester Credit Hour
Pre-requisite(s): MENG 211 and MENG 231
CO-requisite(s): MENG 361
 This course is composed of a set of selected experiments which demonstrate and apply the concepts of thermodynamics and heat transfer. The experiments will be either performed in groups by the students or demonstrated by the instructor. Individual class work will be strongly encouraged as well as teamwork. The lab also includes an open-ended design of experiment.
- MENG 421 Theory of Machines** **(3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): MENG 221
 This course focuses on the kinematic and kinetic analysis of mechanisms. It introduces the fundamental concepts, definitions and terminologies in mechanisms, basic mechanisms and applications, linkages and mobility, dynamic analysis of cams, gears and gear trains, velocity and acceleration analysis in mechanisms, and static and inertia force analysis of machinery.
- MENG 422 Building Utilities II: Illumination, Acoustics, and Electrical Building Services** **(3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): PHYS 110, MENG 468
 This course explores building electrical system and controls, lighting and acoustic design and assist integrating them into architectural design of the building. Evaluates systems types, components, and installation and maintenance procedures as it relates to building regulations and systems impact human's indoor comfort and on environment.
- MENG 441 Turbo Machinery** **(3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): CIEN 251
 Turbomachinery classifications and terminology. Implementation of dimensional analysis for predicting performance of turbomachines and designing engineering systems. Understand the fundamentals of energy transfer between rotating rotors and fluid flow. Demonstrate the ability to construct velocity diagrams for various turbomachines (axial-flow compressors and turbines, radial-flow compressors and turbines, pumps, fans, blowers, hydraulic turbines) and their relation to design. Perform elementary analysis for determining input/output work of various turbo devices. Design and selection of turbomachines for various engineering applications.

- MENG 451 Mechanical Design I (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): CIEN 212, IENG 231
 Introduction to Mechanical Engineering Design, Materials Properties, Load and Stress Analysis, Deflection and Stiffness, Failure Prevention, Fatigue Failure, Design of Mechanical Elements, Screws Fasteners and Nonpermanent Joints, Mechanical Springs.
- MENG 452 Mechanical Design II (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): MENG 451
 This course is a continuation to the machine design I course. Students will be introduced to the analysis and design concepts of various types of machine elements that include: bearings (journal and anti-friction); spur, helical and bevel gears; flexible drives and flywheels; clutches and brakes.
- MENG 453 Computer Aided Design (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): MENG 451
 This is an upper-year mechanical engineering course. It introduces students to the analytical basis to CAD software and the three main ways to represent an entity, namely wireframe, surface and solid modeling. The course aims at introducing the concept and importance of CAD as part of the design process. Also it focuses on mathematical representation and manipulation of geometry. The course introduces students to Computer-Aided Mechanical Design (CAMD) tools and their applications to mechanical systems design.
- MENG 455 Finite Elements in Machine Design (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): MENG 451
 The objective of this course is to learn how to design and analyze structural components of machine system, especially using the finite element method. The course exposes students to analytical and numerical methods for computing stresses and strains in structures, use of finite element software for static structural analysis and the application of design and failure criteria to ensure that mechanical components can carry the design load without failure. Another important area of the course is to make the students recognize the importance of self-education and life learning.
- MENG 461 HVAC and Refrigeration Systems (3:0:0)**
3 Semester Credit Hours -
Pre-requisite(s): MENG 361 and MENG 212
 Review of psychrometry. Air conditioning processes. Thermal comfort, inside and outside design conditions. Ventilation and infiltration. Heating and cooling load calculations. Solar radiation. Water heating systems layout and design. Air systems design. Under floor heating. Review of vapor compression and absorption cycles; compressors, condensers, evaporators, expansion devices; refrigerants (including new ones); cooling towers; components of an absorption cycles, controls.
- MENG 462 Design of Thermal Systems (3:0:0)3**
Semester Credit Hours
Pre-requisite(s): MENG 361 and CIEN 251
 Application of principles of fluid mechanics heat transfer and thermodynamics in the component design of thermal systems. Examples are drawn from power generations, environmental control, and industrial processes such as design and sizing of piping systems, piping networks, pumps sizing and selection, and heat exchangers selection and performance evaluation. Students work individually and on group to conduct assignments and projects for integration of these components in the design of thermal systems.
- MENG 463 Energy Conversion and Management (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): MENG 361
 An introduction to the basic technical and economic criteria for the design of efficient energy conversion systems, including traditional as well as alternative power systems. To discuss strategies for increased energy efficiency and more environmentally sound operation. To assess design alternatives and selection criteria based on long-term economic viability and overall energy management strategies.

- MENG 468 Building Utilities I: HVAC and Mechanical Building Services (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): PHYS 110, ARCH 336
 This course explores building systems such as heating, ventilating and air-conditioning, water supply, drainage, fire safety, vertical transportation, building control systems, and assist integrating them into architectural design of building. Evaluates systems types, components, and installation and maintenance procedures as it relates to building regulations, systems impact on indoor comfort and environment.
- MENG 491 Senior Design Project I (0:6:0)**
2 Semester Credit Hours
Pre-requisite(s): Senior Standing
 The course requires students to work in small design teams to solve a significant engineering problem. Students develop, design, and implement a solution to the engineering problem in conjunction with a faculty advisor. The course reinforces principles of the engineering design process and serves as a capstone for mechanical engineering knowledge obtained in the ME curriculum. The consideration of the ethical and social implications of technology and the basic concepts of business are also aspects of the course.
- MENG 492 Senior Design Project II (0:12:0)**
4 Semester Credit Hours
Pre-requisite(s): MENG 491
 The Mechanical Engineering Senior Design Projects Program coordinates the completion of the second half of the capstone design sequence required of all Mechanical Engineering seniors. Students apply engineering design methodology, using both analysis and synthesis, to solve open-ended problems. The range of design problems considered spans other engineering fields as well as non-engineering disciplines. At the end of the term, each student design team is expected to present information related to their project in both written and oral formats.
- MENG 493 Special Topics in Mechanical Engineering (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): Dept. Approval
 Special up-to-date topic in one of the mechanical engineering streams of applied mechanics or thermal sciences.
- MEPM 511 Project Management Fundamentals (3:0:0)**
3 Semester Credit Hours
Co-requisite: MEPM 532
 The fundamentals of project management including: overview and concepts of project management (principles, body of knowledge, strategies); planning successful projects (defining, specifying, delivery options, scheduling, budgeting); implementing (organizing the team, work assignments, team building, effective leadership); executing (performance measurement, maintaining the schedule, adjustments/mid-course corrections, record keeping, status reporting, communications, managing conflict, time management); and closeout (performance measurement, maintaining the schedule, adjustments/mid-course corrections, record keeping, status reporting, communications, managing conflict, time management).
- MEPM 512 Engineering Contracts and Procurement (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): None
 This course presents fundamental concepts and techniques for project procurement. Students are introduced to the PMBOK Guide four-step procurement process and expected to develop an in-depth understanding of project contracting, negotiation, and procurement execution.
- MEPM 513 Project Implementation and Performance (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): MEPM 511
 This course examines various topics related to project initiation and performance, Key performance indicators and criteria for measuring project success. The course includes using project performance software. Topics include: team building and management, organization structure, performance and success measures and earned-value technique. Other topics include: work breakdown structure, stakeholder management and project communication management.

MEPM 514 Global Projects Management**(3:0:0)****3 Semester Credit Hours****Pre-requisite(s): MEPM 511**

This course provides an overview of the global project management process with specific emphasis on cross-cultural considerations, environmental factors, challenges, benefits & risks of global projects and leadership in global projects. The course also includes best practices that apply to global project management.

MEPM 515 Project Scheduling**(3:0:0)****3 Semester Credit Hours****Pre-requisite(s): MEPM 511**

The course will prepare students to master project scheduling. Project scheduling methods are covered including: network construction, forward pass and backward pass calculations, activity slakes or floats calculations, critical path method (CPM), critical chain scheduling, Program Evaluation Review Technique (PERT), resource loading and resource leveling, Project crashing, and scheduling risk analysis. Students will have the opportunity to learn and use one of the project management software's for project scheduling.

MEPM 516 Project Quality Management**(3:0:0)****3 Semester Credit Hours****Pre-requisite(s): MEPM 511**

This course covers the fundamentals of quality control and management, quality principles, quality control techniques, quality control tools (Check Sheets, Histograms, Pareto Diagram, Cause and Effect Diagram, Scatter Diagram, Flow Process Charts and Control Charts), Control Charts for Variables and for Attributes, Lot-by-Lot acceptance sampling, acceptance sampling system, quantitative techniques, quality decision-making techniques. Examples and case studies on project quality management are presented and discussed from a wide variety of engineering discipline.

MEPM 521 Project Cost Accounting and Finance**(3:0:0)****3 Semester Credit Hours****Pre-requisite(s): None**

This course covers the fundamentals of project cost accounting and finance. It reviews the fundamentals of accounting; examines cost accounting principles, applications, and the impact on profitability. It also examines the principles of project costing and covers the elements and introduces a framework for using an effective project cost system. Moreover, it introduces a framework for how projects are financed.

MEPM 522 Financial Analysis and Decision Making**(3:0:0)****3 Semester Credit Hours**

This course covers some of the theory and practice of decision-making, as well as basic procedures for the application and interpretation of financial statement analysis. Topics covered in this context will include the concept of time value of money, project analysis and evaluation, cost of capital, stock valuation and capital budgeting.

MEPM 531 Operation Research**(3:0:0)****3 Semester Credit Hours**

This course introduces the modeling techniques, theory and computation of linear programming. Topics include: The structure of linear programming, basic feasible solutions, simplex method, sensitivity analysis and linear programming duality. Linear programming modeling techniques for different applications: blending problem, inventory problem, minimum cost network flow problem, transportation problem, capital budgeting and fixed charge problems. Basic integer programming, dynamic programming and queening theory methodologies and techniques will be introduced.

MEPM 532 Engineering Management
3 Semester Credit Hours

(3:0:0)

The fundamentals of engineering management including: overview and concepts of engineering management (Introduction to engineering management, Management cycle includes (Planning, organizing, Leading and controlling. Different aspects of engineering managements will be introduced; like managing research and development, managing engineering design and managing production activities and operations. Market management and service activities for engineers will be clarified as well. Engineers as managers and Engineers code of ethics concepts and elements will be clarified. Finally, the globalization concept and the new challenges for engineering managers in the future will be discussed.

MEPM 533 Information Systems for Project Management
3 Semester Credit Hours

(3:0:0)

Pre-requisite(s): MEPM 511

This course will assist the student in understanding the challenges, opportunities and risks involved in information technology management. It introduces the management information systems foundations; current trends; MIS technology fundamentals; its applications to business functions and management practice and in an ethical manner. The impact of information technology on project management will be stressed since it affects the planning process. This course will also cover various system applications for specific project and business functions and their importance to today's manager.

MEPM 541 Organization Behavior and Project Team Management
3 Semester Credit Hours

(3:0:0)

This course deals with human behavior as individual and as groups in organizations. Individual-level characteristics such as personality, attitudes and values, perceptions and judgment, motivations, career development and ethics are emphasized. Topics include group formation, development, structure, leadership, diversity, and dynamics, as well as the processes of communication, decision making, power, and conflict. Class sessions and assignments are intended to help participants acquire skills and analytic concepts to improve organizational relationships and project team management effectiveness.

MEPM 542 Supply Chain Management for Project Managers
3 Semester Credit Hours

(3:0:0)

This course focuses on management and improvement of supply chain processes and performance. It will be valuable for project managers who would like to pursue a career in consulting or take a position in operations, marketing or finance functions in a manufacturing or distribution firm. We explore important supply chain metrics, primary tradeoffs in making supply chain decisions, and basic tools for effective and efficient supply chain management, production planning and inventory control, order fulfillment and supply chain coordination.

MEPM 543 Risk Management for Project Managers
3 Semester Credit Hours

(3:0:0)

Pre-requisite(s): MEPM 511

This course addresses fundamental issues, principles, and theory of project risk management and planning. It covers the quantitative and qualitative approaches to identifying, analyzing, assessing, and managing risks inherent to engineering projects. Other topics include risk response strategies and planning.

MEPM 590 Thesis I
3 Semester Credit Hours

(0:9:0)

Pre-requisite(s): Department Approval

A research-based thesis course that offers students the opportunity to work on a comprehensive, individual project. The thesis work must make a significant contribution to knowledge in the field of infrastructure engineering. The project will be of suitable complexity for results to be published for an expert audience. **Thesis Research Course**

MEPM 591 Thesis II
3 Semester Credit Hours

(0:9:0)

Pre-requisite(s): MEPM 590

Continuation of the Master Thesis I. A research-based thesis course that offers students the opportunity to work on a comprehensive, individual project. The thesis work must make a significant contribution to knowledge in the field of engineering project management. The project will be of suitable complexity for results to be published for an expert audience. **Thesis Research Course**

- MEST 100 Introduction to Islam in World Culture (3:0:0)**
3 Semester Credit Hours
 This course is the introductory course on Islamic Studies that is required for all majors. The course provides an introduction to the basic sources and historical contexts for the origins of Islam. (*Writing Intensive Course*)
- MGHR 301 Human Resource Management (3:0:0)**
3 Semester Credit Hours
Prerequisites: MGMT 301
 This course is a broad survey of key areas of human resources management, designed to provide line students wishing to become future supervisors and/or staff human resource specialists with a thorough understanding of the role of human resource management and the techniques available to management in making effective use of the human resources of an organization. The material is delivered from the perspective of HRM practice in both the UAE/GCC and internationally.
- MGHR 302 Compensation Management (3:0:0)**
3 Semester Credit Hours
Prerequisites: MGHR 301
 The Compensation Management course focuses on the theories underlying direct compensation and reward systems in organizations, and the practical administrative practices used to implement such systems, with particular reference to the UAE/GCC sub-region. Compensation management practices, including the analysis and evaluation of jobs, criteria and procedures for determining wage levels, individual wage determination, forms of pay, and incentive systems will be covered.
- MGHR 304 Human Resource Recruitment and Selection (3:0:0)**
3 Semester Credit Hours
Prerequisites: MGHR 301
 The Human Resource recruitment and selection course focuses on employee staffing, training, and development processes in organizations. Topics covered include human resource planning, job analysis for selection, multiple recruitment and selection methods are investigated, and the course looks in detail at how these HRM elements are typically implemented in the UAE/GCC region. Through a blend of theory and application, the course introduces students to a wide range of issues, principles, practices and trends in recruitment and selection.
- MGHR 390 Internship in Human Resource Management (0:0:21)**
3 Semester Credit Hours
Prerequisites: Senior Standing
 The Internship is designed for candidates on the undergraduate human resource management major programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "steppingstone" for a career in the field of the student's major. Students will receive orientation, induction, and relevant on-the-job training during their internship. The initial induction and orientation take place at the beginning of the internship, and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.
- MGHR 403 Cross Cultural and International Human Resource Management (3:0:0)**
3 Semester Credit Hours
Prerequisites: MGHR 301
 The course introduces students to the field of international human resource management (I-HRM). Topics covered include: HR planning for international operations, the impact of national cultures on I-HRM, including standardized cultural models; Divergent and Convergent International-HRM practices; Integration versus Differentiation; International people-resourcing; HR cultural adaptation and adoption issues; Organizational Culture and I-HRM practice; comparative I-HRM studies featuring country comparisons. A number of contemporary I-HRM models are also discussed and utilized, with particular reference to I-HRM within and out of the UAE/GCC.

MGHR 431 Employee Relations
3 Semester Credit Hours

(3:0:0)

Prerequisites: MGHR 301

The course is aimed at students who are interested in pursuing a career in human resource management. The course focuses on legal and regulatory issues that affect employee relations in the GCC and UAE. Students will compare and contrast policies and procedures related to employee benefits and possible labor violations. They will also examine the core issues and applications of UAE/GCC labor regulations in managing employee regulations. At the end of this course, students will have developed an understanding of individual and collective labor relations in the UAE/GCC region and the appropriate methods of dealing with employee disputes.

MGHR 444 HRM Project Based Internship
3 Semester Credit Hours

(0:0:21)

Prerequisite(s): MGHR 301

The HRM project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem-solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the HRM specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.

MGHR 461 Diversity in Organizations
3 Semester Credit Hours

(3:0:0)

Prerequisite: MGMT 301

The course enables the student to understand the role of diversity in human affairs particularly in the workplace. The course discusses and analyzes the role of diversity in the workplace, along with the procedures and practices of diversity management. Students will be able to apply these concepts to discussions of specific dimensions of diversity, including gender, race-ethnicity, sexual orientation, religion and disability. It also offers students the opportunity to develop their critical thinking on topics such as identity, relationships across difference and bias, and equality of opportunity in organizations around the world and how they relate to organizational issues of equality of opportunity, inclusion, and effectiveness.

MGHR 462 Strategic Human Resource Management
3 Semester Credit Hours

(3:0:0)

Prerequisite MGHR 301

This course will explore the contribution strategic Human Resource Management (HRM) makes towards the development and support of sustainable organizational strategies. On successful completion of this unit students will have the confidence to contribute to strategic decision-making in an HR context. This will be based on strong theoretical and applied foundations which will add value to an organization's HR function and role.

MGHR 463 Negotiations in Organizations
3 Semester Credit Hours

(3:0:0)

Prerequisite MGHR 301

This course focuses on developing the conflict management and negotiating skills of the learner. By the end of this course, the learner will be able to evaluate negotiation situations, strategize and plan upcoming negotiations, and engage in more effective negotiations. The course content will contextualize conflict management and negotiation to the local UAE/GCC environment, looking at cross-cultural influences. Consequently, to assist the learner in developing effective negotiation and conflict management skills, this course emphasizes experiential learning through student participation in a variety of exercises and role-plays.

- MGHR 464 Training and Development (3:0:0)**
3 Semester Credit Hours
Prerequisite MGHR 301
 The purpose of this course is to provide students a firm understanding of human resource training and development systems in today's business environment. A constant theme setting the foundation for this course will be on the various kinds of change facing organizations, with a focus of UAE/GCC organizations, and how these changes relate to human resource training and development. Topics will cover assessing training needs, developing and delivering training, evaluating training outcomes, career planning and career management. Aspiring HR professionals will gain essential knowledge to effectively manage employee training and development systems in a variety of companies. The course looks in detail as to how these HRM elements are typically implemented in the UAE/GCC region.
- MGMT 301 Principles of Management (3:0:0)**
3 Semester Credit Hours
Prerequisites: Sophomore Standing
 This course presents essential management theories and concepts. It examines the nature of managerial work under a range of business models and under rapidly changing business conditions. Managerial functions and activities such as planning, organizing, leading, and controlling are examined in depth, and in the context of current organizational practices and scenarios.
- MGMT 401 Organizational Behavior (3:0:0)**
3 Semester Credit Hours
Prerequisite Senior Standing
 This course applies essential organizational behavior and leadership theories and concepts to organizational practices. It examines the complexities of human behavior in a range of organizational work settings under rapidly changing business environment. Individual personality, group/team dynamics, motivation, performance, communication, and leadership concepts are examined in depth, and in the context of current organizational practices.
- MIST 301 Introduction to Business Information Systems (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): Sophomore Standing
 This course introduces fundamentals of computer hardware, software, networking, Internet and its technology components. The discussions are centered on the role of technology in contemporary business, and include basic relational storage concepts, with hands-on experience in building business database applications and web sites.
- MKTG 301 Principles of Marketing (3:0:0)**
3 Semester Credit Hours
Prerequisites: ECON 103
 This course examines marketing principles, concepts, strategies, tactics, and analytical tools used by profit and nonprofit organizations to market ideas, products, or services to selected target groups. The course emphasizes how to promote, distribute, and price firm's offerings in dynamic economic, social, political, and international environments.
- MKTG 311 Sales Management (3:0:0)**
3 Semester Credit Hours
Prerequisite(s): MKTG 301
 The course familiarizes students with marketing –sales interfaces including sales force role and capabilities, personal selling strategies, organizational relationships, and responsibilities of sales managers including training, motivating, and evaluating sales force.
- MKTG 312 Consumer Behavior (3:0:0)**
3 Semester Credit Hours
Prerequisites: MKTG 301
 The Consumer Behavior course studies the purchasing behavior of consumers and the exchange process involved in acquiring, consuming and disposing of goods, services, experiences and ideas. In this course students learn to deal with an unstructured situation by the usage of a buyer decision – making model and buyer behavior concepts enabling them to identify important considerations and their possible resolutions.

(3:0:0)

Prerequisites: MKTG 301

MKTG 332 Retailing and E-Commerce

(3:0:0)

Prerequisite MKTG 301

MKTG 351 Marketing Research Techniques & Applications

(3:0:0)

Prerequisites: MKTG 301 & OPMT 311

MKTG 390 Internship in Marketing

(0:0:21)

Prerequisites: Senior Standing

MKTG 407 International Marketing

(3:0:0)

Prerequisite MKTG 301

MKTG 412 Marketing of Services

(3:0:0)

Prerequisites: MKTG 301

293 | Page

- MKTG 444 Marketing Project Based Internship (0:0:21)**
3 Semester Credit Hours
Prerequisites: MKTG 301
The marketing project-based internship course will provide students an experience of a self-employed work environment. The course will imbibe problem- solving skills and creative thinking among students. These life-long learning skills will enhance the employability skills of the students and their ability to become independent learners. The students will integrate the tools and concepts from the marketing specialization courses to develop strategies to solve problems. Through a coaching and mentoring process, the students would produce a project report, which will allow them to showcase their learning acquired during their four-year study.
- MKTG 471 Marketing Strategy (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): MKTG 301
The course emphasizes managerial aspects of marketing, including developing marketing strategies and plans, and integrating specific elements of marketing process. The course additionally places stress on case analysis, as a means for learning the material and for students to develop problem-solving, group work, judgement, and decision-making skills. This is an applied course that utilizes real-world examples, course materials, and project assignments whenever possible in order to allow students to gain practical experience related to the course topics.
- MSRE 511 Energy Conversion (3:0:0)**
3 Semester Credit Hours
Forms of energy. Energy needs and available sources of energy. Conversion of thermal energy into mechanical energy, including power, and heat engine cycles, internal and external combustion systems and turbines. Basic technical and economic criteria for the design of efficient energy conversion systems, including traditional as well as alternative power systems.
- MSRE 512 Energy Systems Modeling and Optimization (3:0:0)**
3 Semester Credit Hours
Review of renewable energy resources; solar energy and photovoltaic, wind energy, wave energy, biomass energy conversion, fuel cells and batteries. Hydroelectric power and geothermal energy. Modeling methodology including system conceptualization. Model construction and validation (computational accuracy). Model evaluation and calibration. Simulation of energy and environmental systems. Optimization techniques; Classical direct search-for-optimum methods, Golden Mean, Conjugate Gradients, Modified Newton Method. Methods for constrained optimization such as Lagrange Multipliers, Search methods, Linear and Dynamic Programming. Use of software packages.
- MSRE 513 Energy Economics (3:0:0)**
3 Semester Credit Hours
Comprehensive understanding of energy markets. Technological, cost, and environmental fundamentals of energy sources and environmental systems. Economic principles underlying the supply and demand for energy in a modern economy, through considerations of topics such as energy demand at the individual and economy-wide level, the supply of renewable and non-renewable energy resources, the rationale for energy policy and energy policy instruments. Advanced tools to analyze how energy and environmental policies affect the demand and supply of different types of energy.
- MSRE 514 Direct Research (3:0:0)**
3 Semester Credit Hours
Pre-requisite(s): Department Approval
Under the guidance of an engineering faculty member, the Directed Research Project provides the student with a meaningful research experience. It requires that the student conduct a research topic or issue of significance to the field of renewable and sustainable energy.
- MSRE 515 Energy Efficiency and Management (3:0:0)**
3 Semester Credit Hours
Techniques and approaches adapted to improve the efficiency of energy generation, utilization, conversion, transport, storage and management. Energy audits. Energy conservation opportunities for efficiency improvements in different sectors: industrial, commercial, transportation and domestic. Economic evaluation of energy conservation opportunities using engineering economic formulas, simple pay-back analysis, and life-cycle cost models. Short- and long-term planning. Restructuring and Privatization. Models of electricity industry, contract issues, markets and transmission pricing.

Deregulation of the energy market around the world. Electrical demand control and power factor correction. Load forecast and generation side management.

MSRE 521 Photovoltaics

(3:0:0)

3 Semester Credit Hours

Introduction to solar radiation and some important related concepts like blackbody, solar spectrum, irradiance, irradiation, air mass and peak sun hour. Semiconductor and P-N junctions. PV cells properties, design, interconnection and module fabrication. Modules and arrays, stand-alone PV schemes with battery energy storage and grid-connected PV schemes.

MSRE 522 Wind Energy Systems

(3:0:0)

3 Semester Credit Hours

Basic characteristics of wind, site characterization, statistical methods of wind analysis, wind resources assessment, fundamental principles of wind energy utilization, aerodynamics, mechanical and electrical design aspects. Wind machine technologies and wind turbines performance analysis. Wind farm planning and design. Wind power integration into the power systems, and the environmental impact of wind power utilization.

MSRE 523 Biofuels

(3:0:0)

3 Semester Credit Hours

Fundamentals and applications of biofuels and bioenergy produced from biomass including processes of production, availability and attribution of biofuel/bioenergy production. Types of biomass raw material and methods of selection. First, second and third generation of biofuels. Methods of biomass conversion to fuel including thermochemical conversion, biochemical conversion and catalytic conversion. Production of hydrogen, biodiesel and microbial fuel cells. Environmental impacts of biofuel production. Economics and life-cycle analysis of biofuel. Value-added processing of biofuel residues.

MSRE 524 Sustainable Desalination

(3:0:0)

3 Semester Credit Hours

Basic science and technology of water desalination to ensure sustainable water supply. Water production via desalination within the water-energy-cost nexus, evaluation of renewable energy- powered desalination processes, power-desalination cogeneration analysis, evaluation and applications of novel desalination systems, such as thermal desalination, membrane distillation and forward osmosis. Recent technological improvements for enhanced desalination processes, and fouling issues in current technologies. Assessing economic feasibility and the environmental impact of new desalination processes.

MSRE 525 Concentrated Solar Power

(3:0:0)

3 Semester Credit Hours

Introduction to solar energy, solar radiation; review of the basics of thermodynamics and heat transfer, power plant technologies; types of CSP systems including CSP parabolic trough systems, CSP dish technology, CSP Fresnel technology and solar tower; heat storage systems; hybridization; secondary use of CSP systems; operation and maintenance of CSP systems; power quality control and grid integration; CSP plant project planning: economic, social and environmental considerations and site assessment.

MSRE 526 Green Buildings

(3:0:0)

3 Semester Credit Hours

Green building initiatives, their origin, characteristics of a green building, certification of green buildings rating systems, criteria for rating. Policies and drivers that are leading to the more widespread uptake of green building technologies; green building codes, policies and planning from the past, present and future from around the globe. Integrated design: urban micro-climate design, passive and active architectural interventions.

MSRE 590 Thesis I

(0:9:0)

3 Semester Credit Hours

Pre-requisite(s): Department Approval

A research-based thesis course that offers students the opportunity to work on a comprehensive, individual project. Topic to be agreed in consultation with a supervisor. The project will be of suitable complexity for results to be published for an expert audience. This is the first part of the master thesis course. **Thesis Research Course**

MSRE 591	Thesis II	(0:18:0)
6 Semester Credit Hours		
Pre-requisite(s): MSRE 590		
Continuation of the Master Thesis I. A research-based thesis course that offers students the opportunity to work on a comprehensive, individual project. Topic to be agreed in consultation with a supervisor. The project will be of suitable complexity for results to be published for an expert audience. Thesis Research Course		
OPMT 301	Operations Management	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): Sophomore Standing		
The course examines principal aspects of organizational operations in various settings. It emphasizes the application of planning and decision-making activities associated with managing operations, with particular focus on production and service operations. It uses analytical models to describe key operations planning and control activities.		
OPMT 311	Methods and Models of Management Science	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): STAT 100		
This is a course in applied operations research for business management. Business situations are represented by analytical mathematical models solved by the effective application of the methods mathematical programming and probabilistic process analysis. This course introduces operation research and management sciences techniques for supporting business management decisions. Specific mathematical programming and probabilistic topics include linear programming, transportation models, network flow models, decision analysis, inventory models, queuing models, and Monte Carlo simulation.		
OPMT 405	Supply Chain Management	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): OPMT 311		
A comprehensive study of the concepts, processes, and strategies used in the development and management of global supply chains. Supply-chain management (SCM) is a systems approach to managing the entire flow of information, materials services from raw material suppliers through factories and warehouses to the final end-customer. SCM represents a philosophy of doing business that stresses processes and integration. Specific topics include global supply chain management, procurement, electronic commerce, information technologies, and logistics activities. This course will be taught through the use of textbook materials, outside readings, and case analysis,		
PENG 101	Physical Geology	(3:0:0)
3 Semester Credit Hours		
Origin of the Earth and its shells. Composition of the Earth's crust and oceans, and their Geological characteristics. Primary and secondary structures. Internal Geological processes. Plate Tectonics and the relation of Geological events to it. External Geological processes. Stratigraphic columns, details of the Geological Time Scale and case studies of Geological ages and their palaeogeographic distribution. Climate. Important biological aspects.		
PENG 202	Intro. to Petroleum Engineering	(1:0:0)
1 Semester Credit Hour		
The course provides an overview and history of the petroleum industry and petroleum engineering, including nature of oil and gas reservoirs, petroleum exploration and drilling, formation evaluation, well completions and production, surface facilities, reservoir mechanics, and improved oil recovery. It introduces the importance of ethical, societal, and environmental considerations and current events on activities in the petroleum industry. It also introduces students to professional society and university resources that aid career development.		
PENG 231	Mechanics of Materials	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): MATH 114		
This course covers external and internal forces in structures and/or machines, including conditions of equilibrium, systems of force, moments of inertia and friction. It also covers an introduction to Petroleum Rock Mechanics: Elasticity and Rock strength, Rock properties from logs, Stresses around wellbore and borehole failure criteria, Hydraulic Fracturing.		

(0:3:0)

1 Semester Credit Hour

Co-requisite(s): PENG 351

This course deals with the measurement of fundamental properties of reservoir rocks and fluids. Rock properties include porosity, irreducible water saturation, residual oil saturation, absolute permeability, capillary pressure, and relative permeability. Fluid properties include oil distillation, oil composition of one of oil fractions, oil density at room conditions and at high pressure and temperature conditions, oil viscosity at high pressure and temperature, surface and interfacial tensions, flash liberation process, estimation of bubble-point pressure at reservoir temperature, and oil-formation-volume factor and solution gas/oil ratio at pressures below the bubble-point pressure.

(0:3:0)

1 Semester Credit Hour

Co-requisite: PENG 361

This course deals with lab measurements of cement and mud properties. Mud preparation, mud rheology, filtration, wall building and resistivity, mud weight control, drilling fluid contamination test, oil well cementing experiment. (Writing Intensive Course)

(3:0:0)

3 Semester Credit Hours

Pre-requisite(s): CHEM 215

Fundamental properties of reservoir rocks, namely porosity, compressibility, permeability, electric conductivity, fluid saturation, surface forces and capillary pressure, effective and relative permeability, and their applications in volumetric and flow calculations in petroleum reservoirs. Fundamental properties of reservoir fluids, namely oil, natural gas, and formation water. The course covers hydrocarbon phase behavior, equations of state, formation volume factors, gas solubility, types of vaporization, K values, and gas separation.

(2:0:0)

2 Semester Credit Hours

Pre-requisite(s): PENG 231

Co-requisite: CIEN 251

This course introduces basic drilling techniques and drilling fluid properties. Topics include rock characteristics, drilling fluids, mud weight calculations, components of rotary drilling rig, drilling hydraulics, drilling bits, factors affecting rate of penetration, cementing operations.

(3:0:0)

2 Semester Credit Hours

Pre-requisite(s): PENG 351

This course deals with material balance (MB) techniques to estimate reserves. Topics include generalized MB equations, fluid drive mechanisms, selection of PVT data, water influx, flow through porous media, displacement of oil, fractional flow, and oil recovery by internal drive.

(3:0:0)

3 Semester Credit Hours

Pre-requisite(s): PENG 101 and PENG 351

Logging is an essential operation to determine the key formation parameters, such as porosity, permeability, reservoir thickness, water saturation. All types of well logging techniques such as Logging while Drilling, Cased Hole Logging, and Production Logging are included. Logs are also run to gain information about well casing and cementing.

(4:0:0)

4 Semester Credit Hours

Pre-requisite(s): CIEN 251 and MENG 211

This course covers basic well performance calculations necessary for the design and analysis of naturally flowing and artificially lifted wells. Topics include Inflow Performance Relationship (IPR), Tubing Performance Relationship (TPR), Flow line Performance Relationship (FPR), Choke Performance Relationship (CPR), Gas-Lift, Electric Submersible Pumps (ESP), and production forecasting.

PENG 403	Petroleum Property Evaluation	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): PENG 371, PENG 382, and IENG 321		
Applications of reservoir engineering techniques, reserve calculations, decline curve analysis, rate of return calculations to project design and evaluation.		
PENG 422	Fluid Flow in Porous Media Lab	(0:3:0)
1 Semester Credit Hour		
Co-requisite(s): PENG 485		
This course deals with the design aspects of oil displacement by another fluid in rock samples. It builds on the experiences of students obtained in lab measurements of individual reservoir rock and fluid properties in PENG 321 to create an integrated lab measurement of all properties needed to analyze oil displacement by a displacing fluid. The displacing fluid can be chosen to study the relative permeability and displacement efficiency of water flooding, gas flooding, or any enhanced oil recovery fluids (acidic water, microbial water, polymer solution, or steam) using cores, fractured cores (sand packs and glass beads may be considered as alternatives) in one-dimensional geometry or packed layers in two-dimensional geometry.		
PENG 442	Transportation and Storage of Petroleum	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): CIEN 251 and PENG 382		
This is an elective and a senior level course on production engineering. The course deals with analysis and design of surface piping and storage facilities of crude oil and natural gas. Topics include fluid flow and pressure losses in pipes, pipeline design, selection and sizing liquid pumps and gas compressors, corrosion in pipes, other transportation methods, and storage of petroleum and its products.		
PENG 443	Separation and Treatment of Petroleum Fluids	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): PENG 382		
This is an elective and a senior level course on production engineering. The course covers nature of fluids produced from oil and gas reservoirs, phase behavior of water/hydrocarbon systems, quantitative prediction of water content in light hydrocarbon systems, oil water separations, oil-water emulsions, treatment of oil field waters, oil and gas separations, flash calculations, separator sizing and design, heater-treater design, oil skimmers selection and design.		
PENG 462	Drilling Engineering II	(2:0:0)
2 Semester Credit Hours		
Pre-requisite(s): PENG 361		
This course deals with additional topics in drilling engineering, namely casing design and landing, directional and horizontal drilling, drilling problems and fishing operations, well control problems and solutions.		
PENG 472	Applied Reservoir Geology	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): PENG 101		
Co-requisite(s): PENG 381		
Oil distribution in the world and in the UAE: geology of reservoirs, which includes the formation of reservoir rocks, cap rocks, source rocks and the environments of depositions: petrophysical parameters of parameters of reservoir fluids: oil field waters, crude oil and natural gas; reservoir conditions: pressure, temperature and their effects on oil maturation, migration and accumulation: oil generation. Oil migration: types of oil traps: methods of exploration.		
PENG 473	Reservoir Simulation	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): PENG 371 and MATH 203		
This course covers fundamental concepts of reservoir simulation to model single-phase flow in petroleum reservoirs. Topics include reservoir engineering concepts, mathematical concepts, derivation of reservoir flow equations, finite difference approximations, and their solutions, and applications to predict reservoir performance.		

PENG 474	Applied Reservoir Simulation	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): PENG 473		
This course covers advanced topics in reservoir simulation. These include reservoir fluid flow equations in multiphase, multidimensional flow, up-scaling of rock properties, pseudo functions, vertical equilibrium, analysis of data for consistency, history matching, and applications to field cases.		
PENG 482	Natural Gas Engineering	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): PENG 371 and PENG 382		
This course introduces the properties of natural gas, covers flow of gas in reservoir, wellbore and surface pipelines. Based on such knowledge, students are able to run nodal analysis for gas production system design. This course also covers gas compression and gas operation issues.		
PENG 483	Well Testing	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): PENG 371 and MATH 214		
This course covers reservoir characterization by pressure test analysis. Topics include fluid flow equations in porous media under transient and pseudo-steady state flow conditions, pressure buildup and pressure drawdown tests, average reservoir pressure, type curve matching, well testing of heterogeneous reservoirs, pressure derivatives analysis technique, multiple well testing, and test design and instrumentation.		
PENG 484	Stimulation and Intervention in Petroleum Production Operations	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): PENG 382		
Topics include well completions, perforations, wellbore damage sources and detection, hydraulic fracturing, fracturing fluids, acid/rock interactions, and acid treatment of oil wells, design and evaluation of treatments, evaluation by nodal system analysis.		
PENG 485	Water Flooding	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): PENG 371		
This is a senior level course on reservoir engineering series courses. The course covers the reservoir engineering aspects of water flooding. Topics include introduction to reservoir forces and concepts of surface and interfacial tension, wettability, capillary pressure, relative permeability; trapping and mobilization of residual oil; fluid distribution and frontal displacement theory, concept of mobility ratio, flood patterns and areal sweep efficiency considerations, pattern injection rates and pressures; characterization of reservoir heterogeneity; vertical and volumetric sweep efficiency, waterflood performance prediction models; waterflood pilot test; designing, monitoring and evaluating a waterflooding operation; water for water flooding, its sources and treatment.		
PENG 486	Enhanced Oil Recovery	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): PENG 485		
This course covers chemical and thermal method of EOR. Specific topics include interfacial tension, entrapment and mobilization of oil in porous media, residual oil, miscibility, adsorption at solid/liquid interfaces, surfactants and micro-emulsions, miscible gas flooding, polymer flooding, thermal methods, and the effect of reservoir heterogeneity.		
PENG 491	Senior Design Project I	(0:9:0)
3 Semester Credit Hours		
Pre-requisite(s): Senior Standing		
A significant design effort in one area of petroleum operations. Design is based on fundamental understanding of petroleum and reservoir engineering concept and a critical review of literature of current state of knowledge for the subject under consideration. Projects may involve analysis and computer program development and/or usage combined with experimental work. Topics for projects may be developed with industry cooperation.		

PENG 492	Senior Design Project II	(0:9:0)
3 Semester Credit Hours		
Pre-requisite(s): PENG 491		
Continuation of phase (1).		
PENG 493	Special Topics in Petroleum Engineering	(3:0:0)
3 Semester Credit Hours		
Pre-requisite(s): Department Consent		
A specific topic in Petroleum Engineering that is not covered in other program courses is presented in a course format.		
PHIL 100	Critical Thinking and Reasoning	(3:0:0)
3 Semester Credit Hours		
This introduction to basic principles of reasoning and critical thinking enhances the learner's abilities to evaluate various forms of reasoning in everyday life and in academic disciplines. The course explores such topics as inductive and deductive reasoning, the nature and function of definitions, fallacy types, statistic use and misuse, and the rudiments of logic. (Writing Intensive Course)		
PHIL 101	Ethics in Today's World	(3:0:0)
3 Semester Credit Hours		
This course addresses theoretical foundations of ethical thought, and background to traditions and movements in the development of ethical theory and methods of reasoning. Students apply ethical principles and perspectives to analyze, compare and critically evaluate relevant personal, social and professional problems and engage in ethical reasoning and decision-making processes.		
PHIL 102	World Philosophies	(3:0:0)
3 Semester Credit Hours		
The course surveys major philosophers from the most important world philosophical traditions. Some of the topics addressed include the internal world of personal identity, the nature of knowledge, the concept of happiness, the nature of reality and the external world, the relation of language to the world, meaning, and truth.		
PHYS 095	Introductory Physics	(3:0:0)
3 Semester Credit Hours		
This course is intended for students who need a general introduction to the fundamental principles of Physics and laws of nature before taking University-level Physics courses required in their major. Topics covered include on Dimensional Motion, Fluids, Heat and Thermodynamics. This course is not for degree credit.		
PHYS 110	University Physics I	(3:0:0)
3 Semester Credit Hours		
Co-requisites: PHYS 111		
This is a calculus-based physics course covering the fundamental principles of mechanics. It concentrates on the conservation of energy, the particle motion, the collisions, the rotation of solid bodies, simple machines and on the fluid mechanics. The focus lies on the resolution of one and two-dimensional mechanical problems.		
PHYS 111	University Physics I Lab	(0:3:0)
1 Semester Credit Hours		
Co-requisites: PHYS 110		
This course is intended to be taken with Physics 110. It primarily includes experiments on classical mechanics. Particular emphasis is placed on laboratory technique, data collection and analysis and on reporting.		
PHYS 220	University Physics II	(3:0:0)
3 Semester Credit Hours		
Prerequisite(s): PHYS 110		
This second calculus-based physics course includes a detailed study of the fundamental principles of classical electricity and magnetism, as well as an introduction to electromagnetic waves. The course's focus targets the resolution of dc- and alternating circuits.		

(0:3:0)

This course is intended to accompany Physics 220. It includes experiments on electricity, magnetism and RLC circuits. Particular emphasis is placed on three aspects of experimentation: laboratory technique, data analysis (including the treatment of statistical and systematic errors) and written communication of experimental procedures and results.

(3:0:0)

Co-requisite: PHYS 223

This course covers important theoretical bases and experiments that support the development of modern physics with applications in various technological and scientific fields. Topics include physical optics, thermodynamics, special relativity, quantum physics, atomic and molecular physics and nuclear physics. The course's focus is on applications that have led to key discoveries.

(0:3:0)

Co-requisite: PHYS 222.

This course is intended to be taken with Physics 222. It includes experiments on optics, thermodynamics and atomic physics. Particular emphasis is placed on three aspects of experimentation: laboratory technique (including both the execution and the documentation of an experiment); data analysis (including the treatment of statistical and systematic errors, as well as computer-aided analysis of experimental data); and written communication of experimental procedures and results.

(3:0:0)

This course gives students a broad, interdisciplinary, comprehensive, and balanced, introduction to key debates, theories, major trends, and topics, relating to Globalization and global developments.

(3:0:0)

The problems of scarcity and security are as much political as they are economic or technological. This course identifies the political aspects of global economic exchange and distribution, flows of labor and capital, and international cooperation, global security and conflict.

(3:0:0)

The course traces the history of the UAE, the establishment of the federation and the development of the UAE as a nation with significant global impact. It covers contemporary life, the economy, society, population, political system, social customs and traditions, and current changes.

(3:0:0)

This course provides an overview of major areas in the field of psychology including: the history of psychology, psychology research methods, organization of human brain and biological basis of behavior, sensation, perception, learning, language, intelligence, emotion, motivation, developmental psychology, personality theories /assessment, stress, abnormal behavior and therapies, social psychology.

(3:0:0)

Study of how society shapes us as individuals and how we as individuals shape society. Explores how we experience life as individuals and as members of cultural systems that shape our personalities, behavior, and perceptions of the world.

(3:0:0)

Introduction to basic sociological concepts. Examines aspects of human behavior in cultural framework including: individual

and group interaction, social mobility and stratification, status and class, race and gender relations, urbanism, crime and criminology, and social change and reform.

SOCI 101 Contemporary Social Issues (3:0:0)
3 Semester Credit Hours

The course is an introduction to basic sociological concepts and examines aspects of human behavior in a cultural framework including: individual and group interaction, social mobility and stratification, status and class, race and gender relations, urbanism, crime and criminology, and social change and reform.

STAT 100 Introductory Probability and Statistics (3:0:0)
3 Semester Credit Hours

This course introduces students to the basics of probability theory and statistical inference with examples and applications in sciences. At the end of this course, students will acquire the necessary quantitative competency in the program.

STAT 210 Probability and Statistics for Life Sciences (3:0:0)
3 Semester Credit Hours - Prerequisite(s): STAT 100

This course involves the use of probability and descriptive and inferential statistical techniques in interpreting biological data.

STAT 346 Probability for Engineers (3:0:0)
3 Semester Credit Hours - Prerequisite(s): MATH 114

The course introduces principles of statistics and probability for undergraduate students in Engineering. The course covers the basic concepts of probability, discrete and continuous random variables, probability distributions, expected values, joint probability distributions, and independence. The course also covers statistical methods and topics including data summary and description techniques, sampling distributions, hypothesis testing, and regression analysis.

UAES 200 Survey of United Arab Emirates Studies (3:0:0)
3 Semester Credit Hours

This course presents the UAE from multiple perspectives in an attempt to expose the students to the distinct qualities of the UAE. The purpose of this course is to give students a broad, interdisciplinary, and comprehensive introduction to key features and issues in the UAE's historical, cultural, political and economic landscapes. **(Writing Intensive Course)**

UNIV 100 University First-Year Transition (1:0:0)
1 Semester Credit Hour

This course is designed to assist students with the transition from secondary school to university life. Students in this course transition to university life by focusing on academic adjustment, by developing decision-making skills, and by learning about services and opportunities for involvement.

UNIV 200 Innovation, Entrepreneurship, and Sustainability (3:0:0)
3 Semester Credit Hours

The course aims at equipping next generation of leaders with an innovative and entrepreneurial mindset and related core skills. The course explains key qualities of entrepreneurs and analyses how entrepreneurs make utilize innovation in creating new ventures. Design-thinking and business model innovation are key frameworks that underpin the course. **(Writing Intensive Course)**

UNIV 390 Internship (0:0:12-20)
3 Semester Credit Hours

Prerequisite(s): Prior completion of 90 credits

The Internship is designed for candidates on the undergraduate finance, marketing, human resource management and accounting programs. It lasts for eight weeks (240 hours), and provides exposure to business practices and issues, in appropriate settings. Candidates will typically opt to carry out internship with local UAE-based companies. The internship program is viewed as a "stepping stone" for a career in the field of the student's major. Students will receive orientation, induction and relevant on-the-job training during their internship. The initial induction and orientation take place at the beginning of the internship and will serve to introduce the student to real-world business practices, culture and management, and provide them with a knowledge base to draw upon throughout the remainder of the internship.

UPAS 100 University Preparation for Academic Success

(1:0:0)

1 Semester Credit Hour

The focus of this course is on the development and application of college-level study skills, personal success strategies, and the use of campus resources that enhance individual student achievement. Designed for matriculated students who are at risk of being placed on Academic Probation 1 due to having a CGPA of less than 2.0. Graded as Satisfactory/Unsatisfactory.



GET IN TOUCH

TOLL FREE 800 AURAK (28725)

☎ +971 7 246 8888

✉ inquiry@aurak.ac.ae

✉ admissions@aurak.ac.ae

📍 P.O.Box : 10021, RAK, U.A.E

Follow Us



 aurak.ac.ae